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Arnaud Trouvé
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ABSTRACT

Title of Dissertation: A PHYSICS-BASED EULERIAN FRAMEWORK FOR MODELING FIREBRAND SHOWERING IN REGIONAL-SCALE WILDLAND AND WILDLAND-URBAN INTERFACE FIRE SIMULATIONS

Yiren Qin
Doctor of Philosophy, 2025

Dissertation Directed by: Professor Arnaud Trouvé
Department of Mechanical Engineering

Landscape-scale fire risk modeling plays a vital role in fire management as global wildfire activity continues to escalate. The objective of this dissertation is to extend the capabilities of an existing landscape-scale wildland fire spread simulation tool, ELMFIRE, to simulate conflagrations that propagate across Wildland–Urban Interface (WUI) communities. ELMFIRE employs a level-set formulation to describe flame spread and accounts for surface and crown fire dynamics in wildland environments. Recent extensions include ignition of structures and structure-to-structure fire propagation in the WUI via thermal radiation and direct flame contact.

Particular attention is devoted to modeling firebrand processes—specifically, generation, atmospheric transport, and spot ignition. The firebrand generation model employs an empirical correlation based on the local heat release rate, with separate generation rates estimated for biomass and structural fuels. A numerically robust algorithm is proposed and verified to ensure that the number of firebrands generated is independent of the simulation time step.

The firebrand transport model uses prescribed statistical distributions for ember flight distance and incorporates a simplified flight time formulation. A series of verification tests using MATLAB were conducted in one- and two-dimensional academic configurations. These tests demonstrate that the firebrand models are numerically converged—i.e., they remain stable under varying grid and time resolutions—provided the wind-based Courant–Friedrichs–Lewy (CFL) number remains below a defined threshold.

The firebrand ignition model assumes two consecutive time delays: the first represents the smoldering-to-flaming transition and is governed by an empirically derived probability of ignition; the second models flame growth using a t-squared law. Additional verification cases were performed to evaluate numerical accuracy. A firebrand consumption model is also proposed to account for the finite lifetime of deposited firebrands, which can influence ignition dynamics.

These firebrand models were coupled with the surface and WUI fire spread models to simulate multi-mechanism fire spread in WUI environments. The combined framework was tested in the MATLAB-based one-dimensional setting to analyze firebrand-driven structure-to-structure fire spread under varying wind speeds, building sizes, and separation distances. The results include a sensitivity analysis of spatial and temporal discretization and criteria for successful firebrand-induced ignitions.

The complete modeling framework—including the proposed firebrand models and WUI fire spread mechanisms—was integrated into ELMFIRE. A comprehensive preprocessing pipeline was developed to generate ELMFIRE-compatible GIS inputs using publicly available datasets and tools, including LANDFIRE, RTMA, the Microsoft Building Footprint dataset, WindNinja for wind field upscaling, and the NFDRS4 model for fuel moisture estimation. This pipeline automates data acquisition and processing to support reproducible simulation workflows.

The full framework was finally applied to reconstruct the 2017 Thomas Fire to evaluate sensitivity to both model improvements and input data quality. Results demonstrate that the proposed model improves predictive fidelity while offering physically interpretable insights. The study highlights the importance of future research on subgrid-scale modeling in WUI areas and the use of higher-resolution, dynamically consistent input datasets.

A PHYSICS-BASED EULERIAN FRAMEWORK FOR MODELING FIREBRAND SHOWERING IN REGIONAL-SCALE WILDLAND AND WILDLAND-URBAN INTERFACE FIRE SIMULATIONS

by

Yiren Qin

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Dedication

To my dear families and friends who provided unwavering support from both sides of the world.

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Chapter 1: Introductions

1.1 Fires in Wildland and Wildland-Urban Interface

The escalating wildfire crisis has garnered global attention due to its devastating impact on human communities and its significant contribution to carbon emissions. According to Ref. [1], in 2023, wildfires consumed 384 million hectares of land globally, the highest since 2017. Canada, in particular, experienced its worst fire season on record, releasing 763 teragrams of carbon—five times more than the previous record set in 2021. The extreme heat of 2023 also fueled catastrophic fires in Hawaii, the Mediterranean, central Amazonia, and central Chile.

While wildfire is a natural component of many ecosystems, human interventions such as fire suppression, coupled with global climate change, have exacerbated the conditions for larger and more intense fires. Regions like coastal California, the Mediterranean, Chile, and Australia are becoming increasingly susceptible to wildfires due to warmer, drier weather, which further perpetuates this destructive feedback loop.

Simultaneously, the expansion of the wildland-urban interface (WUI) has heightened the risk of wildfire impacts. The WUI, broadly defined as areas where human developments intermingle with wildlands such as forests, grasslands, shrublands, and wetlands, now covers over 5% of the world's land area, with nearly half of the global population residing in these regions [2]. In

the U.S. alone, wildfires in WUI areas result in significant economic losses, estimated at $77.4 to $378.7 billion annually. The 2018 Camp Fire, the deadliest and most destructive wildfire in U.S. history, resulted in 88 fatalities and the destruction of over 18,800 structures [3].

This trend is not confined to North America. For instance, Africa has seen a 46% increase in WUI areas from 2005 to 2020, alongside a 41% increase in fire incidents and a 36% rise in the burnt area, suggesting a strong correlation between WUI expansion and wildfire activity. Similar patterns are observed in Europe, South America, and Asia [4].

In summary, climate change is driving more extreme weather, which in turn fuels increasingly severe wildfires, pushing these fires beyond their ecological role. Concurrently, the growing human presence in WUI areas intensifies the risk of catastrophic wildfire events. Addressing this escalating problem requires an interdisciplinary approach that integrates fire science, meteorology, ecology, and other fields to develop effective strategies and tools for mitigating risks in fire-prone WUI communities and restoring fire-adapted ecosystems.

1.2 Landscape-Scale Fire Behavior Modeling

Operational stakeholders involved in managing fire hazards at the landscape scale typically focus on assessing the risk exposure of fire-prone properties and communities, as well as formulating effective strategies to reduce the likelihood of ignition or minimize damages when wildfires occur. While high-resolution, physics-based models can provide detailed and accurate predictions to support long-term planning and infrastructure resilience, time-sensitive scenarios—such as resource deployment, emergency response, and community evacuation—often demand faster, semi-physical or empirical modeling approaches. These simplified models prioritize

computational efficiency and ease of interpretation, enabling stakeholders to make rapid, albeit less precise, decisions during critical wildfire events.

In parallel, given the inherently unpredictable nature of wildland and WUI fires, industry practitioners often aim to quantify the probability of fire exposure and the vulnerability of communities under various fire scenarios. Such risk quantification is typically conducted through large ensembles of simulations—e.g., Monte Carlo methods—to produce spatially resolved risk maps, like effort by Ref. [5]. These simulations are designed to represent similar baseline conditions, while perturbing selected inputs to capture the stochastic nature of wildfire behavior and arrival patterns. In this context, computationally efficient models become essential, as they enable the completion of a sufficient number of realizations to produce statistically meaningful estimates of community-level fire risk.

Generally, the empirical (semi-physical) models summarize the observations from fire accidents and experimental results. They usually solve key parameters such as the Rate of Spread (ROS) of the fire fronts, fire residence time, fire line intensity, etc., take inputs of meteorological and topographical conditions, and categorized fuel properties [6].

From the perspective of the research community, our understanding of the physical mechanisms driving landscape-scale fire propagation remains limited.

Fire propagation is fundamentally governed by a positive feedback loop as is shown in Figure 1.1: burning fuels release heat, which is transferred to adjacent unburnt fuels, elevating their temperature and eventually inducing pyrolysis and combustion. If the transferred heat compensates for the heat deficit—caused by the fuel's heat capacity and heat losses to the surrounding environment—the combustion process becomes self-sustaining and the fire starts to spread. Thus, a natural approach to modeling fire spread is to investigate the thermal exchange between

the flame and the fuel.

In wildland and WUI fires, heat transfer is believed to occur through several key mechanisms, as illustrated in Figure 1.1: (1) direct flame contact, including both convective and radiative heat transfer from the flame to the fuel; (2) long-range thermal radiation from nearby flames not in direct contact with the fuel; and (3) firebrands—burning embers capable of igniting fuels upon deposition. These mechanisms operate across different spatial and temporal scales: direct flame contact typically spans several meters; radiative effects may extend tens of meters; and firebrands can travel distances of up to several kilometers, while inducing ignition at sub-meter or even millimeter scales. Each of these processes is further influenced by multiscale environmental and material factors.

Figure 1.1: Schematic of primary fire spread mechanisms in wildland and WUI fires, adapted from Ref. [7].

Given this complex, multiscale nature of fire behavior, insights derived from laboratory-scale experiments—often conducted at meter-scale—do not directly translate to landscape-scale phenomena occurring over kilometers. As a result, empirical or semi-physical models continue to serve as practical intermediate tools for landscape-scale fire modeling, bridging the gap while fully physics-based approaches remain under development.

The following section provides a brief overview of the evolution of empirical and semi-physical fire spread models, summarizes the current state-of-the-art in model development, and highlights key challenges facing the research community.

In the US, such models can be represented by the Rothermel model [8]. Rothermel did extensive experiments and derived a general expression for the ROS of fires on typical fuel surfaces in the US. Albini [9] further introduced improvements to better incorporate the fuel composition into the model. At this time, the Rothermel model was still incapable of predicting crown fires, which make up a significant part of the wildfires in the US. Later, FARSITE [10, 11] incorporated the study by Wagner [12, 13], which gives the critical condition for a crown fire to occur. Alexander [14] also contributed to the crown fire modeling that helped predict the threshold for the occurrence of an active crown fire, the fire that propagates in the canopy and detached from the floor of a forest. Currently, the improved Rothermel model is adopted in major operational tools, including FARSITE, FlamMap [15, 16], BehavePlus [17, 18] and the WFDSS Interagency Fuel Treatment Decision Support System (IFTDSS) [19]. The models also have been used in other countries such as Portugal, Greece, France, New Zealand, and Italy [20, 21].

Substantial work has also been done in Australia. McArthur led the development of the fire-spreading models for typical vegetation in Australia [22, 23]. Unlike the Rothermel model, these models are, according to Cruz et al. [6], "fuel-specific", which means each model can only be applied to one type of fuel featuring a specific physical structure. The estabilished fuel fire behavior models to date encompass grassland [22, 24, 25], shrubland [26–28], native forests [23,29,30], and pine plantations [31]. These models are integrated into the National Fire Behavior Knowledge Base [32] in Australia.

Note that the above models are built for wildland fires. In recent decades, the fires in the

Wildland-Urban Interface (WUI) have been more of a concern. However, few models are available that specifically target WUI scenarios. Alternatively, models are available for fire spreading in urban areas and are intended to study fires following earthquakes (FFE). In Japan, Hamada [33] led the model development of the FFE. The Hamada model estimates an urban fire spread as a function of the fuel distribution and wind speed. The buildings in the urban area are assumed to have a homogeneous distribution and feature uniform footprint and separation distance. The Hamada model was later improved by Scawthorn [34] and has been adopted in the Hazus [35], a geographic information system-based natural hazard analysis tool developed by the Federal Emergency Management Agency (FEMA). Some other urban fire spreading models consider the fire spread at the structure level. These models describe the fire evolution of each building through empirical (semi-physical) models and consider its impact on adjacent structures. Examples in this category include the TOSHO model [36, 37] developed in 1997, Cousins et al. [38], works by Himoto and Tanaka [39], etc. Lee et al. [40] presented a comprehensive review of the models dedicated to the urban fire propagation. Readers are referred to the article for further information on this issue.

1.3 Fire Tracking Algorithms in Landscape-Scale Fire Risk Models

As mentioned previously, fire behavior models are often compiled into simulation codes to serve operational purposes such as assessing fire-related risks and informing decision-making during active fire events. A key utility of such models is their ability to predict the spatiotemporal distribution of active fires. When implemented effectively, these simulations enable applications such as Monte Carlo-based risk estimation—where multiple realizations are run under varying

conditions to assess the likelihood of fire exposure for a given community—or short-term forecasting of fire spread in the early stages of ignition.

Physics-based modeling approaches have been developed to simulate detailed fire propagation dynamics by coupling heat transfer from flames to solid fuels with pyrolysis-induced vapor generation and subsequent combustion in the presence of ambient oxygen. These models are typically implemented using Computational Fluid Dynamics (CFD) frameworks to resolve the complex interactions between thermal, fluid, and chemical processes.

High-fidelity examples in this category include the Fire Dynamics Simulator (FDS) [41], developed by the National Institute of Standards and Technology (NIST), and PBRFoam, developed by Ahmed et al. [42] at the University of Maryland based on the OpenFOAM solver. FDS supports two schemes for wildland fuel representation: one models vegetation as Lagrangian particles, while the other treats the fuel bed as a porous solid surface. PBRFoam resolves both flaming and post-flame smoldering combustion processes at millimeter-scale resolution, enabling detailed modeling of coupled pyrolysis and combustion dynamics. At Los Alamos National Laboratory, Linn et al. [43] introduced FIRETEC, a multi-physics CFD model designed to capture fire spread phenomena in wildland environments, operating on coarser computational grids with typical resolutions of approximately 2 meters.

While these CFD models offer valuable scientific insights, they are computationally intensive and primarily research-oriented. For example, FIRETEC is capable of simulating kilometer-scale domains at meter-scale resolution, but such simulations require substantial computational resources. These models typically resolve processes at millimeter to centimeter scales, which limits their practicality for real-time or operational landscape-scale fire modeling.

Consequently, the objective of this study is to develop a computational framework that

balances physical realism with computational efficiency—suitable for rapid response applications and ensemble-based fire risk assessments. To this end, we adopt empirical and semi-physical modeling strategies that compute fire rate of spread (ROS) and approximate heat–fuel interactions as functions of meteorological and fuel conditions.

Such modeling strategies have been implemented across a range of computational frameworks. Broadly, these frameworks can be categorized into two principal classes based on their fire propagation algorithms: (a) those relying on statistical descriptions of fire perimeter evolution, and (b) those based on explicit ROS formulations driven by physical or empirical relationships.

A representative of the first approach, utilizing a statistical approach to the fire spread, is the cellular automata-based method. The method usually discretizes the computation field by fuel patches. Compared to the approaches in the second category that usually conduct simulation on Cartesian grids, it calculates and updates the status of a patch according to the status of the surrounding patches and its own properties, in which the patches are not necessarily regular. Application of the cellular automata method to the fire propagation problem can be dated back to Kourtz et al. [44]. Green [45] and Richards [46] then did some similar work with improvements regarding the ignition mechanisms (flame contact, heat accumulation, etc.). Recent efforts by Perryman et al. [47] and Mahmoud and Chulahwat [48], Trucchia et al. [49] used probabilistic descriptions of the status of the patches. The implemented algorithm is more similar to the primitive cellular automata concept brought up by von Neumann [50]. The simulations are performed on regular Cartesian grids, and the status of a cell only depends on the directly adjacent cells. Therefore, the model can model the fire spread through surface spread mechanisms, while additional modifications are required to model the impact of firebrands. Mastorakos et al. [51] presented a new idea of using Lagrangian particles to represent different heat transfer

mechanisms and change the fuel status by considering the contribution of aggregated particles.

The structure-scale fire spread models in urban fire development can readily fit into these algorithms. Therefore, a few models were developed based on these models to tackle the diversity of fuel types in the WUI fires, including Refs. [38, 52–58]. Some of these efforts demonstrate their comparable capability and efficiency in terms of the computational cost compared to the Lagrangian-based fire front-tracking models.

However, to the best of the author's knowledge, the cellular automata models have not been coupled with the dynamic atmosphere, or, no fire-weather interaction is considered. Many such efforts have been made utilizing the ROS-based description to be introduced in the following.

For the second approach, there are mainly two classes of schemes, *i.e.*, the Lagrangian scheme which tracks particle tracers representing fire perimeters, and the Eulerian scheme from a localized point of view. The following provides a detailed review of these schemes.

For the Lagrangian scheme of the fire front-tracking model, one popular type treats a fire perimeter as a discretized point source of new fire. At a particular discretized time, each new fire forms an elliptical perimeter which is parameterized by the corresponding models. The new ensemble of fire perimeters at the next discretized time is then formed by the envelope of the small perimeters. This method was first introduced by Anderson et al. [59]. It was based on the assumption that the fire front originated from a point would evolve in an elliptical shape under the impact of wind. Richards implemented this concept by further parameterizing the ellipse to facilitate the description of the ROS in all directions [60], and provided a numerical scheme to iteratively solve the progression of the perimeter [61]. FARSITE incorporated the method with additional treatment for the perimeter merging problem [10]. Ren and Xie [54] also claimed that they had adopted the method from Richard [60] in their work and simulated FFE for urban areas.

Some other studies applied different perspectives to treat the Lagrangian segments of the firefront. For example, the Coupled Atmosphere-Wildland Fire Environment model (CAWFE) [62,63] employed a different type of Lagrangian front tracking algorithm, which tracks four tracers for each computation cell representing the evolution of the fire perimeter. Compared to the aforementioned simulators, it has the advanced option of simulating fire-atmosphere interaction. The near-surface wind is affected by the heat produced by local fires and vice versa. André et al. [64] provided a comprehensive review of the mathematical formulation of these methods. However, these methods are inevitably complex when dealing with fire fronts with complex geometries, and situations such as fronts merging, spot ignitions, or loss of accuracy with the expansion of the fire front due to a diluted number of Lagrangian tracers along the trajectory.

The Level-Set Method (LSM) was used to tackle the fire front tracking problem by Mallet et al. [65]. The primitive LSM defines a contour on a 2-D plane by the zero-level set of a 3-D function, i.e., the intersecting lines of the x-y plane and a curved surface at $z = 0$. This method avoids the perimeter merging issue that emerged with the Lagrangian-based front-tracking methods. It also has the advantage that the burnt and unburnt areas can be easily distinguished by the value of the level-set variable, *i.e.*, the value of z of the curved surface. Since the LSM essentially solves a partial differential equation (PDE), Rehm and Mcdermott [66] integrate the LSM method using a second-order Runge-Kutta scheme for time-integration and a second-order, non-diffusive, Total Variable Diminishing scheme with a Superbee slope limiter for spatial discretization.

Rehm and Mcdermott [66] adopted the model by Fendell and Wolff [67] to calculate the ROS in all directions. In the meantime, the Rothermel model had already been widely adopted in the US, and a sophisticated database had been established for fuel parameters required by the Rothermel model. Consequently, FDS soon incorporated the Rothermel model in its level-set

scheme [68] and reported comparable capability to FARSITE. Kim [69] also presented a similar effort that integrated the level-set fire propagation algorithm and the Rothermel fire behavior model in the Weather Research & Forecasting Model (WRF), which generated a new branch of WRF called WRF-Fire/SFIRE [70], in which the WRF predictions of meteorological variables was coupled with influence of the fire. The input to some LSM-facilitated frameworks is prescribed before simulations as is done for the Lagrangian and cellular automata methods. Meteorological data such as wind and moisture conditions are taken from external databases, and the simulators cannot extrapolate the weather conditions beyond the given inputs. Examples of this type of simulators include FDS and ELMFIRE [71,72], while these models should be readily enabled to couple with CFD simulations as had been done in FDS.

Some previous work [71,73–75] tried to include calibration algorithms into the LSM frameworks. Methods like parameter regression [75], data assimilation [73, 74], and stochastic optimization [71] are applied to achieve better agreement between the simulation results and field observations.

A few different approaches have been developed for fire front tracking. FlamMap [15, 16], which now integrated FARSITE, applied the minimum travel time (MTT) method [76], which finds the least time of fire spreading between locations based on the graph theory. It is reported to provide more rapid prediction compared to FARSITE, while it is limited to constant ambient weather conditions. Another graph theory-based approach is taken by Mahmoud et al. [77]. The model intends to evaluate the fire risks of buildings and biomass fuels represented by nodes embedded in an interconnected graph. Currently, the model is established for static problems, while the researchers are working on developing its capability to perform dynamic simulations.

Note that the discussed frameworks are mostly built on empirical models that are not rig-

orously validated against real fire events, which is inherently difficult due to lacking accurate and comprehensive datasets. In the meantime, the numerical robustness of these frameworks is to be tested. Among the three main frameworks, implementing the PDE of the LSM is well-studied by the CFD community and the numerical scheme is relatively stable. Therefore, the presented proposal intends to design a robust component based on the level-set numerical framework.

1.4 Firebrand Process and Modeling

The preceding discussion focused on fire propagation mechanisms, yet did not address the role of firebrands. Firebrands, or embers, are hot particles generated from burning vegetation or structural fuels. Carried by fire plumes and ambient winds, these particles can land on receptive fuels at some distance from the main flame front. Upon deposition, they may ignite new fires (spot fires), enabling fire spread across fire breaks. Although the exact fraction of home losses due to firebrand ignitions remains uncertain, estimates cited in Ref. [78] suggest that firebrands were responsible for more than two-thirds of the residential losses during the Grass Valley, Witch Creek, and Guejito fires.

The firebrand propagation process involves three main phases, as illustrated in Figure 1.2: (1) generation by thermally degrading fuels, (2) transport through the atmosphere, and (3) ignition upon deposition. These processes have been studied through both experiments [79–81] and numerical simulations [82–84], but significant knowledge gaps remain.

Among the three, firebrand generation is reported to be the least understood [86]. Early work by Waterman [87] explored firebrand production under different conditions. More recent studies by Manzello et al. [88–90] characterized firebrands generated from vegetation and struc-

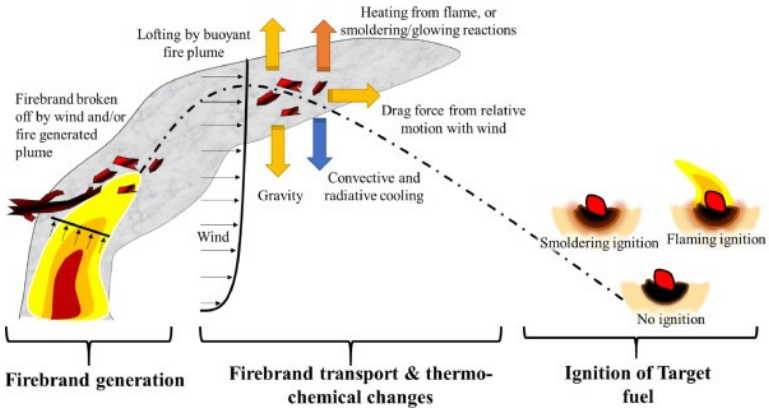


Figure 1.2: Schematic of firebrand processes, excerpted from Ref. [85].

tural components. Blunck and Bahrani [79, 91] provided empirical correlations for firebrands from torched trees and other vegetation types. Full-scale tests by Yoshioka et al. [92] contributed further insights. However, most firebrand counts in these studies were collected indirectly, limiting their predictive capabilities. Numerical models based on first principles [93–95] have also been developed, though validation remains a challenge. Promising experimental efforts like those by Ju et al. [80] offer potential for improved model validation.

Transport modeling is comparatively mature, owing to widespread development of Lagrangian particle tracking methods in CFD. Albini [96, 97] proposed a widely used model for estimating maximum spotting distances. This model assumes firebrands are lofted by fire plumes and then advected by wind at terminal velocities, with properties informed by earlier work by Tarifa et al. [98]. Albini's model is best suited for long-range spotting scenarios. However, most firebrands are of short range and, to the author's knowledge, contribute more critically to WUI fire spread. For such cases, statistical distributions of ground-level deposition, as proposed by Himoto [82, 99], Sardoy [83], and others [100], are more appropriate. A key missing element is the temporal distribution of firebrands. Preliminary simulations show that omitting firebrand travel time can lead to unrealistic fire spread predictions. While temporal information is available

in numerical works [82, 83], it is rarely incorporated into empirical models.

Firebrand ignition depends on various factors, including fuel moisture, ambient wind, firebrand temperature, and mass load. While probabilistic models exist, the mechanisms linking these variables to ignition likelihood are not fully understood. A foundational study by Schroeder [101] proposed a qualitative ignition model based on required potential energy and provided ignition probability as a function of fuel moisture, later integrated into the National Fire Danger Rating System (NFDRS). More recent work by Professor Fernandez-Pello's group [102, 103] examined ignition processes on various fuel beds using different firebrand types. De Beer et al. [104, 105] and Lin et al. [106] developed empirical models relating firebrand pile heat flux to ignition on structural and mulch materials. Yet, the smoldering-to-flaming transition remains poorly understood due to its complexity. Time-to-ignition models [105, 107] offer partial insight but require broader validation. Additionally, the dynamic accumulation of firebrands has been shown to influence ignition [100], though operational models capturing this effect are still lacking.

Collectively, while substantial progress has been made in characterizing the firebrand process, further integration of empirical observations and physically grounded models is necessary to develop predictive, scalable tools suitable for operational fire risk modeling.

1.5 Integration of Firebrands into Landscape-Scale Fire Risk Models

This section reviews current strategies for incorporating the effects of firebrands into landscape-scale fire risk modeling frameworks. Consistent with the structure established in the previous section, the discussion is organized around the three primary phases of the firebrand lifecycle:

generation, transport, and ignition.

Due to limited understanding and predictive capability of firebrand generation, many landscape-scale simulators adopt simplified approaches. For example, tools such as *FlamMap* and *FARSITE* generate a fixed number of representative firebrands at computational nodes that meet predefined spotting criteria [108]. While computationally efficient, these models cannot dynamically resolve variations in firebrand exposure over time and space.

More sophisticated models—such as *PHOENIX* [109], *SWUIFT* [110], and those developed by Himoto et al. [39, 111]—use empirical correlations to estimate firebrand generation. These correlations typically relate the number of firebrands to local variables such as wind speed, roof area, or fuel load. However, a common limitation among these models is that they treat firebrands as Lagrangian particles, tracking each ember individually. This can result in a substantial computational burden when simulating firebrand showers across large domains.

Current practices to resolve the transport in landscape-scale simulations generally follow one of the two approaches. In the first, firebrands are immediately deposited at randomized landing locations drawn from pre-defined statistical distributions. This method is used in some cellular automata models and in *FlamMap*, which also incorporates Albini's maximum spotting distance model [96, 97] to inform distribution parameters.

The second approach involves Lagrangian particle tracking. In this framework, each firebrand is assigned a travel distance—either sampled from a statistical distribution or derived from burnout or free-fall timescales, which are typically treated as inherent properties of the fuel type and assumed to be independent of environmental conditions. The transport process is then resolved by numerically integrating simplified equations of motion for each particle over time. This approach is implemented in models such as *FARSITE*, where firebrands of varying sizes

are generated and their trajectories are simulated using physics-based formulations for particle descent and wind-driven advection.

A major distinction between the two approaches is the consideration of transport time. In the first approach, firebrands are deposited immediately, effectively neglecting their time of flight. This can introduce large numerical errors, particularly when simulations are sensitive to temporal resolution or operate over short time steps.

Ignition modeling is often a weak point in firebrand-capable simulators. Two common formulations exist: (1) assigning a probabilistic ignition value to each deposited firebrand, and (2) defining an ignition threshold based on accumulated firebrand mass or energy.

Simulators such as *FARSITE*, *FlamMap*, and *SWUIFT* adopt a probabilistic approach to model ignition by firebrands. When a firebrand lands on a computational cell representing a fuel patch, the simulator evaluates ignition based on a predefined probability. In many implementations—including *FlamMap* and *FARSITE*—this ignition probability must be manually specified by the user. In the case of *FARSITE*, the ignition probability is often set to 1, while a low firebrand generation probability is assigned to each fuel type. This value defines the likelihood that a given fuel will produce firebrands. Although this strategy accounts for firebrand-driven ignitions in wildland fire scenarios, the selection of the generation probability is largely arbitrary and typically kept low to mitigate computational expense.

In addition, even when fixed ignition probabilities are informed by experimental studies—such as the ignition probability curves presented in Ref. [101]—their application can be problematic. These probabilities remain static throughout the simulation, despite the well-established influence of evolving ambient conditions on ignition likelihood. Moreover, such static formulations are not fully compatible with time-resolved simulation frameworks. The effective igni-

tion probability becomes sensitive to the simulation time step, potentially introducing resolution-dependent biases in fire spread predictions.

Other models—such as *SWUIFT* and *PHOENIX*—implement threshold-based ignition criteria. *SWUIFT* uses a fixed critical firebrand mass, while *PHOENIX* employs a fuel-moisture-dependent ignition threshold. These approaches, while more physically grounded, are often sensitive to the grid resolution and environmental inputs. Furthermore, most current implementations assume no time delay between firebrand arrival and ignition, implicitly ignoring the smoldering-to-flaming transition phase. While *FlamMap* and *FARSITE* allow users to define ignition delays, few predictive models exist to accurately quantify such delays under realistic wildfire conditions.

In summary, although firebrand processes are incorporated into landscape-scale simulators, the treatment of generation, transport, and ignition phases remains simplified or ad hoc. There is a pressing need for physically consistent, resolution-robust formulations that accurately represent these processes while maintaining computational feasibility for large-scale simulations.

1.6 Motivation and Research Plan

Wildland and wildland–urban interface (WUI) fires pose increasing threats to communities, infrastructure, and ecosystems. Operational fire modeling tools are essential for assessing and managing these risks, particularly during active fire events or for long-term hazard mitigation planning. The present thesis is motivated by the need for a physics-based and numerically robust modeling framework that captures firebrand-aided fire spread at the landscape scale—especially within WUI communities, where embers may play a dominant role in structure ignition.

In response to this need, a collaborative NSF-funded initiative—"*PREEVENTS Track 2:*

Fire Spread at the Wildland–Urban Interface (WUI): Modeling and Data Assimilation for Prediction and Risk Assessment (WUI MAPR)"—was launched in 2020 by researchers from CloudFire, Inc., the University of California, Berkeley (UCB), and the University of Maryland (UMD). Within this project, Dr. Chris Lautenberger (CloudFire) led the development of the baseline ELMFIRE simulator. At UCB, Drs. Michael Gollner and Dwi Purnomo contributed semi-empirical WUI fire spread models focused on flame contact and radiative mechanisms. At UMD, Dr. Arnaud Trouvé and the author of this thesis focused on developing a comprehensive firebrand simulation framework.

As is suggested in the literature review, firebrand models in current landscape-scale fire risk tools are oversimplified relative to surface and crown fire models. Specifically, there is a need for physically grounded models describing firebrand generation, transport, and ignition; robust numerical implementations that remain stable and accurate under varying spatial and temporal resolutions; and enhanced treatment of WUI communities, which are increasingly vulnerable and often underrepresented in legacy wildfire simulators.

Accordingly, this thesis sets out the following objectives:

- Develop a physics-controlled and numerically robust framework that simulates firebrand dynamics at the landscape scale;
- Conduct systematic verification tests in canonical configurations to assess numerical stability and accuracy;
- Integrate the developed firebrand simulation framework into ELMFIRE and ensure compatibility with semi-empirical WUI fire spread models;
- Validate the coupled model using historical wildfire events through a structured reconstruc-

tion and analysis pipeline.

As a result, the discussion is structured into nine chapters. Chapter 1 provides background on the growing challenge of wildland and WUI fires, an overview of current fire modeling methodologies, and limitations in the treatment of firebrand-driven spread. This context motivates the development of a numerically accurate, physics-based firebrand model tailored to ELMFIRE. Chapter 2 introduces the core modeling platform—ELMFIRE—and outlines its level-set algorithm for fire perimeter tracking, as well as its surface fire spread models for both wildland vegetation and urban structures.

Chapters 3 through 5 focus on developing a comprehensive model for firebrand processes suitable for landscape-scale simulation. Chapter 3 addresses firebrand generation, Chapter 4 describes firebrand transport models and algorithms, and Chapter 5 presents the ignition modeling framework. Each chapter includes verification tests to demonstrate numerical stability and physical plausibility, with comparisons to existing literature highlighting improvements.

Chapter 6 explores specific modeling considerations in WUI environments, including the representation of structural fuels and the influence of mesh resolution. Demonstration cases are used to propose best practices for WUI applications. Chapter 7 details a robust preprocessing pipeline for generating simulation inputs such as fuels, meteorology, topography, and community layout. Chapter 8 applies the complete modeling framework to historical fire scenarios and compares simulation results to field observations for validation.

Finally, Chapter 9 concludes the thesis with a summary of major findings and a discussion of future research directions needed to advance the modeling of firebrand-driven WUI fire spread.

Chapter 2: Wildland and Wildland–Urban Interface Fire Spread Models

This chapter introduces the fire spread models used in the Eulerian Level-set Method Fire Simulator (ELMFIRE) [71, 72], a landscape-scale fire risk modeling tool. The firebrand simulation framework proposed in this study is developed to operate within ELMFIRE and will be coupled with its existing fire spread algorithms. ELMFIRE currently supports the simulation of wildland fire propagation driven by thermal radiation and flame contact, using a level-set front-tracking method.

For surface and crown fire spread in wildland environments, ELMFIRE implements the Rothermel model with Albini's modifications, combined with the elliptical fire perimeter propagation method developed by Richards [61] and the crown fire initiation criterion introduced by Wagner [13]. More recently, ELMFIRE has been extended to simulate fire spread in urban environments through the integration of the empirical Hamada model [33], as part of the work presented in this thesis. In parallel, researchers at the University of California, Berkeley—working in collaboration with the author—have led the development of physics-based fire spread models specifically designed for wildland–urban interface (WUI) scenarios, with a focus on flame contact and radiation-driven ignition in spatially heterogeneous communities [112, 113].

The firebrand modeling framework introduced in this work is designed to integrate with these fire spread models and to operate seamlessly within ELMFIRE's level-set-based numerical architecture. Accordingly, this chapter provides an overview of the fire spread mechanisms and modeling approaches currently implemented in ELMFIRE, establishing the foundation for the firebrand simulation strategies presented in the subsequent chapters.

2.1 Surface Fire Spreading Models

The current surface fire spread model in ELMFIRE describes fire propagation due to radiation and convection heat transfer for a surface fire fueled by biomass vegetation in the wildland. The model uses a level-set approach and the Rothermel model for the parameterization of the rate of spread, ROS. The following sub-sections will briefly introduce the formulation of the surface fire model, the Hamada model, and the level-set formulation in ELMFIRE.

2.1.1 Wildland Fire Spread Models

The primary variable of interest for the simulation is the rate of spread (ROS) of fire fronts of different fuels in various environmental conditions. ELMFIRE uses the modified Rothermel model to describe the ROS as a function of vegetation type, magnitude of the wind velocity (measured at mid-flame height), and slope angle of the terrain, which can be expressed as follows:

$$ROS_{surface} = V_0(1 + \phi_w + \phi_s) \tag{2.1}$$

Here $ROS_{surface}$ is the maximum ROS of the fire front under the impact of the wind and slope. In the expression, V_0 is the fire spreading rate in a static atmosphere over flat terrain; ϕ_w is

the correction due to wind; ϕ_s is the correction to slope; the factor $(1 + \phi_w + \phi_s)$ assumes both influence factors exert their impact independently. The following discussions will sequentially introduce the proposed Rothermel expressions for calculating V_0, ϕ_w and ϕ_w. The discussion will be formulated using a format that first introduces the main expressions and then dives into the meaning of the parameters and the corresponding submodels.

Baseline spreading rate, V_0:

The Rothermel model assumes that the fire spreading in the wildland is driven by the heat feedback to the fuel bed from the flame. Therefore, the rate of spreading is the ratio of heat release of the fuel $(I_R\xi)$ divided by the heat required to bring the fuel to ignition $(\rho_b\varepsilon Q_{ig})$:

$$V_0 = \frac{I_R\xi}{\rho_b\varepsilon Q_{ig}} \tag{2.2}$$

We now look at each of the parameters that present in Eqn. 2.2, I_R, ξ, ρ_b, ε, and Q_{ig}.

Firstly, we look at the reaction intensity, I_R (unit: $\mathrm{Btu} \cdot \mathrm{ft}^{-2} \cdot \mathrm{min}^{-1}$), which is defined as the energy release rate per unit area of the fire front. It is calculated by the following formula:

$$I_R = \Gamma' h \eta_s (\omega_{n,dead}\eta_{m,dead} + \omega_{n,live}\eta_{m,live}) \tag{2.3}$$

where Γ' (unit: min^{-1}) is the optimum reaction velocity, characterized by the velocity when the fuel was "free of moisture and contained minerals" and "at the same reaction concentration as alpha-cellulose"; h (unit: $\mathrm{Btu} \cdot \mathrm{lb}^{-1}$) is the fuel heat content, which should be provided as an input to the problem by the user. Coefficients η represent the damping effect on the reaction rate due to the presence of minerals and/or moisture in the biomass. The subscripts s and m

represent moisture and mineral respectively. For the mineral-related reaction rate damping, live and dead fuels are accounted separately, denoted by $\eta_{m,live}$ and $\eta_{m,dead}$, to capture the complex composition of fuels in the field. ω_n is the net fuel load (unit: $\mathrm{lb} \cdot \mathrm{ft}^{-2}$). Subscripts $dead$ and $live$ represent the dead and live fuel as well.

The optimum reaction velocity Γ' is calculated by the following expression proposed by Rothermel [8]:

$$\Gamma' = \Gamma'_{peak} \times \left(\frac{\beta}{\beta_{op}}\right)^{A_{coef}} exp\left[A_{coef} \times \left(1 - \frac{\beta}{\beta_{op}}\right)\right] \tag{2.4}$$

where we have:

$$Peak\ reaction\ velocity : \Gamma'_{peak} = \frac{\sigma_{overall}^{1.5}}{495 + 0.0594 \times \sigma_{overall}^{1.5}} \tag{2.5}$$

$$Mean\ packing\ ratio : \beta = \sum_{i=1}^{6} \frac{\omega_0(i)}{\delta \rho_p} \tag{2.6}$$

$$Optimum\ packing\ ratio : \beta_{op} = \frac{3.348}{\sigma_{overall}^{0.8189}} \tag{2.7}$$

δ is the fuel bed depth.

Albini [9] proposed the following expression to calculate the coefficient A_{coef}:

$$A_{coef} = \frac{133}{\sigma_{overall}^{0.7913}} \tag{2.8}$$

In these expressions, σ is the surface-area-to-volume (SAV) ratio. It is taken as a weighted average over a fuel array regarding the proportion of dead and live fuel, *i.e.*:

$$\sigma_{overall} = F_{dead}\sigma_{dead} + F_{live}\sigma_{live} \tag{2.9}$$

In Albini's modification, the fuel is more precisely divided into 6 subclasses by σ. While the model should have 2 (live and dead) by 6 types of fuel, ELMFIRE uses the standard dynamic fuel models [114], which assumes there are 4 subclasses of dead fuel and 2 subclasses of live fuel. Hereby, the dead fuels are represented by indexes $i = 1, 2, 3, 4$, and the live ones are represented by $i = 5, 6$. The SAV ratio for the dead and live fuels is then expressed by the following two weighted averages:

$$\sigma_{dead} = \sum_{i=1}^{4} F(i)\sigma(i),\ \sigma_{live} = \sum_{i=5}^{6} F(i)\sigma(i) \tag{2.10}$$

where the weighting factors $F(i)$ is the ratio of the surface area of the specific class i of fuel to the total surface area of the dead or live fuels (A_{dead} or A_{live}), *i.e.*,

$$F(i) = \frac{A(i)}{A_{dead}}\ i = 1, 2, 3, 4;\ F(i) = \frac{A(i)}{A_{live}},\ i = 5, 6 \tag{2.11}$$

where

$$A_{dead} = \sum_{i=1}^{4} A(i);\ A_{live} = \sum_{i=5}^{6} A(i) \tag{2.12}$$

The surface area of each unique class is determined by the oven-dry fuel load of the size class ($\omega_0(i)$), the SAV ratio ($\sigma(i)$), and the particle density (ρ_p, assumed to be the same for all classes):

$$A(i) = \frac{\sigma(i) \times \omega_0(i)}{\rho_p}, i = 1, 2, 3, 4, 5, 6 \tag{2.13}$$

The left parameters determine the surface area-to-volume ratio in Eqn. 2.9 are the dead and live fuel surface area ratios, F_{dead} and F_{dead}. We have $F_{dead} = A_{dead}/A_{overall}$ and $F_{live} = A_{live}/A_{overall}$, where the total fuel surface area is $A_{overall} = A_{dead} + A_{live}$.

With equations Eqns. 2.4-2.13 one is now able to determine the reaction velocity Γ'. The in-

puts required for the calculation include δ, $\sigma(i)$, $\omega_0(i)$ and ρ_p. Note that $\omega_0(i)$ ($i = 1, 2, 3, 4, 5, 6$) is the initial oven dry fuel loads for the 6 standard type of fuels, with the exception that $\omega_0(4)$ is used in the dynamic fuel model and is a dependent variable, which will be discussed later.

Now we look at the mineral and moisture-damping coefficients that emerged in Eqn. 2.3. They are determined by the following expressions:

$$\eta_s = max\left(\frac{0.174}{S_e^{0.19}},\ 1\right) \tag{2.14}$$

$$\eta_m = 1 - 2.95r_m + 5.11r_m^2 - 3.53r_m^3 \tag{2.15}$$

These empirical expressions come from the original Rothermel model. S_e is the effective mineral content and is assumed constant in ELMFIRE ($S_e = 0.01$). r_m is the ratio of the moisture content of the fuel (m) over the extinction moisture (m_{ex}):

$$r_m = \frac{m}{m_{ex}} \tag{2.16}$$

For dead and live fuels, m and m_{ex} are determined by:

$$m_{dead} = \sum_{i=1}^{4} F(i) \times m(i) \tag{2.17}$$

$$m_{live} = \sum_{i=5}^{6} F(i) \times m(i) \tag{2.18}$$

$$m_{ex,live} = max\left(m_{ex,dead},\ 2.9W\left(1 - \frac{m_{f,dead}}{m_{ex,dead}}\right) - 0.226\right) \tag{2.19}$$

Again the subscripts "$live$" and "$dead$" designate the live and dead fuels respectively. The

subscripts "ex" means the quantity is related to the critical condition of extinction. Note that $m(i)$ $(i = 1, 2, 3, 5, 6)$ and $m_{ex,dead}$ are not specified and should be provided as input by the user. Specifically, $m(1)$, $m(2)$, and $m(3)$ are the 1-hour, 10-hour, and 100-hour dead fuel moisture respectively. $m(5)$ and $m(6)$ are the live herbaceous and woody fuel moisture respectively. $m(4)$ is the dynamic dead fuel moisture which is assumed equal to $m(1)$ during simulation in ELMFIRE.

The left unknown parameters W and $m_{f,dead}$ in Eqn. 2.19 are the dead-live fuel load ratio and fine dead moisture, which are determined using the following expressions:

$$W = \frac{\sum_{i=1}^{4} \omega_0(i)\varepsilon(i)}{\sum_{i=5}^{6} \omega_0(i) exp\left(-\frac{500}{\sigma(i)}\right)} \tag{2.20}$$

$$m_{f,dead} = \frac{\sum_{i=1}^{4} m(i)\omega_0(i)\varepsilon(i)}{\sum_{i=1}^{4} \omega_0(i)\varepsilon(i)} \tag{2.21}$$

where $\varepsilon(i)$ is the effective heating number. It is the portion of fuel per unit volume of the fuel bed raised to ignition ahead of the advancing fire. It represents the efficiency of heating as a function of particle size:

$$\varepsilon(i) = exp\left(\frac{-138}{\sigma(i)}\right), i = 1, 2, 3, 4, 5, 6 \tag{2.22}$$

Now we turn to another set of new parameters that present in Eqn. 2.3, the net fuel load ω_n. The calculation method was modified by Albini [9], which is presented in the form below for the dead and live fuels given the 6 standard fuel models:

$$\omega_n = \omega_0 \times (1 - S_T) \tag{2.23}$$

where S_T is the total mineral content and is assumed to take the constant value of 0.055 in

ELMFIRE simulations; ω_0 is the oven-dry fuel load. For dead and live fuels, ω_0 takes different values:

$$\omega_{0,dead} = \sum_{i=1}^{4} F(i) \times \omega_0(i) \tag{2.24}$$

$$\omega_{0,live} = \sum_{i=5}^{6} F(i) \times \omega_0(i) \tag{2.25}$$

In summary, $\omega_{n,live}$ and $\omega_{n,dead}$ in Eqn. 2.3 are calculated using Eqn. 2.23 by substitute the value of ω_0 by 2.25 and 2.24 respectively.

As a result, Eqns. 2.4-2.25 yields the reaction intensity I_R.

The baseline fire front spreading rate given by Eqn. 2.2 is also dependent on the propagating flux ratio, ξ. The Rothermel model introduces the parameter to "indicates the proportion of the total heat produced in the combustion zone that actually preheats adjacent fuel particles to ignition" [20]. It is found to be a function of the SAV ratio (σ) and the packing ratio(β). Rothermel model uses the following formula to calculate ξ:

$$\xi = \frac{exp\left[(0.792 + 0.681\sqrt{\sigma_{overall}})(0.1 + \beta)\right]}{192 + 0.2595\sigma_{overall}} \tag{2.26}$$

The product of the left 3 parameters in Eqn. 2.2, $(\rho_b \varepsilon Q_{ig})$, represents the energy required for fuel bed ignition. ρ_b is the bulk density of the fuel array; ε is the portion of fuel to be ignited; and Q_{ig} is the ignition heat of the fuel particle. Since the fuel bed is assumed to be composed of multiple classes of fuel particles with different properties, *i.e.*, the oven-dry fuel load $\omega_0(i)$, the SAV ratio $\sigma(i)$, and the moisture content $m(i)$, we again calculate $(\rho_b \varepsilon Q_{ig})$ as a weighted average

of the dead and live fuels:

$$\rho_b \varepsilon Q_{ig} = \rho_b \sum_{i=1}^{6} F(i) \times \varepsilon(i) \times Q_{ig}(i) \tag{2.27}$$

where Q_{ig} uses an empirical expression for "cellulosic fuels by considering the change in specific heat from ambient to ignition temperature and the latent heat of vaporization of the fuel moisture" [20]:

$$Q_{ig}(i) = 250 + 1116 \times m(i), i = 1, 2, 3, 4, 5, 6 \tag{2.28}$$

ρ_b, the average fuel bulk density, is determined using:

$$\rho_b = \frac{1}{\delta} \sum_{i=1}^{6} \omega_0(i) \tag{2.29}$$

In conclusion, Eqns. 2.2-2.29 give the fire front spreading rate under baseline reference conditions (no wind, flat terrain). The inputs required for the calculation include: h, δ, ρ_p, $m(i)$, $\sigma(i)$ and $\omega_0(i)$ $(i = 1, 2, 3, 5, 6)$.

Wind and Slope Adjustment Factor, $(1 + \phi_w + \phi_s)$:

The effects of wind and slope are adjusted by the term $(1 + \phi_w + \phi_s)$ in Eqn. 2.1. ϕ_w represents the influence of wind and ϕ_s the influence of the slope. In ELMFIRE, they are calculated using the following equations taken from the Rothermel model:

$$\phi_{w,surface} = C \times WSMF_{limited}^{B} \frac{\beta}{\beta_{op}}^{-E} \tag{2.30}$$

$$\phi_s = min\left(5.275\beta^{-0.3}(tan\theta)^2, C(0.9I_R)^B(\beta/\beta_{op})^{-E}\right) \tag{2.31}$$

where the empirical coefficients are:

$$B = 0.02526\sigma_{overall}^{0.54} \tag{2.32}$$

$$C = 7.47exp\left(-0.133\sigma_{overall}^{0.55}\right) \tag{2.33}$$

$$E = 0.715exp\left(-0.000359\sigma_{overall}\right) \tag{2.34}$$

θ is the angle of the slope. β/β_{op} is the relative packing ratio as described in the previous section. The input 20-ft wind speed (WS_{20ft}) is adjusted to the mid-flame wind speed using a log profile by Ref. [115] for the atmospheric boundary layer:

$$WSMF = WS_{20ft} \times \frac{5280}{60} \times \begin{cases} \frac{1}{log\left(\frac{20+0.36CH}{0.13CH}\right)} \frac{0.555}{\sqrt{CH \times CC/3}}, & FBH > 0, CC > 0, CH > 0 \\ \frac{1+0.36FBH/H_F}{ln\left(\frac{20+0.36FBH}{0.13FBH}\right)} \left[ln\left(\frac{1+0.36FBH/H_F}{0.13}\right) - 1\right], & FBH > 0, CC = 0, CH = 0 \\ 0, & FBH = 0, CC = 0, CH = 0 \end{cases} \tag{2.35}$$

$$WSMF_{limited} = min(WSMF, 0.9I_R) \tag{2.36}$$

where FBH (unit: ft) is the fuel bed height, which is assumed to be the same as the fuel depth δ (unit: ft). H_F (unit: ft) is the flame height. FBH/H_F is assumed to be 1 for all simulations in ELMFIRE. CH is the canopy height (unit: ft) and CC is the canopy cover in percentage. This adjustment is not presented in the original Rothermel model but is adopted in FARSITE and BeHavePlus. The constant 5280/60 translates WS_{20ft} in mph into $\mathrm{ft} \cdot \mathrm{min}^{-1}$. Note that the

mid-flame wind speed is limited by a maximum value controlled by the reaction intensity from the Rothermel model.

The above calculations for the wind factor are dedicated to surface fires. According to Ref. [71], "if active crown fire occurs, the spread rate is approximated as 3.34 times the spread rate in Fuel Model 10 with a wind reduction factor of 0.4, although Rothermel's correlation factor (3.34) could be treated as an empirical adjustable parameter to be optimized." The original ELMFIRE codes turn out to use the following relationships to calculate ϕ_w when crown fire occurs:

$$\phi_{w,crown} = \begin{cases} min\left(\frac{CROSA}{V_0}, 200\right), & Active \\ min\left(\frac{CROSA \times exp(-CAC)}{V_0} - 1, 200\right), & Passive \end{cases} \tag{2.37}$$

$$CROSA = min\left[11.02 WS_{10m}^{0.9} CBD^{0.19} e^{-0.17m(1)}, CROSA_{max}\right] \tag{2.38}$$

$$CAC = CROSA/R_0 \tag{2.39}$$

where $CROSA$ is the crown fire spread rate. The occurrence of the active crown fire is determined jointly by the value of CAC ($CAC > 1$) and a user-defined critical canopy cover (CC_{crit}) value. Due to the nature of this empirical model, the wind speed is translated into 10-m wind speed in the unit of kilometers per hour.

$$WS_{10m}(\mathrm{km \cdot h^{-1}}) = \frac{1.609 \times WS_{20ft}(\mathrm{mph})}{0.87} \tag{2.40}$$

According to Ref. [13], $R_0 = S_0/CBD$ (unit: $\mathrm{m \cdot s^{-1}}$) is the critical spread rate that must be achieved by surface fire to make crown spread possible. In ELMFIRE, the mass flow rate

of fuel into the crown space (S_0, in terms of mass per unit cross-sectional area per unit time) is assumed constant (0.05), and the canopy bulk density CBD is provided as an input.

It should be noted that this wind factor will only be calculated when the fire line intensity of the surface fire reaches a value that is higher than the following critical value:

$$I_{f,critical\,crown\,fire} = [0.01CBH \times (460 + 26FMC)]^{1.5} \tag{2.41}$$

where CBH is the canopy base height and FMC is the foliar moisture content in percentage. The fireline intensity should be calculated using the result from the Rothermel model:

$$I_B = I_R t_r ROS_{surface} \tag{2.42}$$

where the residence time t_r (unit: minute) is defined by the time for the fire front to pass a given point. ELMFIRE uses the correlation proposed for the flaming time of fuel particles burning in a uniform fuel array, which is a function of the fuel particle size d (unit: inch), $t_r = 8d$. In Rothermel's model, the particle size d is characterized by the overall SAV ratio $\sigma_{overall}$ that $d = \frac{48}{\sigma_{overall}}$. Therefore, the residence time expressed by the SAV ratio is:

$$t_r = \frac{384}{\sigma_{overall}} \tag{2.43}$$

Considering the situation when the wind factor is calculated using both Eqns. 2.30 and 2.37, we use:

$$\phi_w = \max\left(\phi_{w,surface}, \phi_{w,crown}\right) \tag{2.44}$$

as the final wind factor applied in the simulation.

In conclusion, Eqns. 2.30 - 2.44 give the wind and slope coefficients in Eqn. 2.1 for $ROS_{surface}$. Newly introduced input parameters in addition to calculating V_0 include: CC, CC_{crit}, $CROSA_{max}$, CH, CBD, CBH, and FMC.

Dynamic Fuel Model:

As mentioned above, ELMFIRE utilizes a standard fuel model to facilitate the calculation of the Rothermel expressions. It is also a common practice that has been adopted since the model was developed. The dynamic fuel model is introduced to reflect the seasonal changes. The fuel load is transferred from the live to the dead category to represent curing [20], which is usually in the form of fuel moisture. Recall that indexes $i = 1, 2, 3, 4$ represent the dead fuel and $i = 5, 6$ represent the live fuels, where fuels 1 - 3 represent those are dead for 1 hour, 10 hours and 100 hours, respectively. Fuels 5 and 6 represent the live herbaceous and woody fuels. Fuel 4 represents the fuels in the intermediate state, but its load and SAV ratio are adjusted to 0 and a large value (e.g. 9999), respectively, after being used to update the loads and moisture contents of the other 5 classes. Therefore, its presence is not considered when calculating the heat release and absorption. In ELMFIRE, the moisture and the load are updated using the following algorithm:

$$\omega_0(4)|_{MLH(t+\Delta t)} = (1 - T) \times \omega_0(5)|_{MLH(t)} \tag{2.45}$$

$$\omega_0(5)|_{MLH(t+\Delta t)} = T \times \omega_0(5)|_{MLH(t)} \tag{2.46}$$

$$\sigma(4)|_{MLH(t+\Delta t)} = \sigma(5)|_{MLH(t)} \tag{2.47}$$

$$\sigma(1)|_{MLH(t+\Delta t)} = \frac{[\omega_0(1)\sigma^2(1)]_{MLH(t)} + [\omega_0(4)\sigma^2(4)]_{MLH(t+\Delta t)}}{[\omega_0(1)\sigma(1)]_{MLH(t)} + [\omega_0(4)\sigma(4)]_{MLH(t+\Delta t)}} \tag{2.48}$$

$$\omega_0(1)|_{MLH(t+\Delta t)} = \omega_0(1)|_{MLH(t)} + \omega_0(4)|_{MLH(t+\Delta t)} \tag{2.49}$$

$$\omega_0(4)|_{MLH(t+\Delta t)} = 0 \tag{2.50}$$

$$\sigma(4)|_{MLH(t+\Delta t)} = 9999 \tag{2.51}$$

T is the percentage of the left live herbaceous load not transferred to dead. It is a function of $m(5)$:

$$T = min\left[max\left(\frac{m(5)-30}{120-30}, 0\right), 1\right] \tag{2.52}$$

The initial value of $m(4)$ is forced to take the value of $m(1)$ in ELMFIRE.

ROS Transform in All Directions:

The $ROS_{surface}$ above predicts the fire spread rate in the direction where the maximum spreading rate is achieved, *i.e.*, in the direction of the wind or the direction of the upslope). To extrapolate the result to all directions, ELMFIRE employs the method from FARSITE, which assumes that the fire perimeter evolves in an elliptical shape. The relationship developed by Anderson [116] predicts the length-to-breadth ratio (LB) of the single ellipse (to be distinguished from a double ellipse):

$$\begin{aligned} LB =& 0.936exp\left(0.2566WSMF_{eff} \times \frac{60}{5280}\right) + \\ & 0.461exp\left(-0.1548WSMF_{eff} \times \frac{60}{5280}\right) - 0.397 \end{aligned} \tag{2.53}$$

Assuming the rear focus of the ellipse to be the fire origin [117], designated by point O in

Figure 2.1a. the head-to-back ratio is described as:

$$HB = \frac{LB + (LB^2 - 1)^{0.5}}{LB - (LB^2 - 1)^{0.5}} \tag{2.54}$$

The dimensions of the ellipse are:

$$a = ROS_{surface} \frac{1 + 1/HB}{2.0 \times LB} \tag{2.55}$$

$$b = ROS_{surface} \frac{1 + 1/HB}{2.0} \tag{2.56}$$

$$c = b - \frac{ROS_{surface}}{HB} \tag{2.57}$$

where dimensions a, b, and c denote the semi-minor axis length, semi-major axis length, and linear eccentricity, as is shown in 2.1a. The ellipse drawn in the red solid curve represents the magnitude of $\overrightarrow{ROS}_{surface}$ in all directions.

It should be noted that an effective mid-flame wind speed is used to construct the ellipse in ELMFIRE, which is:

$$WSMF_{eff} = min\left(\frac{|\phi_s + \phi_w|^{1/B}}{[C(\beta/\beta_{op})^{-E}]^{1/B}},\ 0.9I_R\right) \tag{2.58}$$

Therefore, in 1-D simulations, with wind and over flat terrain, the fire spreading rate opposite to the wind direction should be:

$$ROS_{surface,back} = b - c = \frac{ROS_{surface}}{HB} \tag{2.59}$$

In a 2-D ELMFIRE simulation, the ellipse model should predict the components of the

vector $\overrightarrow{ROS}_{surface} = (ROS_{surface,x}, ROS_{surface,y})$ in a Cartesian coordinate system $x - y$ for arbitrary directions as is shown in Figure 2.1b.

Take the $\overrightarrow{ROS}_{surface}$ pointing from point O to P for example, as is shown in Figure 2.1b. For now, assuming that there are no effects due to slope, we denote the direction of $\overrightarrow{ROS}_{surface}$ by a unit vector $\overrightarrow{n} = (n_x, n_y)$ in the $x-y$ coordinates, which is represented by the dark red arrow in Figure 2.1a. Meanwhile, the unit vector denoting the direction where the maximum $\overrightarrow{ROS}_{surface}$ is observed by $\overrightarrow{n}_{max} = (n_{x,max}, n_{y,max})$, represented by the dark red arrow in Figure 2.1b. By construction, we have:

$$n_x^2 + n_y^2 = n_{x,max}^2 + n_{y,max}^2 = 1 \tag{2.60}$$

Denote the angle between $\overrightarrow{n}_{max}$ and $\overrightarrow{n}$ by $(\theta - \theta_w)$ and according to the dimensions of the ellipse, we have the magnitude of $\overrightarrow{ROS}_{surface}$, designated by the line segment OP, projected to a local coordinate system $x' - y'$ where the x' (y')-axis is parallel (normal) to the vector $\overrightarrow{n}$:

$$\frac{dx}{dt} = \frac{1}{\Delta t}\left[\frac{b^2 cos(\theta - \theta_w)}{\sqrt{(b \times cos(\theta - \theta_w))^2 + (a \times sin(\theta - \theta_w))^2}} + c\right] \tag{2.61}$$

$$\frac{dy}{dt} = \frac{1}{\Delta t}\frac{a^2 sin(\theta - \theta_w)}{\sqrt{(b \times cos(\theta - \theta_w))^2 + (a \times sin(\theta - \theta_w))^2}} \tag{2.62}$$

where dx/dt and dy/dt are the projections of $\overrightarrow{ROS}_{surface}$ parallel to the x'-axis and y'-axis, respectively. Therefore, components of the vector $\overrightarrow{ROS}_{surface}$ in the $x - y$ coordinates can be obtained by projecting dx/dt and dy/dt to the $x - y$ coordinates:

$$ROS_{surface,x} = n_x \frac{dx}{dt} + n_y \frac{dy}{dt} \tag{2.63}$$

$$ROS_{surface,y} = n_y \frac{dx}{dt} - n_x \frac{dy}{dt} \tag{2.64}$$

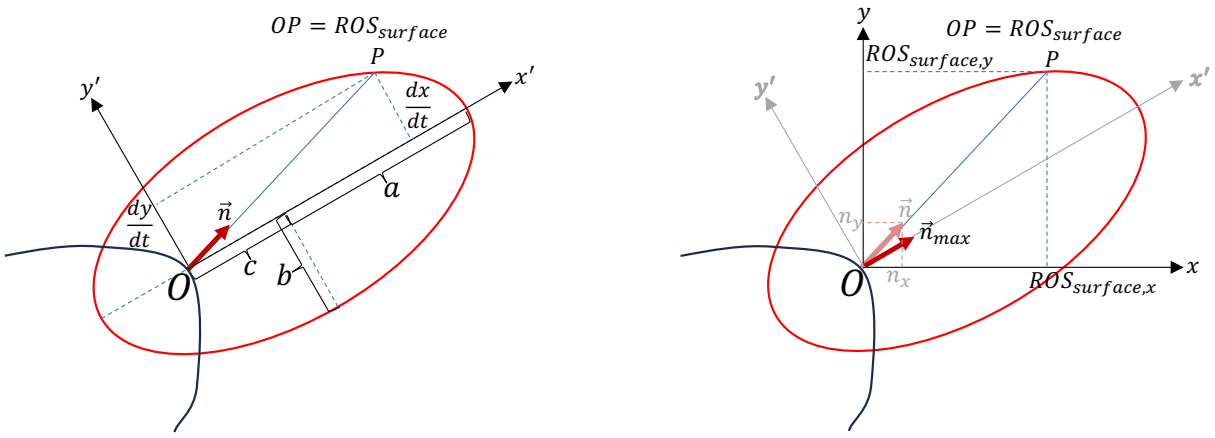


Figure 2.1: Schematic of the ROS projection algorithm. (a) shows the projections in a local coordinate system $x' - y'$; (b) shows the projections in the global coordinate system $x - y$. The schematic investigates the $\overrightarrow{ROS}_{surface}$ in every direction at point O on the black curve, which designates the previous fire front. Point O is the rear focus of the ellipse drawn by the red solid curve. The distances between O and points on the solid red curve designate the magnitude of $\overrightarrow{ROS}_{surface}$.

When a slope is present, the velocity will further be projected to a horizontal plane, as is shown in Figure 2.2 by Ref. [71]. The ratio of the projected velocity (V_s^*) to the velocity in the sloped terrain (V_s) is given by:

$$\frac{V_s^*}{V_s} = 1 - |cos(\theta_\alpha - \theta)|\,(1 - cos\gamma) \tag{2.65}$$

where θ is the angle normal to the fire front measured clockwise from the positive direction of the y-axis; θ_α is the azimuth angle of the slope also measured clockwise from the positive direction of the y-axis; γ is the angle of the slope.

As a result, the input parameters for the wildland surface and crown fire spread models based on the Rothermel model are summarized in Table 2.1.

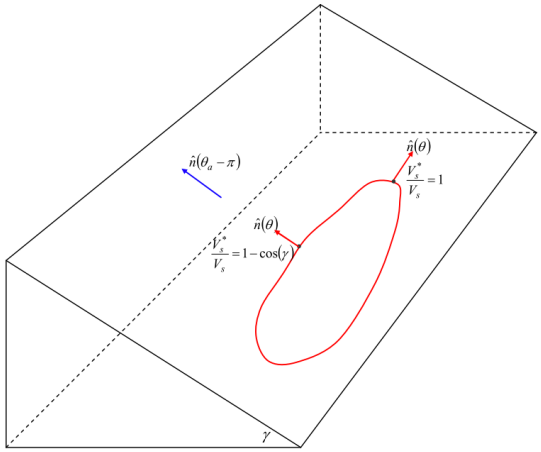


Figure 2.2: Fire perimeter in sloped terrain. For fire spreading upslope (and downslope) the spread rate projected to map coordinates (V_s^*) is reduced from the spread rate parallel to the slope (V_s) by a factor of $1 - cos(\gamma)$. The projected spread rate perpendicular to the slope is unaffected [71].

2.1.2 Urban Fire Spread Models

The original version of ELMFIRE (published in 2013) lacked the ability to simulate fire spread in urban environments, treating urban cells as inert and non-combustible. In an early phase of the current project, the author collaborated with Dr. Chris Lautenberger from CloudFire Inc. to implement the Hamada model within ELMFIRE's level-set framework. The following sections introduce the original Hamada model and describe the adaptations necessary for its integration into a level-set-based fire spread simulator.

The original Hamada model, proposed in 1951 [33], estimates the fire perimeter evolution from a single burning building within a homogeneous community. The fire perimeter is approximated as an ellipse, whose dimensions depend on time, wind speed, and the community's structural configuration (i.e., building footprint size, separation distance, and combustibility fraction).

Table 2.1: Overview of the input variables to the Rothermel Model

Input variables	**Notations**
Fuel properties:	
1-hour dead fuel moisture	m(1)
10-hour dead fuel moisture	m(2)
10-hour dead fuel moisture	m(3)
Live herbaceous fuel moisture (MLH)	m(5)
Live woody fuel moisture (MLW)	m(6)
Initial oven-dry fuel load ($\mathrm{lb} \cdot \mathrm{ft}^{-2}$)	$\omega_0(1)$, $\omega_0(2)$, $\omega_0(3)$, $\omega_0(5)$, $\omega_0(6)$
Initial SAV ratio (ft^{-1})	$\sigma(1)$, $\sigma(2) = 109$, $\sigma(3) = 30$, $\sigma(5)$, $\sigma(6)$
Fuel particle density	$\rho_p = 32\ \mathrm{lb} \cdot \mathrm{ft}^{-3}$
Total mineral content	$S_T = 0.055$
Effective mineral content	$S_e = 0.01$
Fuel heat content	$h = 8000\ \mathrm{Btu} \cdot \mathrm{lb}^{-1}$
Dead fuel moisture of extinction	$m_{ex,dead}$
Fuel bed depth	δ, ft
Canopy Cover	CC
Critical Canopy Cover	CC_{crit}
Maximum crown fire spread rate	$CROSA_{max}$, $\mathrm{ft} \cdot \mathrm{min}^{-1}$
Canopy height	CH, m
Canopy bulk density	CBD, $\mathrm{kg} \cdot \mathrm{m}^{-3}$
Canopy base height	CBH, m
Foliar moisture content	FMC
Meteorology:	
20-ft wind speed	WS_{20ft}, mph
Topography:	
Slope angle	γ
Slope azimuth angle	θ_α

The ellipse dimensions are defined by:

$$K_d = \frac{(a_0 + d)t}{T_d} \tag{2.66}$$

$$K_s = \frac{a_0}{2} + d + \frac{(a_0 + d)(t - T_s)}{T_s} \tag{2.67}$$

$$K_u = \frac{a_0}{2} + d + \frac{(a_0 + d)(t - T_u)}{T_u} \tag{2.68}$$

where K_d and K_u are the distances from the ignition point to the downwind and upwind fire

perimeters, respectively, and K_s is the crosswind (minor axis) extent. a_0 denotes average building footprint dimension, d is average building separation, and t is the simulation time. The times T_d, T_s, and T_u represent the durations required for fire to reach adjacent buildings in each direction, calculated using:

$$T_i = (1 - f_b)\left(3 + 0.375a_0 + \frac{8d}{c_{4i} + c_{5i}V}\right) + \frac{f_b}{C(V)}\left(5 + 0.625a_0 + \frac{16d}{c_{4i} + c_{5i}V}\right) \tag{2.69}$$

$$C(V) = c_{1i}(1 + c_{2i}V + c_{4i}V^2) \tag{2.70}$$

where i indexes wind directions (downwind, crosswind, upwind), f_b is the combustible fraction, and V is wind speed. Empirical constants c_{ki} are tabulated in Table 2.2.

Table 2.2: Empirical coefficients of the Hamada model

i	c_{1i}	c_{2i}	c_{3i}	c_{4i}	c_{5i}
d	1.6	0.1	0.007	25.0	2.5
s	1.0	0.0	0.005	5.0	0.25
u	1.0	0.0	0.002	5.0	0.2

Scawthorn [34, 35] noted that the original model overestimates the downwind extent of fire spread when wind speed is zero. To correct this, Hazus introduced a normalized form for K_d, K_s, and K_u:

$$K'_d = K_d\frac{V}{10} + \left(1 - \frac{V}{10}\right)\sqrt{\left(\frac{K_d + K_u}{2}\right)K_s} \tag{2.71}$$

$$K'_s = K_s\frac{V}{10} + \left(1 - \frac{V}{10}\right)\sqrt{\left(\frac{K_d + K_u}{2}\right)K_s} \tag{2.72}$$

$$K'_u = K_u\frac{V}{10} + \left(1 - \frac{V}{10}\right)\sqrt{\left(\frac{K_d + K_u}{2}\right)K_s} \tag{2.73}$$

To incorporate these dynamics into the level-set framework, we require the time derivatives

of each ellipse dimension:

$$V_d = \frac{dK'_d}{dt} = \frac{V}{10}\frac{dK_d}{dt} + \frac{1}{4}\left(1 - \frac{V}{10}\right)\sqrt{\frac{2}{(K_d + K_u)K_s}}\left[\frac{dK_s}{dt}(K_d + K_u) + K_s\left(\frac{dK_d}{dt} + \frac{dK_u}{dt}\right)\right] \tag{2.74}$$

$$V_s = \frac{dK'_s}{dt} = \frac{V}{10}\frac{dK_s}{dt} + \frac{1}{4}\left(1 - \frac{V}{10}\right)\sqrt{\frac{2}{(K_d + K_u)K_s}}\left[\frac{dK_s}{dt}(K_d + K_u) + K_s\left(\frac{dK_d}{dt} + \frac{dK_u}{dt}\right)\right] \tag{2.75}$$

$$V_u = \frac{dK'_u}{dt} = \frac{V}{10}\frac{dK_u}{dt} + \frac{1}{4}\left(1 - \frac{V}{10}\right)\sqrt{\frac{2}{(K_d + K_u)K_s}}\left[\frac{dK_s}{dt}(K_d + K_u) + K_s\left(\frac{dK_d}{dt} + \frac{dK_u}{dt}\right)\right] \tag{2.76}$$

The temporal derivatives of K_d, K_s, and K_u are:

$$\frac{dK_d}{dt} = \frac{a_0 + d}{T_d} \tag{2.77}$$

$$\frac{dK_s}{dt} = \frac{a_0 + d}{T_s} \tag{2.78}$$

$$\frac{dK_u}{dt} = \frac{a_0 + d}{T_u} \tag{2.79}$$

These results yield the fire spread rates along the major and minor axes. As in standard elliptical models for wildfire propagation, the spreading velocities must be projected directionally. The length-to-breadth ratio (L/B) of the fire ellipse is given by:

$$LB = \frac{V_u + V_d}{V_s} \tag{2.80}$$

With LB, ellipse parameters are computed using Eqns. 2.54-2.57, and the directional com-

ponents ($ROS_{surface,x}$, $ROS_{surface,y}$) of the surface rate of spread are derived using Eqns. 2.61-2.64. In this case, $ROS_{surface}$ is replaced by V_d as given in Eqn. 2.77.

2.1.3 Semi-Physical WUI Fire Spread Models (WU-E Model)

(This work is performed at UC Berkeley)

Recent work by Purnomo et al. [112, 113] developed a semi-physical model that simulates fire spread using empirical correlations for heat transfer and thermal accumulation. In this framework, both vegetative and structural fuels are treated as heat sources that impose thermal impacts on neighboring cells. The primary distinction between the two lies in their combustion characteristics. This unified treatment allows for flexible application in wildland–urban interface (WUI) scenarios where vegetation and structures are intermingled. As the model is specifically designed for WUI fire simulations and integrated into ELMFIRE, it is referred to as the WU-E model (Wildland–Urban—ELMFIRE).

The model focuses on heat transfer mechanisms that promote continuous fire spread, specifically radiation from nearby flames and direct flame contact. The influence of firebrands is not considered at this stage, as firebrand-driven ignition introduces discontinuities and stochastic behavior that are conceptually incompatible with the deterministic nature of the rate-of-spread (ROS) formulation in the level-set framework.

Radiative heat flux is modeled using a classical point-source approximation, in which the flux received at a target location is proportional to the heat release rate (HRR) of the source and inversely proportional to the square of the separation distance. The expression is given by:

$$\dot{q}_r'' = \frac{0.3 \cdot HRR}{4\pi R^2} \tag{2.81}$$

where HRR is the heat release rate of the source (kW), and R is the distance from the source to the target (m). To maintain computational efficiency and physical realism, the radiative contribution is truncated for $R > 100\,\mathrm{m}$, under the assumption that heat fluxes at such distances are negligible.

The HRR of structural fuels is parameterized based on a design fire curve from Ref. [118], incorporating a growth phase, steady burning, and decay. Vegetation heat release is estimated using the fireline intensity I_B from the Rothermel model (Eqn. 2.42) as

$$HRR = I_B \cdot \Delta x, \tag{2.82}$$

where Δx is the cell size. Note that in the current implementation, this HRR from vegetation is held constant once ignition occurs—implying that the growth and decay phases have zero duration, and the steady burning phase is assumed to persist indefinitely.

Direct flame contact heat transfer rate is estimated based on the portion of a cell affected by the flame, and is expressed as:

$$\dot{q}_c = \frac{HRR \cdot A_t}{\Delta x^2}, \tag{2.83}$$

where Δx^2 is the surface area of the source cell, and A_t is the fire-affected area of the receiving cell. This formulation assumes uniform heat flux distribution from the source over its surface.

The fire-affected area A_t is determined using an elliptical flame perimeter model. The ellipse dimensions in the downwind (K_d), crosswind (K_s), and upwind (K_u) directions are em-

pirically correlated to the local wind speed v, structure footprint dimension d, and inter-building spacing s as follows:

$$K_d = \begin{cases} \alpha_1 v^2 + \alpha_2 v + \alpha_3, & 10 \leq v \leq 17.3\,\mathrm{m \cdot s^{-1}} \\ \alpha_1 v + \alpha_2, & \text{otherwise} \end{cases} \tag{2.84}$$

$$K_s = \begin{cases} \beta_1 v^2 + \beta_2 v + \beta_3, & 10 \leq v \leq 17.3\,\mathrm{m \cdot s^{-1}} \\ \beta_1 v + \beta_2, & \text{otherwise} \end{cases} \tag{2.85}$$

$$K_u = \begin{cases} \gamma_1 v^2 + \gamma_2 v + \gamma_3, & 10 \leq v \leq 17.3\,\mathrm{m \cdot s^{-1}} \\ \gamma_1 v + \gamma_2, & \text{otherwise} \end{cases} \tag{2.86}$$

The coefficients α_i, β_i, and γ_i ($i = 1, 2, 3$) are empirical functions of d and s, and their full formulations are listed in Table 2.3.

Table 2.3: Empirical coefficients for the semi-physical ellipse model [113]. Coefficients α_i, β_i, and γ_i correspond to the downwind, crosswind, and upwind fire spread directions, respectively, and are functions of wind speed v (in $\mathrm{m \cdot s^{-1}}$), structure footprint a, and separation distance d.

Coefficient	$v < 10\,\mathrm{m \cdot s^{-1}}$	$10 \leq v \leq 17.3\,\mathrm{m \cdot s^{-1}}$	$v > 17.3\,\mathrm{m \cdot s^{-1}}$
α_1	$1.68 - 0.12a + 0.31d$	$4.09 - 0.001a + 0.13d$	$-7.16 - 0.04a - 0.15d$
α_2	$78.63 + 1.54a - 0.57d$	$-94.27 - 0.001a - 3.05d$	$394.49 + 0.72a + 11.42d$
α_3	–	$615.19 + 0.31a + 19.34d$	–
β_1	$-2.92 - 0.06a + 0.02d$	$0.44 + 0.008a - 0.003d$	$-0.58 - 0.2a + 0.1d$
β_2	$39.31 + 0.77a - 0.28d$	$-10.14 - 0.19a + 0.07d$	$38.11 + 0.81a - 0.41d$
β_3	–	$66.32 + 1.28a - 0.48d$	–
γ_1	$-6.29 - 0.12a + 0.04d$	$0.53 + 0.01a - 0.004d$	$-1.09 - 0.03a + 0.02d$
γ_2	$78.63 + 1.54a - 0.57d$	$-12.41 - 0.25a + 0.11d$	$52.39 + 1.11a - 0.57d$
γ_3	–	$84.64 + 1.73a - 0.81d$	–

For a computation cell within the elliptical fire-active region described by parameters K_d, K_s, and K_u, the affected area A_t of the cell is approximated by the product of the cell size (Δx)

and the distance from the furthest intersection point between the ellipse and the cell edge.

The total heat transfer rate at a target cell is then defined as:

$$\dot{q}_t = \alpha_c \dot{q}_c + \alpha_r \dot{q}''_r \cdot \Delta x^2 \tag{2.87}$$

Two empirical coefficients are introduced: α_c and α_r. The coefficient $\alpha_c = (S_b + S_v)/S_t$ represents the ratio of combustible surface area (structural materials S_b and biomass fuels S_v) to total cell area $S_t = \Delta x^2$. The radiation-related coefficient $\alpha_r = \epsilon \cdot \alpha_c$, where ϵ is the absorptivity of the target material.

During simulation, for any time interval Δt, the total heat flux $\dot{q}_t$ received by each cell from all active fire sources is computed, and the accumulated heat over the interval is denoted by $\sum \dot{q}_t \cdot \Delta t$. Physically, ignition is expected to occur when the accumulated heat exceeds a critical threshold defined by a material-sensitive quantity known as the flux-time product (FTP). Assuming constant-rate heating, the ignition time t_{ign} can be estimated as:

$$t_{\text{ign}} = \frac{FTP \cdot \Delta x^2}{\sum \dot{q}_t} \tag{2.88}$$

Here, FTP has units of energy per unit area, so $FTP \cdot \Delta x^2$ represents the total heat required to ignite the entire cell. A uniform heat flux distribution across the cell is assumed, meaning the entire cell ignites simultaneously once this threshold is reached.

The average fire spread rate U across the cell can then be approximated by:

$$U = \frac{\Delta x}{t_{\text{ign}}} = \frac{\sum \dot{q}_t}{FTP \cdot \Delta x} \tag{2.89}$$

However, U represents a scalar rate of spread and is not directly compatible with the level-set model, which requires directional rates of spread to construct the front propagation velocity vector $\overrightarrow{ROS}_{\text{surface}}$. To address this, we assume the spread rate distribution follows an elliptical perimeter based on parameters K_d, K_s, and K_u defined in Eqns. 2.84–2.86.

The semi-major axis a, semi-minor axis b, and eccentricity c of the ellipse are calculated as:

$$a = \frac{K_d + K_u}{2} \tag{2.90}$$

$$c = \min\left(\frac{a}{2}, a - K_u\right) \tag{2.91}$$

$$b = \frac{K_s}{\sqrt{1 - \left(\frac{c}{a}\right)^2}} \tag{2.92}$$

Using these parameters, we assign directional rates of spread:

$$V_d = U \cdot \frac{(a+c)\Delta x}{2(a+b)} \tag{2.93}$$

$$V_s = U \cdot \frac{2b\Delta x}{2(a+b)} \tag{2.94}$$

$$V_u = U \cdot \frac{(a-c)\Delta x}{2(a+b)} \tag{2.95}$$

The resulting rates V_d, V_s, and V_u represent the spread velocities in the downwind, cross-wind, and upwind directions, respectively. They serve the same functional role as in the Hamada model (cf. Eqn. 2.80) and are used to reconstruct the ROS ellipse. Applying Eqns. 2.54–2.57, we compute the components $(ROS_{\text{surface},x}, ROS_{\text{surface},y})$ of $\overrightarrow{ROS}_{\text{surface}}$ using Eqns. 2.61–2.64, substituting ROS_{surface} with V_d from Eqn. 2.93.

It is worth noting that both the Hamada model and the current semi-physical model neglect

the influence of topography. This simplification assumes flat terrain for urban communities and should be addressed in future research.

2.2 Eulerian Level-Set Method

Surface fire spread models provide analytical expressions for calculating fire spread rates in all directions from a burning source. To implement these models within a numerical simulation framework, the level-set method offers a robust approach for continuously tracking the evolution of fire perimeters over a discretized Cartesian mesh.

Unlike particle-based methods such as *FARSITE*, which represent fire perimeters as a collection of discrete marker particles that must be regularly re-sampled or re-initialized when spread leads to sparse coverage, the level-set method maintains a continuous signed distance function whose zero level-set defines the fire front. This Eulerian approach avoids complications associated with particle management and naturally handles topological changes, such as merging or splitting fire fronts.

The sections that follow introduce the fundamentals of the level-set model as applied to fire perimeter tracking, and subsequently describe the incorporation of firebrand effects into the level-set framework. While the level-set method is inherently designed for continuous front propagation, firebrands introduce nonlocal, discontinuous ignition mechanisms that must be integrated carefully into the otherwise smooth evolution of the fire front.

2.2.1 Original Level-Set Method

In the baseline level-set formulation used to describe fire spread in ELMFIRE [71, 72], the fire is modeled as a thin interface separating unburnt and burnt regions. This interface propagates into unburnt areas with a known rate of spread, $ROS_{surface}$, due to surface or crown fire behavior. The evolution of the fire front is tracked using a scalar field variable ϕ: the fire perimeter corresponds to the isocontour $\phi = 0$, the unburnt region is defined by $\phi > 0$, and the burnt region by $\phi < 0$. The propagation of this front is governed by the level-set equation:

$$\frac{\partial \phi}{\partial t} = -ROS_{surface} \left(\left(\frac{\partial \phi}{\partial x} \right)^2 + \left(\frac{\partial \phi}{\partial y} \right)^2 \right)^{1/2} = -ROS_{surface,x} \frac{\partial \phi}{\partial x} - ROS_{surface,y} \frac{\partial \phi}{\partial y} \tag{2.96}$$

where (x, y) is a Cartesian coordinate system. For horizontal terrain, this coordinate system coincides with the ground plane, while for complex terrain it represents a projected 2D horizontal map. The propagation speed $ROS_{surface}$ is defined in the direction normal to the fire front, which is given by the unit vector $\vec{n} = \nabla\phi/|\nabla\phi|$. The fire spread vector $\vec{ROS}_{surface}$ has components $(ROS_{surface,x}, ROS_{surface,y})$ and satisfies $\vec{ROS}_{surface} = ROS_{surface} \cdot \vec{n}$.

In this formulation, $ROS_{surface}$ is evaluated using one of the fire spread models introduced in Section 2.1, such as the Rothermel model, the Hamada model, or the semi-physical WUI fire spread model.

The level-set equation is solved in ELMFIRE using a second-order Runge–Kutta method for time integration and a second-order, non-diffusive, Total Variation Diminishing (TVD) scheme with a Superbee slope limiter for spatial discretization [66]. A 1-D MATLAB implementation of this framework, developed by the author, is used to support analysis and verification. This simulator is capable of modeling one-dimensional fire spread across flat terrain with spatially and

temporally varying wind and spatially varying fuel conditions. The source code is available in the accompanying GitHub repository [119].

In ELMFIRE, spatial resolution is typically 30 m but we assume in the following, without loss of generality, a 10-m resolution, $\Delta x = \Delta y = 10$ m. Time discretization is determined using a Courant-Friedrichs-Lewy (CFL) condition that limits the magnitude of the time step $\Delta t_n = (t_{n+1} - t_n)$ to $\Delta t_n = CFL_{t_n} \times (\Delta x / \max(ROS_{surface}(x, y\,, t_n))$, where $\max(Q(x, y, t_n))$ designates the maximum value of Q across the computational domain at time t_n, and where CFL_{t_n} is a number that takes values below 1; in ELMFIRE, the baseline choice is $CFL_{t_n} = 0.5$.

In the following, we limit our discussion to cases with uniform vegetation and a flat terrain. The vegetation corresponds to grassland fuel model #102 (GR2) in the nomenclature of Ref. [114]; the fuel bed height is 0.3048 m (1 ft); the surface-to-volume ratio is 5971 m^{-1}; the packing ratio is 0.00158; the fuel load is 0.247 $\mathrm{kg} \cdot \mathrm{m}^{-2}$; the moisture content is 0%. The fire is driven by wind; the wind velocity is 6.71 $\mathrm{m} \cdot \mathrm{s}^{-1}$ (15 mph) at 6.1 m (20 feet) elevation and 2.43 $\mathrm{kg} \cdot \mathrm{m}^{-2}$ at mid-flame height (the flame height is treated as equal to the fuel bed height). The rate-of-spread of the headfire (backfire), *i.e.*, in the downwind (upwind) direction, is $ROS_{surface} = 0.71$ $\mathrm{kg} \cdot \mathrm{m}^{-2}$ (0.014 $\mathrm{kg} \cdot \mathrm{m}^{-2}$). The corresponding fireline intensity of the headfire (also an output of the Rothermel model) is $\dot{Q}'_{fire} = 3189\ \mathrm{kW} \cdot \mathrm{m}^{-1}$.

Figure 2.3 shows a representative example of instantaneous spatial variations of the level-set variable ϕ in the simplified context of a surface fire that propagates as a linefire in a one-dimensional test configuration. The simulation is performed using the baseline ELMFIRE level-set model, recoded in MATLAB for ease of analysis. The simulation is here run without a firebrand algorithm. The numerical configuration corresponds to $\Delta x = 10$ m, $\Delta t_n = 7.04$ s ($CFL_{t_n} = 0.5$); the simulated fire is initialized by setting $\phi = -1$ at the center of the first grid

cell (*i.e.*, at $x = 5$ m) and $\phi = 1$ everywhere else. Figure 2.3 shows the solution after 7 time steps, at time $t_n = 49.3$ s; the fireline has spread from $\phi = 0$ at $x = 10$ m to $\phi = 0$ at $x \approx 46.7$ m. As seen in Figure 2.3, the level-set variable ϕ features a simple monotonic profile characterized by sharp variations between $\phi = -1$ and $\phi = 1$, and a single point where $\phi = 0$.

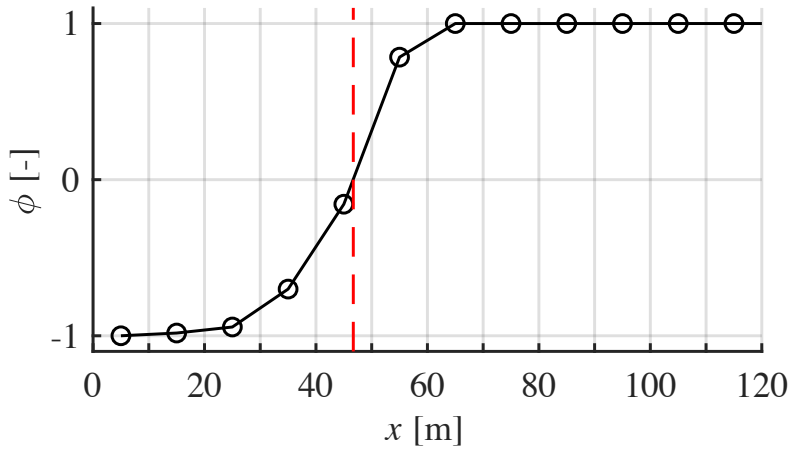


Figure 2.3: Instantaneous spatial variations $\phi(x, t)$ in a one-dimensional test configuration corresponding to a surface fire propagating (from left to right) across grassland fuel in the presence of wind. The dashed vertical line marks the position of the (single) fireline.

2.2.2 Level-Set Model with Firebrand

The level-set model used to simulate fire spread in ELMFIRE can be readily extended to incorporate the effects of firebrands. The firebrand algorithm computes the landing locations, landing times, and total number of deposited firebrands. The formulation adopted in ELMFIRE is consistent with that used in other fire spread models and serves as the basis for integration into the level-set framework.

As an intermediate step, we assume in this section that all firebrands ignite fires instantaneously upon landing, at all locations. While this assumption is not physically realistic—since only a fraction of firebrands typically result in successful spot ignitions, and ignition requires

finite time—it allows us to demonstrate the structural integration of firebrand effects into the level-set formulation. A more realistic, physics-based ignition model will be introduced and discussed in Chapter 5.

In the present formulation, we assume a spot ignition probability of unity, $P_{\text{ign}} = 1$. Under this assumption, only the spatial and temporal distributions of firebrand landings influence fire spread in the level-set model. The number of accumulated firebrands, although calculated, is not utilized in this simplified framework. This parameter will be incorporated in subsequent model development that accounts for partial ignition likelihood and ignition delay.

Spot ignition is readily implemented in the level-set model by setting the local value of ϕ to (-1) at the center of the landing cells and at the first discrete time following landing. In essence, the right-hand side of the level-set equation is augmented of new terms that may be written as:

$$\int_x \int_y \int_t \delta(x - x_{Ck})\delta(y - y_{Ck})\delta(t - TOA_k - t_{ignition,k}) \left[\frac{\phi - (-1)}{\Delta t_n}\right] dxdydt \tag{2.97}$$

where δ is the Dirac delta function, (x_{Ck}, y_{Ck}) the coordinates of the center of a given landing cell k, TOA_k the corresponding time of arrival (*i.e.*, the landing time), $t_{ignition,k}$ the time of ignition, Δt_n the time step of the level-set algorithm, and where the ratio in the integrand is a relaxation term that forces ϕ to take the value (-1) on a very short time scale of order Δt_n. The ignition time $t_{ignition,k}$ is comprised of the time to establish a small flame due to the smoldering-to-flaming transition (SFT), and the time to grow that small flame into a fully-developed fire at the scale of a cell. In the following example, this time delay is ignored (i.e., set to zero) and will be discussed in detail in the following sections.

Figure 2.4 illustrates a representative example of the instantaneous spatial distribution of

the level-set variable ϕ for a configuration similar to that shown in Figure 2.3, now incorporating a test implementation of the firebrand algorithm. In this test case, firebrands are not considered until time $t_n = 49.3$ s. At that time, a single firebrand is emitted from the fireline located at $x = x_{Ck} = 45$ m, corresponding to the center of the cell that contains the active flame front (see Figure 2.3).

The firebrand is transported by the ambient wind at a velocity of $u_{\text{wind}} = 6.71\ \text{m} \cdot \text{s}^{-1}$. In this simplified model, the firebrand trajectory is linear and does not account for vertical lofting or gravitational settling (a limitation discussed further in Chapter 4). The firebrand travels a horizontal distance $X = 40$ m, landing at $x = x_{Ck} + X = 85$ m.

To accurately capture the influence of firebrand deposition on fire spread, the simulation adopts a modified Courant–Friedrichs–Lewy (CFL) condition based on wind speed:

$$\Delta t_n = CFL_{t_n} \cdot \left(\frac{\Delta x}{u_{\text{wind}}} \right) = 0.5 \cdot \left(\frac{10}{6.71} \right) \approx 0.745 \text{ s}. \tag{2.98}$$

This shorter timestep is activated at $t_n = 49.3$ s. Given the flight time of the firebrand is $X/u_{\text{wind}} \approx 5.96$ s (i.e., 8 discrete time steps), the firebrand arrives at the target cell centered at $x = 85$ m at time $TOA_k = 49.3 + 5.96 = 55.3$ s.

Figure 2.4 shows the spatial profile of ϕ at $t = TOA_k$. Under the assumption of instantaneous and guaranteed ignition ($P_{\text{ign}} = 1$), the presence of the firebrand introduces a more complex fire front structure: in addition to the original surface fireline (now located at $x = 50.7$ m, marked by the red dashed line), two new spot firelines appear at $x = 80.0$ m and $x = 90.0$ m, respectively (marked by black dashed lines). After ignition, each fireline propagates independently at a rate determined by the Rothermel model. Specifically, the firelines at $x = 50.7$ m and $x = 90.0$ m

(propagating downwind) spread at $ROS_{\text{surface}} = 0.71\ \text{m} \cdot \text{s}^{-1}$, while the fireline at $x = 80.0$ m (propagating upwind) spreads at the slower rate of $0.014\ \text{m} \cdot \text{s}^{-1}$.

It is important to note that in this formulation, the influence of fire-induced wind alterations is not considered; that is, local modifications to wind velocity due to multiple firelines are neglected.

In summary, the firebrand algorithm will lead to more complex fire front shapes featuring multiple firelines and alternating patches of burnt and unburnt vegetation. In the following, the structure of the fire is characterized by global features such as a leading edge coordinate, defined as the most downwind burning location, $x_{LE} = \max(x|\phi(x) = 0)$, a trailing edge coordinate, defined as the most upwind burning location, $x_{TE} = \min(x|\phi(x) = 0)$, and a fire depth, defined as the thickness of the burning region, $d_{fire} = (x_{LE} - x_{TE})$. In Figure 2, $x_{LE} = 90.0$ m ; $x_{TE} = 50.7$ m; and $d_{fire} = 39.3$ m.

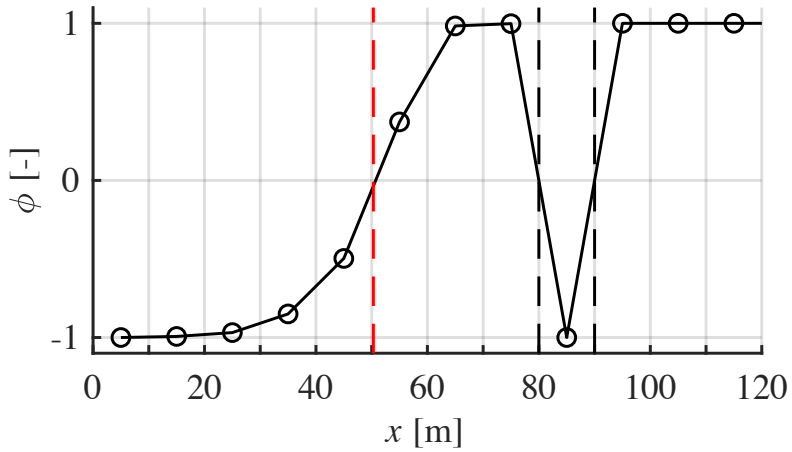


Figure 2.4: Instantaneous spatial variations $\phi(x, t)$ in a one-dimensional test configuration corresponding to a surface fire propagating across grassland fuel in the presence of wind and featuring one spot fire. The dashed vertical lines mark the position of the (multiple) firelines.

2.3 Conclusions

In this chapter, an overview of surface fire spread models and the level-set method was provided. In the context of ELMFIRE, the Rothermel model for the surface fire spread rate of biomass fuels in wildland environments—with Albini's modification—and the elliptical fire perimeter assumption proposed by Richards [60] were originally implemented.

To extend the model's capability to simulate fire spread in urban areas, an empirical formulation, the Hamada model, was first incorporated. While the Hamada model is purely empirical, a physics-based fire spread model—referred to as the WU-E model—was later introduced. The name WU-E reflects its intended application to WUI (wildland–urban interface) environments and its integration into ELMFIRE. The WU-E model accounts for heat flux accumulation in wildland–urban interface (WUI) environments. The WU-E model enables prediction of heat-flux-dependent fire spread rates within the existing level-set framework.

The numerical integration of the level-set method was also briefly discussed in this chapter. In particular, the approach for coupling firebrand effects into the level-set equation was described. Both the original and modified versions of the level-set method were demonstrated using a MATLAB surrogate model. This MATLAB model is capable of simulating one dimensional surface fire spread over flat terrain under varying wind fields and heterogeneous fuel maps.

This MATLAB-based tool will continue to be used in the following chapters to perform verification tests for the proposed firebrand generation, transport, and ignition models, owing to its ease of analysis and transparency in numerical performance evaluation.

Chapter 3: Landscape-Scale Firebrand Modeling: Generation

This chapter describes the methods used to model firebrand generation at the landscape scale. Firebrands are assumed to originate from both biomass fuels and structural fuels. Separate formulations are used to quantify the rate of firebrand generation from each source, based on experimental studies and empirical correlations from the literature.

We first define firebrand generation in the present application as the number of firebrand particles produced from the fuel source. For simplicity, we assume that these particles are uniform in size and mass. In reality, firebrands follow statistical distributions in mass and projected area [88, 90], which may influence ignition behavior. However, we believe this variability can be incorporated in the next chapter (Chapter 4) through the probability density functions and ultimately used during firebrand ignition modeling (Chapter 5). The generated firebrand count facilitates the estimation of the cumulative firebrand mass deposited at target fuel cells, which is assumed to be a key variable for ignition prediction in the present study.

3.1 Firebrand Generation Rate as a Function of Fuel Heat Release Rate

Firebrand generation refers to the detachment of combustible particles from the fuel source. This process is primarily driven by two mechanisms: (1) lifting due to the convective plume generated by the fire and (2) aerodynamic drag from the ambient wind [95]. Since the convective

plume strength correlates directly with the heat release rate of the fire (or fireline intensity in wildland contexts), and the ambient wind speed also strongly influences fire intensity, it is reasonable to assume that firebrand generation is strongly correlated with fire power.

While additional fuel-dependent factors—such as fuel type, moisture content [80], and porosity—may influence firebrand generation, they are not yet included in the present implementation due to limited data availability. However, the modular structure of the modeling framework enables easy integration of such parameters when validated datasets become available.

As a result, we propose estimating firebrand generation as a linear function of the heat release rate of the fuel source. Specifically, a firebrand generation rate per unit power (GR^*) (in units of pieces per megawatt per second) is introduced. Separate parameterizations are adopted for vegetation and structural fuels. The following sections will present these estimates based on laboratory measurements and published analyses.

In the simulation, this generation rate is incorporated into the level-set algorithm by coupling with the heat release rate extrapolated from the surface fire spread models. At a given time t and Cartesian coordinate (x, y), the number of generated firebrand particles can then be written as:

$$GR(x, y, t) = \begin{cases} GR^*(x, y) \times HRR(t), & t_{ign} \leq t < t_{ign} + t_{res} \\ 0, & \text{otherwise} \end{cases} \tag{3.1}$$

where t_{ign} is the time when the fireline arrives at (x, y), and t_{res} is the fireline residence time. This time scale is important because in a temporally discretized scheme, the number of firebrands emitted at each time step Δt is typically given by $GR \times \Delta t$. For firelines with relatively long residence times ($\Delta t \ll t_{res}$), this treatment is valid and does not introduce significant error when

calculating the total number of firebrands generated. However, when the residence time is short (as is often the case in wildland fire scenarios), the estimated number of firebrands should be adjusted to prevent overestimation. A more accurate expression for the number of firebrands generated at each time step is:

$$GR \times \min\left[\Delta t, t - t_{ign}\right] \text{ if } t < t_{ign} + t_{res} \tag{3.2}$$

where t is the current simulation time. The equation ensures that the number of emitted firebrands does not exceed what would be produced during the remaining portion of the fireline residence time. Given Eqn. 3.1, the generation of firebrands can be coupled directly within the level-set model's time-stepping iteration.

The following presents an implementation example of the firebrand generation scheme. In the tests, we impose a constant generation rate $GR = 10\,\mathrm{pcs}\cdot\mathrm{s}^{-1}$ over a residence time $t_{res} = 10\,\mathrm{s}$. A 1-D mesh grid with 10 m resolution is used, and firebrands are assumed to be emitted from the first computational grid cell. The firebrands are then distributed across the domain according to a prescribed probability density function (a truncated log-normal distribution up to the 99th percentile, X_{99th}), starting from the adjacent downwind cell:

$$f(x) = \begin{cases} \frac{1}{\sigma x\sqrt{2\pi}} \exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right] / 0.99, & 0 \le x < X_{99th} \\ 0, & \text{otherwise} \end{cases} \tag{3.3}$$

where the parameters are $\mu = 2.18$ and $\sigma = 1.23$. The total number of firebrands distributed should match the expected emission count $GR \times t_{res}$. Simulations are conducted with two dif-

ferent time step sizes, $\Delta t = 0.13$ s and $\Delta t = 12.7$ s.

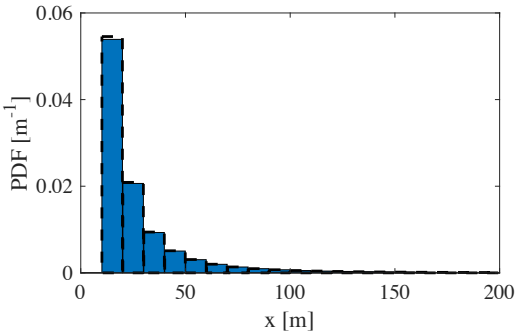
Figures 3.1a and 3.1b show the results when using the simple approach of emitting $GR \times \Delta t$ firebrands at each time step. For $\Delta t = 0.13$ s, the simulation captures the total firebrand count with a discrepancy of only 0.2% relative to the analytical solution. However, for $\Delta t = 12.7$ s, the algorithm fails to capture any firebrand emissions, because the time step exceeds the fireline residence time, causing the emission process to terminate prematurely.

Figures 3.1c and 3.1d show the results using the corrected scheme defined in Eqn. 3.2. Under this treatment, both cases ($\Delta t = 0.13$ s and 12.7 s) match the analytical reference perfectly, demonstrating the robustness of the corrected algorithm. This corrected emission strategy is therefore adopted in subsequent firebrand generation modeling to ensure numerical consistency regardless of the time step size.

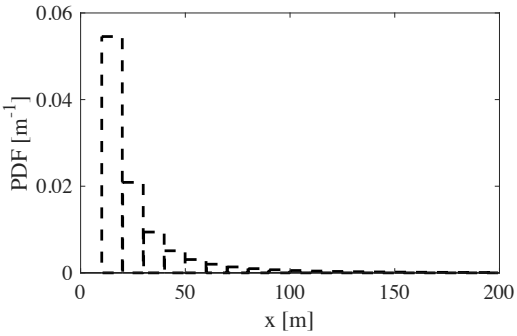
3.2 Firebrand Generation from Biomass Fuels

Given the numerically accurate scheme coupled with the level-set algorithm, we now turn to physics-based parameterization for different fuels. Two major new parameters introduced in this formulation are the generation rate per unit fire intensity (GR^*) and the fireline residence time (t_{res}). This section first presents the parameterization for biomass fuels, followed by considerations for structural fuels.

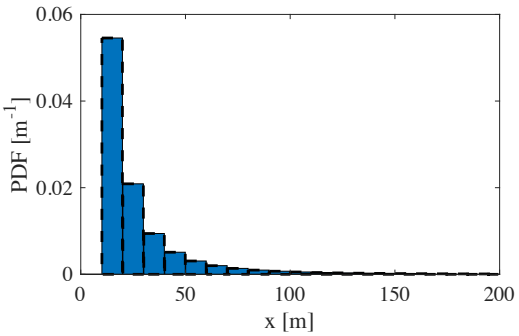
As noted in the introduction, significant efforts have been made in the literature to characterize firebrand generation from fuel sources. These studies typically rely on indirect measurements, counting firebrands collected post-event using water pans or similar methods to stabilize reactive particles. A recent study by Ju et al. [80] provided a more direct approach: they em-



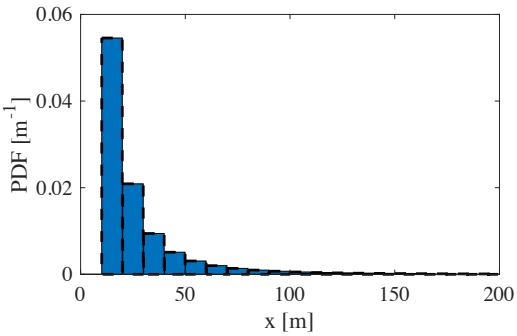
(a) Simple emission scheme ($GR \times \Delta t$), $\Delta t = 0.13$ s.



(b) Simple emission scheme ($GR \times \Delta t$), $\Delta t = 12.7$ s.



(c) Corrected emission scheme (Eqn. 3.2), $\Delta t = 0.13$ s.



(d) Corrected emission scheme (Eqn. 3.2), $\Delta t = 12.7$ s.

Figure 3.1: Verification of the firebrand generation algorithm. Panels (a) and (b) use the simple time-stepping method where $GR \times \Delta t$ firebrands are emitted per step; discrepancies emerge when Δt is large compared to the fireline residence time. Panels (c) and (d) apply the corrected scheme defined by Eqn. 3.2, achieving excellent agreement with the analytical distribution for both small and large Δt values. The blue bars show the simulated histogram, the black solid lines shows the discretized verision of the analytical PDF.

ployed a bench-scale wind tunnel experiment and measured the number of firebrands generated from various tree twig species through video analysis immediately after particle detachment.

Although efforts are underway to extend such measurements to tree-scale samples [120], detailed data for large-scale experiments have not yet been published. Thus, in the present study, a first-order estimation based on the bench-scale data reported by Ju et al. [80] is adopted.

Figure 3.2 presents results from Ju et al. [80], showing the yield of firebrands, defined as the mass of firebrands generated per unit mass of burnt fuel, plotted against the total dry

mass lost during combustion experiments. The experimental conditions, including twig diameter (D_{twig}), fuel moisture content (FMC), and freestream wind speed (U_∞), are annotated in the figure. Various parameters were investigated, such as moisture content, wind speed, and twig trunk diameter.

The data show that most points fall within the range of 5% to 10% yield, indicating that between 5% and 10% of the dry mass loss is converted into firebrands under a range of conditions. To avoid underestimating firebrand production and thus the associated ignition risk, the present study adopts a conservative yield of 10% for biomass fuels.

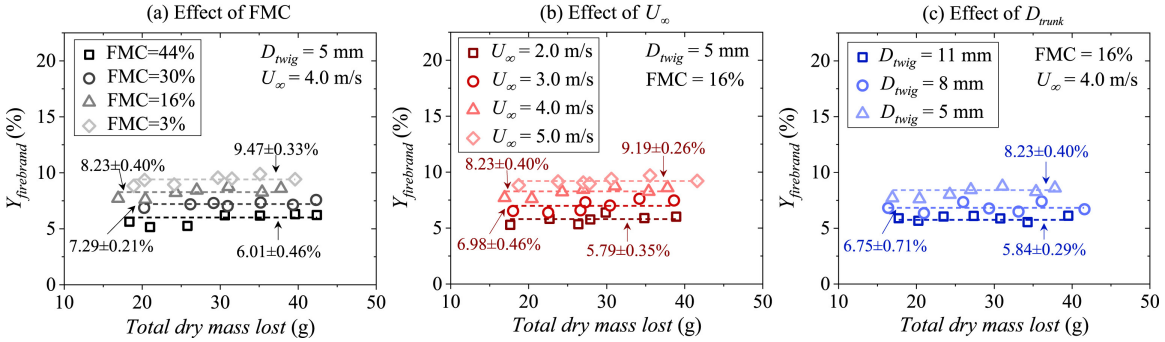


Figure 3.2: Firebrand yield for Douglas Fir fuels under varying fuel moisture content, crosswind velocity, twig diameter, and total dry mass loss. Figure adopted from bench-scale experiments conducted by Ju et al. [80].

Given the above assumptions, we now calculate the number of firebrands emitted from a single computational grid cell. In the previous section, the residence time was defined as the physical timescale during which a flame front of finite depth passes a location (x, y). However, in the present simulation framework, the characteristic depth of a surface flame is typically on the order of meters, while the computational grid cell size is generally 10 meters or larger—thus exceeding the characteristic flame thickness.

Because firebrand generation is not resolved at subgrid scales, all firebrands are assumed to be emitted from the center of the burning computational grid cell. Consequently, the relevant

timescale for emission becomes the time required for the fireline to traverse an entire grid cell, hereafter referred to as the emission time $t_{\text{Emission}} = \Delta x / ROS_{surface}$.

Given local wind speed u_{wind} and terrain conditions, the Rothermel model provides the surface fire rate of spread ROS_{surface} and fireline intensity $\dot{Q}'_{\text{fire}}$. The emission time is approximated as:

$$t_{\text{Emission}} = \frac{\Delta x}{ROS_{\text{surface}}} \tag{3.4}$$

The heat release rate of a burning grid cell is then:

$$\dot{Q}_{\text{fire}} = \dot{Q}'_{\text{fire}} \times \Delta y \tag{3.5}$$

where Δy is the cell size in the direction perpendicular to fire spread. This directional assumption does not affect the final result, as the grid dimensions cancel during normalization.

Assuming a typical heat of combustion for biomass fuels, $\Delta h_c = 15\ \text{MJ} \cdot \text{kg}^{-1}$, the corresponding mass loss rate of fuel is:

$$MLR = \frac{\dot{Q}_{\text{fire}}}{\Delta h_c} \tag{3.6}$$

Adopting a mean mass per firebrand of $m_{\text{firebrand}} = 0.2\ \text{g}$ and assuming that 10% of the combusted dry mass is converted into firebrands, the firebrand generation rate GR becomes:

$$GR = \frac{0.10 \times MLR}{m_{\text{firebrand}}} = \frac{0.10 \times \dot{Q}_{\text{fire}}}{m_{\text{firebrand}} \times \Delta h_c} \tag{3.7}$$

Normalizing by the cell heat release rate $\dot{Q}_{\text{fire}}$, we obtain the firebrand generation rate per

megawatt of fire power:

$$GR^* = \frac{GR}{\dot{Q}_{\text{fire}}} = \frac{0.10}{m_{\text{firebrand}} \times \Delta h_c} \approx 33.3 \text{ pcs} \cdot \text{s}^{-1} \cdot \text{MW}^{-1} \tag{3.8}$$

Although the derivation initially assumes fire spread along the x-axis, the final normalized generation rate is direction-independent, as it is expressed relative to the local heat release rate.

It is important to note that this correlation depends only on two fuel-specific parameters: the average mass of an individual firebrand and the characteristic heat of combustion of the fuel. Consequently, Eqn. 3.8 is easily extrapolated to other biomass species once appropriate experimental data become available. As mentioned earlier, the firebrand yield of 10% per unit mass of dry fuel could, in future applications, be modeled as a function of additional environmental parameters, such as fuel moisture content, ambient wind speed, and fuel source size. However, the present estimation is considered adequate as a first step to establish a functional landscape-scale firebrand generation framework. Detailed parameterization of fuel- and condition-dependent firebrand production is left to future studies and is beyond the scope of the present work.

The following presents an example showing that the spatial-resolution-dependent scheme for the emission time produces resolution-insensitive results in terms of the accumulated firebrands on the ground. This, in turn, demonstrates that the generation of firebrands in the wildland is correctly resolved in the present formulation.

Consider a similar configuration as presented in Figures 2.3 and 2.4, where a fuel of type FBFM102 is uniformly distributed over flat terrain in a one-dimensional configuration. The fire is initiated at the left end of the simulation domain and allowed to spread under a wind blowing from left to right. The ambient wind speed is 15 mph ($6.71 \text{ m} \cdot \text{s}^{-1}$). All ignited cells emit

firebrands according to the model developed in this section, using $GR^* = 33.3\ \mathrm{pcs} \cdot \mathrm{MW}^{-1} \cdot \mathrm{s}^{-1}$ and $t_{\text{Emission}} = \Delta x / ROS_{\text{surface}}$.

Once generated, the firebrands are transported using the Eulerian algorithm to be developed in Section 4.2. At this stage, it is noted that the algorithm distributes the generated firebrands according to a prescribed probability density function (PDF), namely a truncated lognormal distribution whose parameters are functions of the fireline intensity and the wind speed. As a uniform fuel type and wind field are used here, the distribution parameters are fixed at $\mu = 2.18$ and $\sigma = 1.23$, truncated at $X_{\text{max}} = 150$ m (given $I_B = 3.19\ \mathrm{MW} \cdot \mathrm{m}^{-1}$ for FBFM102 under $U_{\text{wind}} = 6.71\ \mathrm{m} \cdot \mathrm{s}^{-1}$).

Here, μ and σ are the logarithms of the mean and standard deviation of the spotting distance of the firebrands, respectively. The origin of the PDF and the definition of X_{max} account for the location of the emission source.

In this test, it is expected that the firebrand distribution on the ground should not be affected by different grid resolutions. In the test configuration, an analytical solution exists for the number of accumulated firebrands per unit length, $N'(x)$, at a sufficiently far location x_0 (where $x_0 > X_{\text{max}}$ such that the cell collects the maximum possible amount of firebrands):

$$N'(x_0) = \int_0^{t_\infty} \int_0^{x_0} GR'(x,t) \times pdf(x - x_0)\, \mathrm{d}x\, \mathrm{d}t, \tag{3.9}$$

where $GR'(x,t)$ is the firebrand generation rate per unit length at coordinate x, and $pdf(x - x_0)$ represents the probability that a firebrand generated at x lands at x_0.

Since the fuel and wind fields are assumed uniform, $GR'(x,t)$ and $pdf(x)$ are independent

of space and time, leading to:

$$N'(x_0) = \left(\int_0^{t_\infty} GR'(t)\ \mathrm{d}t \right) \times \left(\int_0^{x_0} pdf(x - x_0)\ \mathrm{d}x \right). \tag{3.10}$$

Here, the first integral represents the total number of firebrands generated per unit length over time, while the second integral captures the total probability that firebrands reach x_0 from any upstream location.

Noting that $\int_0^{x_0} pdf(x_0 - x)\ \mathrm{d}x = 1$ for $x_0 > X_{\text{max}}$ — because the PDF is truncated at X_{max} — the expression simplifies further. Thus, the total number of accumulated firebrands per unit length at x_0 becomes simply the total number of firebrands generated per unit length:

$$N'(x_0) = GR^* \times \dot{Q}'_{\text{fire}} \times t_{\text{res}}, \tag{3.11}$$

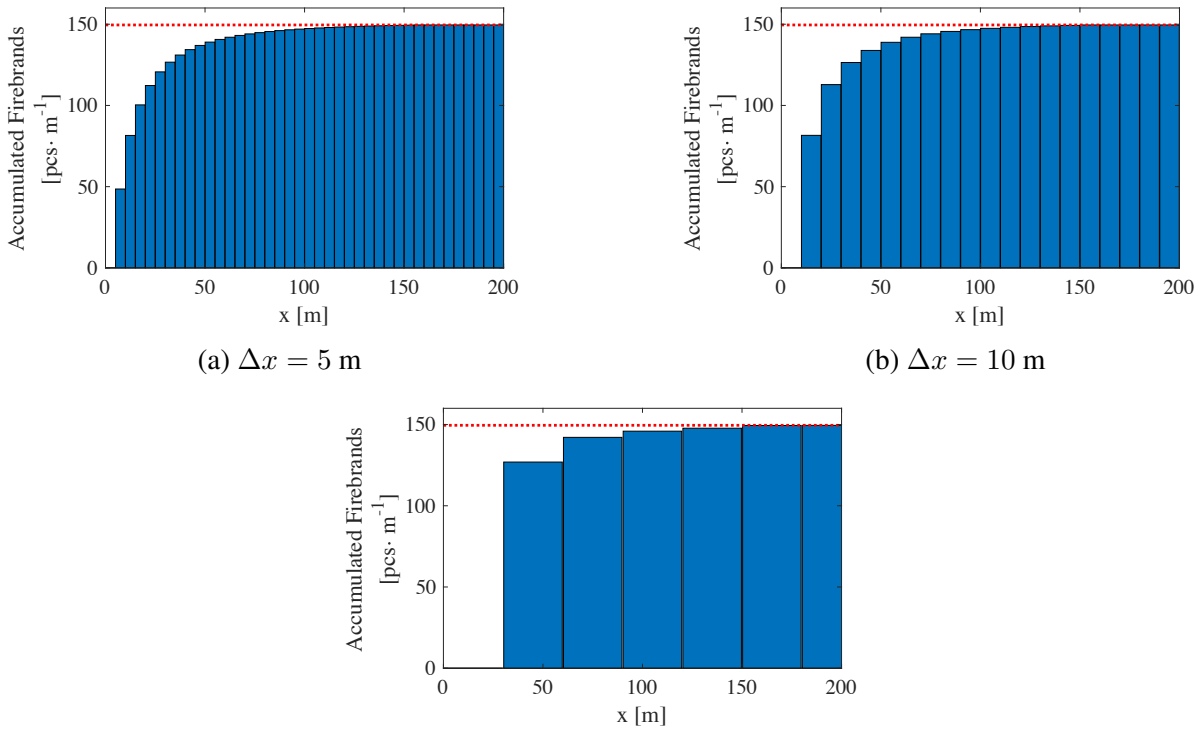
where GR^* is the normalized firebrand generation rate (per unit fire heat release rate), $\dot{Q}'_{\text{fire}}$ is the fireline intensity, and t_{res} is the local residence time of the flame, which can be represented by the unit length divided by the fire spread rate, $t_{\text{res}} = 1/ROS_{\text{surface}}$. Substituting numerical values for the current test case:

$$N'(x_0) = 33.3\ \text{pcs} \cdot \text{MW}^{-1} \cdot \text{s}^{-1} \times 3.19\ \text{MW} \cdot \text{m}^{-1} \times \left(\frac{1}{0.71} \right)\ \text{s} = 149.54\ \text{pcs} \cdot \text{m}^{-1}.$$

This theoretical value serves as the reference for verifying the simulation results at large distances.

This result is verified in Figure 3.3. The accumulated number of firebrands is shown as bar plots, expressed in pieces of firebrands per unit length along the x-axis (total number of firebrands

accumulated in each cell divided by Δx). When $x > X_{\max} = 150$ m, the accumulated number of firebrands approaches the predicted analytical value, independent of the mesh resolution.



(a) $\Delta x = 5$ m

(b) $\Delta x = 10$ m

(c) $\Delta x = 30$ m

Figure 3.3: Verification of the emission time scheme. The red dotted line indicates the analytical result predicted by Eqn. 3.11. Panels (a), (b), and (c) show results for $\Delta x = 5$ m, 10 m, and 30 m, respectively.

3.3 Firebrand Generation from Structural Fuels

For structural fuels, we first consider the estimation of the normalized firebrand generation rate GR^*. To the best of the author's knowledge, direct experimental measurements of firebrand generation rates, comparable to those available for vegetation fuels, are even more limited for structural fuels. However, several empirical correlations have been proposed and incorporated into practice-oriented modeling efforts [110, 111].

Ref. [110] adopted a correlation developed by Lee et al. [121], which in turn was based on experimental data from Waterman and Takata [87]. In this approach, the firebrand generation rate per unit structure footprint is modeled as a function of wind speed:

$$\frac{GR}{A_{\text{structure}}} = 306.77e^{0.1879u} \tag{3.12}$$

where u is the wind speed in meters per second.

Alternatively, Himoto [111] proposed a correlation based on crib fire experiments conducted by Miura et al. [122], expressed as:

$$\frac{GR}{A_{\text{structure}}} = 3.11 \times 10^5 \cdot MLRPUA^{3.57} \tag{3.13}$$

where $MLRPUA$ is the mass loss rate per unit area ($\mathrm{kg \cdot m^{-2} \cdot s^{-1}}$).

It is important to note that Eqn. 3.12 is based on burner-driven roof assembly tests, where wind speed is the sole independent variable, whereas Eqn. 3.13 is formulated based on intrinsic burning characteristics, namely mass loss rate and heat release rate. For consistency with the biomass fuel firebrand generation models—where GR is correlated to the local fire power—we adopt Himoto's correlation (Eqn. 3.13) in this study.

Given that $GR^* = GR/\dot{Q}_{\text{fire}}$, and noting that $\dot{Q}_{\text{fire}} = HRRPUA \times A_{\text{structure}}$, we derive:

$$GR^* = \frac{GR}{\dot{Q}_{\text{fire}}} = \frac{3.11 \times 10^5 \cdot MLRPUA^{3.57}}{HRRPUA} \tag{3.14}$$

Assuming again that the characteristic heat of combustion for structural fuels is Δh_c (in

$\mathrm{MJ} \cdot \mathrm{kg}^{-1}$), we have:

$$HRRPUA = MLRPUA \times \Delta h_c \tag{3.15}$$

Thus, substituting into the expression for GR^*:

$$GR^* = \frac{3.11 \times 10^5 \cdot MLRPUA^{2.57}}{\Delta h_c} \tag{3.16}$$

The derived expression shows that the normalized firebrand generation rate depends primarily on the mass loss rate per unit area ($MLRPUA$) and the characteristic heat of combustion (Δh_c). Both parameters are known to vary with ambient conditions, such as ventilation, external wind speed, building configuration, and fire load. For instance, the ventilation conditions during a structural fire can significantly alter burning behavior, affecting both $MLRPUA$ and Δh_c.

For the purposes of this initial framework development, representative values for $MLRPUA$ and Δh_c are adopted. More detailed parameterizations could be incorporated as more comprehensive experimental data become available.

Specifically, Yoshioka et al. [92] reported a typical value of $MLRPUA \approx 0.045\,\mathrm{kg} \cdot \mathrm{m}^{-2} \cdot \mathrm{s}^{-1}$ for a full-scale shed. In addition, assuming typical American residential construction wood, we adopt $\Delta h_c = 20\ \mathrm{MJ} \cdot \mathrm{kg}^{-1}$ [123].

Substituting into Eqn. 3.16, we obtain:

$$GR^* \approx 5.68\ \mathrm{pcs} \cdot \mathrm{s}^{-1} \cdot \mathrm{MW}^{-1} \tag{3.17}$$

This value provides a practical basis for estimating firebrand production from structural fuels under the conditions considered in the referenced experiments. It is acknowledged that

these estimates are derived from idealized crib fires and that real structure fires, involving diverse construction materials and ventilation regimes, may deviate from this approximation. Nevertheless, the proposed correlation offers a reasonable first-order approach suitable for landscape-scale firebrand generation modeling.

In addition to GR^*, the current framework requires the specification of the firebrand emission time t_{Emission}. For wildland fuels, t_{Emission} was approximated as the flame residence time in a computation cell. However, for structural fuels—such as buildings in the wildland–urban interface (WUI)—the burning time is significantly longer, often sustained for hours. Therefore, it is inappropriate to estimate t_{Emission} using $\Delta x / ROS_{\text{surface}}$ for structures.

Instead, in the present framework, t_{Emission} for structural fuels is derived from design fire curves. Specifically, the emission duration is governed by the timespan during which the structure is actively burning. In WU-E-type models (see Section 2.1.3), the HRR follows a three-phase curve (growth, steady, decay) between 0 and the maximum HRR. Since firebrand generation is assumed to be proportional to the instantaneous HRR, firebrand emission is activated when HRR becomes greater than zero and ceases when HRR returns to zero.

In models such as the Hamada implementation(see Section 2.1.2), where no specific design fire curve is implied for structural fuels, an additional design fire curve is assigned solely for the purpose of modeling firebrand generation. Figure 3.4 presents a example design fire curve, featured by a linear increasing and decreasing phases and a steady phase the the maximum heat release rate. This supplementary design curve governs the temporal evolution of heat release rate (HRR) associated with each structural cell, independent of the surface fire spread dynamics. In other words, while the Hamada model determines the perimeter evolution, the firebrand generation process relies separately on a prescribed HRR curve to simulate the firebrand emission rate

over time. This treatment ensures that firebrand generation reflects realistic burning durations and intensities without modifying the behavior of the surface fire spreading model.

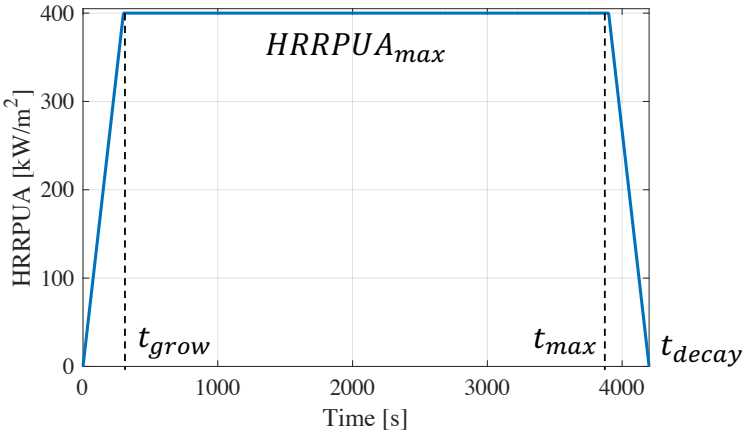


Figure 3.4: Example design fire curve. The fire growth profile follows three linear phases: growth ($0 \leq t < 300\,\mathrm{s}$), steady burning ($300 \leq t < 3900\,\mathrm{s}$), and decay ($3900 \leq t < 4200\,\mathrm{s}$). The peak heat release rate per unit area is set to $HRRPUA_{\mathrm{max}} = 400\,\mathrm{kW} \cdot \mathrm{m}^{-2}$.

3.4 Conclusions

In this chapter, a numerical framework has been developed to simulate firebrand generation within level-set or similar landscape-scale fire spread models. Firebrand generation is defined through a particle number rate correlated with the local fuel heat release rate, leading to the introduction of a normalized firebrand generation rate per unit fuel heat release rate. This formulation is motivated by experimental findings in the literature suggesting a strong physical correlation between fire intensity and firebrand production. Based on this foundation, a numerical scheme was proposed to ensure mathematically consistent firebrand generation, independent of the temporal resolution used in the time-integration of the level-set-like algorithms.

Although direct experimental measurements of firebrand generation rates remain limited, first-order estimations were made based on existing experimental studies and empirical correla-

tions. Characteristic values for the normalized firebrand generation rate were established separately for vegetation and structural fuels. Additionally, the emission duration—or effective firebrand residence time—associated with each computational cell during simulation was parameterized.

For vegetation fuels, the normalized generation rate was estimated based on bench-scale experiments, assuming relatively short residence times characteristic of wildland surface fires. A grid-resolution-dependent scheme was employed to determine the emission duration, ensuring that firebrand generation remains consistent across varying numerical setups. As demonstrated in the exmaple cases in Section 3.2, when coupled with the proposed transport model (see the next Chapter), this approach ensures both correct total firebrand production and minimal distortion of the deposition distribution.

For structural fuels, an empirical correlation available in the literature was used to estimate the normalized generation rate. Unlike vegetation fuels, structures exhibit long-duration burning behavior, and therefore, the emission duration is determined based on a design fire curve. It should be noted that the treatment of emission duration may depend sensitively on how structures are represented within the simulation (e.g., discretized segments versus integrated units), a topic that will be discussed in Chapter 6.

Chapter 4: Landscape-Scale Firebrand Modeling: Transport

In this chapter[1], we focus on the transport of short-distance firebrands, here loosely defined as firebrands with travel distances shorter than 100 meters. These firebrands may land in a flaming or smoldering state and represent a high spot fire risk due to both their significant probability of ignition and the potential for accumulation at the landing site. Although short-distance firebrands have a limited impact on large-scale wildland fire spread, they are believed to play a significant, and often dominant, role in wildland-urban interface (WUI) fire scenarios, such as fire jumping over roads and structure-to-structure fire spread.

The transport of firebrands is described using a stochastic model based on a prescribed probability density function (PDF) for downwind travel distances, combined with a simple flight time model. While this framework differs from deterministic approaches that directly compute travel distances and times based on physics-based models, we consider the stochastic treatment sufficiently general for landscape-scale modeling, and many of the discussions here are applicable to deterministic frameworks as well.

The objective of this chapter is to review the state of the art in firebrand transport modeling used in landscape-scale fire spread simulators and to establish a mathematical and numerical framework capable of achieving numerically converged results. Chapter 2 previously reviewed

[1] Parts of this chapter are taken from the publication by Qin and Trouvé [124]

the level-set model employed in ELMFIRE for surface and crown fire spread; while the present chapter uses ELMFIRE as a reference, the modeling challenges associated with firebrand transport are largely independent of the specific surface fire spread model employed. Models based on Huygens' principle or Cellular Automata would face similar challenges.

We consider both a discrete particle (Lagrangian) firebrand transport model, representative of approaches used in simulators such as FARSITE [10,11] and FlamMap [15,16], and a Eulerian firebrand transport model, which is found to be computationally faster and numerically more accurate under many conditions.

Results from several verification tests are presented to evaluate the numerical performance of both the Eulerian and Lagrangian formulations. The sensitivity of each model to changes in numerical parameters—specifically, spatial and temporal resolution in the fire spread solver, and the generation rate of computational particles in the Lagrangian formulation—is assessed. The study establishes a mathematical framework that achieves numerically accurate or converged results, meaning that the solutions are primarily controlled by physical parameters rather than numerical configurations. The results may still feature physical modeling errors, but the impact of numerical errors is reduced to suitably small levels.

4.1 Probability Density Function of Flight Distance

The firebrand algorithm adopted in ELMFIRE is based on a prescribed statistical distribution for ember flight distance combined with a simple model for flight time. The algorithm assumes that embers are emitted from all fire locations (*i.e.*, from all computational cells where $\phi(x) = 0$) at a certain rate and for a certain duration. In the following, the emission rate of embers

and the duration of the emission phase are either user-prescribed or given by a physical model.

Let us consider a given location that emits firebrands; this location is the origin of a two-dimensional coordinate system (X, Y) where X (Y) designates flight distance in the downwind (cross-wind) direction. The probability density function of ember flight distance is a function of (X, Y) that can be expressed as the product of $pdf_X(X)$ times $pdf_Y(Y)$, where $pdf_X(X)$ ($pdf_Y(Y)$) is the PDF of landing locations in the downwind (cross-wind) direction.

We first consider the formulation of the probability density function $pdf_X(X)$ for firebrand flight distance. In ELMFIRE, $pdf_X(X)$ is implemented as a discretized and truncated version of an analytical distribution drawn from the literature. Specifically, we adopt the lognormal distribution proposed by Sardoy et al. [83], hereafter denoted as $pdfS_X(X)$. This distribution features a peak near $X = 0$ and a gradual decay at larger distances, consistent with the expected behavior of short-range firebrands. The parameters of $pdfS_X(X)$ are modeled as functions of the fireline intensity and the ambient wind velocity measured at 10-meter elevation (U_{wind}, $\mathrm{m \cdot s^{-1}}$):

$$pdfS_X(X) = \frac{1}{X \cdot \sigma\sqrt{2\pi}} \exp\left[-\frac{(\ln X - \mu)^2}{2\sigma^2}\right], \tag{4.1}$$

where X is the downwind travel distance of firebrands, and μ and σ are the mean and standard deviation of the natural logarithm of X, respectively. These parameters depend on the fireline intensity I_B ($\mathrm{MW \cdot m^{-1}}$) and ambient wind speed, and are determined based on the Froude number:

$$Fr = \frac{U_{\text{wind}}}{\sqrt{gL_c}}, \quad L_c = \left(\frac{1000 \times I_B}{\rho_\infty c_{p,g} T_\infty \sqrt{g}}\right)^{2/3}, \tag{4.2}$$

with constants: air density $\rho_\infty = 1.1\,\mathrm{kg \cdot m^{-3}}$, specific heat capacity of air $c_{p,g} = 1.0\,\mathrm{kJ \cdot kg^{-1} \cdot K^{-1}}$,

ambient air temperature $T_\infty = 300\,\mathrm{K}$, and gravitational acceleration $g = 9.81\,\mathrm{m}\cdot\mathrm{s}^{-2}$.

The deposition parameters are then given as:

If $Fr \leq 1$:

$$\mu = 1.47 \cdot I_B^{0.54} \cdot U_{\text{wind}}^{-0.55} + 1.14, \tag{4.3}$$

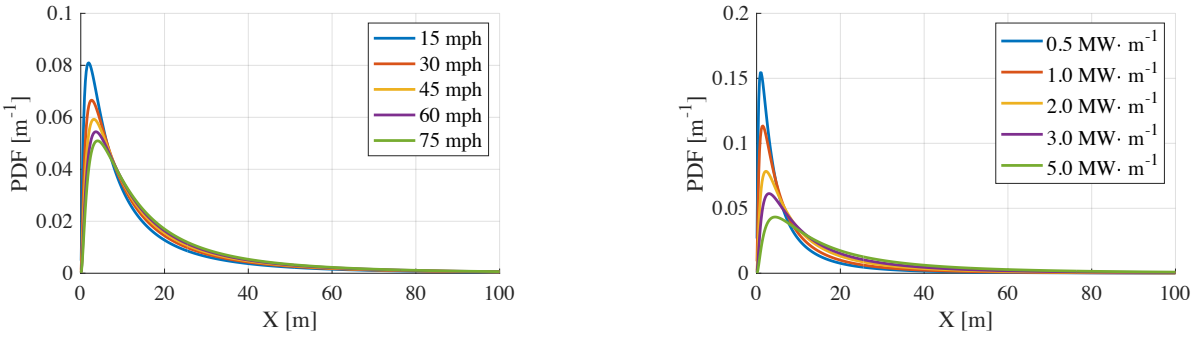
$$\sigma = 0.86 \cdot I_B^{-0.21} \cdot U_{\text{wind}}^{0.44} + 0.19. \tag{4.4}$$

If $Fr > 1$:

$$\mu = 1.32 \cdot I_B^{0.26} \cdot U_{\text{wind}}^{0.11} - 0.02, \tag{4.5}$$

$$\sigma = 4.95 \cdot I_B^{-0.01} \cdot U_{\text{wind}}^{-0.02} - 3.48. \tag{4.6}$$

Figure 4.1 shows the dependence of the PDF on wind speed (Figure 4.1a) and fireline intensity (Figure 4.1b). Firebrands are more widely dispersed when the wind speed or fireline intensity increases, resulting in greater downwind transport distances.



(a) (b)

Figure 4.1: Examples of Sardoy's model for firebrand travel distance distribution. (a) Dependence on wind speed ($I_B = 3\ \mathrm{MW}\cdot\mathrm{m}^{-1}$); (b) Dependence on fireline intensity ($U_{\text{wind}} = 40\ \mathrm{mph} = 17.88\ \mathrm{m}\cdot\mathrm{s}^{-1}$).

As discussed in Sardoy et al. [83], $pdfS_X(X)$ is designed to represent short-range firebrands that remain sufficiently hot upon landing to ignite spot fires. It should be noted that while Sardoy's lognormal distribution is selected for the present analysis, other candidate distributions can easily be substituted as needed. For instance, it will be demonstrated in Chapter 6 that distributions proposed by Himoto et al. [82] are more suitable when modeling firebrand transport originating from structural fuels in urban areas.

Next we consider a discretized version of the PDF. We assume a spatial resolution Δx and decompose $pdfS_X(X)$ into k_{max} bins, centered on the discrete values $X_{k+1/2} = (k+1/2)\Delta x$, and define $pdfS_X(X_{k+1/2})$ as follows:

$$\begin{aligned} pdfS_X(X_{k+1/2}) \times \Delta x &= Probability(k\Delta x \leq X < (k+1)\Delta x) \\ &= \int_{k\Delta x}^{(k+1)\Delta x} pdfS_X(X')dX', 0 \leq k \leq \infty \end{aligned} \tag{4.7}$$

We then modify the PDF by introducing a maximum value of the flight distance, noted X_{max}. We first define X_{max} as the downstream edge of the bin where the value of the probability that the ember flight distance is less than X_{max} takes an arbitrary high value, or more precisely where the value of the cumulative distribution function (CDF) takes an arbitrary high value $(1-\varepsilon)$ $(\varepsilon = 0.01)$:

$$cdfS_X(X_{max}) = \int_0^{X_{max}} pdfS_X(X')dX' = 1-\varepsilon \tag{4.8}$$

where $cdfS_X(X)$ is the CDF of $pdfS_X(X)$, which is a monotonically increasing function of X.

We then define $X_{k_{max}} = (k_{max}\Delta x)$ as the nearest discretized value of X_{max}: $k_{max} = \mathrm{round}(X_{max}/\Delta x)$, where "round" designates a function that finds the nearest integer of a given real number. We have by construction $cdfS_X(X_{k_{max}}) = \sum_{k=0}^{(k_{max}-1)} pdfS_X(X_{k+1/2})\Delta x \approx (1-$

$\varepsilon) = 0.99$.

We finally introduce a truncated PDF, noted $pdf_X(X)$, as follows:

$$pdf_X(X) = \frac{pdfS_X(X)}{\int_0^{X_{max}} pdfS_X(X')dX'}, \qquad 0 \leq X < X_{max} \tag{4.9}$$

$$pdf_X(X) = 0, \qquad X_{max} \leq X \tag{4.10}$$

or, in discretized form:

$$pdf_X(X_{k+1/2}) = \frac{pdfS_X(X_{k+1/2})}{\sum_{k=0}^{k_{max}-1} pdfS_X(X_{k+1/2})\Delta x}, \qquad 0 \leq k \leq (k_{max}-1) \tag{4.11}$$

$$pdf_X(X_{k+1/2}) = 0, \qquad k_{max} \leq k \tag{4.12}$$

We have by construction: $\sum_{k=0}^{k_{max}-1} pdf_X(X_{k+1/2})\Delta x = 1$.

We now consider the formulation of $pdf_Y(Y)$. In ELMFIRE, pdf_Y is treated as a discretized truncated version of the normal distribution proposed by Himoto and Tanaka [82]: the distribution is symmetric with respect to $Y = 0$, and features a peak value at $Y = 0$ and a rapid decay at large values of Y; the parameter of the distribution is a function of the fireline intensity. Following arguments that are similar to those presented above, we decompose $pdf_Y(Y)$ into $(2k'_{max} + 1)$ bins of size Δy, centered on the discrete values $Y_{k'} = (k'\Delta y)$, $(-k'_{max}) \leq k' \leq k'_{max}$, and define $(pdf_Y(Y_{k'}) \times \Delta y)$ as the probability of $((k' - 1/2)\Delta y) \leq Y < ((k' + 1/2)\Delta y)$. Note that the model proposed by Himoto and Tanaka [82] predicts relatively narrow distributions of firebrands in the cross-wind direction (*i.e.*, $Y_{max} = Y_{k'_{max}} \leq \Delta y, k'_{max} = 0$ or 1) and that in the present study, the impact of $pdf_Y(Y)$ remains limited.

In conclusion, using the present discretized framework, the PDF of ember flight distance is

expressed as: $pdf(X_{k+1/2}, Y_{k'}) = pdf_X(X_{k+1/2}) \times pdf_Y(Y_{k'})$.

4.2 Presentation of the Eulerian Model

We now consider a two-dimensional planar domain discretized into an array of uniform computational cells arranged in an $(x - y)$ rectangular Cartesian coordinate system, as illustrated in Figure 4.2. For simplicity, we assume square cells with $\Delta x = \Delta y$; these are the same Δx and Δy values used in the discretization of $pdf_X(X)$ and $pdf_Y(Y)$ introduced earlier.

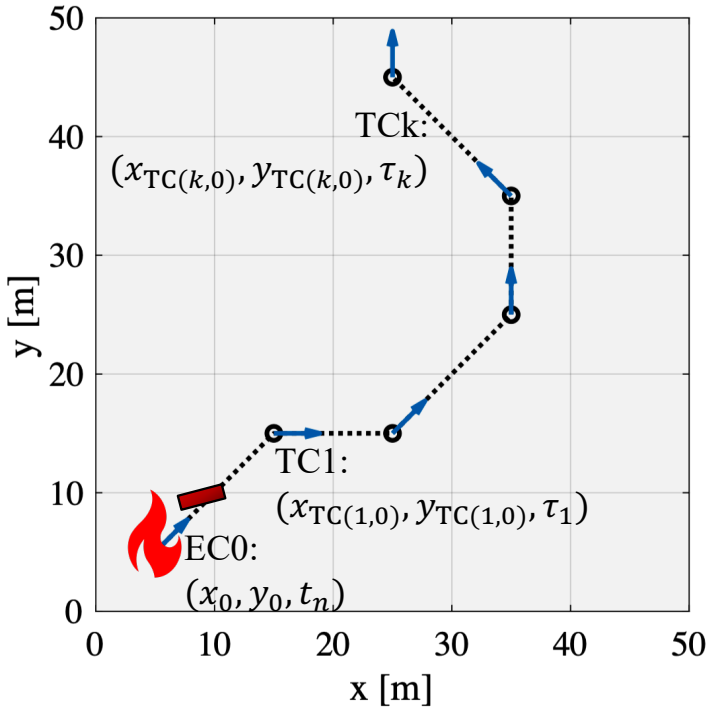


Figure 4.2: Schematic of the two-dimensional planar domain discretized into an array of uniform computational cells in an $(x - y)$ Cartesian coordinate system ($\Delta x = \Delta y = 10$ m). At time t_n, the fire reaches cell EC0 located at (x_0, y_0) and begins emitting firebrands. Firebrands travel a distance of Δx at each firebrand trajectory solver timestep $\Delta\tau$ (reaching cell TC1 at τ_1, cell TCk at τ_k, etc.). Blue arrows indicate the local wind direction at each cell center.

We focus on a specific cell in the domain, denoted as EC0, which emits firebrands between times $t = t_{\text{Start}}$ and $t = t_{\text{End}}$. The generation rate of firebrands, $GR(t)$ (units: $\text{pcs} \cdot \text{s}^{-1}$), is

time-dependent and assumed known—either user-prescribed or derived from a physical firebrand generation model as described in Chapter 3.

The objective is to compute the transport and deposition rates of these firebrands onto other cells within the domain. In addition to $GR(t)$, the input data for the model include the spatial and temporal variations of the wind velocity field $(u_{\text{wind}}(x, y, t), v_{\text{wind}}(x, y, t))$.

In the two-dimensional computational mesh, wind velocities are stored at the centers of cells, with the center coordinates of a generic cell Ck denoted by (x_{Ck}, y_{Ck}). Temporal discretization is applied consistently across all fields, and GR, u_{wind}, and v_{wind} are provided at discrete simulation times t_n, resulting in quantities $GR(t_n)$ and $(u_{\text{wind}}(x_{Ck}, y_{Ck}, t_n), v_{\text{wind}}(x_{Ck}, y_{Ck}, t_n))$.

To ensure numerical stability and accuracy in the firebrand transport simulation, the time step Δt_n is constrained by a wind-based Courant–Friedrichs–Lewy (CFL) condition, formulated as:

$$\Delta t_n = CFL_{t_n} \times \left(\frac{\Delta x}{\max\left(u_{\text{wind}}(x_{Ck}, y_{Ck}, t_n), v_{\text{wind}}(x_{Ck}, y_{Ck}, t_n)\right)} \right) \tag{4.13}$$

where CFL_{t_n} is the CFL number and is set to 0.5 in the present framework.

We now consider a certain time interval Δt_n starting at time t_n and during which firebrands are emitted from cell $EC0$, $t_{Start} \leq t_n < (t_n + \Delta t_n) \leq t_{End}$. The total number of firebrands emitted by cell $EC0$ during Δt_n is $(GR(t_n) \times \Delta t_n)$. We want to calculate the landing locations, landing times and number distribution of this group of firebrands. The sequence of steps used in the algorithm discussed below is presented in Figure 4.3 (left panel).

The calculation of the firebrand trajectories uses a special temporal discretization that is different from the one that corresponds to the values t_n. The new temporal discretization is designed based on a specific condition that provides control of flight distance during each time

Eulerian model

Loop on discretized time

Update time t_n

Calculate time increment Δt_n *based on* CFL_{t_n} *condition*

Loop on cells that emit firebrands

Update coordinates of emitting cell EC0

Calculate PDFs $pdf_X(X)$, $pdf_Y(Y)$

Loop on trajectory of firebrands

- Update time $\tau = \tau_{k+1}$
- Calculate coordinates of target cell TC$(k+1, k')$ according to Eqns. 4.15, 4.16, 4.18, 4.19
- Record time of arrival $TOA_k = t_n + \tau_{k+1}$
- Update number of deposited firebrands in cell TC$(k+1, k')$ according to Eq. (15): $N_{t_n+\tau_{k+1}} = N_{t<t_n+\tau_{k+1}} + \Delta N_{t_n+\tau_{k+1}}$
- Stop loop if $k = k_{max}$

Lagrangian model

Loop on discretized time

Update time t_n

Calculate time increment Δt_n *based on* CFL_{t_n} *condition*

Loop on cells that emit firebrands

Update coordinates of emitting cell EC0

Calculate PDFs $pdf_X(X)$, $pdf_Y(Y)$

Loop on computational particles

- Randomly select (X_p, Y_p), the flight distances of the computational particle in the downwind and crosswind directions; the selection process uses an inverse transform sampling technique [20] and is based on $pdf_X(X)$, $pdf_Y(Y)$ Calculate coordinates of target cell TC$(k+1, k')$ according to Eqns. 4.15, 4.16, 4.18, 4.19
- **Loop on trajectory of firebrands**
  - *Calculate the trajectory of the computational particle from EC0 to cell TC corresponding to* (X_p, Y_p)
  - *Record time of arrival* $TOA = t_n + \tau_p$
  - *Update number of deposited firebrands in cell TC according to Eq. (16):* $N_{t_n+\tau_p} = N_{t<t_n+\tau_p} + \Delta N_{t_n+\tau_p}$

Figure 4.3: Sequence of steps used in the Eulerian (left figure) and Lagrangian (right figure) versions of the firebrand transport algorithm.

interval. Note that we adopt here a simple two-dimensional model for firebrand trajectory that assumes that firebrands move horizontally at the speed of the wind evaluated at 6.1 m (20 feet) elevation. The model does not consider variations in elevation due to lofting or gravity. While this is clearly a crude treatment, we note that more elaborate descriptions of the transport of firebrands could readily be introduced into the present framework; we consider this issue beyond the scope of the present paper (it will be addressed in future work).

Let us illustrate the algorithm by considering the kth-time step in the calculation of the firebrand trajectories, $1 \leq k \leq (k_{max} - 1)$. At time $\tau = \tau_k$, the firebrands are located inside cell $TC(k, 0)$, characterized by a cell center with coordinates $(x_{TC(k,0)}, y_{TC(k,0)})$. The duration of the time step is based on the following condition that uses local wind information at cell $TC(k, 0)$

obtained at time t_k:

$$\Delta\tau_k = \frac{CFL_\tau \times \Delta x}{\sqrt{u^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k) + v^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k)}} \tag{4.14}$$

where t_k is the closest time to $\tau = \tau_k$ in the discretization of the time variations of the wind field, and where CFL_τ is the Courant-Friedrichs-Lewy number of the firebrand trajectory algorithm. The equations of firebrand motion can now be integrated as follows:

$$x_p(\tau = \tau_{k+1}) = x_p(\tau = \tau_k) + u_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k) \times \Delta\tau_k \tag{4.15}$$

$$y_p(\tau = \tau_{k+1}) = y_p(\tau = \tau_k) + v_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k) \times \Delta\tau_k \tag{4.16}$$

where $\tau_{k+1} = \tau_k + \Delta\tau_k$, which corresponds to a transport of firebrands by a distance $\Delta s_{k\to k+1} = (u^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k) + v^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k))^{1/2} \times \Delta\tau_k = CFL_\tau \times \Delta x$ in the direction parallel to the wind velocity vector $(u_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k), v_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k))$. The coordinates $(x_p(\tau = \tau_{k+1}), y_p(\tau = \tau_{k+1}))$ belong to a computational cell called target cell $TC(k+1,0)$ (note that cell $TC(k+1,0)$ may be the same cell as, or may be a different cell from cell $TC(k,0)$).

We now evaluate the deposition of firebrands in a sequence of two steps. In the first step, we neglect the cross-wind firebrand distribution and assume $k'_{max} = 0$ and $(pdf_Y(0) \times \Delta y) = 1$. At arrival time $\tau = \tau_{k+1}$, the firebrands have been transported to cell $TC(k+1,0)$, the downwind travel distance is $S_{k+1} = S_k + CFL_\tau \times \Delta x$, and the total number of firebrands emitted by cell EC0 during Δt_n and deposited in cell $TC(k+1,0)$ can be approximated as $(GR(t_n) \times \Delta t_n) \times (pdf_X(S_{k+1/2}) \times \Delta x)$, where the product $(pdf_X(S_{k+1/2}) \times \Delta x)$ is the probability

that the firebrands land at a downwind distance $S_k \leq X < S_(k+1)$ from the center of cell $EC0$. To avoid repeated interpolations of stored values of $pdf_X(X)$, we use $CFL_\tau = 1$. Under those conditions, the total number of firebrands emitted by cell $EC0$ during Δt_n and deposited in cell $TC(k+1,0)$ can be approximated as:

$$(GR(t_n) \times \Delta t_n) \times (pdf_X(X_{k+1/2}) \times \Delta x) \tag{4.17}$$

In the second step, we account for the cross-wind firebrand distribution and assume $k'_{max} \geq 0$. At arrival time $\tau = \tau_{k+1}$, the firebrands are assigned to $(2k'_{max}+1)$ positions, located along a straight line passing through the central point $(x_p(\tau = \tau_{k+1}), y_p(\tau = \tau_{k+1}))$ and normal to the local wind direction; the corresponding coordinates are:

$$x_p(Y_{k'}) = x_p(\tau = \tau_{k+1}) - Y_{k'} \times \frac{v_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k)}{\sqrt{u^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k) + v^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k)}} \tag{4.18}$$

$$y_p(Y_{k'}) = y_p(\tau = \tau_{k+1}) + Y_{k'} \times \frac{u_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k)}{\sqrt{u^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k) + v^2_{wind}(x_{TC(k,0)}, y_{TC(k,0)}, t_k)}} \tag{4.19}$$

where $Y_{k'} = (k'\Delta y)$, $(-k'_{max}) \leq k' \leq k'_{max}$. The coordinates $(x_p(Y_{k'}, y_p(Y_{k'})$ belong to a computational cell called target cell $TC(k+1,k')$. At arrival time $\tau = \tau_{k+1}$, the firebrands have been transported to cell $TC(k+1,k')$, the downwind travel distance is S_{k+1}, and the total number of firebrands emitted by cell EC0 during Δt_n and deposited in cell $TC(k+1,k')$ can be

approximated as:

$$\Delta N_{t_n+\tau_{k+1}} = (GR(t_n) \times \Delta t_n) \times (pdf(X_{k+1/2}, Y_{k'}) \times \Delta x \times \Delta y) \tag{4.20}$$

where $(pdf(X_{k+1/2}, Y_{k'}) \times \Delta x \times \Delta y) = ((pdf_X(X_{k+1/2}) \times \Delta x) \times (pdf_Y(Y_{k'}) \times \Delta y))$ is the probability that the firebrands land at a downwind distance $S_k \leq X < S_(k+1)$ from the center of cell $EC0$, and at a cross-wind distance $((k'-1/2)\Delta y) \leq Y < ((k'+1/2)\Delta y)$ from the center of cell $TC(k+1, 0)$.

This calculation is repeated until $S_(k+1) \geq X_{max}$, *i.e.*, until $k = (k_{max} - 1)$.

Thus, for each emitting cell $EC0$ and for each time interval $t_n \leq t < (t_n + \Delta t_n)$, the firebrand algorithm saves the coordinates of the target cells $TC(k, k')$ (*i.e.*, the landing locations), the times of arrival $TOA_k = \tau_k$ (*i.e.*, the landing times), and the total number of firebrands deposited, $\Delta N_{t_n+\tau_k}$ (*i.e.*, the firebrand accumulation).

4.3 Presentation of the Lagrangian (discrete particle) model

The Lagrangian version—or more accurately, the discrete particle version—of the firebrand transport algorithm is conceptually similar to the Eulerian formulation discussed earlier. The sequence of steps used in the Lagrangian algorithm is illustrated in Figure 4.3 (right panel).

The principal difference compared to the Eulerian algorithm described in Section 4.2 is that the probability density function (PDF) of flight distance, $\text{pdf}(X, Y) = \text{pdf}_X(X) \times \text{pdf}_Y(Y)$, is now represented by a statistical ensemble of computational particles. The firebrand algorithm is executed independently for each particle, sampling its landing distance and flight time according to the prescribed PDFs.

It is worth noting that the original firebrand transport model implemented in ELMFIRE is conceptually close to the Lagrangian model presented here. However, one key difference is that the original model does not account for flight time delay: in ELMFIRE's earlier implementation, the time of arrival (TOA) for a firebrand at a target cell is assumed to be the emission time t_n, i.e., $\text{TOA}_k = t_n$.

The total number of firebrands emitted by cell $EC0$ during Δt_n is $n = (GR(t_n) \times \Delta t_n)$. The total number of computational particles emitted by cell $EC0$ during Δt_n is $n_p = (GR_p(t_n) \times \Delta t_n)$, where GR_p is the generation rate of computational particles. We typically choose $n_p \ll n$ and each computational particle p in the ensemble represents a group of (n/n_p) physical firebrands. Each particle p, $1 \leq p \leq n_p$, is associated with randomly-selected values of downwind and cross-wind travel distance, noted X_p and Y_p; the random selection of X_p and Y_p is informed by the presumed PDFs of flight distance, *i.e.*, values of X_p and Y_p that correspond to high (low) values of $pdf_X(X)$ and $pdf_Y(Y)$ are more (less) likely to be selected, respectively; the trajectory of the particle p is calculated following the algorithm described above; the particle is deposited on a target cell $TC(k+1, k')$ located at a downwind distance X_p from cell $EC0$ and at a cross-wind distance Y_p from cell $TC(k+1, 0)$, at the time of arrival $TOA_k = \tau_p$, and the total number of firebrands deposited is:

$$\Delta N_{t_n+\tau_p} = (n/n_p) = (GR(t_n) \times \Delta t_n) \times (1/(GR_p(t_n) \times \Delta t_n)) \tag{4.21}$$

Thus, comparing the expressions in Eqns. 4.20 and 4.21, it is seen that the factor $(1/(GR_p(t_n) \times \Delta t_n))$ used in the discrete particle version of the firebrand algorithm plays the role of the weight $((pdf_X(X_{k+1/2}) \times \Delta x) \times (pdf_Y(Y_(k')) \times \Delta y))$ used in the Eulerian version. At

large values of $GR_p(t_n)$, there will be multiple particles emitted by cell $EC0$ and landing on any target cell, and one can expect the two expressions 4.20 and 4.21 to be equivalent. In contrast, at low values of $GR_p(t_n)$, there will be at most one particle emitted by cell $EC0$ and landing on any target cell, and one can expect the two expressions 4.20 and 4.21 to be quite different. In other words, the discrete particle version of the firebrand algorithm is characterized by a greater computational cost (the algorithm is run n_p times instead of once) and a lower level of accuracy due to statistical noise at low values of GR_p. Surprisingly, despite its lower performance, the discrete particle version is the dominant approach in current fire risk models.

4.4 Verification Tests

This section presents results that show the accuracy and robustness of the proposed framework. Verification tests are first conducted in MATLAB for a 1-D wildland fire configuration with ignition probability of 1 and $t_{ignition} = 0$. Then a 2-D demonstration using MATLAB is presented to further illustrate the model's performance and the significance of the ignition model.

4.4.1 Numerical Robustness of Discretization Schemes (Test 4-1)

We now turn to a simple one-dimensional verification test, referred to as Test 4-1, conducted using the modified ELMFIRE level-set model, implemented in MATLAB for ease of analysis. The Test 4-1 configuration is similar to that already considered in Figures 2.3 and 2.4, but now utilizes the full version of the firebrand transport algorithm: firebrands are emitted from all active fire locations at a fixed rate for a fixed emission duration, $t_{\text{Emission}} = (t_{\text{End}} - t_{\text{Start}}) = \Delta t_n$.

The numerical setup for Test 4-1 adopts a spatial resolution of $\Delta x = 10$ m and $\Delta y = 1$

m. The time step is $\Delta t_n \approx 0.745$ s, corresponding to a wind-based Courant–Friedrichs–Lewy (CFL) number of $CFL_{t_n} = 0.5$. The fuel model used is grassland fuel model #102 (GR2). The wind speed is set as $u_{\text{wind}} = 6.71$ m·s^{-1}. The surface fire spread rate is artificially set to zero, $ROS_{\text{surface}} = 0$ m·s^{-1}, to facilitate interpretation by isolating the firebrand transport process. The fireline intensity is specified as $\dot{Q}'_{\text{fire}} = 3189$ kW·m^{-1}.

The firebrand generation rate is fixed at $GR = 100 \text{ pcs} \cdot \Delta \text{t}_\text{n}^{-1}$. This choice is based on the estimates derived in Chapter 3, where the normalized firebrand generation rate per unit fire power was estimated as $GR^* = 33.3 \text{ pcs} \cdot \text{MW}^{-1} \cdot \text{s}^{-1}$. Given the fireline intensity and $\Delta y = 1$ m, the heat release rate of a burning cell becomes $\dot{Q}_{\text{fire}} = \dot{Q}'_{\text{fire}} \times \Delta y = 3189$ kW. This yields a corresponding firebrand generation rate of approximately $GR = GR^* \times \dot{Q}_{\text{fire}} \approx 33.3 \times 3.189 \approx 106$ pcs·s^{-1}, or approximately 79 pcs·Δt_n^{-1}.

The configuration for Test 4-1 is schematically illustrated in Figure 4.4. At the initial time step t_1, the fireline is located at $x = 5$ m, emitting firebrands that are advected by the wind and land downwind to ignite new spot fires. The leading edge of the fire, denoted $x_{\text{LE}}(t_1)$, corresponds to the farthest spot fire created by firebrands (top panel of Figure 4.4). At the next time step t_2, newly ignited cells begin to emit firebrands themselves, and the fire leading edge advances to a new location $x_{\text{LE}}(t_2)$ (bottom panel of Figure 4.4). The ambient wind is assumed to blow uniformly from left to right.

The configuration is designed to provide simple, analytically verifiable results. Beyond a certain distance from the ignition location, specifically at $x \geq (X_{\text{max}} + 5)$ m, where ignition initially occurs at $x = 5$ m, the cumulative number of firebrands deposited should reach $N_\infty = GR \times \Delta t_n = 100$ pcs. The time of arrival (TOA) of firebrands at a given location x can be

predicted as:

$$TOA(x) = \frac{x}{u_{\text{wind}}}, \tag{4.22}$$

and the effective fire rate of spread induced by successive spotting events equals the ambient wind speed,

$$ROS = u_{\text{wind}}. \tag{4.23}$$

The leading edge of the fire front evolves according to

$$x_{\text{LE}}(t) = u_{\text{wind}} \times t. \tag{4.24}$$

Furthermore, in this configuration featuring a uniform and steady wind field, all firebrands emitted from upwind locations land at downstream cells simultaneously. Consequently, the fire front has no depth: the trailing edge coincides with the leading edge, $x_{\text{TE}} = x_{\text{LE}}$, and the fire depth is zero, $d_{\text{fire}} = 0$.

Figure 4.5 presents the PDF of the firebrands landing locations in the wind direction, $pdf_X(X)$. With $\Delta x = 10$ m and $\varepsilon = 0.01$, the PDF is truncated at $X_{max} = 150$ m and discretized into 15 bins, $k_{max} = 15$.

Figure 4.6 presents the main features of the numerical solution of Test 4-1 obtained with the Eulerian version of the firebrand algorithm. Figure 4.6a presents the spatial variations of the total number of accumulated embers on the ground and shows that the numerical results are in excellent agreement with the exact solution, $N_\infty = 100\, pcs$ for $x \geq 150\, m$ (N takes lower values at smaller x-locations because of the missing contributions of cells located at $x < 0$).

Figure 4.6b presents the spatial variations of the landing time of the firebrands, TOA, and

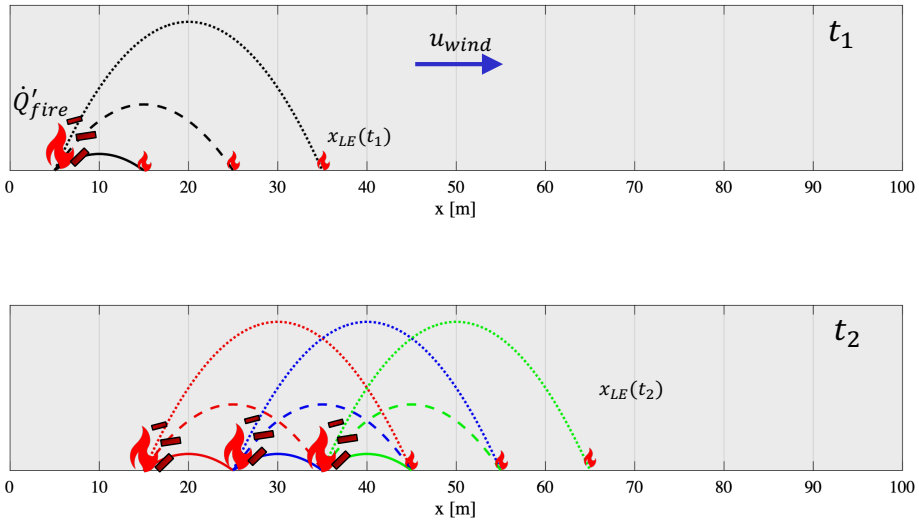


Figure 4.4: Schematic configuration for the 1-D verification tests of the firebrand transport scheme. **Top panel**: Fireline location at time t_1, where the primary surface fireline (fireline intensity $\dot{Q}'_{\text{fire}}$) is located at $x = 5$ m, emitting firebrands (brown rectangles) that land downwind and ignite new spot fires. The leading edge of the fire, $x_{\text{LE}}(t_1)$, corresponds to the farthest ignited spot fire. **Bottom panel**: Fire evolution at the next time t_2, where newly ignited cells begin to emit firebrands, further advancing the leading edge to $x_{\text{LE}}(t_2)$. The ambient wind direction is left to right.

shows that the numerical results are in excellent agreement with the exact solution, $TOA(x) = (x/u_{wind}) = [x/(u_{wind}\Delta t_n)] \times \Delta t_n$. The exact solution is plotted as a solid line in Figure 4.6b.

A different perspective is adopted in Figure 4.6c which presents the temporal variations of the fire leading edge position, $x_{LE}(t)$. The numerical results are again in excellent agreement with the exact solution, $x_{LE}(t) = (u_{wind} \times t)$, but the figure now shows the effect of discretization: in a sequence that repeats itself, during a first time step, the fire remains within the same computational cell and the simulated fire does not spread; during the next time step, the fire propagates to the downwind adjacent cell and the fire spreads by a distance Δx. This binary behavior is a direct consequence of the time resolution scheme: with the choice $CFL_{t_n} = 0.5$, during one time step Δt_n, firebrands move over a distance $(u_{wind} \times \Delta t_n) = (CFL_{t_n} \times \Delta x) = (\Delta x/2)$.

Figure 4.6d presents the corresponding temporal variations of the rate of spread of the fire,

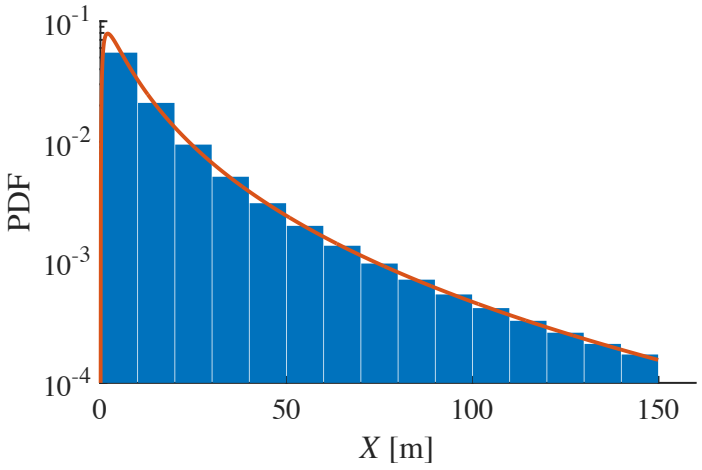


Figure 4.5: Probability density function of ember flight distance in the downwind direction, $pdf_X(X)$. The histogram corresponds to the present numerical model; the solid line corresponds to the analytical function due to Sardoy et al. [83].

$ROS(t)$. The rate of spread can be directly extracted from the results presented in Figure 4.6c and calculated as the time derivative of the variations $x_{LE}(t)$, $ROS = d(x_{LE})/dt$. Figure 4.6d shows that while due to discretization errors, the instantaneous values of simulated ROS fluctuate between 0 and $(\Delta x/\Delta t_n) \approx 13.4\ m \cdot s^{-1}$, the average value is very close to the exact solution, $ROS = u_{wind}$.

We now turn to the numerical solution of Test 4-1 obtained with the Lagrangian version of the firebrand algorithm. Figure 4.7 presents numerical results obtained with a large value of the generation rate of computational particles, $GR_p = 10\ pcs \cdot \Delta t_n^{-1}$ (each computational particle p in the ensemble represents a group of 10 physical firebrands). Under those conditions, it is found that the discrete particle model provides very good results, *i.e.*, results that are similar to those obtained with the Eulerian version of the firebrand algorithm. The main difference is seen in Figure 4.7a that shows significant errors in the prediction of the total number of accumulated embers on the ground. In the present case where $P_{ign} = 1$, these errors have no consequence on the simulated rate of spread of the fire (but we anticipate that in the future, a physics-based ignition

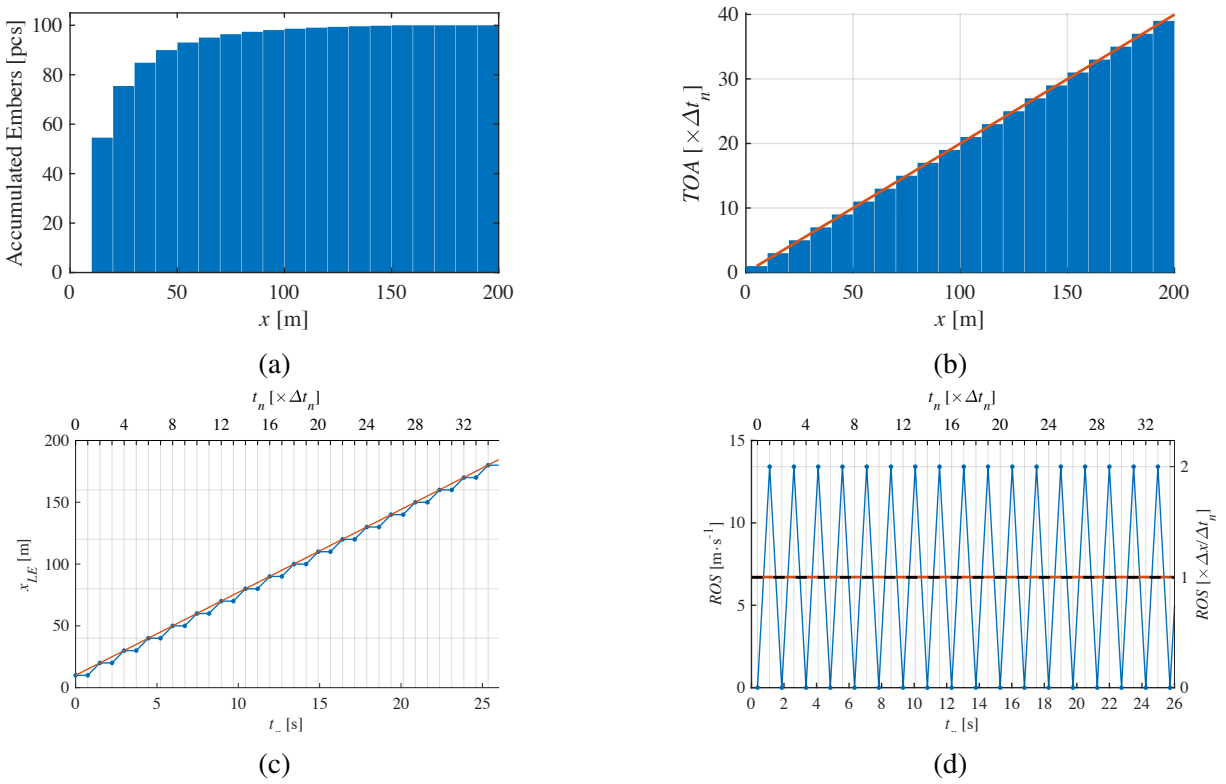


Figure 4.6: Simulation of Test 4-1 performed with the Eulerian version of the firebrand algorithm. Spatial variations of: 4.6a the total number of accumulated embers on the ground; 4.6b the time of arrival of the firebrands, TOA (TOA is plotted in units of number of time steps). Temporal variations of: 4.6c the fire leading edge position, x_{LE}; 4.6d the fire rate of spread, ROS. In figures 4.6c and 4.6d, the solid line with (without) symbols corresponds to the numerical (exact) solution.

model will use the information on firebrand accumulation and errors in firebrand accumulation may negatively impact the accuracy of the simulated spread rate).

Figure 4.8 presents numerical results obtained with a smaller value of the generation rate of computational particles, $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$(each computational particle p in the ensemble represents a group of 100 physical firebrands). Under those conditions, it is found that the discrete particle model provides results that are inaccurate due to statistical noise. Figure 4.8a shows significant errors in the prediction of the total number of accumulated embers on the ground with many computational cells collecting no firebrand. Figures 4.8b and 4.8c show that even though

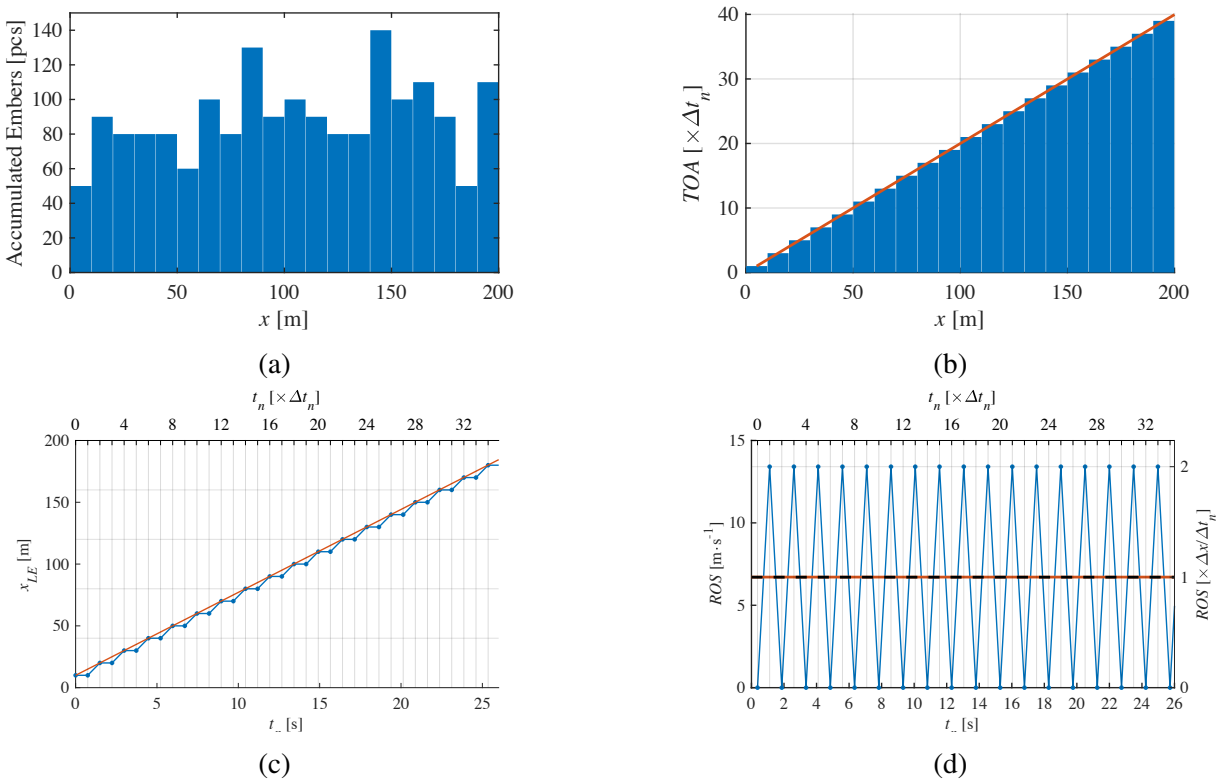


Figure 4.7: Simulation of Test 4-1 performed with the Lagrangian version of the firebrand algorithm and $GR_p = 10\ pcs \cdot \Delta t_n^{-1}$. See the caption of Figure 4.6

the firebrands are not correctly distributed, they are transported at the correct wind velocity and the corresponding landing times and locations remain accurate. Figure 4.8d shows that the instantaneous values of simulated ROS are 0, $(\Delta x/\Delta t_n) \approx 13.4\ m \cdot s^{-1}$ or $(2 \times \Delta x/\Delta t_n) \approx 26.8\ m \cdot s^{-1}$; the time-averaged value is close to $2\ m \cdot s^{-1}$ and is in significant error (this value is visualized as a horizontal dashed line in Figure 4.8d). In addition, the fire is artificially thickened by numerical errors and the global structure of the fire is incorrect: $x_{TE} = 10\ m$ and d_{fire} increases without bounds.

The negative impact of the statistical noise observed in the Lagrangian version of the firebrand algorithm is greatly decreased when considering scenarios in which the duration of firebrand emission is longer than that assumed until now in Test 4-1. A possible expression

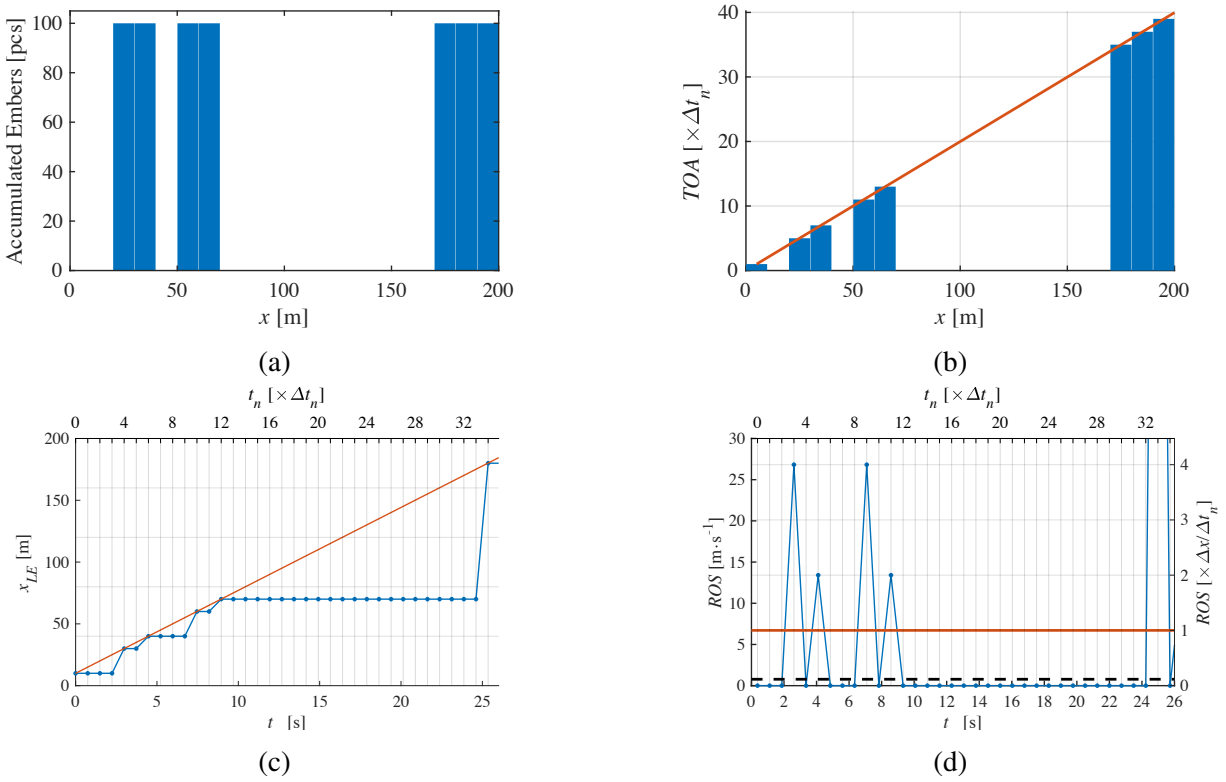


Figure 4.8: Simulation of Test 4-1 performed with the Lagrangian version of the firebrand algorithm and $GR_p = 1 \text{ pcs} \cdot \Delta \text{t}_\text{n}^{-1}$. See the caption of Figure 4.6

for the emission duration is obtained by assuming that it is equal to the fire residence time inside a given computational cell, $t_{Emission} \approx (\Delta x / ROS_{surface})$, which in Test 4-1, leads to $t_{Emission} \approx (10/0.71) \approx 14\ s$. In order to study the effect of $t_{Emission}$, we now turn to a modified version of the Test 4-1 configuration with $t_{Emission} = 10\ s$. Under those modified conditions, it is found that the discrete particle model provides much-improved results (Figure 4.9). Figure 4.9a shows the presence of residual errors in the prediction of the total number of accumulated embers on the ground; the exact value (at $x \geq 155\ m$) is $N_\infty = (GR \times t_{Emission}) = 1340\ pcs$ and the results feature simulated values with relative errors up to 50% in magnitude. But we note that all computational cells collect firebrands. In addition, the landing times and locations of firebrands are correctly simulated (Figures 4.9b and 4.9c). Figure 4.9d shows that the instantaneous values

of the simulated rate of spread of are 0, $(\Delta x/\Delta t_n) \approx 13.4\ m \cdot s^{-1}$ or $(2 \times \Delta x/\Delta t_n) \approx 26.8\ m \cdot s^{-1}$; and the time-averaged value is close to the exact value, $ROS = u_{wind}$ (this value is visualized as a horizontal dashed line in Figure 4.9d).

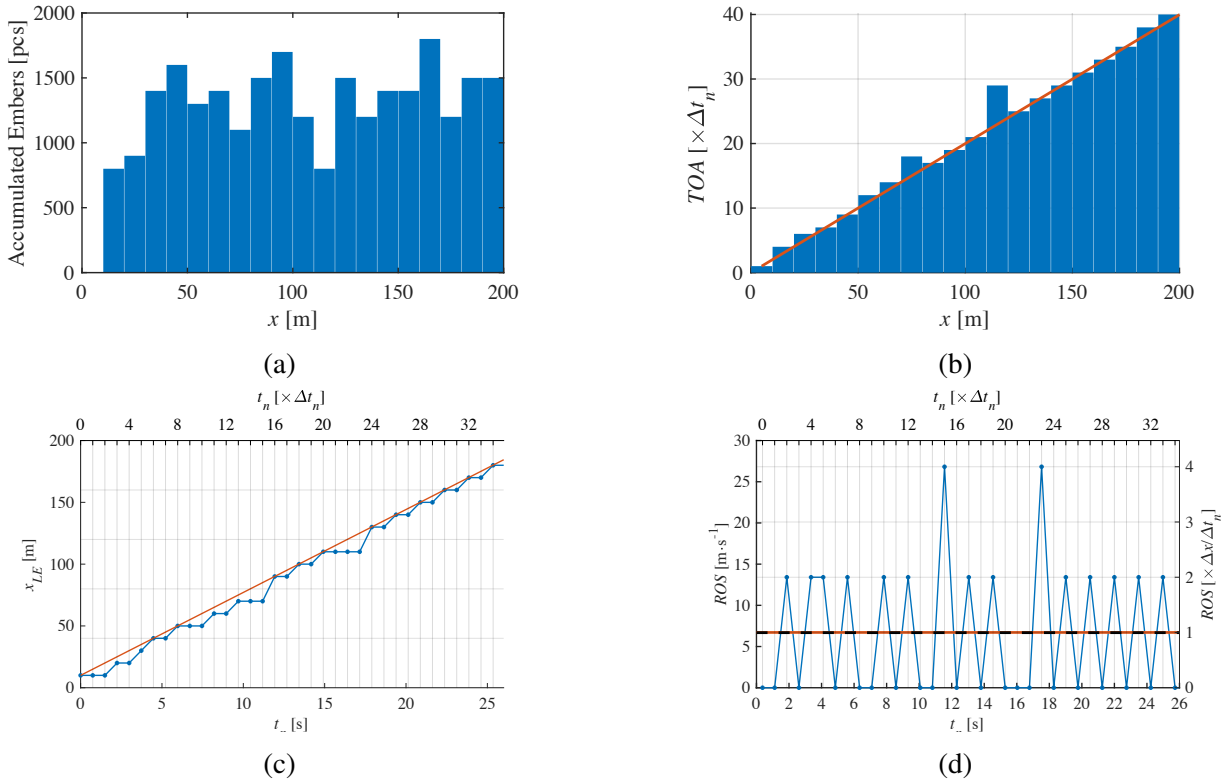


Figure 4.9: Simulation of modified Test 4-1 ($t_{Emission}$=10 s) performed with the Lagrangian version of the firebrand algorithm and $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$. See the caption of Figure 4.6

Thus, the errors in the Lagrangian version of the firebrand algorithm are decreased in simulations featuring longer durations of firebrand emission, and can be further decreased by selecting larger values of the generation rate of computational particles GR_p.

Lastly, we consider a variant of the Lagrangian firebrand transport model, corresponding to the default implementation used in earlier versions of ELMFIRE and FlamMap, where firebrands are instantaneously allocated to target locations at the next time step when the source cell ignites. Figure 4.10 presents the corresponding simulation results for this variant, with a particle

generation rate $GR_p = 1 \text{ pcs} \cdot \Delta \text{t}_\text{n}^{-1}$ and an emission duration $t_{\text{Emission}} = 10$ s.

As shown in Figure 4.10a, the total number of firebrands deposited across the domain is not accurately resolved, consistent with the statistical noise issues discussed previously. However, in contrast to the time-resolved Lagrangian version demonstrated earlier, the arrival time of firebrands is now significantly underestimated, as illustrated in Figure 4.10b. Firebrands are assumed to arrive immediately after the ignition of the source cell, regardless of the physically required flight time.

Consequently, as shown in Figures 4.10c and 4.10d, the fire leading edge rapidly advances across the simulation domain within just a few seconds, resulting in an effective rate of spread (ROS) on the order of 100 m·s^{-1}. Such a value is unrealistically high and physically meaningless in typical wildland or WUI fire scenarios.

Importantly, this systematic error cannot be corrected by increasing the emission duration t_{Emission} or the particle generation rate GR_p. The problem arises fundamentally from the lack of firebrand flight time modeling. In this variant, firebrands are deposited instantaneously over large distances based solely on their sampled travel distances, without accounting for the realistic time it would take to traverse those distances under the influence of wind.

Thus, although this variant may appear computationally simpler, it introduces substantial biases in both the spatial and temporal evolution of the fire spread. These deficiencies reinforce the necessity of incorporating explicit firebrand travel time tracking in order to produce numerically and physically accurate simulations of firebrand-driven spotting fires.

Given the observations above, we now proceed to a discussion of numerical errors. Both the Eulerian and Lagrangian (firebrand travel time accounted) formulations contain numerical errors associated with spatial and temporal discretization. These errors are controlled by suitable

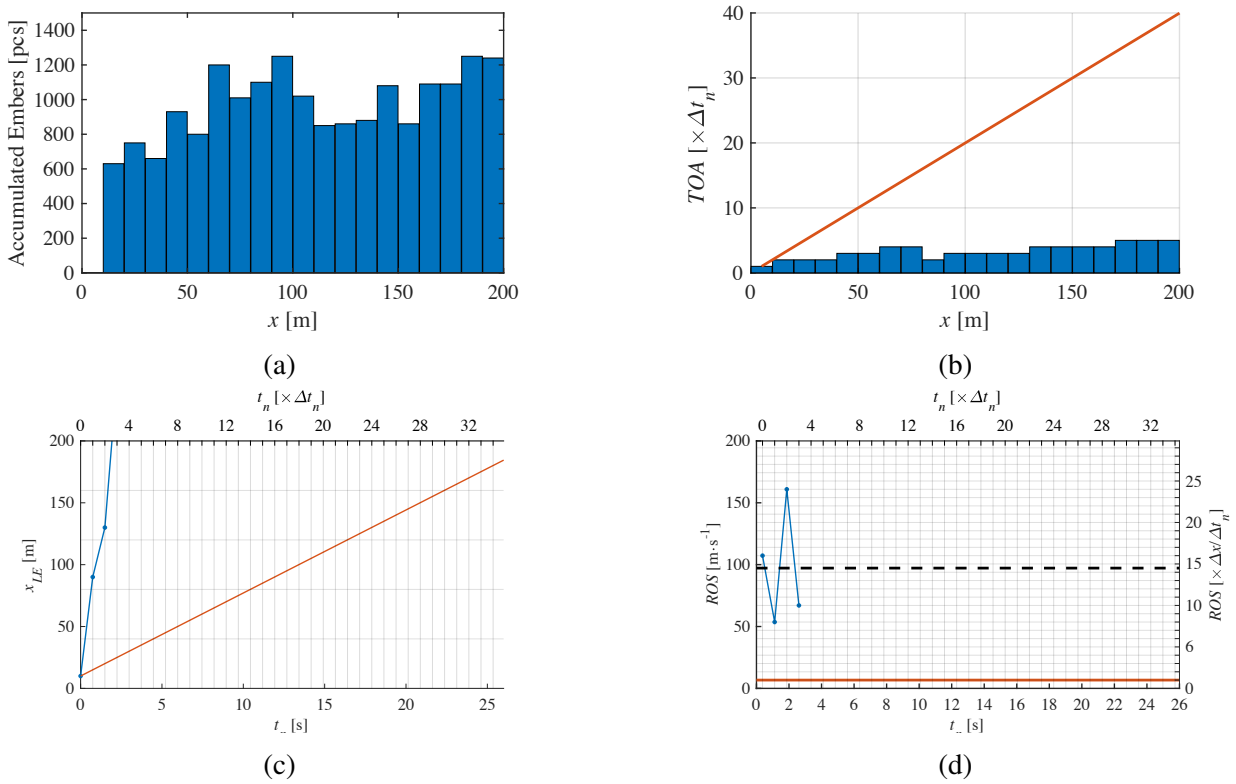


Figure 4.10: Simulation of modified Test 4-1 ($t_{\text{Emission}} = 10$ s) performed with the Lagrangian version of the firebrand algorithm similar to the original implementation in ELMFIRE and FlamMap, with $GR_p = 1\ \text{pcs} \cdot \Delta\text{t}_\text{n}^{-1}$. (a) Total number of accumulated embers on the ground; (b) Time of arrival of firebrands; (c) Fire leading edge position; (d) Rate of spread (ROS). See the caption of Figure 4.6 for additional simulation settings.

choices of spatial and temporal resolution. In Test 4-1, the choice of spatial resolution is guided by the need to resolve the maximum flight distance characterized by X_{max}, $\Delta x < X_{max}$; the choice of temporal resolution is guided by the wind-based CFL condition, $CFL_{t_n} = (u_{wind} \times \Delta t_n/\Delta x) < 1$. Figure 4.11 presents the mean (time-averaged) value of the rate of spread of the fire obtained in a series of simulations of the modified Test 4-1 configuration ($t_{Emission} = 10$ s) performed with the Eulerian firebrand model and with different choices of $\Delta x(\Delta x < X_{max})$ and $\Delta t_n(CFL_{t_n} \leq 1.2)$. It is found that the simulated ROS is within 10% of the exact value ($ROS = u_{wind}$) provided that $CFL_{t_n} \leq 0.8$. Consistent with these results, our baseline choice in ELMFIRE is $CFL_{t_n} = 0.5$.

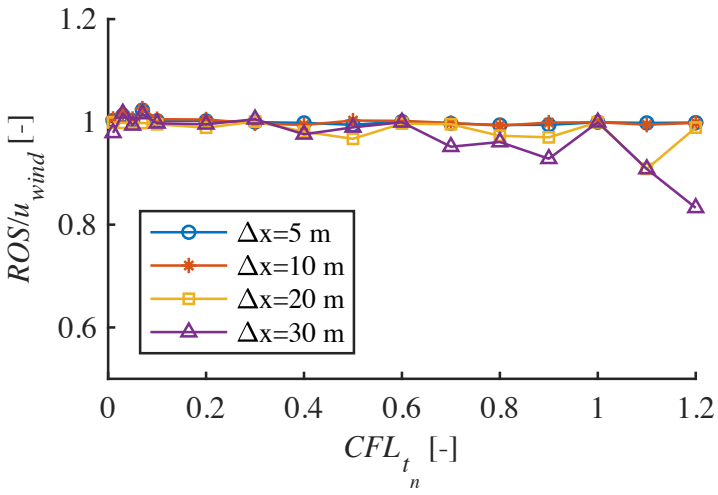


Figure 4.11: Simulations of modified Test 4-1 ($t_{Emission} = 10\ s$) performed with the Eulerian version of the firebrand algorithm. Variations of the time-averaged value of the simulated rate of spread of the fire (made non-dimensional by the exact solution, $ROS = u_{wind}$) with different choices of spatial and temporal resolution: $\Delta x = 5\ m$ (circle symbols); $\Delta x = 10\ m$ (asterisk symbols); $\Delta x = 20\ m$ (square symbols); $\Delta x = 30\ m$ (triangle symbols).

Similar to Figure 4.11, Figure 4.12 presents the variations of mean ROS with Δx and CFL_{t_n} for the Lagrangian firebrand model and $GR_p = 1\ pcs{\cdot}\Delta t_n^{-1}$. It is found that the numerical errors in the Lagrangian model are larger than those observed in the Eulerian model, and that for $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$, the simulated ROS is within 20% of the exact value provided that $CFL_{t_n} \leq 0.8$. Consistent with these results, our baseline choice in ELMFIRE is $CFL_{t_n} = 0.5$.

As pointed out earlier, the magnitude of the numerical errors in the Lagrangian version of the firebrand algorithm is sensitive to the choice of the generation rate of computational particles, GR_p. Figure 4.13 illustrates this point and shows that accurate simulations of the rate of fire spread require $GR_p \geq 1\ pcs{\cdot}\Delta t_n^{-1}$. These results are consistent with the discussion of Figures 4.7, 4.8 and 4.9. Our baseline choice in ELMFIRE is $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$.

However, the conclusions drawn earlier regarding numerical accuracy primarily apply to the Lagrangian version of the model where the flight time of firebrands is accounted for during

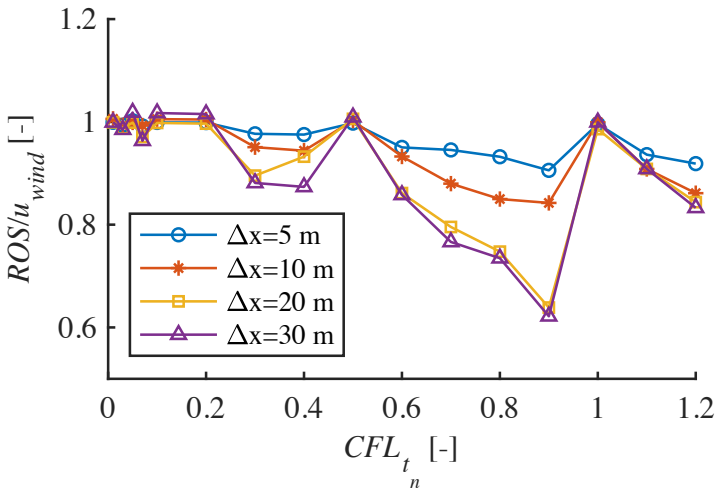


Figure 4.12: Simulations of modified Test 4-1 ($t_{Emission} = 10\ s$) performed with the Lagrangian version of the firebrand algorithm and $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$. See the caption of Figure 4.11

transport. Figure 4.14 shows results simulated under the same Test 4-1 conditions, but using a Lagrangian scheme without explicitly tracking the firebrand flight time delay. In this version, the arrival of firebrands at landing sites is assumed instantaneous, as originally implemented in earlier versions of ELMFIRE and FlamMap.

As illustrated in Figure 4.14, the resulting rate of spread (ROS) becomes unrealistically high and is bounded by the characteristic scale $X_{\text{max}}/\Delta t_n$, where X_{max} is the maximum travel distance defined by the truncated probability density function (PDF). Since firebrands can traverse up to X_{max} within a single time step, this scheme leads to erroneously accelerated fire spread.

This issue is particularly problematic in configurations where $P_{\text{ign}} = 1$ and no ignition delay is included, as spot fires are initiated immediately upon firebrand landing. While the numerical error introduced by this scheme may be less severe if the ignition delay is large compared to the firebrand flight time, significant inaccuracies persist in spatially and temporally resolving the deposition of firebrands. Consequently, predictions of ignition behavior—and by extension, fire spread dynamics—can be substantially distorted.

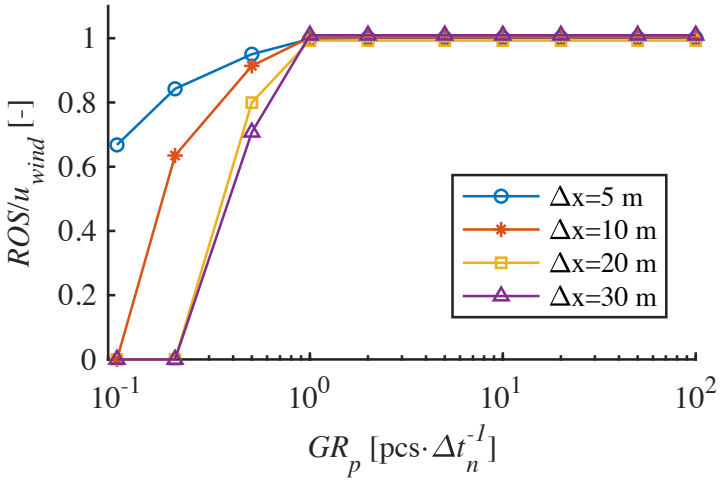


Figure 4.13: Simulations of modified Test 4-1 ($t_{Emission} = 10\ s$) performed with the Lagrangian version of the firebrand algorithm and $CFL_{t_n} = 0.5$. Variations of the time-averaged value of the simulated rate of spread of the fire with different choices of spatial and particle resolution. See the caption of Figure 4.11.

In summary, it is found that the Eulerian firebrand model is accurate provided that $CFL_{t_n} \leq 0.8$, and that the Lagrangian firebrand model is also accurate provided that $CFL_{t_n} \leq 0.8$ and that $GR_p \geq 10pcs \cdot \Delta t_n^{-1}$ for $t_{Emission} = \Delta t_n$, or that $GR_p \geq 1\ pcs \cdot \Delta t_n^{-1}$ for $t_{\text{Emission}} \gg \Delta t_n$ when the firebrand travel time is accounted. It is worth mentioning that current fire risk models, generally based on a Lagrangian formulation, do not necessarily satisfy these conditions. For instance, in its baseline version, FARSITE [10, 11] uses $GR_p \approx 1\ pcs \cdot \Delta t_n^{-1}$ (the exact value will depend on a user-defined spotting probability) but also uses $t_{\text{Emission}} = \Delta t_n$; therefore, simulations of firebrands with FARSITE exhibit significant errors due to insufficient statistical sampling. Similarly, in its baseline version, FlamMap [15, 16] uses $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$ but also uses $t_{Emission} = \Delta t_n$; FlamMap also ignores the delay associated with firebrand flight; therefore, simulations of firebrands with FlamMap lead to incorrect estimates of the rate of spread due to spotting fires.

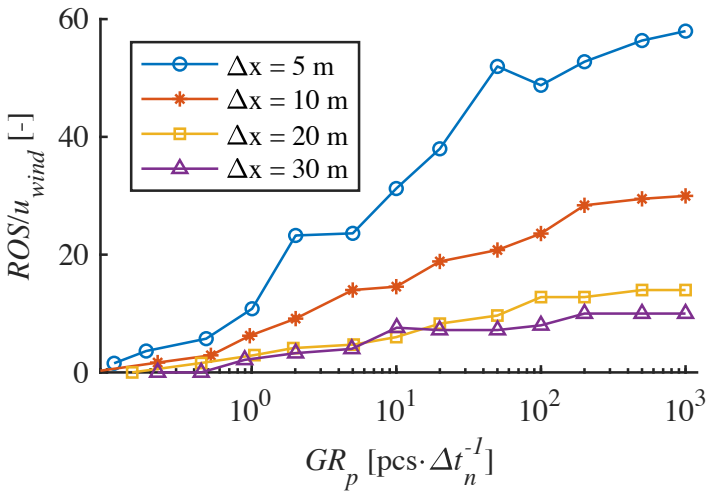


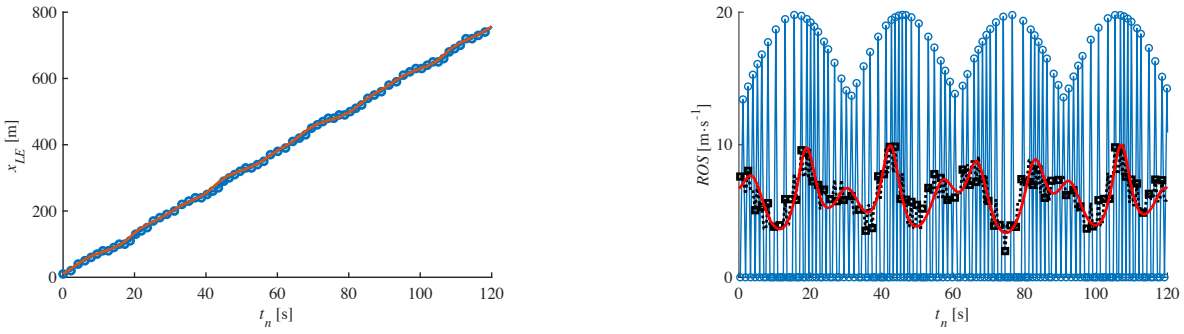
Figure 4.14: Simulations of modified Test 4-1 ($t_{\text{Emission}} = 10$ s) performed with the Lagrangian version of the firebrand algorithm without flight time tracking, similar to the original implementations in ELMFIRE and FlamMap. $CFL_{t_n} = 0.5$. The figure shows variations in the time-averaged rate of spread (ROS) as a function of the particle generation rate, under different choices of spatial and particle resolution. See the caption of Figure 4.11 for more details on simulation settings.

4.4.2 Numerical Accuracy of in the Time-Dependent Wind Field (Test 4-2&3)

We now consider a second one-dimensional verification test, called Test 4-2. The Test 4-2 configuration is identical to modified Test 4-1 ($t_{Emission} = 10\ s$) except for the use of a variable wind field: $u_{wind}(x,t) = \overline{u}_{wind} + A \times sin(2\pi x/\lambda_x) \times sin(2\pi t/T)$, where $\overline{u}_{wind} = 6.71\ m \cdot s^{-1}$, $A = 3.355\ m \cdot s^{-1}$, $\lambda_x = 100\ m$, and $T = 60\ s$. This test is intended to check the ability of the firebrand algorithms to correctly simulate spatio-temporal variations in the wind-field. The simulations are performed in MATLAB with $CFL_{t_n} = 0.5$ (note that Δt_n is now changing during the simulation, $\Delta t_n = CFL_{t_n} \times \Delta x/max(u_{wind}(x_{Ck}, t_n))$).

Figure 4.15 presents the main features of the numerical solution of Test 4-2 obtained with the Eulerian version of the firebrand algorithm. Figure 4.15a compares the simulated and exact temporal variations of the leading edge position of the fire; the exact solution is obtained by

numerical integration of the governing ordinary differential equation: $dx_{LE}/dt = u_{wind}(x_{LE}, t)$. Figure 4.15a shows that the numerical results are in excellent agreement with the exact solution. Figure 4.15b presents the corresponding variations of the rate of spread, ROS(t). The figure compares instantaneous values (which fluctuate between 0 and $(\Delta x/\Delta t_n)$), running-averaged values (calculated over a 9-point stencil, *i.e.*, over a duration of approximately 6 s) and the exact solution (obtained by numerical integration). Figure 4.15b shows that the simulated running average is in very good agreement with the exact solution.

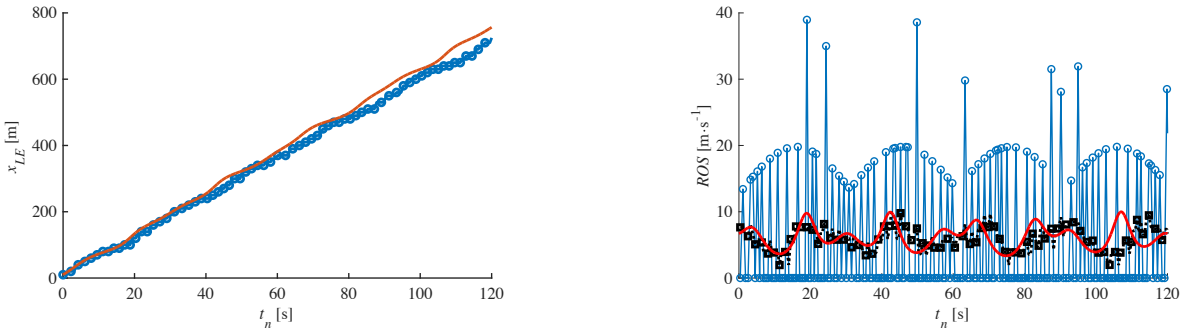


(a) (b)

Figure 4.15: Simulation of Test 4-2 performed with the Eulerian version of the firebrand algorithm. Temporal variations of: 4.15a the fire leading edge position, x_{LE}; 4.15b the fire rate of spread, ROS. In figure 4.15a, the solid line with (without) symbols corresponds to the numerical (exact) solution. In figure 4.15b, the solid line without symbols is the exact solution; the solid (dotted) line with symbols correspond to the instantaneous (running-averaged) values of simulated ROS.

Figure 4.16 presents similar results obtained with the Lagrangian version of the firebrand algorithm and $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$. Figure 4.16a shows that the numerical results are in fair agreement with the exact solution, but a discrepancy is clearly visible. Figure 4.16b shows that the simulated running average is in fair agreement with the exact solution but the details of the time variations of $ROS(t)$ are not captured correctly. As already noted in the discussion of Test 4-1, these errors can be decreased by selecting a larger value for GR_p, albeit at a higher

computational cost.



(a) (b)

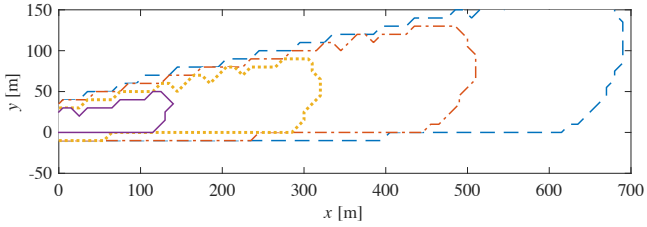
Figure 4.16: Simulation of Test 4-2 performed with the Lagrangian version of the firebrand algorithm and $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$. See the caption of Figure 4.15.

Regarding computational cost, as mentioned earlier, the firebrand algorithm is run multiple times in the Lagrangian formulation, once for each individual computational particle. Therefore, the cost of the Lagrangian model increases linearly with GR_p. It is found that for $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$, the cost of the Lagrangian model is similar to that of the Eulerian model, and that for larger values of GR_p, the Lagrangian model becomes computationally more expensive and therefore less competitive.

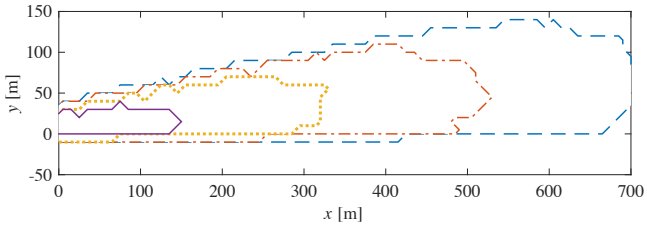
We now consider a two-dimensional demonstration test, called Test 4-3. The Test 4-3 configuration is an extension of Test 4-2 where $u_{wind}(x, y, t) = \overline{u}_{wind} + A \times sin(2\pi x/\lambda_x) \times sin(2\pi t/T)$, $v_{wind}(x, y, t) = \overline{v}_{wind} + B \times sin(2\pi x/\lambda_y) \times sin(2\pi t/T)$, with $\overline{u}_{wind} = 6.71\ m \cdot s^{-1}$, $A = 3.355\ m \cdot s^{-1}$, $\lambda_x = 100\ m$, $T = 60\ s$, $\overline{v}_{wind} = 1\ m \cdot s^{-1}$, $B = 0.5\ m \cdot s^{-1}$, and $\lambda_y = 100\ m$. In addition to a two-dimensional framework, Test 4-3 differs slightly from Test 4-2 as it considers a surface headfire with $ROS_{surface} = 0.71\ m \cdot s^{-1}$, and treats the duration of firebrand emission as $t_{Emission} = (\Delta x/ROS_{surface})$. Also, Test 4-3 uses a PDF of cross-wind flight distance, $pdf_Y(Y)$, but the PDF has no impact since it is a narrow distribution with

$k'_{max} = 0$. This new test is intended to check the ability of the firebrand algorithms to simulate two-dimensional configurations. The simulations are performed in MATLAB.

Figure 4.17 presents the successive positions of the simulated fire perimeters (identified by the condition $\phi(x, y, t) = 0$) at times $t = 30, 60, 90, 120\ s$. The fire is initiated at $(x, y) = (0, 0)$ and spreads in the positive x- and y-directions. Figure 4.17a presents the fire contours predicted by the Eulerian firebrand model, whereas Figure 4.17b presents the fire contours predicted by the Lagrangian firebrand model with $GR_p = 1\ pcs \cdot \Delta t_n^{-1}$. The two simulations are found to be in good agreement, which confirms that the two models provide comparable options for the simulation of short-distance firebrand transport.



(a)



(b)

Figure 4.17: Simulation of Test 4-3 performed with: 4.17a the Eulerian firebrand model; 4.17b the Lagrangian firebrand model. Comparison of the simulated fire perimeters obtained at times $t = 30, 60, 90, 120\ s$.

4.5 Conclusions

This chapter presents an evaluation of the numerical performance of the firebrand transport models used in landscape-scale fire spread simulators. The study focuses on short-distance firebrands that may land in a flaming or smoldering state and pose a high spot fire risk. To simplify the analysis, it is assumed that all landing firebrands ignite spot fires instantaneously (i.e., the spot ignition probability is taken as unity), thereby avoiding the complexities associated with ignition dynamics. This assumption enables verification against exact or analytical reference solutions.

Both an Eulerian version and a discrete particle (Lagrangian) version of the firebrand transport model are considered. While the Lagrangian version corresponds to the preferred approach in many current fire spread simulators, it is found that the Eulerian model offers advantages in numerical performance, achieving higher accuracy and/or lower computational cost for comparable configurations.

The sensitivity of both formulations to numerical parameters—specifically, spatial and temporal resolution, and in the case of the Lagrangian model, the particle generation rate—is evaluated through a series of one-dimensional verification tests. Results show that:

- The Eulerian firebrand model achieves numerically accurate results provided that the wind-based Courant-Friedrichs-Lewy (CFL) number satisfies $CFL_{t_n} \leq 0.8$.

- The Lagrangian firebrand model achieves comparable numerical accuracy under the same CFL condition, $CFL_{t_n} \leq 0.8$, provided that the particle generation rate is sufficiently high: $GR_p \geq 10 \text{ pcs} \cdot \Delta t_n^{-1}$ when the emission duration is short ($t_{\text{Emission}} = \Delta t_n$), or $GR_p \geq 1 \text{ pcs} \cdot \Delta t_n^{-1}$ when the emission duration is long ($t_{\text{Emission}} \gg \Delta t_n$).

Finally, while the present study focuses on providing solutions with firebrand effects that are "numerically accurate", it is worth emphasizing that the solutions may still feature significant physical modeling errors. Sources of physical modeling errors include simplifications in the description of firebrand travel distance, firebrand travel time, and spot fire ignition conditions. Current work is aimed at providing "physically accurate" descriptions of these phenomena.

Chapter 5: Landscape-Scale Firebrand Modeling: Ignition

Building on the empirical expressions for firebrand generation based on local fuel heat release rate developed in Chapter 3, and the Eulerian transport model coupled with a prescribed probability density function (PDF) for flight distance and a simple flight time model developed in Chapter 4, this chapter completes the firebrand modeling framework by focusing on firebrand ignition.[1]

The firebrand ignition model proposed in the present study draws on the existing literature, incorporates key aspects of the smoldering-to-flaming transition (SFT)—recognized as a critical process in ignition—and adopts a probabilistic approach parametrized by relevant physical quantities. Two consecutive steps are distinguished in the ignition process: first, the occurrence of a small flame due to SFT; and second, the growth of this small flame into a self-sustaining flame capable of propagating fire spread. While the first step has been the primary focus in previous studies, the second step is often overlooked; here, it is incorporated through a simplified empirical approach borrowed from compartment fire modeling literature.

The first step is modeled using the probability of small-flame ignition recently developed by De Beer et al. [104, 105]. This model is based on experimental data involving horizontal piles of smoldering firebrands accumulated on flammable substrate materials exposed to crosswinds,

[1] Parts of this chapter are taken from the publication by Qin and Trouvé [125]

complemented by pyrolysis simulations using ThermaKin [126, 127]. In this framework, the probability of ignition is parametrized as a function of the accumulated mass of firebrands per unit ground surface and the crosswind velocity.

While the full firebrand generation, transport, and ignition model has been integrated into the ELMFIRE simulator with the goal of modeling WUI fire scenarios, the verification tests presented in this chapter primarily focus on wildland fire configurations. (The case of WUI fire configurations will be addressed in Chapter 6.) The tests are conducted in MATLAB and in one-dimensional geometries, allowing a detailed evaluation of the sensitivity of the model performance to spatial and temporal resolution.

5.1 Firebrand Ignition: Ignition of a Small Flame

As discussed in section 2.2, the cells receiving firebrands are treated as igniting ($\phi = -1$ at time $t = TOA_k + t_{ignition,k}$) when certain conditions are met. We consider that the ignition process consists of two consecutive steps: first, the ignition of a local small flame (due to SFT); and second, the transition to a fully-developed fire. In this framework, we define the ignition time as:

$$t_{ignition} = t_{ign,small} + t_{ign,large} \tag{5.1}$$

where $t_{ign,small}$ is the time to establish a small flame, and $t_{ign,large}$ the time to transition from a small to a large flame, and where in order to simplify notations, the subscript k has been dropped. We now turn to the model description of $t_{ign,small}$. In the present formulation, we treat $t_{ign,small}$ as a random variable that is calculated based on a probability of ignition. We define $P_{ign,small}$ as the probability that firebrands ignite a small flame on a flammable substrate within a certain

duration $\tau_{ign,small}$:

$$P_{ign,small} = Probability(t_{ign,small} \leq \tau_{ign,small}) \tag{5.2}$$

In the literature, the parameters $P_{ign,small}$ and $\tau_{ign,small}$ have been studied using experimental and/or numerical approaches, and have been reported as functions of the firebrand characteristics, environmental conditions, and the flammable substrate properties [104, 105]. A model for estimating these parameters will be presented in Section 5.2 and we assume in the remainder of this section that $P_{ign,small}$ and $\tau_{ign,small}$ are known.

We first discuss here the numerical algorithm required to implement the probabilistic description formulated in Eqn. 5.2 into ELMFIRE. Suppose that at time step t_n, a target cell TC starts receiving firebrands. As soon as the firebrands are detected in cell TC, the numerical algorithm will start ignition attempts with a chance of success that is controlled by $P_{ign,small}$, $\tau_{ign,small}$ and elapsed time. To explain the implementation of Eqn. 5.2 in a numerical time-dependent framework, let us consider that the local ignition parameters, $P_{ign,small}$, $\tau_{ign,small}$, and the time step, Δt_n, are all constant; we also assume for simplicity that $\tau_{ign,small} = N \times \Delta t_n$, where N is an integer. During the duration $\tau_{ign,small}$, a total of N independent ignition attempts will be made, and the probability of ignition during each one of those N attempts can be related to the probability of ignition over the entire duration $\tau_{ign,small}$. This relationship is readily established by considering the probability of no ignition: the probability of no ignition during the time interval $0 \leq t \leq \tau_{ign,small}$ can be calculated as the product of the probabilities of no ignition during the N time intervals $t_n \leq t \leq t_{n+1} = (t_n + \Delta t_n)$:

$$(1 - P_{ign,small}) = (1 - P_{ign,small,\Delta t_n})^N \tag{5.3}$$

where $P_{ign,small,\Delta t_n}$ is the probability of ignition during Δt_n. We finally get:

$$P_{ign,small,\Delta t_n} = 1 - (1 - P_{ign,small})^{1/N} = 1 - (1 - P_{ign,small})^{\Delta t_n / \tau_{ign,small}} \tag{5.4}$$

Eqn. 5.4 provides the basis to simulate the stochastic process of small flame ignition by firebrands:

1. First, for each unignited computational cell loaded with firebrands at time t_n, calculate the small-duration ignition probability $P_{ign,small,\Delta t_n}$ using Eqn. 5.4.

2. Then, draw a random number R_0 between 0 and 1 from a uniform distribution (R_0 is the output of a standard random computer-based number generator). If $R_0 \leq P_{ign,small,\Delta t_n}$, treat the cell as successfully igniting ($t_{ign,small} = t_n - TOA_k$). Otherwise, treat the cell as unignited and proceed to the next time step $t_{n+1} = (t_n + \Delta t_n)$.

3. Finally, repeat steps (1) and (2) for subsequent time steps until $R_0 \leq P_{ign,small,\Delta t_n}$ is satisfied (*i.e.* until ignition is successful), or until the simulation is terminated.

This algorithm guarantees that the probability of ignition continuously increases in time and is equal to $P_{ign,small}$ after time $\tau_{ign,small}$ has elapsed. Figure 5.1 presents an example of the $Probability(t_{ign,small} \leq t_n)$ history. In this case, we prescribe the $P_{ign,small} = 0.9$ and $\tau_{ign,small} = 10\ s$. Given the $\Delta t_n = 0.5$ s, $P_{ign,small,\Delta t} = 0.11$ by Eqn. 5.4, and the height of the bars in the plot is calculated using $(1 - P_{ign,small,\Delta t_n})^{t_n/\Delta t_n}$. It is verified that at $t_n = \tau_{ign,small}$, the height of the bar matches the prescribed value of $P_{ign,small}$ (red dashed line). Note that Eqn. 5.4 and the algorithm described above can readily be extended to the general case with time-dependent ignition parameters, $P_{ign,small}$ and $\tau_{ign,small}$, and time-dependent time step, Δt_n.

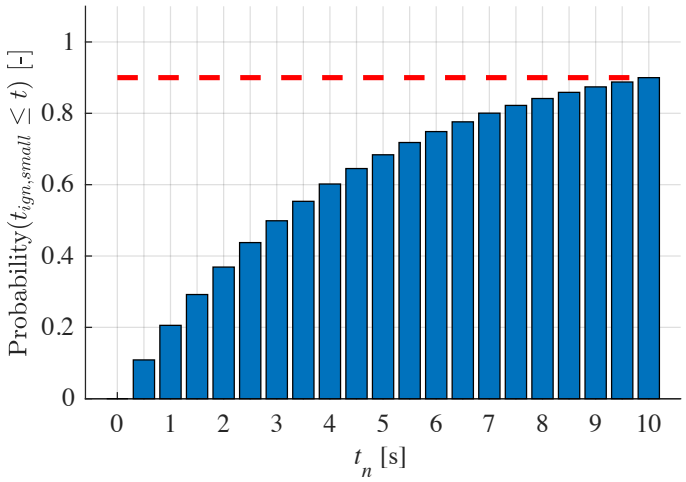


Figure 5.1: Time evolution of the probability of a small flame ignition by firebrands, $Probability(t_{ign,small} \leq t_n)$, versus time. Case with $P_{ign,small} = 0.9$, $\tau_{ign,small} = 10$ s, and $\Delta t_n = 0.5$ s. The blue bars show the values of the probability. The red dashed line marks the prescribed value of $P_{ign,small}$

5.2 Parameterization of the Probability of Small Flame Ignition

We now turn to the description of the model parameters $P_{ign,small}$ and $\tau_{ign,small}$. Recent studies by De Beer et al. [104, 105] reported a series of experiments to characterize the heat flux induced by a horizontal pile of smoldering firebrands to flammable substrate materials. The experiments were performed for a range of firebrand mass loads (ψ, $\mathrm{g \cdot cm^{-2}}$), a range of air flow velocities (v_{air}, $\mathrm{m \cdot s^{-1}}$), a selection of representative structural materials found in WUI communities, and for several crosswind widths of the firebrand piles. The experimental database was first used to develop empirical correlations to predict the time variations of the external heat flux induced on the substrate given v_{air} and ψ (and given also the crosswind width of the firebrand pile). The heat flux correlations were then applied as boundary conditions to simulate the thermal degradation of the substrate material using a pyrolysis model and a one-dimensional solver called ThermaKin [126, 127]. The simulation results provided a description of the fuel mass loss rate

of the material and were finally used to provide an estimate of the probability of (small flame) ignition as a function of v_{air} and ψ, based on an empirical criterion that compares the fuel mass loss rate (or more precisely the theoretical heat release rate) to the rate of convective cooling provided by the ambient air flow [104, 105].

5.2.1 Pyrolysis Modeling of Typical Structural Materials

Built upon the numerical model proposed by De Beer et al. [104, 105], a series of one-dimensional pyrolysis simulations were conducted to analyze typical materials encountered at the wildland-urban interface (WUI). In this study, we utilized ThermaKin to simulate the thermal decomposition of common structural materials, including Western Red Cedar (WRC), Oriented Strand Board (OSB), and Pressure Treated Wood (PTW). Below, we briefly outline the mathematical formulations implemented in ThermaKin, with additional details available in Ref. [126].

The ThermaKin model employs a one-dimensional numerical solver that simultaneously solves mass and energy conservation equations (the solver has been extended to two dimensions in the form of ThermaKin2Ds [127]). The 1-D species mass conservation equation is expressed as:

$$\frac{\partial \xi_j}{\partial t} = \sum_{i=1}^{N_r} \theta_i^j r_i - \frac{\partial J_j}{\partial x} + \frac{\partial}{\partial x}\left(\xi_j \int_0^x \frac{1}{\rho}\frac{\partial \rho}{\partial t}\, dx\right) \tag{5.5}$$

where ξ_j is the mass concentration of component j, t is time, θ_i^j is the stoichiometric coefficient of species j for reaction i indicating production (positive) or consumption (negative), and r_i is the reaction rate for reaction i, modeled using Arrhenius-type formulations. The term J_j denotes gas-phase mass flux and applies only to gaseous species. The spatial coordinate x represents position within the one-dimensional Cartesian domain, and ρ is the density of species ξ_j. The three terms

on the right-hand side of Equation 5.5 describe chemical reactions, gas-phase transport driven by concentration gradients, and thermal expansion, respectively.

The energy conservation equation is given by:

$$\sum_{j=1}^{N} \xi_j c_j \frac{\partial T}{\partial t} = \sum_{i=1}^{N_r} h_i r_i - \frac{\partial q_x}{\partial x} - \frac{\partial I_{ex}}{\partial x} + \frac{\partial I_{rr}}{\partial x} - \sum_{g=1}^{N_g} c_g J_g^x \frac{\partial T}{\partial x} + c\,\rho\,\frac{\partial T}{\partial x}\int_0^x \frac{1}{\rho}\frac{\partial \rho}{\partial t}\,dx \quad (5.6)$$

where T is the system temperature, dependent on time t and spatial coordinate x. The heat capacity of species j is denoted by c_j, h_i represents the heat of reaction for reaction i, and q_x is conductive heat flux. The terms I_{ex} and I_{rr} represent external radiation flux and internal radiative heat exchange, respectively. Equation 5.6 accounts for reaction heat generation, conductive and radiative heat transfers, and thermal expansion effects.

As is suggested, an essential input to the simulation is the reaction kinetics that govern the production and consumption of species and energy. We consider typical structural materials found in residential buildings in the US. As a result, the reaction kinetics for Western Red Cedar (WRC), Oriented Strand Board (OSB), and Pressure Treated Wood (PTW) are summarized in Table 5.1. The kinetic parameters for WRC are adopted from Ref. [105], for OSB from Refs. [128, 129], and for PTW from Ref. [130].

In addition to the reaction kinetics, Table 5.2 presents the consolidated physical properties of the key species involved in the reactions, including density, heat capacity, thermal conductivity, and emissivity.

To validate the accurate implementation of the reaction mechanisms in ThermaKin, a Thermogravimetric analysis(TGA)-like scenario is used. In this validation, a sufficiently small sample (with a thickness of $5 \times 10^{-5}\,\mathrm{m}$, corresponding to one computational grid cell) is subjected

Reactions	**A**	**E**	**Heat of Reaction**
	$[\mathrm{s}^{-1}]$	$[\mathrm{J}\cdot\mathrm{mol}^{-1}]$	$[\mathrm{J}\cdot\mathrm{kg}^{-1}]$
WRC $\rightarrow$ 0.952WRC1 + 0.048GAS1	1.40×10^{8}	1.05×10^{5}	-2.60×10^{4}
WRC1 $\rightarrow$ 0.627WRC2 + 0.373GAS2	2.57×10^{6}	1.01×10^{5}	-7.70×10^{4}
WRC2 $\rightarrow$ 0.523WRC3 + 0.477GAS3	5.70×10^{18}	2.55×10^{5}	-2.80×10^{5}
WRC3 $\rightarrow$ 0.721WRC4 + 0.279GAS4	5.48×10^{-2}	1.89×10^{4}	5.00×10^{5}
WRC4 $\rightarrow$ 0.483CHAR + 0.517GAS5	3.67×10^{-4}	5.47×10^{3}	3.00×10^{5}
PTW $\rightarrow$ 0.73PTW1 + 0.27GAS1	1.92×10^{7}	1.06×10^{5}	-6.82×10^{3}
PTW1 $\rightarrow$ 0.43PTW2 + 0.57GAS2	1.86×10^{11}	1.62×10^{5}	-1.67×10^{5}
PTW2 $\rightarrow$ 0.82PTW3 + 0.18GAS3	1.79×10^{5}	1.01×10^{5}	1.82×10^{5}
PTW3 $\rightarrow$ 0.77CHAR + 0.23GAS4	0.21	3.44×10^{4}	2.02×10^{5}
OSB $\rightarrow$ 0.72OSB1 + 0.28GAS1	1.56×10^{7}	1.04×10^{5}	-6.82×10^{3}
OSB1 $\rightarrow$ 0.45OSB2 + 0.55GAS2	2.65×10^{12}	1.74×10^{5}	-1.37×10^{5}
OSB2 $\rightarrow$ 0.77OSB3 + 0.23GAS3	8.93×10^{3}	8.37×10^{4}	2.90×10^{5}
OSB3 $\rightarrow$ 0.77CHAR + 0.23GAS4	4.40×10^{-1}	3.86×10^{4}	2.32×10^{5}
Water $\rightarrow$ Water Vapor	1.55×10^{4}	4.35×10^{4}	-2.78×10^{6}

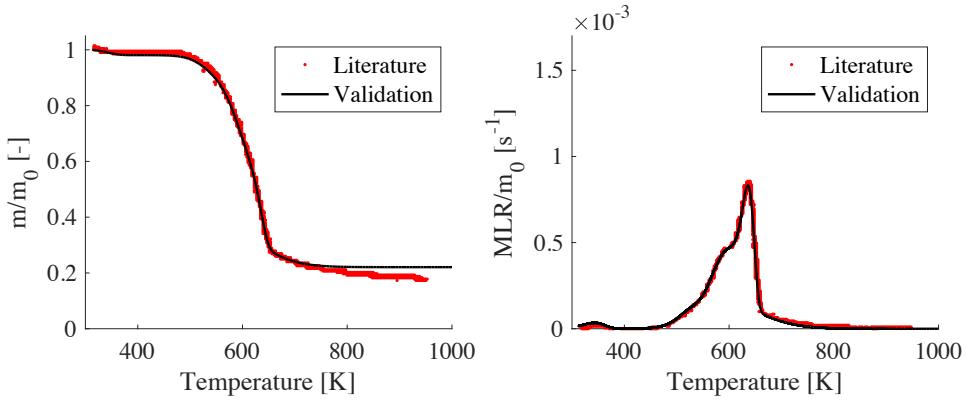
Table 5.1: Consolidated reaction kinetics parameters and corresponding heats of reaction for WRC, PTW, and OSB. Data compiled from Refs. [105, 128–130].

to a constant heating process. The temperature is increased at constant rates of 5 $\mathrm{K}\cdot\mathrm{min}^{-1}$, 10 $K\cdot min^{-1}$, and 10 $\mathrm{K}\cdot\mathrm{min}^{-1}$ for WRC, OSB, and PTW, respectively. The material is assumed to be thermally thin, such that its temperature closely follows that of the ambient air. A large forced convective heat transfer coefficient of $1\times10^{5}\,\mathrm{W}\cdot\mathrm{m}^{-2}\cdot\mathrm{K}^{-1}$ is applied to enforce rapid thermal equilibration.

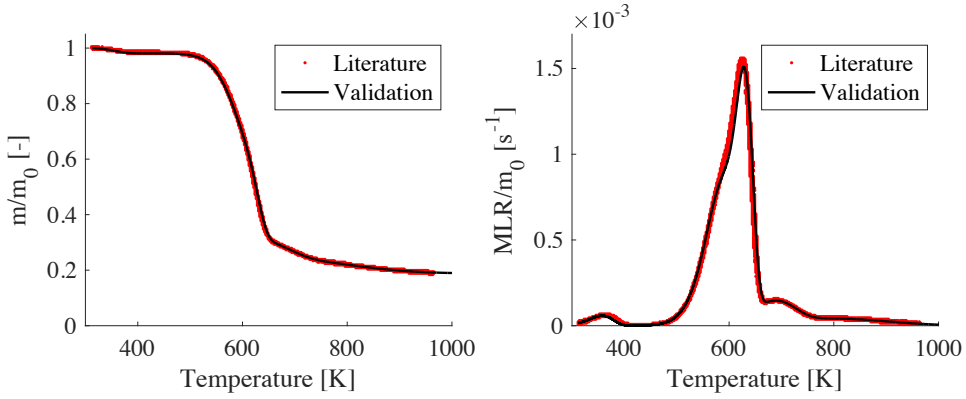
The simulation results (black curves) closely reproduce the original data (red dots) obtained from ThermaKin simulations reported in Refs. [105, 128–130]. Figure 5.2 presents the validation results, where the initial temperature of both the ambient air and the sample is set to 313 K, and all samples contain 1.9% moisture at the initial state. The strong agreement between the simulation and reference data confirms the validity of the reaction kinetics and material properties, supporting their use in subsequent firebrand ignition simulations.

Species	**Density** $[\mathrm{kg \cdot m^{-3}}]$	**Heat Capacity** $[\mathrm{J \cdot (kg \cdot K)^{-1}}]$	**Conductivity** $[\mathrm{W \cdot (m \cdot K)^{-1}}]$	**Emissivity**
Paint	376.8	$-496.5 + 5.16T$	0.128	0.94
Water	1000	$5200 - 6.7T + 0.011T^2$	0.083	0.81
Water Vapor	-	$2400 - 1.6T + 0.002T^2$	0.083	0.81
WRC	376.8	$-496.5 + 5.16T$	0.128	0.81
WRC1	347.95	$-214 + 4.22T$	0.085	0.79
WRC2	217.99	$69.4 + 3.29T$	0.085	0.77
WRC3	114.09	$350.9 + 2.35T$	0.305	0.74
WRC4	82.26	$633 + 1.42T$	0.58	0.72
CHAR(WRC)	39.73	$915.8 + 0.48T$	1.20	0.70
GAS1-5(WRC)	372	2100	0.09	0.81
PTW	498	$-302 + 4.93T$	0.15	0.81
PTW1	364	$-71 + 4.00T$	0.15	0.78
PTW2	156	$161 + 3.07T$	0.26	0.76
PTW3	128	$392 + 2.15T$	0.80	0.73
CHAR(PTW)	98	$624 + 1.22T$	1.00	0.70
GAS1-4(PTW)	372	2100	0.09	0.81
OSB	652	$-159 + 4.53T$	0.13	0.81
OSB1	468	$197 + 3.40T$	0.06	0.78
OSB2	184	$553 + 2.27T$	0.42	0.76
OSB3	142	$909 + 1.13T$	0.39	0.73
CHAR(OSB)	108	1270	0.53	0.70
GAS1(OSB)	372	2100	0.09	0.81
GAS2-4(OSB)	652	2100	0.09	0.81

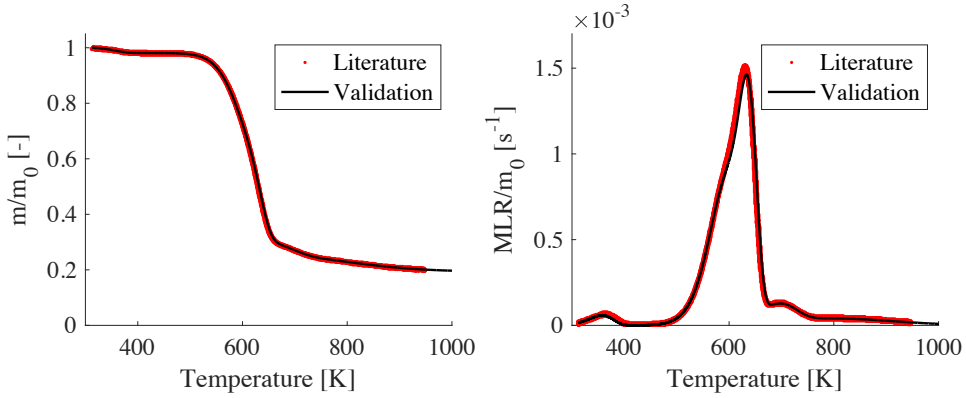
Table 5.2: Consolidated physical properties of key species involved in the pyrolysis reactions of WRC, PTW, and OSB. Data compiled from Refs. [105, 128–130].



(a) Validation results for WRC. Left: Mass history; Right: Mass loss rate (MLR)



(b) Validation results for OSB. Left: Mass history; Right: Mass loss rate (MLR)



(c) Validation results for PTW. Left: Mass history; Right: Mass loss rate (MLR)

Figure 5.2: Thermogravimetric analysis (TGA) validation results comparing the simulation output (black curves) reproduced in this study with the original ThermaKin simulation data (red dots) reported in [105, 128–130].

5.2.2 Thermal Ignition Model under Firebrand Exposure

The ignition of selected structural materials by firebrands was modeled based on the numerical approach described previously, and illustrated schematically in Figure 5.3. The substrate was represented as a horizontally oriented panel with a specified thickness. Following experimental setups reported by De Beer et al. [105], a paint layer was attached beneath the test panel, creating two distinct solid layers: the primary substrate and the paint layer. Firebrands were placed on the upper surface of the substrate panel.

The top boundary, exposed to the firebrand pile, was subjected to a time-dependent heat flux following the model proposed by De Beer et al., combined with convective heat losses due to ambient wind. According to Ref. [104], the heat flux from firebrands initially increases, peaks, subsequently decreases, and eventually stabilizes. The heat flux evolution is given by:

$$HF(t) = \begin{cases} HF_{Rise} \times (t + 12), & 0 \leq t < t_{rise} \\ HF_{Rise} \times (t_{rise} + 12) - HF_{Decay} \times (t - t_{rise}), & t_{rise} \leq t < t_{rise} + t_{decay} \\ HF_{Final}, & t \geq t_{rise} + t_{decay} \end{cases} \tag{5.7}$$

where individual terms are calculated as:

$$\begin{aligned} t_{rise} &= \max(-0.58 v_{air}^2 - 0.50 v_{air} + 43,\ 0) \\ HF_{Rise} &= \max\left[(-0.29 v_{air}^2 + 1.49 v_{air} + 0.1) \times \tanh(12 \times \psi),\ 0\right] \\ HF_{Decay} &= \min\left[(-0.02 v_{air}^2 - 0.13 v_{air} - 0.01) \times \tanh(12 \times \psi),\ 0\right] \\ HF_{Final} &= 8 \times \tanh(12 \times \psi) \end{aligned} \tag{5.8}$$

v_{air} is the velocity of the ambient air in unit of meters per second, and ψ is the firebrand mass load defined as the total mass of local firebrand particles per unit area, measured in unit of grams per squared centimeter.

Forced convective heat loss at the top boundary was modeled as per Ref. [131]:

$$\begin{aligned} h &= \frac{Nu \cdot k}{L_{local}} \\ Nu &= 0.332 Re^{1/2} Pr^{1/3} \\ Re &= \frac{v_{air} L_{local}}{\nu} \end{aligned} \tag{5.9}$$

Air properties at the film temperature ($T_{film} = 493$ K), including kinetic viscosity $\nu = 36.94 \times 10^{-6}\,\mathrm{m^2 \cdot s^{-1}}$, thermal conductivity $k = 39.5 \times 10^{-3}\,\mathrm{W \cdot m^{-1} \cdot K^{-1}}$, and Prandtl number $Pr = 0.698$, were applied. The characteristic length $L_{local} = 0.125$ m corresponds to the length measured from the wind tunnel's test section edge to the firebrand pile's leading edge center.

The bottom boundary was exposed to ambient radiation ($0.43\,\mathrm{kW \cdot m^{-2}}$) and natural convective cooling ($h = 4.82\,\mathrm{W \cdot m^{-2} \cdot K^{-1}}$).

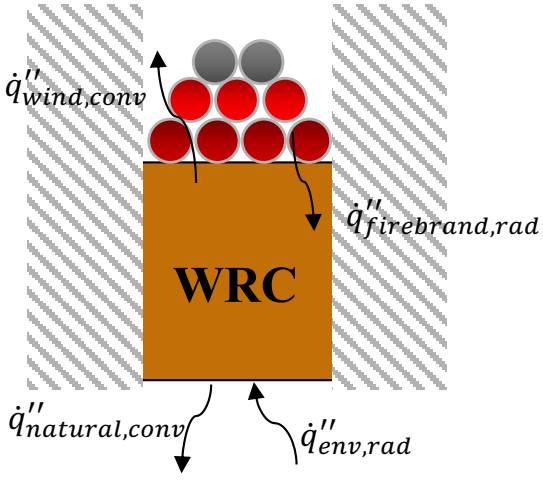


Figure 5.3: Schematic of the 1-D ThermaKin simulation setup illustrating substrate and boundary conditions.

Simulation results illustrating the ignition process for Western Red Cedar (WRC) are presented below. In this example, a firebrand pile with a mass load of $0.16\,\mathrm{g}\cdot\mathrm{cm}^{-2}$ was deposited onto the substrate surface under an ambient wind speed of $2.4\,\mathrm{m}\cdot\mathrm{s}^{-1}$. Figure 5.4 displays the mass concentration history of solid and gaseous species during thermal decomposition, clearly highlighting the consumption of the solid fuel (solid lines) and the generation of gaseous products (dashed lines).

Assuming complete combustion of these gaseous species, the heat release rate (HRR) was calculated and is illustrated in Figure 5.5a. The heats of combustion for each gaseous species derived from the pyrolysis of WRC, OSB, and PTW are summarized in Table 5.3. To quantify the ignition process, De Beer et al. introduced a dimensionless number, $\Omega(t)$, representing the ratio between the HRR and the convective cooling:

$$\Omega(t) = \frac{HRRPUA(t)}{h \times (T_f - T_\infty)} \tag{5.10}$$

where $T_f = 1573$ K is the flame quenching temperature, and $T_\infty = 293$ K is the ambient air temperature [132]. The convective heat transfer coefficient h is calculated using Eqn. 5.9 with air properties evaluated at the film temperature $0.5(T_f + T_\infty)$, yielding thermal conductivity $k = 64.26 \times 10^{-3}\,\mathrm{W}\cdot(\mathrm{m}\cdot\mathrm{K})^{-1}$, Prandtl number $Pr = 0.73$, and kinematic viscosity $\nu = 105.25 \times 10^{-6}\,\mathrm{m}^2\cdot\mathrm{s}^{-1}$.

The ignition probabilities for firebrand piles of varying sizes were estimated using the empirical relationship:

$$P_{ign,10cm} = 1 - (1 - P_{ign,5cm})^{4.4}, \quad \text{with} \quad P_{ign,5cm} = \frac{1}{1 + \exp\left(\frac{\mu - \Omega^*_{peak}}{s}\right)} \tag{5.11}$$

where $\mu = 10.8$ and $s = 1$ are logistic model parameters obtained through fitting experimental data. The parameter Ω^*_{peak} represents the maximum 60-s moving average of $\Omega(t)$.

Species	**WRC-derived species**	**OSB-derived species**	**PTW-derived species**
		ΔH_c [kJ $\cdot$ kg^{-1}]	
GAS1	11.5	12.5	14.8
GAS2	14.8	15.2	16.5
GAS3	13.6	11.5	13.0
GAS4	18.0	7.5	5.0
GAS5	5.0	-	-
Water Vapor	0	0	0

Table 5.3: Heats of combustion for gaseous products generated from the decomposition of WRC, OSB, and PTW.

For the presented case, the ignition probabilities for 5-cm and 10-cm piles are 97.6% and 100%, respectively. Using a critical threshold of $\Omega = 5.9$, ignition time is predicted at 31 s after firebrand deposition.

Figure 5.5b illustrates the temporal evolution of the substrate's top surface temperature. At the ignition time, $t_{ign} = 31$ s, the temperature reaches approximately 716.9 K. According to Ref. [133], typical ignition temperatures for wood materials range between 500 K and 900 K. This temperature range confirms that the conditions at the calculated ignition time are indeed favorable for the flaming ignition of WRC. Furthermore, it should be noted that the gaseous pyrolysis products in the simulation were instantaneously removed from the substrate surface, whereas in practical scenarios, the combustion of these gases near the surface would likely result in an even higher local temperature, creating more favorable conditions for ignition.

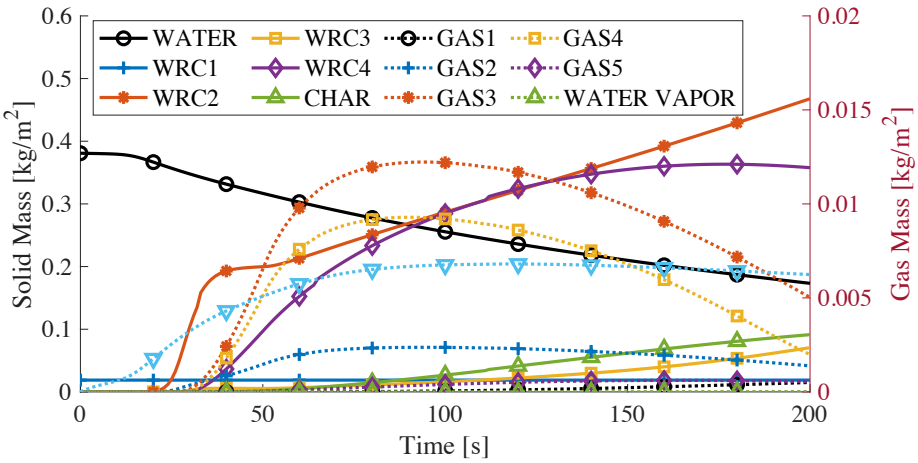
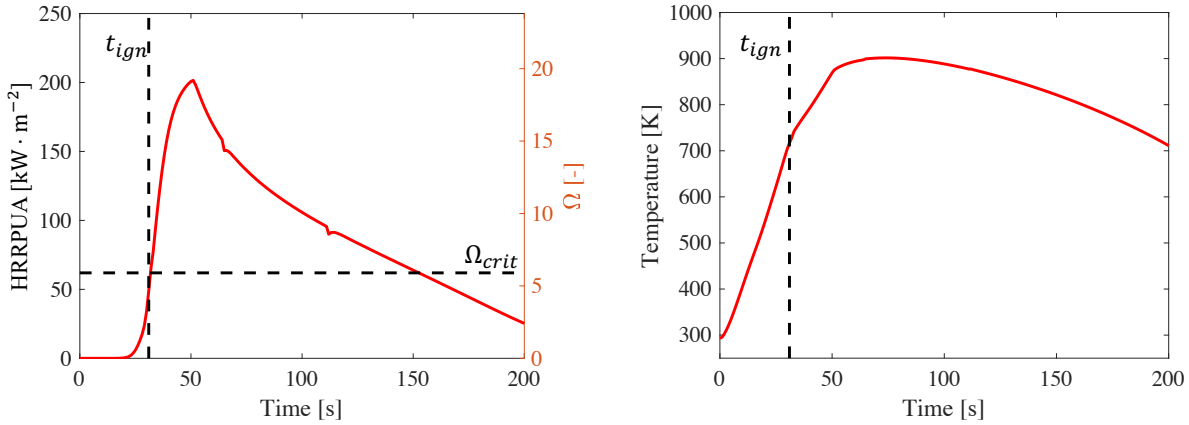


Figure 5.4: Mass concentration history of solid and gaseous species during pyrolysis.



(a) HRRPUA and dimensionless Ω over time.

(b) Surface temperature evolution.

Figure 5.5: Heat release rate per unit area and surface temperature profiles relevant to ignition prediction.

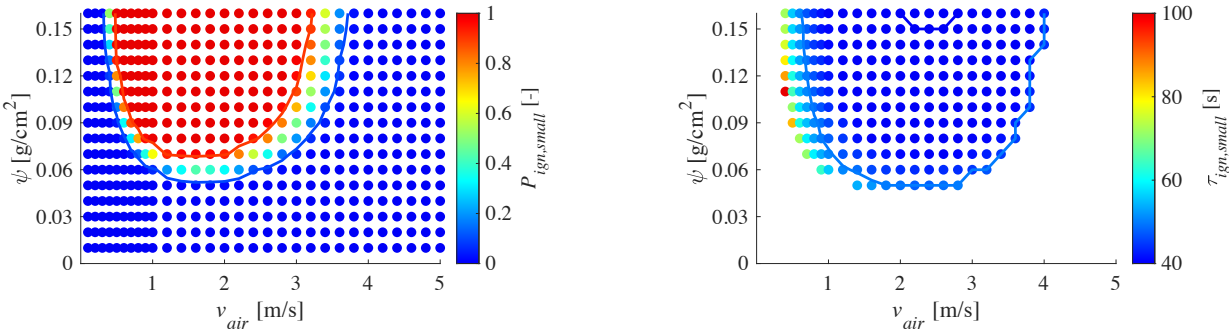
5.2.3 Development of an Empirical Firebrand Ignition Behavior Model

Based on the previously described definitions, we adopted the small-flame ignition model and conducted a parametric study considering variations in wind speed, firebrand pile mass load, fuel moisture content (FMC), and different structural materials. Hereafter, for simplicity, we define $P_{ign,small} = P_{ign,10cm}$, since according to Ref. [105], ignition probabilities for piles with a leading edge length greater than 10 cm can be approximated as combinations of multiple firebrand

piles each having a 10 cm leading edge.

Simulations were carried out for Western Red Cedar (WRC), Oriented Strand Board (OSB), and Pressure Treated Wood (PTW) under varying wind conditions ($0.1 \leq v_{air} \leq 5\ \mathrm{m \cdot s^{-1}}$) and firebrand mass loads ($0.01 \leq \psi \leq 0.2\ \mathrm{g \cdot cm^{-2}}$). Using Eqn. 5.11, we calculated the ignition probability and corresponding ignition time based on a critical dimensionless value of $\Omega = 5.9$.

Representative simulation results for WRC with an initial FMC of 4.6% are presented in Figure 5.6, depicting variations in $P_{ign,small}$ and ignition time $\tau_{ign,small}$. To maintain clarity, data points with $P_{ign,small} = 0$ are omitted from Figure 5.6b. Results indicate that ignition is essentially binary; ignition probabilities are either close to 0 (no ignition) or close to 1 (successful ignition), with a narrow intermediate region that can generally be neglected.

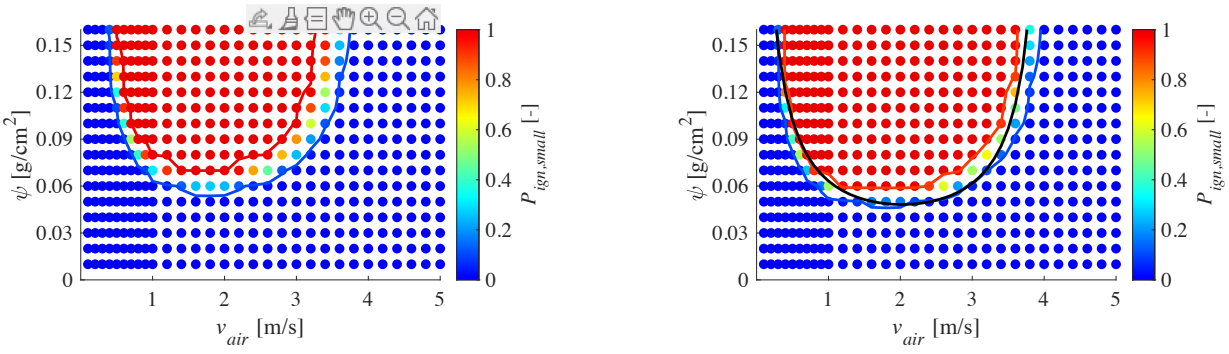


(a) Probability of ignition, $P_{ign,small}$

(b) Time to ignition, $\tau_{ign,small}$

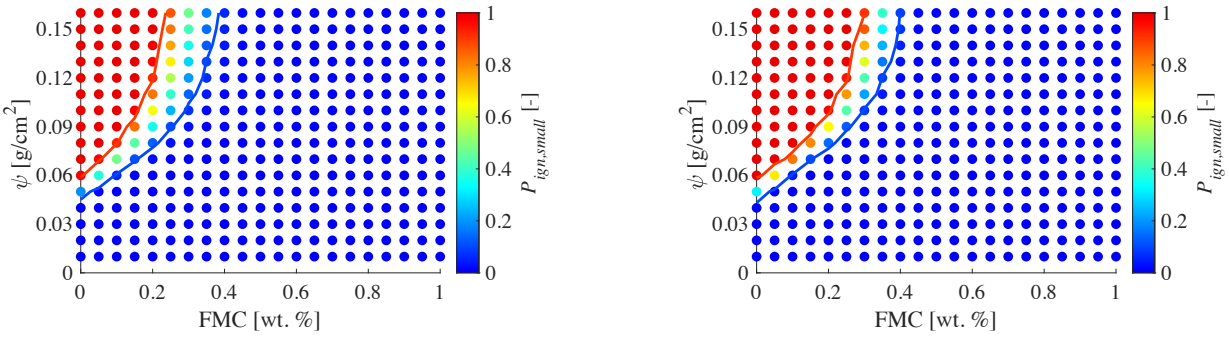
Figure 5.6: Representative ignition results from simulations of WRC. In panel (a), blue and red solid lines represent $P_{ign,small} = 0.1$ and 0.9, respectively.

The binary nature of ignition is further confirmed for OSB (initial FMC = 7%) and PTW (initial FMC = 1.9%), as shown in Figure 5.7. Notably, similar qualitative behavior is observed across all materials. Additionally, ignition probability as a function of FMC (Figure 5.8) under a fixed wind speed ($v_{air} = 1.6\,\mathrm{m \cdot s^{-1}}$) exhibits binary behavior, with ignition probabilities either close to 1 for dry conditions or close to 0 for moist conditions.



(a) OSB (b) PTW

Figure 5.7: Ignition probability as a function of ψ and v_{air} for OSB and PTW under fixed initial FMC (4.6%). The black line is a fitted regression curve from Eqn. 5.12.



(a) WRC (b) OSB

Figure 5.8: Ignition probability versus fuel moisture content (FMC) for WRC and OSB under fixed wind speed ($v_{air} = 1.6\,\mathrm{m} \cdot \mathrm{s}^{-1}$).

Figure 5.9 examines the influence of material hardening, depicting ignition probabilities for PTW protected by a 1 mm Kaowool insulation layer ($\rho = 232\,\mathrm{kg \cdot m^{-3}}$, $c_p = 1070\,\mathrm{J \cdot (kg \cdot K)^{-1}}$, $k = 0.052 - 4 \times 10^{-5}T + 1 \times 10^{-7}T^2\,\mathrm{W \cdot m^{-1} \cdot K^{-1}}$, $\varepsilon = 0.64$). The insulated PTW demonstrates significantly reduced ignition potential, only igniting under optimal wind and high firebrand loads.

These results enable the construction of a simplified analytical ignition model. Considering only dry, unprotected materials—and specifically Pressure-Treated Wood (PTW), commonly found in modern wildland-urban interface areas—a regression curve fitted to the $P_{ign,small} = 0.5$ contour is presented in Figure 5.7b. The fitted analytical curve is expressed as:

$$\psi(v_{air}) = \frac{C}{(v_{air} - v_{min})(v_{max} - v_{air})}, \tag{5.12}$$

with parameters: $v_{min} = -0.07\,\mathrm{m \cdot s^{-1}}$, $v_{max} = 4.11\,\mathrm{m \cdot s^{-1}}$, and $C = 0.21$ ($R^2 = 0.95$). Successful ignition is therefore predicted when $\psi \geq \psi^(v_{air})$.

Currently, the model does not explicitly account for material variability, as the tested materials exhibited similar ignition behavior. Additionally, the influence of fuel moisture content (FMC) is not explicitly considered, as field conditions typically involve FMC values below 20%. The impact of material hardening (e.g., protective insulation layers) is also beyond the scope of this study. However, it is acknowledged that similar empirical models can be derived using the described approach: first generating ignition probability maps via ThermaKin simulations, and subsequently fitting analytical curves to the $P_{ign,small} = 0.5$ contour. Such empirical curves or more comprehensive ignition condition tables could be incorporated into future modeling frameworks but are beyond the scope of the present work.

Reflecting observed binary behavior, the ELMFIRE algorithm assigns $P_{ign,small} = 0.9$ for successful ignition and $P_{ign,small} = 0$ otherwise, avoiding singularities. Typical ignition times

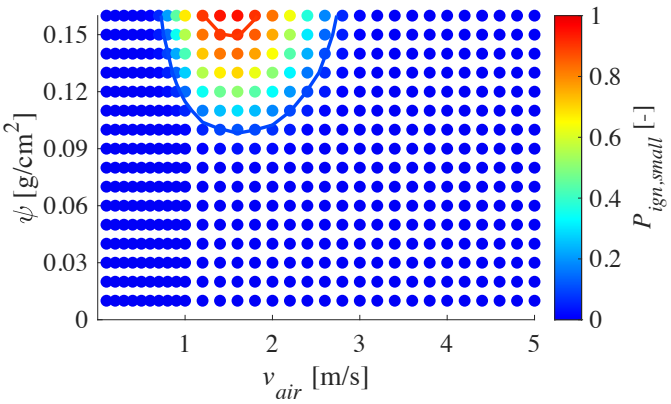


Figure 5.9: Ignition probability of PTW protected by a 1 mm Kaowool insulation layer.

range between 30 and 50 s (Figure 5.10), with an adopted average ignition time of $\tau_{ign,small} = 47\,s$.

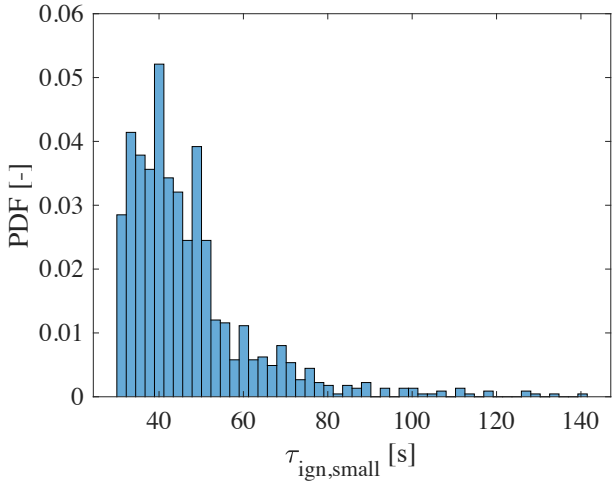


Figure 5.10: Histogram of ignition times across all simulated cases.

Finally, ground-level wind velocity v_{air} required by the model is estimated from higher-

elevation wind velocities U_{H+20} using Albini's logarithmic profile [115]:

$$\frac{v_{air}}{U_{H+20}} = \frac{0.555}{\sqrt{fH}\ln\left(\frac{H+20-0.64H}{0.13H}\right)} \tag{5.13}$$

where H is canopy height (ft) and f is canopy aspect ratio.

5.3 Firebrand Ignition: Transition to a Large Flame

We now turn to a description of $t_{ign,large}$. Not much is known on the time required for a small flame to grow into a fully-developed fire in outdoor environments. In a wildland fire configuration, $t_{ign,large}$ designates the time required to establish a surface fire capable of propagating to the surrounding vegetation; from a modeling viewpoint, $(t_{ign,small} + t_{ign,large})$ is the total delay between the arrival time of firebrands and the activation time of the Rothermel model. In a WUI fire configuration (dominated by structure-to-structure fire spread), $t_{ign,large}$ designates the time required to establish a large enough fire capable of burning the entire target structure; from a modeling viewpoint, $(t_{ign,small} + t_{ign,large})$ is the total delay between the arrival time of firebrands and the activation time of a design fire curve for the target structure.

Because of the large uncertainties associated with the description of $t_{ign,large}$, we choose to treat this time delay using a classical approach adopted in compartment fires: we first estimate the size of the initial flame produced after SFT as having a heat release rate of 10 W; we then estimate the size of a fully-developed fire (that can spread as a surface fire in a wildland configuration or burn a structure in a WUI configuration) as having a heat release rate of 1 MW; and we finally estimate the growth of the heat release rate through a t-square law, $HRR = (t/t_{1MW})^2$, where t is the time elapsed since SFT, t_{1MW} the time required to reach a 1-MW fire, and HRR the

heat release rate of the fire in units of MW. As discussed in the SFPE handbook [134], biomass and building materials can be classified into several categories depending on the rapidity of fire growth and can be characterized by different values of t_{1MW}. We adopt here this approach and use $t_{ign,large} = t_{1MW}$.

5.4 Verification Tests

We present below the results from a series of one-dimensional verification simulations. These simulations use the same fire spread model implemented in ELMFIRE as described in Chapter 2 and are again performed using the one-dimensional MATLAB solver for ease of analysis, consistent with the verification methodology presented in Section 4.4.1.

Four verification scenarios are considered, corresponding to limiting regimes with known exact solutions for the rate of fire spread or relevant statistics:

- **Test 5-1**: Simulation that revisits the baseline performance of the firebrand generation model developed in Chapter 3 and the firebrand transport model developed in Chapter 4, in which the delay for firebrand ignition is ignored ($t_{\text{ignition}} = 0$) and fire spread is controlled purely by firebrand transport. The expected mean rate of spread of the fire leading edge is $\overline{ROS}_{LE} = u_{\text{wind}}$.

- **Test 5-2**: A simulation in which the time scale associated with the transition to a full-scale fire is ignored. The numerical accuracy of the computational representation of the smoldering-to-flaming transition (SFT) time scale as described in Section 5.1 is evaluated.

- **Test 5-3**: A simulation using the full version of the ignition model (SFT and transition to a fully developed fire), but fire spread is controlled by the surface fire, and the expected

mean rate of spread is $\overline{ROS}_{LE} = ROS_{\text{surface}}$.

- **Test 5-4**: An intermediate demonstration test corresponding to a mixed-mode fire spread regime, controlled jointly by both the surface fire and firebrand-induced ignitions.

These verification cases are configured similarly to the test setup already presented in Figures 2.3 and 2.4, except additional comments are made. The numerical configuration adopts a uniform grassland fuel (FBFM#102), a flat terrain, and a uniform and constant wind field. The ambient wind speed is $u_{\text{wind}} = 6.71\ \text{m} \cdot \text{s}^{-1}$ (measured at 6.1 m elevation). The rate of spread of the surface headfire is $ROS_{\text{surface}} = 0.71\ \text{m} \cdot \text{s}^{-1}$, while the backfire rate of spread is $ROS_{\text{back}} = 0.014\ \text{m} \cdot \text{s}^{-1}$. The headfire fireline intensity is $\dot{Q}'_{\text{fire}} = 3189\ \text{kW} \cdot \text{m}^{-1}$. The simulations are performed with $\Delta x = 10$ m and a corresponding time step of $\Delta t_n \approx 0.745$ s ($CFL_{t_n} = 0.5$).

Firebrands in these tests are emitted from all computational cells where heat is released during the residence time of the fireline in a computation cell ($t_{\text{Emission}} = \Delta x / ROS_{surface}$). The prescribed probability density function (PDF) for firebrand downwind travel distance, $pdf_X(X)$, is based on the lognormal distribution proposed by Sardoy et al. [83]. Given the present wind speed and fireline intensity, the parameters of the distribution are $\mu = 2.18$ and $\sigma = 1.23$ for the natural logarithm of the travel distance. The distribution is truncated at the 99th percentile of the cumulative distribution function (CDF), resulting in a maximum spotting distance of $X_{k_{\max}} = 150$ m. Note again that in the current one-dimensional configuration, the crosswind travel distance PDF, $pdf_Y(Y)$, is not utilized.

5.4.1 Fire Spread in the Wildland without Firebrand Ignition Delay (Test 5-1)

In Test 5-1, firebrands are assumed to ignite immediately upon landing on the ground surface, *i.e.*, at time $t = TOA$ ($t_{\text{ign,small}} = t_{\text{ign,large}} = 0$). The configuration is designed to provide simple, verifiable results: for instance, beyond a certain distance from the ignition location (i.e., at $x \geq (X_{k_{\text{max}}} + 5)$ m, where the fire is first ignited at $x = 5$ m), the total number of accumulated embers on the ground should equal $N_\infty = GR \times t_{\text{Emission}} = 1497.2$ pcs; the time of arrival (TOA) of firebrands is predicted as $TOA(x) = x/u_{\text{wind}}$; the rate of spread due to successive spot ignitions equals the wind speed, $ROS_{LE} = u_{\text{wind}}$; and the position of the leading edge of the fire evolves as $x_{LE}(t) = u_{\text{wind}} \times t$.

Figure 5.11 presents the main features of the Test 5-1 solution. Figure 5.11a shows the spatial variation of the total number of accumulated embers on the ground (assuming $\Delta y = 1$ m) and demonstrates excellent agreement between the numerical results and the exact solution, with $N_\infty = 1497.2$ pcs for $x \geq X_{k_{\text{max}}}$ (lower values occur at smaller x due to missing contributions from cells at $x < 0$).

Figure 5.11b presents the spatial variation of the arrival time of the fire at the center of each cell, $TOA(x)$, which also matches the exact solution $TOA(x) = x/u_{\text{wind}}$ very closely. The solid line in the figure represents the exact solution.

Figure 5.11c adopts a different perspective and presents the temporal variation of the fire leading edge position, $x_{LE}(t)$. The numerical results again show excellent agreement with the exact solution $x_{LE}(t) = u_{\text{wind}} \times t$, while also highlighting the effects of discretization: in a repeating sequence, during one time step, the fire remains within the same computational cell and spreads at a rate close to ROS_{surface}; during the next time step, the fire propagates to the

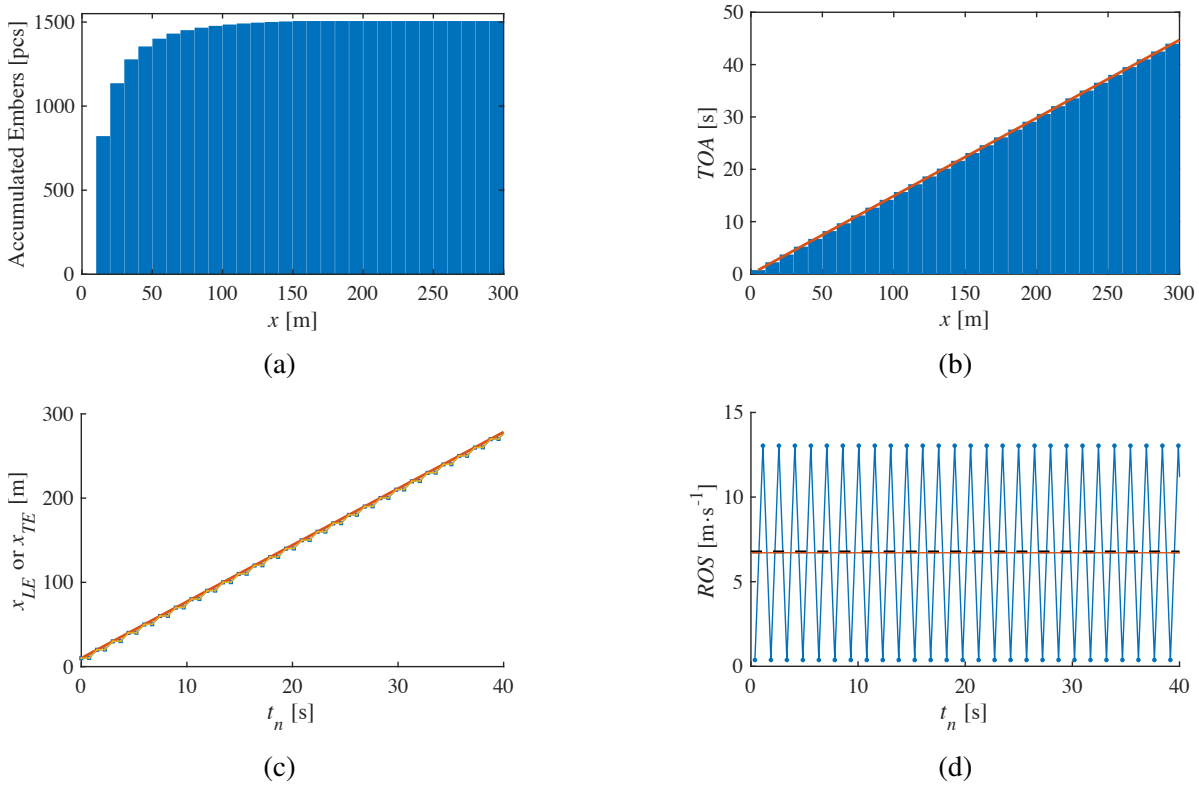


Figure 5.11: Test 5-1 simulation with $t_{\text{ignition}} = 0$. Spatial variations of: (a) the total number of accumulated embers on the ground; (b) the time of arrival of the fire, TOA. Temporal variations of: (c) the fire leading edge position, $x_{LE}(t)$; (d) the fire rate of spread, $ROS_{LE}(t)$. In figures (c) and (d), the solid line with (without) symbols corresponds to the numerical (exact) solution.

adjacent downwind cell by approximately Δx. This binary behavior results from the temporal discretization: with $CFL_{t_n} = 0.5$, firebrands travel $(u_{\text{wind}} \times \Delta t_n) = (CFL_{t_n} \times \Delta x) = \Delta x/2$ during each time step.

Finally, Figure 5.11d presents the corresponding temporal variations of the rate of spread of the fire, $ROS_{LE}(t)$. The rate of spread is computed as the time derivative of $x_{LE}(t)$, $ROS_{LE} = dx_{LE}/dt$. Due to alternating effects of surface fire and spotting ignition, the instantaneous values of ROS fluctuate between approximately $\alpha \times ROS_{\text{surface}}$ (with $\alpha \approx 0.53$ determined by the TVD scheme used) and a value close to u_{wind}. However, the present firebrand transport model guarantees that the time-averaged value of ROS_{LE} matches the exact solution, $\overline{ROS}_{LE} = u_{\text{wind}}$.

5.4.2 Numerical Robustness of the Smoldering-to-Flaming Transition Time Scale (Test 5-2)

We now investigate the effects of activating the ignition process. Figure 5.12 presents a verification test where the local ignition model is activated, with $P_{\text{ign,small}} = 0.9$, $\tau_{\text{ign,small}} = 10$ s, and $t_{\text{ign,large}} = 0$ s. This case emphasizes the impact of the stochastic ignition framework on the simulated fire spread behavior and the surface fire spread mechnisms are deactivated.

Under these conditions, Figure 5.12a shows that the spatial accumulation of firebrands remains similar to that shown in Figure 5.11a, as all cells ignite and emit firebrands at the same rates and same duration. However, the time of arrival (TOA) profile displayed in Figure 5.12b is significantly altered by the stochastic ignition delays: some cells that receive firebrands earlier ignite later than cells deposited with firebrands at later times.

Figures 5.12c further illustrate that the stochastic ignition delays lead to a gradual increasing of distance between the leading and trailing edges of the fire, and Figures 5.12d shows that the stochastic ignition delay causes a slower advancement of the fire leading edge comparing to the baseline case presented in Figure 5.11. Note that the ROS presented in Figure 5.12d oscillates with a minimum value of 0, due to the imposed condition of no surface fire spreading mechanisms. The resulting mean rate of spread of the fire leading edge is approximately $ROS = 5.36\ \text{m} \cdot \text{s}^{-1}$.

To verify that the simulated stochastic ignition delays correctly represent the prescribed ignition model, we examine the distribution of ignition delays. Figure 5.13 shows a histogram of ignition delays, where the delay is defined as the difference between the ignition time and the first arrival of firebrands at a given cell, and corresponds to the variable $t_{\text{ign,small}}$ introduced in

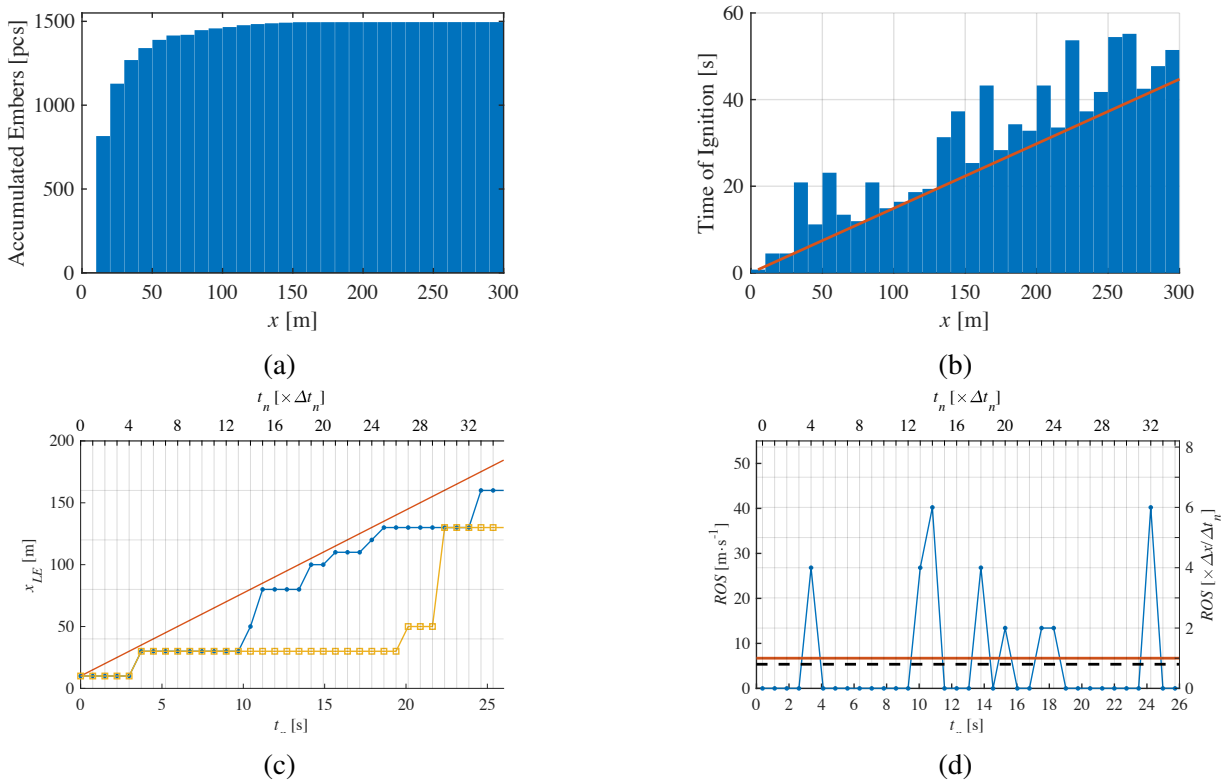


Figure 5.12: Verification simulation with the local ignition model activated ($P_{\text{ign,small}} = 0.9$, $\tau_{\text{ign,small}} = 10$ s, $t_{\text{ign,large}} = 0$ s). Spatial variations of: (a) the total number of accumulated embers on the ground; (b) the time of arrival (TOA) of the firebrands (TOA plotted in units of time steps). Temporal variations of: (c) fire leading edge position, x_{LE}; (d) fire leading edge rate of spread, ROS_{LE}. In (c) and (d), the solid line with (without) symbols corresponds to the numerical (reference) solution.

Section 5.1. The bin width of the histogram is set to $\Delta t_n = 0.745$ s.

The red solid line in Figure 5.13 marks the theoretical 90th percentile value, corresponding to $\tau_{\text{ign,small}} = 10$ s, while the black dashed line shows the simulated 90th percentile obtained from the numerical results. The agreement between the simulated and theoretical values confirms the proper implementation of the stochastic ignition model. A discrepancy of approximately 2 seconds between the analytical value and the simulated results are attributed to limited sample size. In the current simulation, a domain of 3000 m was used, discretized into 300 computational cells with a grid size of $\Delta x = 10$ m. Thus, the ignition delay statistics are based on 300 samples.

It is observed that increasing the simulation domain length or performing multiple realizations of the simulation reduces sampling variability and leads to convergence of the simulated 90th percentile value toward the theoretical value of 10 s.

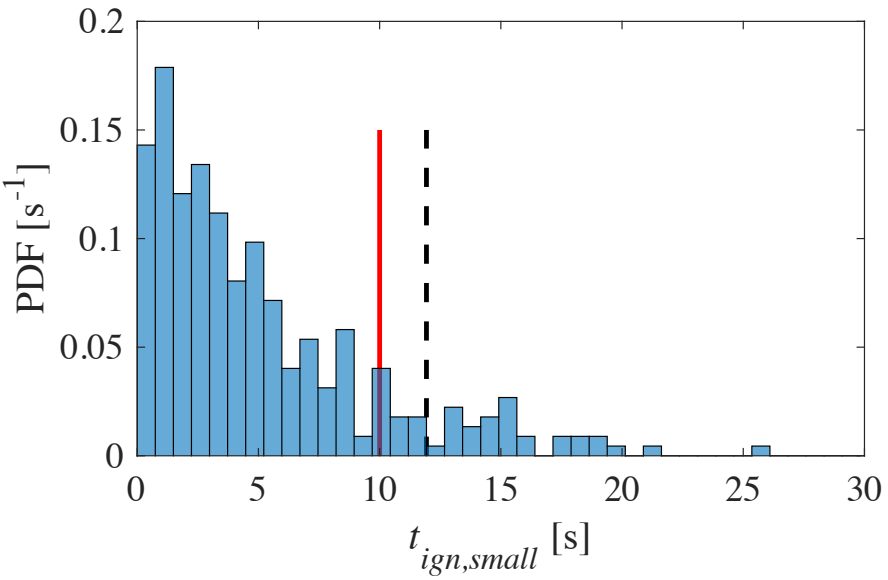


Figure 5.13: Histogram of simulated ignition delays for the verification simulation ($P_{\text{ign,small}} = 0.9$, $\tau_{\text{ign,small}} = 10$ s, $t_{\text{dev}} = 0$). The red solid line marks the theoretical 90th percentile delay; the black dashed line shows the simulated 90th percentile.

Finally, Figure 5.14 presents the mean (time-averaged) values of ROS obtained from a series of simulations performed with different spatial resolutions ($\Delta x = 5$ m, 10 m, 20 m, and 30 m) and various values of $P_{\text{ign,small}}$ ranging from 0 to 1. The time step is consistently chosen to satisfy $CFL_{t_n} = 0.5$.

It is found that the simulated ROS increases with increasing $P_{\text{ign,small}}$ and asymptotically approaches the prescribed wind speed u_{wind} as $P_{\text{ign,small}} \to 1$. This behavior is expected: a higher ignition probability accelerates the ignition of deposited firebrands, thus increasing the fire spread rate. Importantly, the observed trends are consistent across different spatial resolutions, demonstrating the robustness of the ignition model with respect to grid refinement.

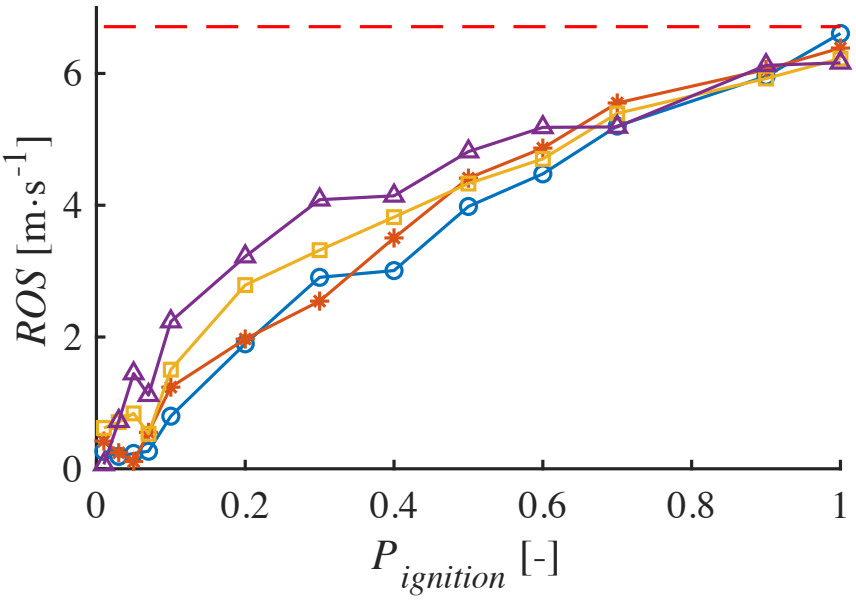


Figure 5.14: Simulations of the verification test with the activated ignition model. Variation of the time-averaged fire leading edge rate of spread, ROS_{LE}, made non-dimensional by the exact wind speed u_{wind}, for different values of $P_{\text{ign,small}}$. Results are shown for four spatial resolutions: $\Delta x = 5$ m (circles), $\Delta x = 10$ m (asterisks), $\Delta x = 20$ m (squares), and $\Delta x = 30$ m (triangles).

5.4.3 Fire Spread in the Wildland with Firebrand Ignition Delays (Test 5-3)

Next, we consider Test 5-3, which features a configuration similar to that of Test 5-1, except that the ignition delay for firebrands is nonzero and is described using the ignition models presented in Sections 5.2–5.3. The firebrand ignition model parameters are specified as follows: $P_{\text{ign,small}} = 0.9$ when $\psi \geq \psi^*(v_{\text{air}})$ and $P_{\text{ign,small}} = 0$ otherwise (see the discussion of Eqn. 5.12; $\tau_{\text{ign,small}} = 42$ s; and $t_{\text{ign,large}} = 300$ s. The value of $t_{\text{ign,large}}$ is based on a medium growth rate scenario, consistent with compartment fire models [134].

It is found that, under these conditions, firebrand ignition does not contribute to fire spread in Test 5-3 for two primary reasons.

First, the accumulated mass load of firebrands is insufficient to allow smoldering-to-flaming

transition (SFT). Assuming an average firebrand mass of $m_{\text{firebrand}} = 0.2$ g, the maximum firebrand mass load per unit ground surface area is

$$\psi = \frac{N_\infty \times m_{\text{firebrand}}}{\Delta x \times \Delta y} = \frac{1497.2 \times 0.2}{1000 \times 100} \approx 0.003 \text{ g} \cdot \text{cm}^{-2}. \tag{5.14}$$

For a wind speed $u_{\text{wind}} = 6.71 \text{ m} \cdot \text{s}^{-1}$, a flame height $H = 1$ ft, and $f = 1$, Eqn. 5.13 first gives $v_{\text{air}} \approx (0.11 \times u_{\text{wind}}) \approx 0.74 \text{ m} \cdot \text{s}^{-1}$, and Eqn. 5.12 yields $\psi^*(v_{\text{air}}) \approx 0.08 \text{ g} \cdot \text{cm}^{-2}$. Since $\psi < \psi^*(v_{\text{air}})$, the model sets $P_{\text{ign,small}} = 0$, resulting in $t_{\text{ign,small}} = \infty$. Thus, no spot ignition events are triggered by firebrands under these conditions.

Second, even if firebrands were capable of immediate ignition, the large value of $t_{\text{ign,large}}$ would prevent spot fire formation before the arrival of the surface fire. Consider a firebrand transported the maximum spotting distance $X_{k_{\max}} = 150$ m. The firebrand lands at

$$TOA = \frac{X_{k_{\max}}}{u_{\text{wind}}} = \frac{150}{6.71} \approx 22.4 \text{ s} \tag{5.15}$$

after emission. Assuming immediate flaming ignition ($t_{\text{ign,small}} = 0$), the spot fire would grow to a full-scale flame at

$$t = TOA + t_{\text{ign,large}} \approx 322.4 \text{ s}. \tag{5.16}$$

In comparison, the surface fire would reach $X_{k_{\max}}$ at

$$t = \frac{X_{k_{\max}}}{ROS_{\text{surface}}} = \frac{150}{0.71} \approx 211.3 \text{ s}. \tag{5.17}$$

Thus, the surface fire would overtake the firebrand landing site before the firebrand-induced ig-

nition could develop, confirming that fire spread remains surface fire-dominated in Test 5-3.

This result is expected, as firebrands typically play a minor role in wildland fire spread where fuels are sparse and residence times are short. Current work is focused on applying the new firebrand ignition model to WUI fires, where t_{Emission} is large and firebrand effects are more significant.

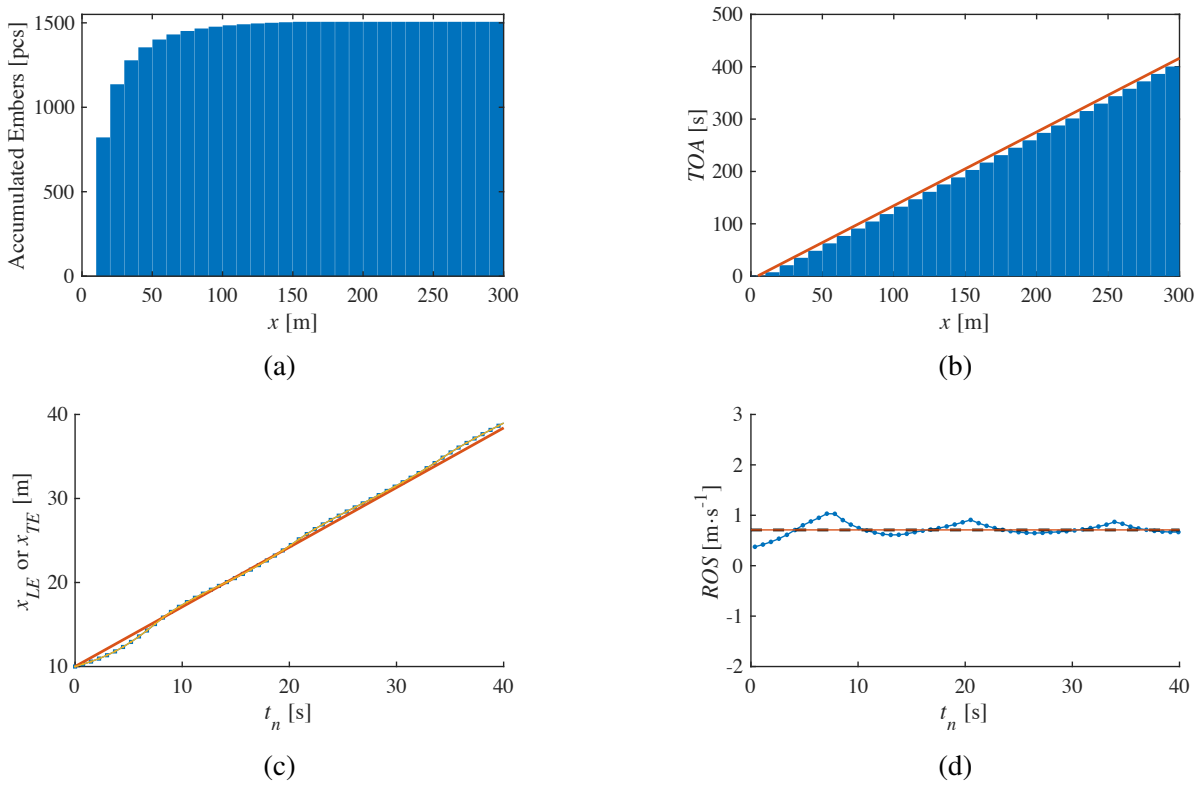
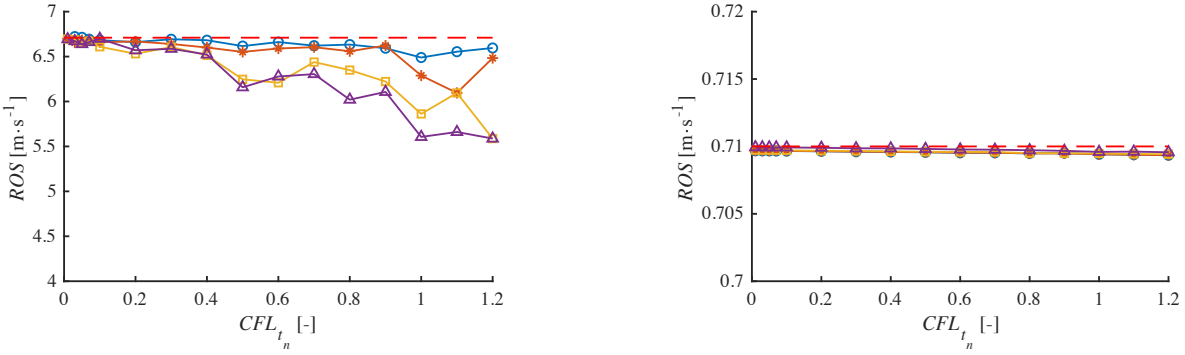


Figure 5.15: Test 5-3 simulation with large values of t_{ignition}. (a) Total number of accumulated embers on the ground; (b) Time of arrival (TOA) of the fire; (c) Fire leading edge position x_{LE}; (d) Fire leading edge rate of spread ROS_{LE}. Solid lines with (without) symbols correspond to the numerical (exact) solutions.

Figure 5.15 presents the main features of the Test 5-3 simulation. Similarly to Test 5-1 (Figure 5.11a), Figure 5.15a shows that for $x \geq X_{k_{\max}}$, the cumulative number of embers reaches $N_\infty = 1497.2$ pcs. However, unlike Test 5-1, Figures 5.15b and 5.15d demonstrate that the fire spread rate is now much slower, consistent with surface fire spread only: $TOA(x) =$

x/ROS_{surface}, $x_{\text{LE}}(t) = ROS_{\text{surface}} \times t$, and $\overline{ROS}_{LE} = ROS_{\text{surface}}$.

Although clearly idealized, Test 5-1 and 5-3 serve as valuable verification tests, as they feature exact solutions for $\overline{ROS}_{LE}$, providing convenient benchmarks to quantify numerical errors. Figure 5.16 compares the time-averaged simulated ROS_{LE} to the exact solutions for different spatial and temporal resolutions.



(a) Test 5-1 ($\overline{ROS}_{LE} = u_{\text{wind}}$)

(b) Test 5-3 ($\overline{ROS}_{LE} = ROS_{\text{surface}}$)

Figure 5.16: Variations of the time-averaged simulated fire rate of spread $\overline{ROS}_{LE}$ with different choices of spatial and temporal resolution. Symbols: $\Delta x = 5$ m (circle), $\Delta x = 10$ m (asterisk), $\Delta x = 20$ m (square), $\Delta x = 30$ m (triangle). The horizontal dashed lines mark the corresponding exact solutions.

These results show that the firebrand ignition model exhibits good numerical convergence: the simulated $\overline{ROS}_{LE}$ remains within 10% of the exact solution provided that $CFL_{t_n} \leq 0.8$. Accordingly, the baseline configuration in ELMFIRE adopts $CFL_{t_n} = 0.5$ to ensure numerical robustness. It should be noted that Figure 5.16a is distinct from the results presented in Figure 4.11, as different surface fire spreading rates are applied in each case. Figure 5.16a further demonstrates that the proposed framework couples well with the surface fire spread algorithms, accurately capturing the interaction between firebrand ignition and surface fire propagation across varying spatial and temporal resolutions.

5.4.4 Mixed-Mode Fire Spread (Test 5-4)

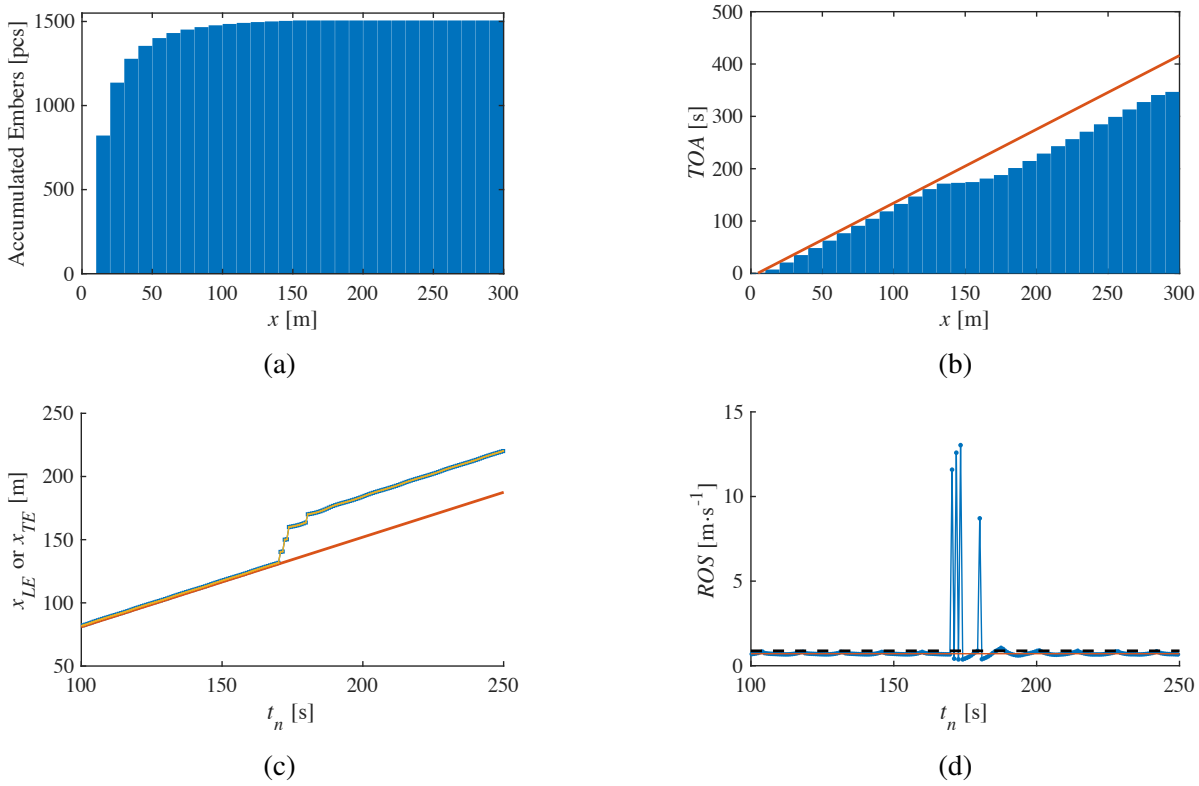


Figure 5.17: Test 5-4 simulation with a finite ignition delay ($t_{\text{ign,large}} = 150$ s). Spatial variations of: (a) total number of accumulated embers on the ground; (b) time of arrival of fire; Temporal variations of: (c) fire leading edge position, x_{LE}; (d) fire leading edge rate of spread, ROS_{LE}. See the caption of Figure 5.11 for additional details.

Finally, we consider an intermediate case, called Test 5-4, in which the delay for firebrand ignition is set to a finite arbitrary value, selected here for demonstration purposes: $t_{\text{ign,small}} = 0$ s and $t_{\text{ign,large}} = 150$ s. With these settings, firebrands landing at locations close to the maximum spotting distance $X_{k_{\text{max}}} = 150$ m have sufficient time to ignite spot fires before the arrival of the surface fire front.

Figure 5.17 presents the main features of the Test 5-4 solution. As expected, Figures 5.17b, 5.17c, and 5.17d show an acceleration of fire spread at around $t \approx 170$ s, attributed to the ignition

and development of spot fires ahead of the primary fire front. Thus, fire spread in Test 5-4 corresponds to a mixed-mode regime, influenced by both the surface fire and firebrand-induced spot fires.

This case serves to illustrate the capability of the proposed firebrand model to accurately simulate complex fire dynamics in wildland environments where spot fire ignition plays a significant role.

5.5 Consumption of Deposited Firebrands

5.5.1 Presentation of the Firebrand Consumption Model

At the end of this chapter, for completeness of the discussion, we address the consumption of active firebrands following deposition. This topic is important because the current modeling framework has thus far considered only the accumulation of deposited firebrand particles, without accounting for their limited lifespans. Although the ignition events predicted by the model occur on characteristic timescales of several minutes, the accumulation of sufficient firebrand mass to meet ignition thresholds may take tens of minutes, as will be demonstrated in Chapter 6 and discussed in Section 5.2. In contrast, the lifetime of a typical firebrand—measured from deposition on the substrate to the complete consumption of its combustible mass—has been observed in wildland fire events to last only a few minutes. Once consumed, a firebrand can no longer deliver heat to the target surface and therefore ceases to contribute to ignition.

To address this limitation, we propose a first-order approximation of firebrand consumption after deposition. In this formulation, deposited firebrands are categorized into *active* and *inactive* particles, with only active firebrands contributing to the heat flux imparted to the substrate.

Instead of directly tracking the consumption of firebrand number, a more straightforward approach is to model the decay of firebrand mass. As a first-order approximation, we assume that the local mass of active firebrands decreases linearly over their lifetime. Thus, the temporal variation of the local firebrand mass load is expressed as:

$$\frac{\partial \psi}{\partial t} = -\frac{\psi}{t_{\text{fb,live}}}, \tag{5.18}$$

where ψ is the local firebrand mass per unit area (in $\mathrm{g} \cdot \mathrm{cm}^{-2}$, consistent with previous sections), and $t_{\text{fb,live}}$ is the firebrand lifetime. Since ψ is the evolving quantity, an appropriate parameterization for $t_{\text{fb,live}}$ must be introduced.

Section 5.2 introduced an empirical correlation for the incident heat flux generated by a pile of deposited firebrands, expressed as a function of ψ and the ambient air velocity v_{air}. Given that only firebrands capable of providing effective heat release are considered active, we propose to use a *critical incident heat flux*, $HF_{\text{fb,live}}$, to distinguish between active and inactive firebrands. Using this threshold, the effective heating duration of a firebrand pile can be retrieved and interpreted as the firebrand lifetime.

Based on the empirical model for incident heat flux presented in Equations 5.7 and 5.8, the firebrand lifetime is estimated as:

$$t_{\text{fb,live}} = \tag{5.19}$$
$$\min\left[(t_{\text{decay}} + t_{\text{rise}}), \frac{HF_{\text{Rise}} \times (t_{\text{rise}} + 12) - HF_{\text{fb,live}}}{HF_{\text{Decay}}} + t_{\text{rise}}\right] - \left(\frac{HF_{\text{fb,live}}}{HF_{\text{Rise}}} - 12\right),$$

where the minimum function ensures a finite lifetime, given that the empirical heat flux profile

features a nonzero final heat flux, HF_{Final}.

Figure 5.18 illustrates the correlation between the predicted firebrand lifetime $t_{\text{fb,live}}$ and two variables: ambient air velocity ($\mathrm{m} \cdot \mathrm{s}^{-1}$)—distinct from wind speed—and local firebrand mass loading ψ ($\mathrm{g} \cdot \mathrm{cm}^{-2}$). It suggests that firebrand lifetime initially increases and then decreases with increasing air velocity, consistent with known convective enhancement and burnout effects. In addition, firebrand lifetime decreases with decreasing firebrand mass loading, as lower ψ values are insufficient to sustain heat flux over time.

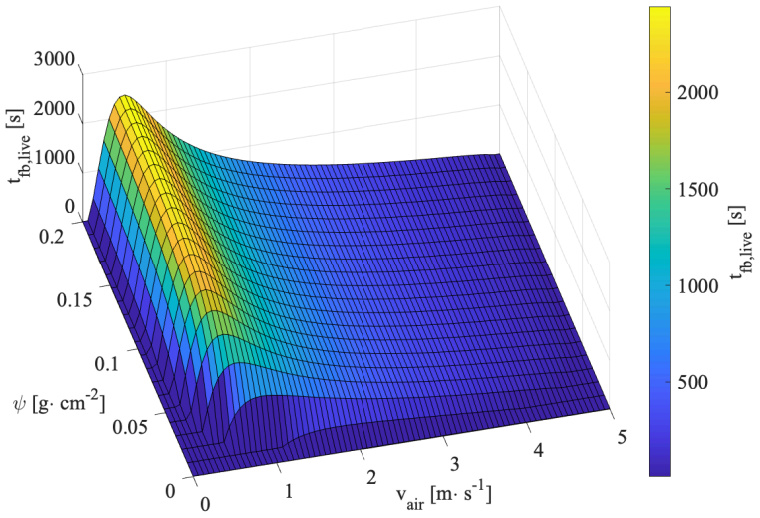


Figure 5.18: Predicted firebrand lifetime $t_{\text{fb,live}}$ as a function of ambient air velocity and local firebrand mass load ψ. ($HF_{\text{fb,live}} = 10\ \mathrm{kW} \cdot \mathrm{m}^{-2}$ and a minimum $t_{\text{fb,live}} = 10$ s are assumed)

To integrate this formulation in ELMFIRE, a simple first-order explicit scheme is applied, updating the local firebrand mass simultaneously with the level-set variable ϕ:

$$\psi(t + \Delta t_n) = \psi(t) - \frac{\psi(t)}{t_{\text{fb,live}}\left(\psi(t), v_{\text{air}}(t)\right)} \Delta t_n \tag{5.20}$$

5.5.2 Verification of Firebrand Consumption Modeling

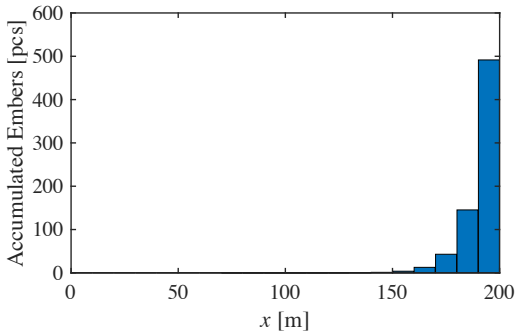
Figure 5.19 presents a verification test using the MATLAB version of ELMFIRE, demonstrating the impact of the firebrand consumption model. The test replicates the configuration used in Test 5-3 from Section 5.4.3, except that the firebrand consumption model is now applied. The critical heat flux threshold is set to $HF_{\text{fb,live}} = 10\ \text{kW} \cdot \text{m}^{-2}$.

Figures 5.19b, 5.19c, and 5.19d show that fire spread remains dominated by the surface fire mechanism. However, significant discrepancies appear in the firebrand accumulation profiles. Figure 5.19a shows the spatial distribution of active firebrands at $t = 260$ s after ignition (noting that the last cell near $x = 200$ m ignites at approximately 258.76 s). It is observed that most firebrands deposited upstream of $x = 200$ m have been consumed.

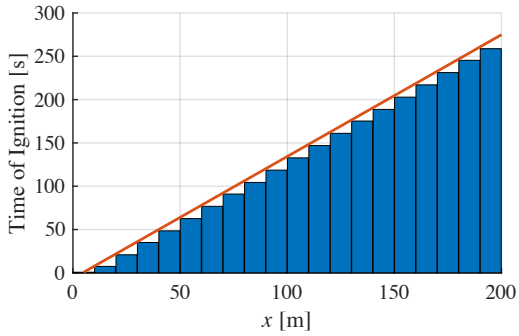
Since $t_{\text{fb,live}}$ depends on the local mass load ψ and the ambient air velocity v_{air}, very short lifetimes are predicted as ψ becomes small. Equation 5.8 indicates that incident heat fluxes decay rapidly with small ψ. To avoid numerical instabilities, a minimum lifetime $t_{\text{fb,live}} = 10$ s is imposed. As a result, more than 96% of firebrands are consumed before $t = 260$ s, compared to Test 5-3 without consumption. These findings indicate that better lifetime parameterizations are needed for accurate active firebrand population estimates.

The firebrand consumption model demonstrates numerical convergence. Figure 5.20 presents results for the same configuration at different grid resolutions, while maintaining $CFL_{t_n} = 0.5$.

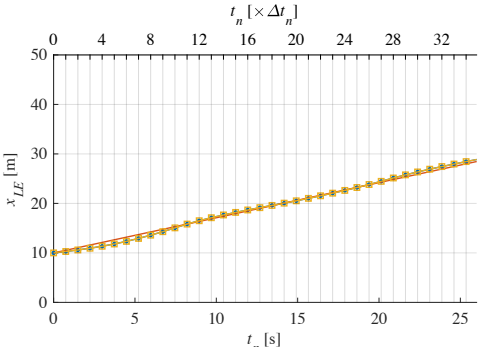
Figures 5.20a and 5.20b show that both the temporal history of accumulated firebrands at $x = 200$ m and the spatial profile at $t = 250$ s converge as the grid is refined from $\Delta x = 30$ m to $\Delta x = 1$ m. However, significant discretization errors occur when coarse grids ($\Delta x = 30$ m) are used, primarily due to inaccuracies in deposition and local mass decay.



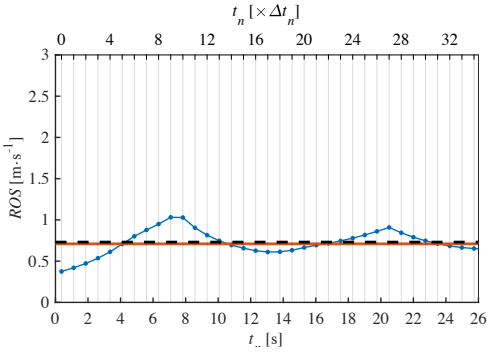
(a) Accumulated active firebrands at $t = 260$ s.



(b) Time of arrival (TOA) of the firebrands.



(c) Fire leading edge position, x_{LE}.

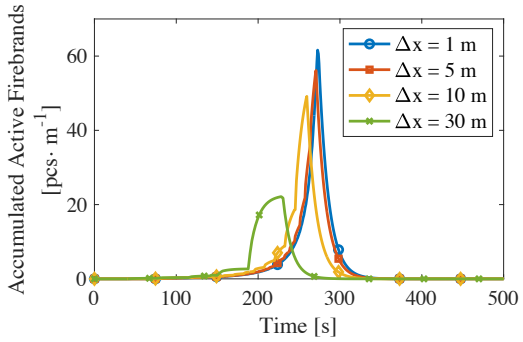


(d) Fire leading edge rate of spread, ROS_{LE}.

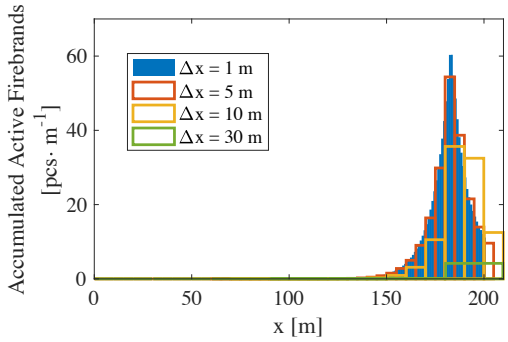
Figure 5.19: Modified Test 5-3 simulation with firebrand consumption model applied. Solid lines with (without) symbols correspond to the numerical (exact) solutions.

While the firebrand consumption model is numerically robust under sufficiently fine spatial resolution, the present framework still exhibits several limitations. Verification tests indicate that even in the absence of a consumption model, a significant accumulation of firebrands is generally required to achieve ignition. Moreover, the current assumption of uniform deposition within each computational cell represents a major simplification: in reality, firebrands tend to accumulate preferentially around vulnerable features such as corners, crevices, and other surface irregularities.

Given that ELMFIRE simulations typically operate on relatively coarse spatial grids (e.g., $\Delta x = 30$ m), the absence of a subgrid-scale model to represent firebrand redistribution further exacerbates discrepancies, in addition to the limitations already observed in coarse-resolution



(a) Temporal evolution of accumulated active firebrands at $x = 200$ m.



(b) Spatial profile of active firebrands at $t = 250$ s.

Figure 5.20: Convergence tests for the firebrand consumption model under varying spatial resolutions. Finer grids lead to improved consistency in both temporal and spatial distributions.

simulations. Future work must incorporate subgrid-scale treatments that capture localized firebrand clustering to more realistically represent ignition dynamics in WUI fires.

At present, the firebrand consumption model has not been incorporated into the default ELMFIRE implementation, pending the development of a better numerical scheme for coarse grids and physically realistic subgrid-scale redistribution framework.

5.6 Conclusions

This chapter aimed at developing a formulation of a firebrand ignition model for use in landscape-scale fire spread simulators, with an emphasis on achieving numerically accurate results—that is, results controlled by physics-based model parameters rather than by numerical resolution parameters.

The firebrand ignition model assumes two consecutive time delays: first, a delay associated with the occurrence of a small flame after firebrand landing, governed by the smoldering-to-flaming transition (SFT) and based on an empirically derived probability of ignition parameterized by the accumulated mass of firebrands per unit ground surface and the wind velocity;

and second, a delay associated with the growth of a small flame into a flame sufficiently large to sustain fire spread, described by a t-square growth law. The treatments of these time delays should be viewed as work-in-progress. They are expected to evolve as new observational and validation data are collected and as a more complete understanding of spot ignition processes becomes available. Nonetheless, the basic framework proposed in the present study is flexible enough to accommodate future improvements.

The firebrand modeling capability has been incorporated into ELMFIRE. Simulations were performed using a MATLAB version of ELMFIRE for ease of analysis. The model was tested through a series of one-dimensional verification cases, including: Test 5-1: a baseline case without ignition delays where fire spread is dominated by firebrand transport; Test 5-2: a case where the statistical ignition algorithm for the SFT process was verified, demonstrating accurate representation of the prescribed mathematics and insensitivity to spatial resolution; Test 5-3: a case with long ignition delays where fire spread is governed by surface fire propagation; and Test 5-4: an intermediate case featuring mixed-mode spread controlled by both firebrand ignition and surface fire dynamics.

Tests 5-1 and 5-3 have exact solutions for the rate of fire spread, thereby enabling convenient estimates of numerical errors. The results demonstrate that the firebrand model is numerically converged, meaning it is insensitive to changes in spatial and temporal resolution, provided that the wind-based Courant–Friedrichs–Lewy (CFL) number remains sufficiently low. These findings provide a solid mathematical foundation for simulating firebrand effects in landscape-scale fire spread models.

While the scope of this chapter is limited to idealized one-dimensional wildland fire configurations, the ultimate goal is to develop a physics-based model capable of describing firebrand

effects in two-dimensional WUI fire configurations. Current work is focused on extending the model to WUI fires, which will be discussed in Chapter 6.

Finally, a firebrand consumption model was proposed to account for the finite lifetime of deposited firebrands. Although the proposed scheme exhibits numerically converged behavior at fine spatial resolutions, it fails to accurately capture firebrand dynamics at coarse resolutions typically used in ELMFIRE (e.g., $\Delta x = 30$ m). Moreover, the current assumption of uniform firebrand distribution across computational cells leads to an unrealistic requirement for excessive firebrand accumulation to trigger ignition. This observation highlights the need for a subgrid-scale model to capture the preferential clustering of firebrands after deposition. As such, the firebrand consumption model has not yet been incorporated into the default ELMFIRE implementation pending the development of appropriate subgrid-scale treatments.

Chapter 6: Simulation Framework for Wildland–Urban Interface Fire Spread Modeling

In the previous chapters, we developed a comprehensive firebrand model suitable for integration into landscape-scale fire spread simulators. This model incorporates key phases of the firebrand lifecycle—including generation, transport, and ignition—and has been numerically verified for stability and consistency. It is now coupled with surface fire spread models, enabling simulation of firebrand-driven ignition processes in complex environments.

As it has been recognized, wildland–urban interface (WUI) fire spread introduces unique challenges and carries significant implications for community-level fire risk assessment. As fire hazards in WUI regions directly threaten lives and infrastructure, this chapter intends to evaluate the performance of the firebrand–surface spread modeling framework in representative WUI scenarios.

To support systematic evaluation and isolate specific modeling behaviors, we adopt idealized community layouts throughout this chapter. These configurations are designed to enable controlled comparisons and facilitate model interpretation. The shared simulation assumptions and configurations are outlined in the first section.

Subsequent sections will first implement the one-dimensional MATLAB surrogate of the

ELMFIRE framework to explore key numerical modeling decisions. Specifically, we investigate:

- The numerical treatment of structures in the WUI—whether they should be represented as integrated units or discretized over individual mesh cells;
- The ability of the model to simulate fire propagation across the wildland–urban interface;
- The influence of spatial resolution and time step size on predicted fire spread behavior and firebrand accumulation.

The goal of these tests is to establish best practices for applying the simulation framework in WUI settings and to inform model users about sensitivity to numerical resolution.

Finally, we perform a parametric analysis using the validated 1D configuration to assess the model's response to key variables in urban fire spread scenarios including wind speed, structure separation distance, and firebrand generation rate. These tests aim to reveal threshold conditions for sustained fire spread through firebrand ignition.

6.1 Design of 1-D WUI Fire Spread Simulation

As presented in previous chapters, the key inputs to the firebrand model include the heat release rate (HRR) of the fire sources, which governs both firebrand generation and deposition behavior; the firebrand deposition model, which may differ for biomass and structural fuels; and the wind conditions, which affect firebrand model behaviors directly by changing deposition distribution and indirectly by influencing the fire intensity.

In earlier sections, we assumed that the firebrand particle generation rate is proportional to the HRR, and most tests were performed using uniform wildland fuel configurations. A constant

generation duration and rate were applied, based on the estimations in Chapter 3. In the present study, we fully couple the firebrand generation process to the dynamically evolving HRR of the fire source.

For vegetation fuels, the HRR is computed from the Rothermel fireline intensity, which is given by Eqn. 2.42. The local heat release rate is then calculated as:

$$HRR_{\text{vegetation}} = I_B \times \Delta y \tag{6.1}$$

where I_B is the fireline intensity from the Rothermel model, and Δy is the mesh resolution in the direction perpendicular to fire propagation. In the one-dimensional simulations presented here, Δy is assumed to be constant and is selected to match the footprint dimension of a structure for simplicity.

For structural fuels, a prescribed design fire curve is used to model fire growth, consistent with the assumptions made in the ignition model. While design fire curves for structural fires are well established in the literature, they are typically derived from scenarios where fires originate indoors. In contrast, structure ignition during WUI fires often occurs externally. Although this difference may lead to some underprediction of fire behavior in WUI contexts, the adopted design fire curve is deemed sufficient to represent general fire development trends.

Following the implementation in the WU-E surface fire spread model, the design fire curve for structural burning is simplified into three linear phases: a growth phase in which the heat release rate per unit area (HRRPUA) increases linearly from zero to a peak value, a fully developed steady phase at the peak HRRPUA, and a linear decay phase in which the HRRPUA returns to

zero. The time-dependent HRRPUA for a burning structure is given by:

$$HRRPUA_{\text{structure}} = \begin{cases} \dfrac{HRRPUA_{\text{max}}}{t_{\text{grow}}} \cdot t, & 0 \le t < t_{\text{grow}} \\ HRRPUA_{\text{max}}, & t_{\text{grow}} \le t < t_{\text{max}} \\ HRRPUA_{\text{max}} - \dfrac{HRRPUA_{\text{max}}}{t_{\text{decay}} - t_{\text{max}}} \cdot (t - t_{\text{max}}), & t_{\text{max}} \le t < t_{\text{decay}} \\ 0, & t \ge t_{\text{decay}} \end{cases} \tag{6.2}$$

where $HRRPUA_{\text{max}}$ denotes the peak heat release rate per unit area, t_{grow} is the duration of the growth phase, t_{max} marks the end of the steady burning period, and t_{decay} denotes the end of combustion.

Typical values for $HRRPUA_{\text{max}}$, t_{grow}, t_{max}, and t_{decay} are informed by findings in the literature. Studies of fire loads in typical residential structures report fire load densities (i.e., energy per unit area) ranging from 400 to 600 $\text{MJ} \cdot \text{m}^2$. Additionally, peak heat release rates for residential occupancy fires typically range from 300 to 600 $\text{kW} \cdot \text{m}^{-2}$ [135–137].

However, Yoshioka et al. [92]reported in a real-scale experiment that a model house measuring approximately 3 m by 3 m burned for about 50 minutes until all combustibles were consumed. Notably, Ref. [135] did not account for the structural frame in its fire load estimates. Therefore, to conservatively reflect the contribution of structural elements, we assume a total burning duration of approximately 1 hour. For simplicity, and without loss of generality, we adopt an idealized design fire curve with linearly increasing and decreasing phases, each lasting 5 minutes. This curve features values representative of medium-growth fires in residential-type

structures [134]. The resulting heat release rate (HRR) profile is visualized in Figure 6.1:

$$HRRPUA_{\text{max}} = 400\,\text{kW}\cdot\text{m}^{-2}, \quad t_{\text{grow}} = 300\,\text{s}, \quad t_{\text{max}} = 3900\,\text{s}, \quad t_{\text{decay}} = 4200\,\text{s}.$$

When implementing this design fire curve in simulations, the heat release rate per unit area is multiplied by the structure's footprint area to compute the total HRR for each structure.

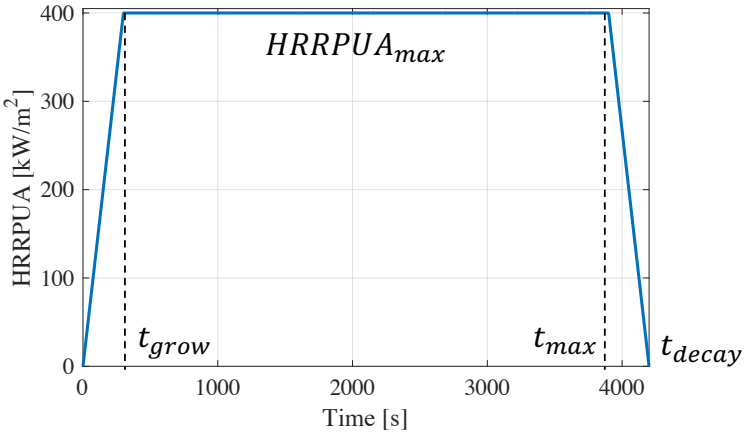


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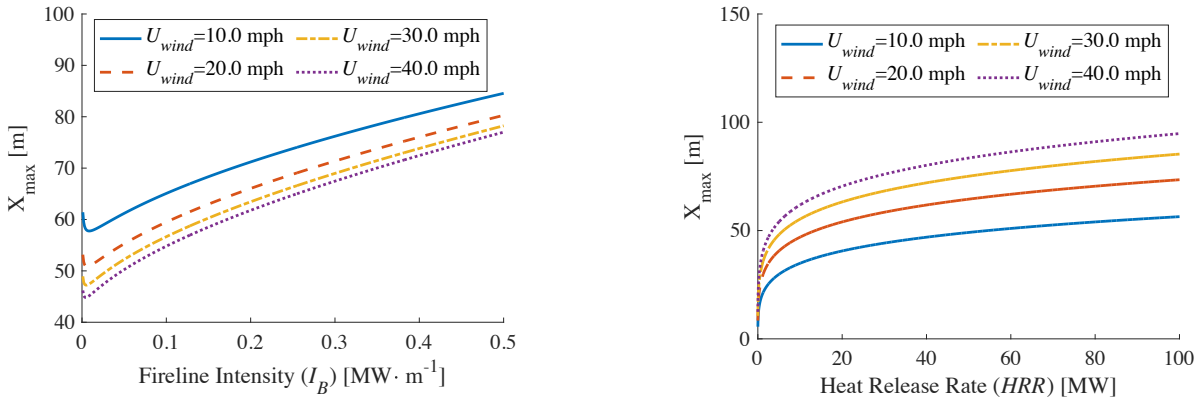
Figure 6.1: Design fire curve used in the 1-D MATLAB WUI configuration tests. The fire growth profile follows three linear phases: growth ($0 \leq t < 300\,\text{s}$), steady burning ($300 \leq t < 3900\,\text{s}$), and decay ($3900 \leq t < 4200\,\text{s}$). The peak heat release rate per unit area is set to $HRRPUA_{\text{max}} = 400\,\text{kW}\cdot\text{m}^{-2}$.

For firebrand generation rates, we adopt the baseline values established in Chapter 3: $GR^*_{\text{vegetation}} = 33.3\,\text{pcs}\cdot\text{MW}^{-1}\cdot\text{s}^{-1}$ for vegetative fuels and $GR^*_{\text{structure}} = 10.0\,\text{pcs}\cdot\text{MW}^{-1}\cdot\text{s}^{-1}$ for structural fuels. It is noted that the structural generation rate used here is higher than the reference value presented in Chapter 3 ($GR^*_{\text{structure}} = 5.68\,\text{pcs}\cdot\text{MW}^{-1}\cdot\text{s}^{-1}$). This adjustment is made based on observations of model behavior, to be discussed in Section 6.5, and is intended to represent a critical condition for sustained firebrand-driven fire propagation in idealized WUI configurations under moderate wind speeds and typical structure separation distances.

For firebrand deposition modeling, Sardoy's distribution [83] is implemented for wild-

land fuels, as described in Section 4.1 through Equations 4.1 to 4.5, which define the distance-dependent deposition probability for firebrands released from flaming wildland fuels under varying wind and fireline intensity conditions. However, the Sardoy model performs poorly at low fireline intensities—particularly in structural fire contexts, where the initial heat release rate (HRR) may approach zero.

As shown in Figure 6.2a, the model predicts a maximum spotting distance (defined as the 99th percentile of the distribution) that remains elevated—above 40 meters—and even increases as fire intensity diminishes. This behavior is physically unrealistic, as firebrand spotting should vanish in the absence of active combustion.



(a) Sardoy model

(b) Himoto model

Figure 6.2: Maximum spotting distance (99th percentile of the deposition distance distribution) as a function of fireline intensity. (a) and (b) show results using Sardoy and Himoto models, respectively. Different line styles indicate different wind speeds: solid (10 mph), dashed (20 mph), dash-dotted (30 mph), and dotted (40 mph).

To address this limitation, we adopt the model developed by Himoto et al. [82], originally formulated for firebrand transport in urban environments. Himoto's model defines a lognormal deposition distribution, where the parameters depend on wind speed, fireline intensity, and geometric characteristics of the fuel source. Specifically, the mean (μ') and standard deviation (σ')

of the downwind travel distance are given by:

$$\mu'/D = 0.47\, B^{*2/3}, \qquad \sigma'/D = 0.88\, B^{*1/3} \tag{6.3}$$

where D is the characteristic source dimension (e.g., pool diameter). In this study, we adopt $D = 10\,\mathrm{m}$ as a representative structure size for typical WUI configurations. The parameter B^* is a non-dimensional buoyancy-based number defined as:

$$B^* = \frac{U_{\text{wind}}}{\sqrt{gD}} \left(\frac{\rho_p}{\rho_\infty}\right)^{-3/4} \left(\frac{d_p}{D}\right)^{-3/4} \left(\frac{HRR}{\rho_\infty c_p T_\infty \sqrt{g} D^{2.5}}\right)^{1/2} \tag{6.4}$$

where HRR is the heat release rate of the entire fire source, ρ_p and d_p are the density and diameter of the firebrand particles. We assume $\rho_p = 100\,\mathrm{kg}\cdot\mathrm{m}^{-3}$ and $d_p = 5\,\mathrm{mm}$. The other parameters are identical to the ones in Sardoy's model.

This model has been successfully integrated into prior urban fire spread simulations [39] and, as illustrated in Figure 6.2b, provides physically realistic predictions of spotting behavior—particularly, reduced firebrand transport under low fire intensity conditions. Accordingly, we adopt the Himoto model for structural fuels in the present simulation framework. For biomass fuels, we retain the Sardoy model, as no prescribed design fire curve is defined for wildland fuels (where fireline intensity is constant and derived from the Rothermel model upon ignition), and because the Sardoy model is specifically developed for line fire configurations typically found in wildland fire contexts.

The ignition models applied in this study follow the formulations described in Chapter 5. At this stage, we assume that the transition from smoldering to flaming, as well as the response

to accumulated firebrands and ambient wind conditions, are identical across all fuel types. This behavior is parameterized using the ignition characteristics of pressure-treated wood (PTW), as detailed in Section 5.2.1.

This simplification introduces a known limitation: prior studies [107, 138] have shown that ignition time scales differ significantly between porous wildland fuels (on the order of a few seconds) and solid structural materials (tens of seconds). However, it is important to note that in wildland fuels, the total ignition time scale (i.e., the combined duration of $t_{\text{ign,small}}$ and $t_{\text{ign,large}}$) is typically much longer than the surface fire spread time scale, meaning that potential spot fires are frequently overrun by the advancing surface fire front. Therefore, while this assumption may warrant refinement in future work, it is considered adequate for capturing the qualitative fire spread behavior in the present simulation scenarios.

Despite the use of a unified ignition model, the ignition behavior of the two fuel categories is still differentiated in two key ways. First, the nominal wind speed at the fuel surface differs due to varying surface roughness, which depends on topographic and fuel characteristics. Second, we assign different durations for the transition from ignition to a fully developed fire: structural fuels are assumed to exhibit a medium fire growth rate with $t_{\text{ign, large}} = 300\,\text{s}$, while vegetation is treated as an ultra-fast growing fire with $t_{\text{ign, large}} = 75\,\text{s}$.

Next we describe the baseline configuration for the simulator. As previously mentioned, the simulations are conducted within an idealized artificial WUI community. In the one-dimensional framework, the community is characterized by uniformly distributed structures with constant building footprint and spacing, denoted by a and d, respectively.

Figure 6.3 presents statistics from a preliminary survey of a representative WUI community affected by the Thomas Fire. The results indicate that the average building footprint dimension

is approximately $a = 16.24\,\mathrm{m}$, and the average separation distance between structures is $d = 7.69\,\mathrm{m}$. For simplicity, we use $a = d = 10$ m as a representative case, without loss of generality. These values are adopted as baseline parameters for the simulation cases. A schematic layout of this idealized community configuration is shown in Figure 6.4. For numerical consistency and ease of implementation, both a and d are chosen as integer multiples of the grid resolution ($\Delta x = 10\,\mathrm{m}$ for the baseline case) to ensure accurate spatial representation on the computational mesh.

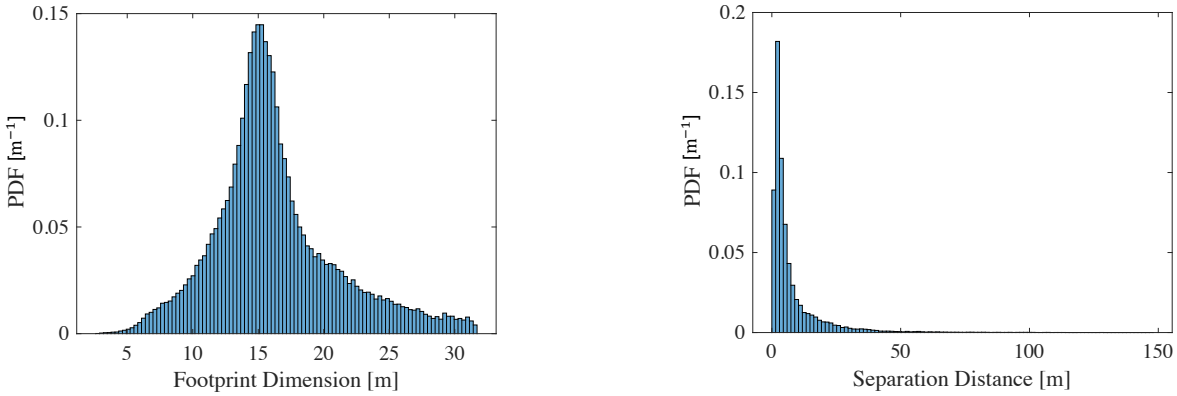


Figure 6.3: Statistics of fire-affected WUI communities during the Thomas Fire. (a) Distribution of structure footprints; the average footprint size is 16.24 m (standard deviation 4.72 m). (b) Distribution of building separation distances; the average spacing is 7.69 m (standard deviation 11.20 m).

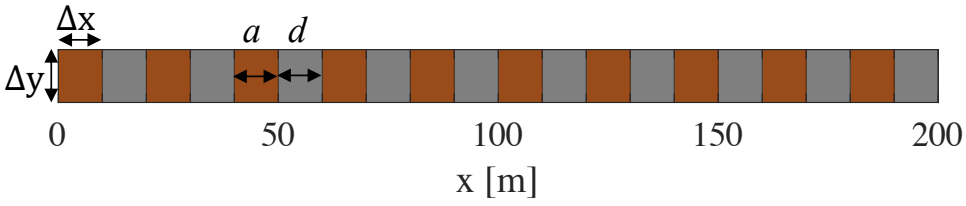


Figure 6.4: Baseline 1-D WUI configuration implemented in MATLAB. Brown cells represent structures, and gray cells represent firebreaks. Δx is the grid resolution, Δy is the grid size perpendicular to the fire propagation direction, a is the structure footprint dimension, and d is the building separation distance. In this example, $\Delta x = 10\,\mathrm{m}$, $a = d = \Delta y = 10\,\mathrm{m}$.

A baseline wind speed of 40 mph is selected to represent a relatively high wind scenario during a WUI fire event. This choice facilitates effective firebrand-driven spread in the simulation and aligns with observed meteorological conditions during historical fires. Figure 6.5 presents a histogram of gust wind speeds recorded over a 38-hour period during the Thomas Fire in December 2017, confirming that 40 mph constitutes a high but realistic wind condition for this case study.

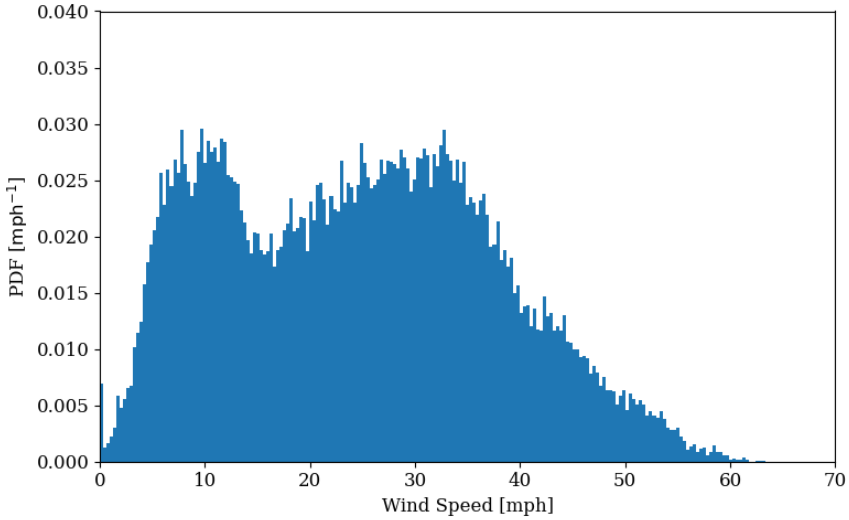


Figure 6.5: Histogram of gust wind speeds during the Thomas Fire event from 2017/12/05 UTC 02:00 to 2017/12/06 UTC 15:00. Data are extracted from the RTMA dataset [139] for the Thomas Fire region with latitude/longitude bounds: $(-119.8406, -118.7603, 34.0160, 34.7793)$. The observed distribution supports the selection of 40 mph as a representative high-wind condition.

In summary, the baseline simulation configuration is presented in Table 6.1. It represents an idealized wildland–urban interface (WUI) scenario, characterized by structures with square footprints of dimension $a = 10$ m, separated by inert firebreaks of width $d = 10$ m, and subjected to a temporally constant and spatially uniform wind of 40 mph. The numerical resolution is set to $\Delta x = 10$ m, with a Courant–Friedrichs–Lewy (CFL) number of 0.5 to ensure accurate resolution of structural features and firebrand transport dynamics.

The normalized firebrand generation rates (GR^*) are prescribed as 33.3 and 10.0 $\mathrm{pcs}\cdot$

$\mathrm{MW}^{-1} \cdot \mathrm{s}^{-1}$ for vegetation and structural fuels, respectively. The emission durations for vegetation and structural fuels follow the formulations described in Sections 3.2 and 3.3. Firebrand travel distances are modeled using lognormal distributions, with parameters derived from Sardoy's model (Equations 4.3 and 4.5) for vegetation fuels and Himoto's model (Equation 6.3) for structural fuels. The time to reach a fully developed fire is assumed to be 75 s for vegetation and 300 s for structural fuels.

Table 6.1: Summary of baseline simulation parameters.

Parameter	**Vegetation**	**Structure**
GR^* [$\mathrm{pcs} \cdot \mathrm{MW}^{-1} \cdot \mathrm{s}^{-1}$]	33.3	10.0
Firebrand emitting duration	$\Delta x / ROS_{\text{surface}}$	By design fire curve (Figure 6.1)
Firebrand travel distance PDF	Sardoy model (Eqns. 4.3, 4.5)	Himoto model (Eqn. 6.3)
Time to fully developed fire [s]	75	300
Global Settings		
Structure footprint size, a [m]		10
Firebreak width, d [m]		10
Wind speed, U_{wind} [$\mathrm{m} \cdot \mathrm{s}^{-1}$]		17.8 ($\approx$ 40 mph)
x-Grid resolution, Δx [m]		10
y-Grid resolution, Δy [m]		10
CFL number ($\frac{U_{\text{wind}} \Delta t}{\Delta x}$)		0.5

6.2 Spatial Resolution Sensitivity in Firebrand-Driven Spread Configurations

In this section, we examine the numerical treatment of structures within wildland–urban interface (WUI) regions when performing fire spread simulations using tools such as ELMFIRE. The previously introduced MATLAB surrogate model and idealized WUI configuration are used to illustrate key considerations.

In landscape-scale fire simulations, computational grids typically have spatial resolutions on the order of 10 meters or more. For example, ELMFIRE commonly utilizes inputs at 30-meter resolution. While such resolution is relatively fine for most GIS-based environmental modeling,

it remains insufficient to capture important subgrid-scale phenomena—particularly in heterogeneous areas such as WUI communities. In some instances, a single structural unit (e.g., a residential building) may span multiple computational cells, while in others, several smaller structures may be aggregated into a single cell.

These discretization practices introduce several modeling challenges and can lead to misrepresentation of structure-level fire behavior. In particular, errors arise in the application of urban fire spread models—both surface fire models and firebrand models—due to underlying assumptions that are not always compatible with coarse grid-based representations.

For example, the Hamada model computes fire spread rates based on a uniform community with identical building footprints and separation distances. When implemented within a level-set framework, the model does not distinguish between fire spreading within a structure and between structures. Consequently, if a building is partially resolved across multiple cells, the model treats fire propagation within that structure as equivalent to inter-structure spread—violating the model's assumptions.

Similarly, the WU-E model encounters limitations when applied to discretized structures. In addition to spatial misrepresentation, the model relies on a prescribed heat release rate (HRR) for each structure to compute ignition and fire spread. In grid-based implementations, HRR is typically assigned per unit area, implicitly assuming that the total HRR of a structure is the linear sum of its constituent cells. However, classical scaling theory and experimental evidence suggest that whole-building fire behavior does not scale linearly with surface area. Ventilation, compartmentalization, and fuel distribution all contribute to nonlinear combustion dynamics at the structure level.

Moreover, the design fire curves and radiative heat transfer models used in the WU-E

framework have not been rigorously validated for substructure fire spread. When a structure is split across multiple cells, discontinuities in thermal feedback and combustion behavior introduce uncertainty. This can lead to erroneous predictions in flame development, spread timing, and firebrand generation—particularly in WUI scenarios where structure-level fidelity is critical.

The firebrand generation and deposition submodels face similar concerns. Models developed by Sardoy [83] and Himoto [82] assume that firebrands are generated from integral, fully developed fire sources—vegetative or structural. It remains an open question whether these correlations apply when a structure is fragmented across a discretized grid.

Given these considerations, we investigate in this section the compatibility of the modeling framework developed in Chapters 3 through 5 when applied to structure fuels treated as discrete entities in simulation frameworks like ELMFIRE. This issue is addressed through a series of verification tests that systematically vary the mesh resolution thus the spatial representation of structural fuels.

6.2.1 Baseline Case: Firebrand-Driven Spread Verification (Test 6-1)

We begin with a baseline case to verify the numerical accuracy of the firebrand transport and ignition models. The test configuration, illustrated in Figure 6.4, follows the idealized WUI layout summarized in Table 6.1. In this simulation, fire spread is driven solely by firebrands, with the surface fire spread rate explicitly set to zero ($ROS_{\text{surface}} = 0\ \text{m} \cdot \text{s}^{-1}$). At the given resolution, the first structure ($0 \leq x \leq 10\ \text{m}$) is initialized as ignited ($\phi = -1$) and begins emitting firebrands in accordance with the prescribed design fire curve (Figure 6.1).

Figure 6.6a presents the accumulated firebrand number load at each structure—defined

as the total number of deposited firebrands normalized by the cell area ($N/(\Delta x \times \Delta y)$)—and compares it with the analytical prediction.

Given the prescribed heat release rate (HRR), normalized firebrand generation rate (GR^*), and wind speed ($U_{\text{wind}} = 40$ mph), the number of firebrands deposited in cell k, bounded by $x_k = k\Delta x$ and $x_{k+1} = (k+1)\Delta x$, is calculated as:

$$N_k = \sum_{i=1}^{k-1} \int_0^{\infty} \int_{x_k}^{x_{k+1}} [GR_i^* \cdot HRR_i(t)] \cdot f\left(x - i\Delta x \,|\, \mu_i(t), \sigma_i(t)\right) \, \mathrm{d}x \, \mathrm{d}t, \tag{6.5}$$

where $f(\cdot)$ is the lognormal probability density function defined in Eqn. 4.1, with parameters $\mu_i(t)$ and $\sigma_i(t)$ given by Eqn. 6.3, which are functions of the local HRR and wind speed at emission location i and time t.

This formulation accounts for firebrands emitted from all upwind source cells $i = 1, 2, \ldots, k-1$ and deposited in the downwind target cell k. The analytical predictions, shown as black dotted lines in Figure 6.6a, exhibit excellent agreement with the numerical results, with relative errors below 1×10^{-3}. These results confirm the numerical accuracy of the firebrand generation and deposition submodels under the prescribed simulation conditions.

Figure 6.6b presents the time of arrival (TOA) of the first firebrand in each computational cell. For the first few structures, TOA follows the wind-driven trajectory:

$$TOA(x) = \frac{x + \Delta x}{U_{\text{wind}}} + TOA_2,$$

where TOA_2 denotes the arrival time at the second structure. This behavior is consistent with the verification tests in Section 4.4.1. The figure is plotted on a logarithmic scale for improved

visualization. Rather than using the simple reference $TOA = x/U_{\text{wind}}$, the offset accounts for the delay in reaching the HRR sufficient to produce spotting distances beyond Δx.

The maximum spotting distance—defined as the 99th percentile of the lognormal distribution at peak HRR—is approximately 80 m. Due to the time-dependent nature of the design fire curve, which requires around 300 s to reach maximum HRR, firebrand transport behavior deviates from the wind trajectory at longer distances. Beyond $x \approx 80$ m, TOA becomes nonlinear as ignition of new structures is required to emit firebrands capable of reaching farther cells. This validates the coupled behavior of the transport and fire growth model.

To verify ignition modeling, we calculate the critical firebrand number load required to exceed ψ_{crit}, the mass-based threshold for smoldering-to-flaming ignition, as defined in Eqn. 5.12. At a wind speed of 40 mph, the corrected near-pile air velocity is $v_{\text{air}} \approx 1.14\ \text{m} \cdot \text{s}^{-1}$, based on Eqn. 5.13 using a correction factor of 0.064. This yields:

$$\psi_{\text{crit}} = \frac{0.21}{(v_{\text{air}} + 0.07)(4.11 - v_{\text{air}})} = \frac{0.21}{(1.14 + 0.07)(4.11 - 1.14)} \approx 0.059\ \text{g} \cdot \text{cm}^{-2}. \tag{6.6}$$

Assuming an average firebrand mass of 0.2 g, the corresponding number load is:

$$N_{\text{crit}} = \frac{\psi_{\text{crit}} \cdot 10^4\ \text{cm}^2 \cdot \text{m}^{-2}}{0.2\ \text{g}} = 2928.9\ \text{pcs} \cdot \text{m}^{-2}.$$

Figure 6.6c shows the firebrand accumulation history at the cell spanning $20 \leq x \leq 30$ m. The ignition threshold (N_{crit}) is reached at $t \approx 2496.9$ s (marked with a black dotted line). Figure 6.6d presents the corresponding ignition time ($t \approx 2858.8$ s), confirming the expected time lag due to the ignition model. The delay includes the 42 s smoldering-to-flaming transition time

($\tau_{\text{ign,small}}$) and a 300 s delay ($t_{\text{ign,large}}$) to establish a fully developed fire. These results verify the ignition model's fidelity and consistency with the prescribed probabilistic ignition behavior.

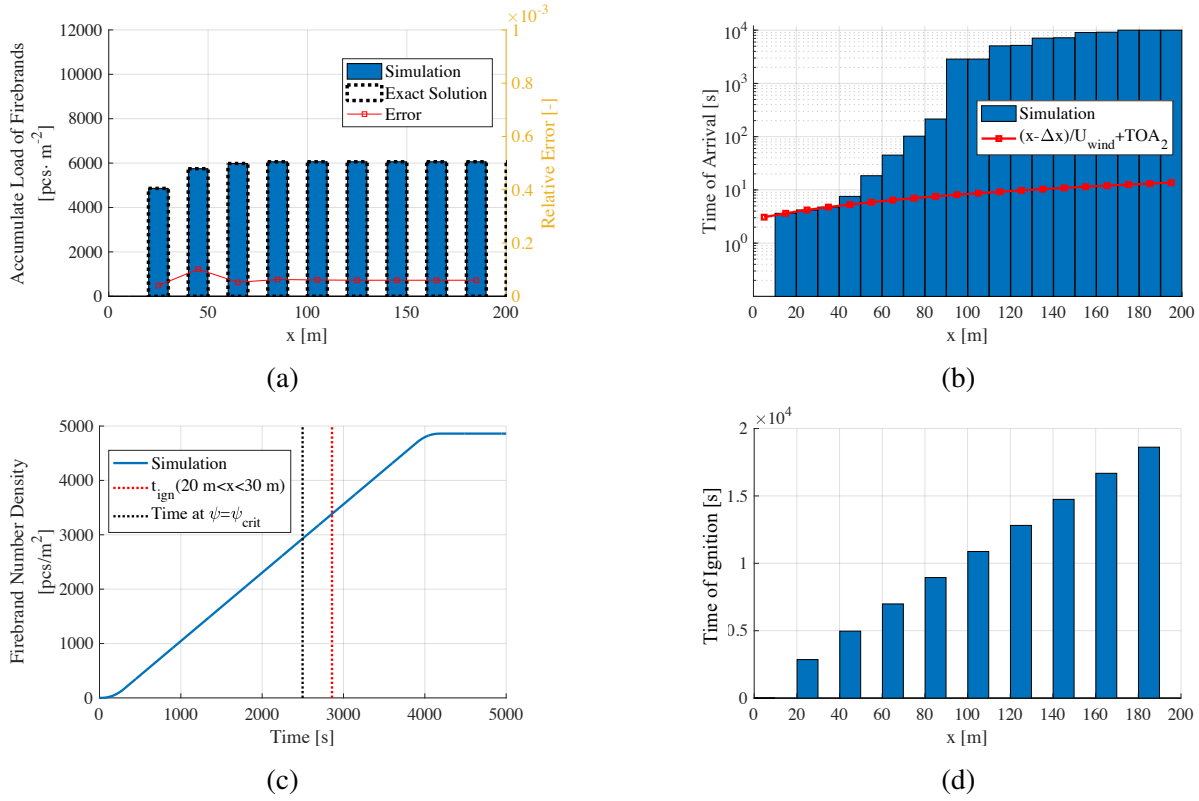
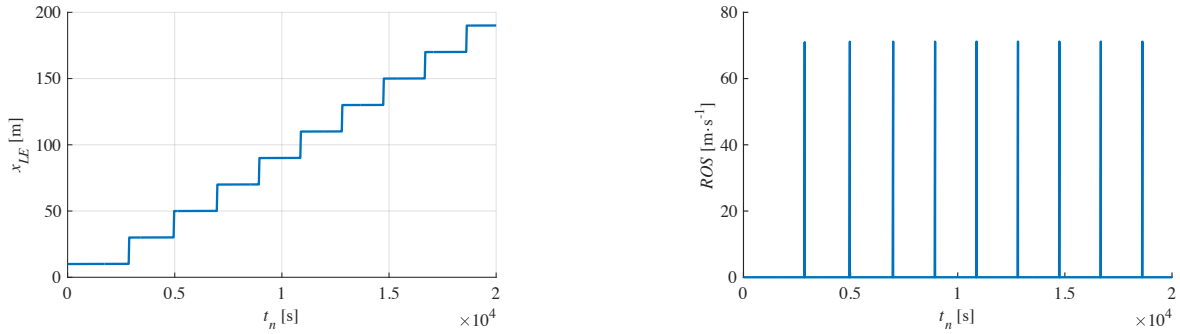


Figure 6.6: Baseline WUI verification test for firebrand-driven spread with surface fire disabled. (a) Accumulated firebrand load (numerical vs analytical); (b) Time of first firebrand arrival (TOA); (c) Accumulated number of firebrands at $x = 20{-}30$ m; (d) Ignition time of each cell.

As in the previous verification tests in Chapters 4 and 5, the fire leading edge is evaluated by tracking the zero contour of the level set function ($\phi = 0$), and the instantaneous rate of spread (ROS) is computed as its time derivative. Figure 6.7a shows the evolution of the fire front location over time, while Figure 6.7b presents the corresponding ROS. A stepwise progression of the fire front is observed, with the ROS oscillating between zero and high values due to the time integration scheme: ROS remains zero during inactive phases and spikes sharply when ignition occurs, as ϕ is updated in a single time step, yielding $ROS = (a + d)/\Delta t \approx 71.5\ \text{m} \cdot \text{s}^{-1}$. Although the stochastic ignition model introduces some variability, the intervals between ignition

events are relatively uniform, reflecting the relative scale between the time for smoldering-to-flaming transition (SFT, on the order of tens of seconds) and fire development (300 s).

The resulting average ROS is approximately $0.006\ \mathrm{m} \cdot \mathrm{s}^{-1}$, which is significantly lower than the value predicted by the Hamada model ($0.34\ \mathrm{m} \cdot \mathrm{s}^{-1}$) for a similar configuration ($a = d = 10$ m, fully combustible structures). However, the difference is expected: the current case includes only firebrand-driven ignition, whereas the Hamada model accounts for all modes of fire spread. Thus, while conservative, the predicted ROS remains within a physically reasonable range given the limited ignition mechanisms represented in this test.



(a) (b)

Figure 6.7: WUI test results—fire front dynamics under firebrand-driven ignition. (a) Fire front location ($\phi = 0$); (b) Instantaneous rate of spread (ROS).

6.2.2 Impact of Structure Discretization (Test 6-2)

We now examine the effects of mesh resolution first by applying a finer grid resolution of $\Delta x = 2.5$ m, such that $\Delta x < a$ as illustrated in the bottom panel of Figure 6.8. In this configuration, the computational cells still align exactly with the prescribed edges of structures and firebreaks. The time step is controlled by a CFL number of 0.5, giving $\Delta t = 0.5\Delta x / U_{\text{wind}} = 0.07$ s. Other parameters are identical to the baseline configuration in Test 6-1. The grid spacing

in the y direction remains $\Delta y = 10$ m to retain consistent structure footprint dimensions (10 m width).

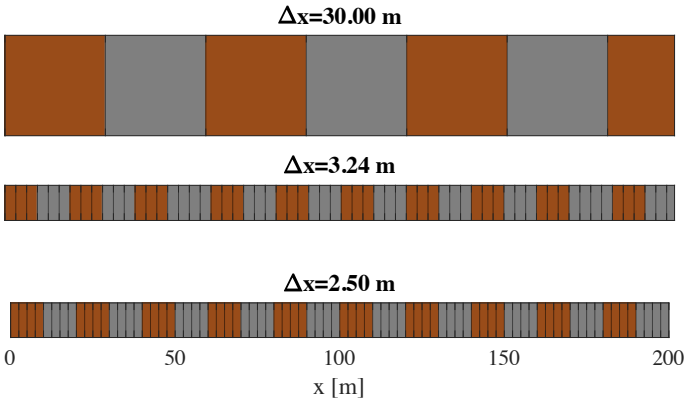


Figure 6.8: Schematic illustration of discretized academic WUI configurations. Top: $\Delta x = 30$ m; Middle: $\Delta x = 3.24$ m; Bottom: $\Delta x = 2.5$ m. Gray areas represent firebreaks, and brown areas denote structures. The baseline layout matches the configuration shown in Figure 4.4, where $a = d = 10$ m. For cases where $\Delta x \leq a$, the cross-grid dimension is fixed at $\Delta y = 10$ m; for $\Delta x > a$, we set $\Delta y = \Delta x$ to preserve a square footprint.

As the structures are now resolved by multiple grid cells, each structural cell behaves independently: every ignited cell emits firebrands according to its own heat release rate, and ignition occurs locally based on firebrand deposition. This behavior is illustrated in Figure 6.9.

Figure 6.9a presents the accumulated firebrand number load per unit area. The black dotted line represents the analytical solution derived in Section 6.2.1. Unlike the coarse-grid case (Test 6-1), cells comprising the same structure accumulate different numbers of firebrands. Notably, the downwind cells of each structure receive more firebrands, as they benefit from firebrands emitted by upstream cells within the same structure.

This results in a total firebrand accumulation per structure that exceeds the analytical prediction, for two primary reasons:

1. Self-originated firebrands are now included in the deposition calculation, whereas the pre-

vious analysis only considered external sources.

2. The heat release rate (HRR) used for transport modeling is now applied on a per-cell basis: $HRR_i = HRRPUA \cdot \Delta x \cdot \Delta y$. Although the total firebrand output per structure remains the same—$N \cdot HRRPUA \cdot \Delta x \cdot \Delta y \cdot GR^*$, where N is the number of cells per structure—the per-cell HRR used to determine transport distance is lower.

This behavior is further explained in Figure 6.10, which compares the firebrand transport probability density functions (PDFs) for the same wind speed (40 mph) but with three different HRRs: 40 MW (for $\Delta x = 10$ m), 10 MW (for $\Delta x = 2.5$ m) and 120 MW (for $\Delta x = 30$ m). Due to the lognormal parameterization of the PDF, the lower HRR yields a distribution skewed toward shorter distances, thereby increasing local deposition near the source.

Figure 6.9b shows the ignition time of each cell. Due to local variability in firebrand accumulation, ignition occurs at significantly different times across the structure footprint. Comparing the ignition time of the first ignited cell (measured from the upwind to the downwind side) to the ignition time predicted in the case with $\Delta x = 10$ m, the former case exhibits a longer ignition delay. This is because firebrands accumulate at a lower rate due to the altered probability density function (PDF), as shown in Figure 6.9c, where the firebrand accumulation rate at the cell located within $20 < x < 22.5$ m is lower than that in Test 6-1 (where accumulation occurs over $20 < x < 30$ m). As the fire spreads through the community, these timing discrepancies accumulate and delay the progression of the fire front. The resulting average rate of spread (ROS) is approximately $0.001\ \mathrm{m \cdot s^{-1}}$, significantly lower than the value obtained in Test 6-1 ($\sim$0.006 $\mathrm{m \cdot s^{-1}}$) and much lower than predictions from the Hamada model ($\sim$0.34 $\mathrm{m \cdot s^{-1}}$ for $a = d = 10$ m).

However, we caution against interpreting this ROS value as physically meaningful. The firebrand transport model used in this simulation is calibrated to represent transport from an entire structural fire source. Applying the same formulation to individual subgrid cells misrepresents the associated HRR, resulting in a physically inconsistent estimation of firebrand travel distance. Consequently, while the results of Test 6-2 are explainable based on the numerical setup, the simulated ROS should not be interpreted as a reliable prediction.

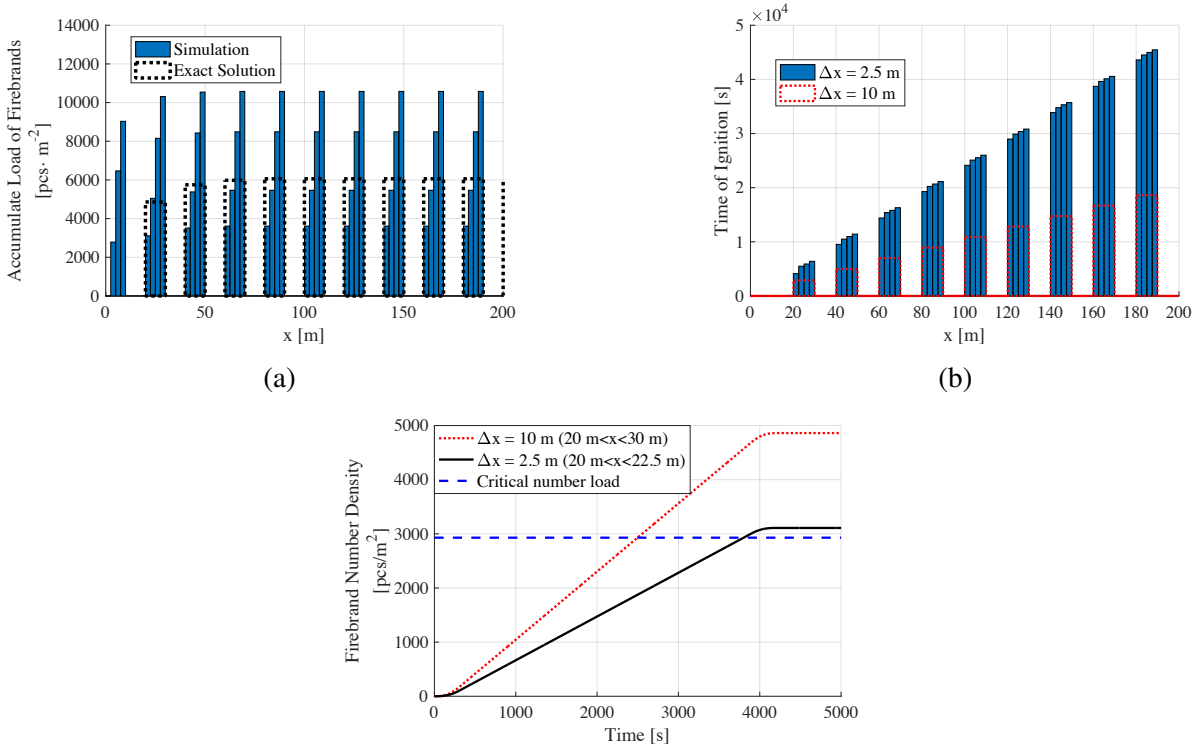


Figure 6.9: Test 6-2: WUI simulation with refined grid resolution ($\Delta x = 2.5$ m). (a) Accumulated firebrand load (numerical vs. analytical); (b) Ignition time of each structural cell; (c) Accumulated number of firebrands for cell $20 < x < 30$ m ($\Delta x = 10$ m) and cell at $20 < x < 22.5$ m ($\Delta x = 2.5$ m).

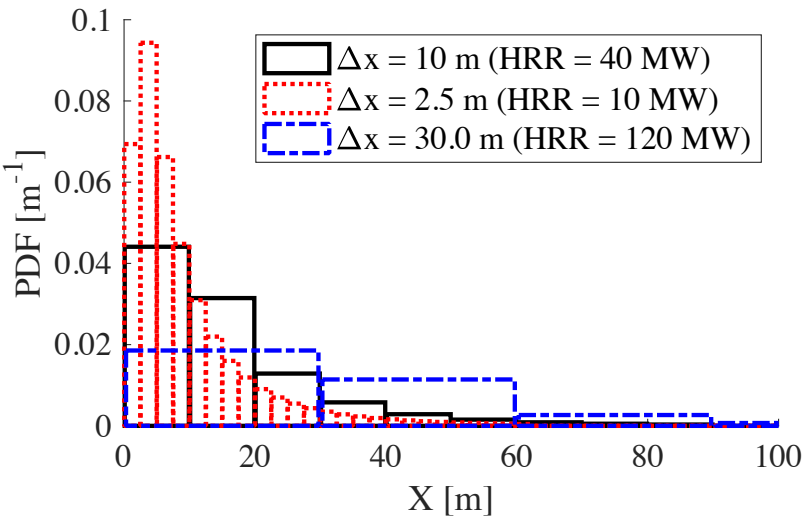


Figure 6.10: Comparison of firebrand travel distance PDFs for $U_{\text{wind}} = 40$ mph and HRR = 10 MW vs 40 MW.

6.2.3 Impact of Coarse Spatial Resolution (Test 6-3)

A third variation of Test 6-1 investigates the case where the grid resolution is coarser than the structure footprint, a condition frequently encountered in practical ELMFIRE applications. This case, referred to as Test 6-3, applies a spatial resolution of $\Delta x = 30$ m—consistent with ELMFIRE's typical operational scale. The top panel of Figure 6.8 shows the domain layout.

In ELMFIRE, structural representation in coarsened fuel maps is determined through down-sampling, implemented using the GDAL software package [140]. The case presented here uses `mode` resampling, which assigns each coarse cell the most frequent class value (mode) from the fine-resolution grid it overlaps. This process effectively results in a new grid where both structure and firebreak dimensions are $a = d = 30$ m, with Δy also set to 30 m to retain square structure footprints.

Figure 6.11a shows the accumulated number load of firebrands per unit area. The observed firebrand loads are close to the ignition threshold of $2928.9\ \text{pcs} \cdot \text{m}^{-2}$, but generally reduced

compared to the baseline (Test 6-1). Two main factors contribute to this reduction:

1. The separation distance between structures has increased from $d = 10$ m to $d = 30$ m. As shown in Figure 6.10, the probability of firebrand deposition at the target decreases with distance. The maximum probability density for deposition drops from 0.039 (for $\Delta x = 10$ m) to 0.014 (for $\Delta x = 30$ m).

2. The downsampled structure has a larger footprint area. Because the deposited firebrands are normalized by area, increasing Δy from 10 m to 30 m results in a factor-of-three dilution of the firebrand load per unit area.

Despite these reductions, the total number of firebrands collected remains comparable to the analytical prediction due to a compensating factor: the effective HRR is overestimated during downsampling (this treatment of HRR is consistent with the way in the current implementation of the WU-E model in ELMFIRE). When two smaller structures are merged into one coarse cell, the footprint dimension triples (area increases from 100 m^2 to 900 m^2), so the total number of generated firebrands increases proportionally, as the HRR scales accordingly. As a result, the total number of firebrands deposited at one cell can be estimated as:

$$N''_{\text{firebrand}} \propto \frac{\text{PDF}(x)\Delta x \cdot (\text{HRRPUA} \cdot \Delta x \Delta y \cdot N_{\text{cell}})}{\Delta x \Delta y} \tag{6.7}$$

where $\text{PDF}(x)\Delta x$ corresponds to the proportion of firebrands deposited at the target; $\text{HRRPUA} \cdot \Delta x \Delta y \cdot N_{\text{cell}}$ represents the effective number of firebrands generated from upstream cells located prior to $x = 60$ m; and N_{cell} is the number of contributing upstream cells (equal to 1 for $\Delta x = 30$ m and 2 for $\Delta x = 10$ m). The value $N_{\text{cell}} = 2$ for $\Delta x = 10$ m is used because the first

emitting cell ($0 < x < 10$ m) contributes a negligible number of firebrands to the structure located at $60 < x < 70$ m. The entire numerator is normalized by the area of each receiving cell.

Therefore, when comparing the results at two spatial resolutions:

$$\begin{aligned}\frac{N''_{\text{firebrand}}(\Delta x = 10\text{ m})}{N''_{\text{firebrand}}(\Delta x = 30\text{ m})} &= \frac{[\text{PDF}(x)\Delta x]_{10\text{ m}}}{[\text{PDF}(x)\Delta x]_{30\text{ m}}} \cdot \frac{N_{\text{cell, 10 m}}}{N_{\text{cell, 30 m}}} \\ &= \frac{0.039}{0.014} \cdot \frac{10}{30} \cdot \frac{2}{1} = 1.86\end{aligned} \tag{6.8}$$

which aligns with the simulation result shown in Figure 6.11a.

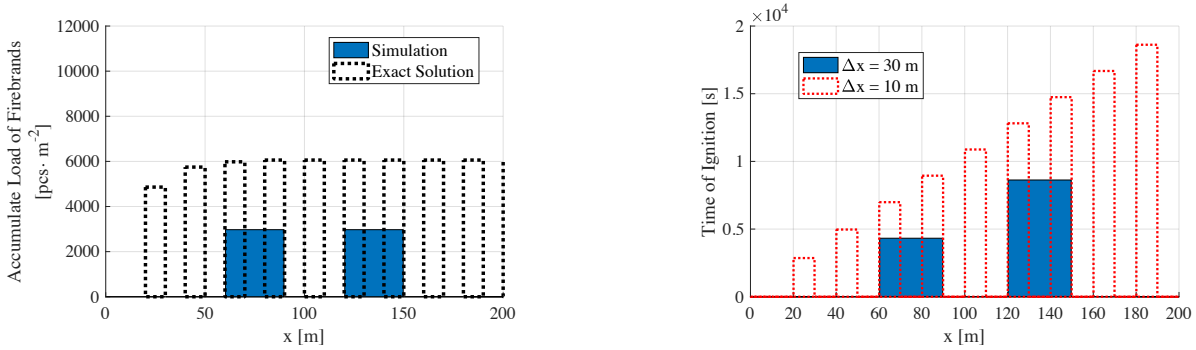
Figure 6.11b presents the ignition time for each structure. Fire spread appears accelerated compared to finer-resolution tests, but this is primarily an artifact of the inflated HRR values applied to the coarse-grid structures. The increase in firebrand generation leads to earlier accumulation of the critical ignition load at downstream cells, thus reducing ignition delays.

However, this faster spread should not be interpreted as a physically meaningful result. The overestimation of HRR and the artificial homogenization of structure footprints introduce significant errors. The findings suggest two key needs for ELMFIRE:

- Implementation of a subgrid model that accounts for realistic HRR distribution and firebrand clustering within coarse grid cells.
- A revised meshing strategy that preserves structure size and spacing during resampling to prevent distortion of WUI layout fidelity.

6.2.4 Mitigating Discretization Errors via Surface Fire Coupling (Test 6-4)

Given the discrepancies identified in Tests 6-2 and 6-3, we now investigate a potential solution aimed at mitigating the numerical issues related to firebrand-driven spread under fine or



(a) (b)

Figure 6.11: Test 6-3: WUI simulation with coarse resolution ($\Delta x = 30$ m). (a) Accumulated firebrand load (numerical vs. analytical); (b) Ignition time of each structure.

coarse resolution. Two primary issues are identified:

1. Misrepresentation of the heat release rate (HRR) when spatial discretization alters the perceived footprint size of a structure.

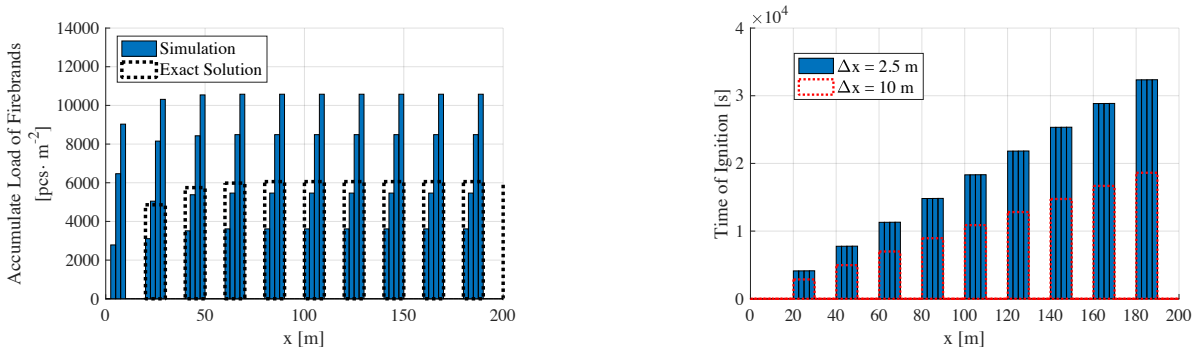
2. Uneven distribution of deposited firebrands across structural cells, which leads to inconsistent ignition times within a single structure and propagates errors as the fire spreads throughout the community.

The first issue affects the firebrand generation rate and distorts the firebrand transport distribution by altering the shape parameters of the lognormal PDF of travel distance. The second issue introduces artificial delays in firebrand ignition, as different parts of the same structure ignite at different times. This, in turn, affects downstream firebrand emissions and leads to compounding errors in long-range WUI spread predictions.

A practical workaround is to activate the surface fire spread mechanism. If the surface rate of spread (ROS_{surface}) is large enough, the entire structure is ignited quasi-simultaneously, thereby reducing intra-structure ignition time offsets.

Figure 6.12 shows results from Test 6-4, which is a variation of Test 6-2, now including a

non-zero surface fire spread rate. Specifically, $ROS_{\text{surface}} = 0.34\ \text{m} \cdot \text{s}^{-1}$ is applied, consistent with predictions from the Hamada model for $a = d = 10$ m.



(a) (b)

Figure 6.12: Test 6-4: Firebrand-driven WUI spread with surface fire spread activated ($ROS_{\text{surface}} = 0.34\ \text{m} \cdot \text{s}^{-1}$). (a) Accumulated firebrand load; (b) Ignition time of each structure.

Figure 6.12b shows that structure ignition occurs more uniformly and earlier compared to Test 6-2, thereby accelerating fire propagation through the WUI. However, Figure 6.12a reveals that the misrepresentation of firebrand accumulation persists—i.e., the number load is still distorted due to uneven generation and transport assumptions at the substructure level. Thus, even though surface fire spread improves synchronization of structure ignition, it does not fully resolve the root issue of spatial under-resolution in firebrand modeling.

Moreover, in some cases—particularly when HRR is low—the heat flux from a burning structure may not be sufficient to ignite adjacent structures via radiation or flame contact. In such situations, models like WU-E predict very low ROS_{surface}, which means surface spread is ineffective at synchronizing ignition. Therefore, while activating surface spread helps under some conditions, a more robust solution requires:

1. Subgrid models for firebrand redistribution and ignition,

2. HRR reassignment based on original (pre-downsampled) structural geometries,

3. Adaptive meshing strategies to preserve footprint-scale fidelity.

6.3 Resolution-Independent Framework for Discretized Structures

6.3.1 Proposed Method

(The methods proposed in this section have not yet been implemented in ELMFIRE. However, they could be incorporated in future work when additional input data become available to distinguish cells originating from individual structures—a capability currently under development by colleagues at UC Berkeley.)

To address resolution-induced artifacts in the firebrand transport and ignition model, we propose a simplified, resolution-independent treatment for discretized structures. This strategy ensures consistent firebrand deposition and ignition behavior across a range of mesh resolutions and discretizations.

1. Uniform Redistribution of Deposited Firebrands. Let a structure be represented by a contiguous set of k computational cells indexed by $i = 1, 2, \ldots, k$ in the mesh. The total number of firebrands deposited across the structure is aggregated and then uniformly distributed over the footprint area. The effective firebrand number load for each cell is defined as:

$$N'' = \frac{\sum_{i=1}^{k} N_i}{A_{\text{structure}}}, \tag{6.9}$$

where N_i is the number of firebrands deposited in cell i, and $A_{\text{structure}}$ is the total footprint area of the structure.

This averaging prevents uneven firebrand distribution that arises when different portions of a structure accumulate firebrands at different rates. For example, in Figure 6.9, the most upwind cell of a structure ignites earlier than others due to receiving more firebrands. In contrast, the case presented in Figure 6.6 assumes uniform deposition and produces consistent ignition behavior. Uniform redistribution across the structure footprint addresses this inconsistency and improves resolution independence.

This treatment is supported analytically in one-dimensional configurations. For both coarse and fine meshes, the number of firebrands deposited in a cell is the integral of the probability density function (PDF) over that cell's spatial interval. Provided that the structure boundaries align with the cell edges, the total deposited firebrand load remains the same. Even when the cell width Δx is not a divisor of the structure footprint (e.g., $\Delta x = 3.24$ m for a 10 m-wide structure), the long-tailed nature of the lognormal distribution ensures that small shifts in edge position produce negligible effects on firebrand accumulation. Moreover, the near-source region—where the PDF has its steepest gradient—is typically overrun by surface fire spread, limiting the impact of deposition errors on the overall spread dynamics.

2. Consistent Evaluation of HRR and Firebrand Generation. To eliminate the resolution dependence of firebrand generation, the heat release rate (HRR) for each structure is evaluated based on the total structure footprint area, rather than summing HRR from all occupied cells:

$$HRR_{\text{structure}} = HRRPUA \times A_{\text{structure}}, \tag{6.10}$$

where $HRRPUA$ is the heat release rate per unit area. This ensures that the total HRR, and thus the total firebrand generation, remains invariant with respect to mesh resolution. Since firebrand

generation depends on $GR^* \cdot HRR$, this treatment eliminates artificial variations in firebrand quantity and transport characteristics introduced by over- or underestimation of HRR in coarse or fine meshes. As a result, the firebrand travel distance PDF—parameterized by HRR—is also preserved across resolutions.

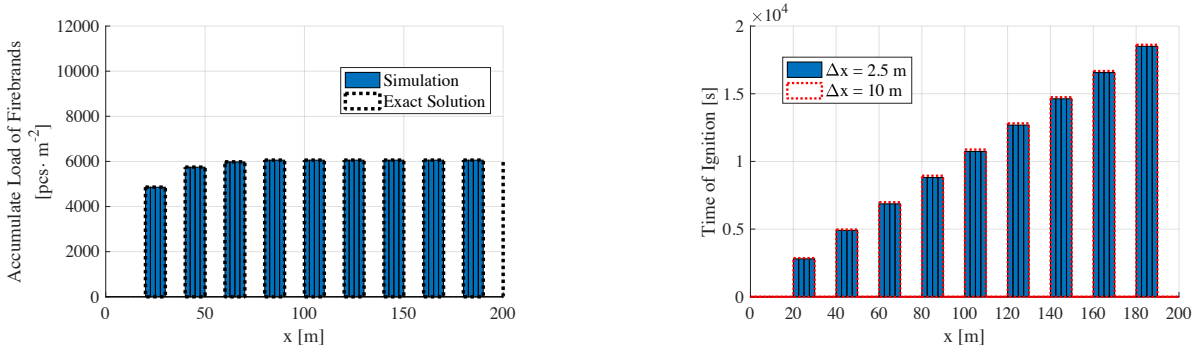
3. Centralized Emission and Downwind Deposition. Finally, we assume that all firebrands are emitted from the geometric center of each structure and are deposited only outside the emitting footprint. This eliminates artificial self-deposition of firebrands, which can arise in fine meshes when structures span multiple cells. By forcing firebrands to land outside the source structure, the overall deposition profile retains a consistent shape, irrespective of resolution.

Note that the proposed resolution-independent treatments do not resolve the broader issue of geometric misrepresentation of WUI community layout under coarse mesh resolutions. Specifically, coarse grids may inaccurately represent structural footprints and separation distances, thereby introducing systematic bias into fire spread predictions. This limitation is inherent in uniform-grid approaches and may only be adequately addressed through adaptive mesh strategies. However, due to time constraints, such developments are deferred to future work. Subsequent investigations could explore: (1) simulation sensitivity to WUI layout fidelity, (2) adaptive gridding algorithms tailored for structural layouts, and (3) availability of higher-resolution input data products. Notably, as discussed in Chapter 7, current publicly available WUI datasets are gridded uniformly, thus limiting near-term implementation of mesh refinement strategies.

6.3.2 Verification and Robustness Tests (Tests 6-5 & 6)

To assess the effectiveness of the proposed structure treatment, we present two example tests below.

The first example replicates the conditions of Test 6-2, with $\Delta x = 2.5$ m, $a = d = \Delta y = 10$ m, and $ROS_{\text{surface}} = 0$. Since the structure footprints align exactly with the computational mesh, we expect the results to match those from the baseline Test 6-1. Figures 6.13a and 6.13b confirm this expectation: the firebrand deposition profile matches the analytical reference, and the ignition timing matches the baseline simulation, demonstrating numerical correctness.



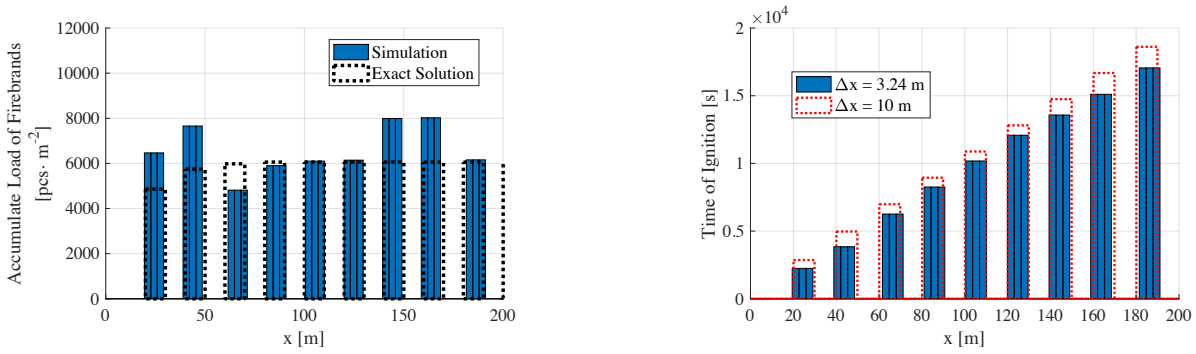
(a) (b)

Figure 6.13: Example 1: Reproduced Test 6-2 using discretized structure treatment ($\Delta x = 2.5$ m). (a) Accumulated firebrand load; (b) Ignition time of each structure.

The second example demonstrates the robustness of the treatment under imperfect discretization. Here, $\Delta x = 3.24$ m, representing a case where structure footprints cannot be perfectly aligned with grid cells. The fuel map is generated using mode sampling (as described earlier), and the resulting WUI layout is visualized in Figure 6.8 (middle panel). Despite slight misalignment, the layout visually preserves the original community design.

Figures 6.14a and 6.14b show that while deviations up to 33% exist in accumulated firebrand loads due to integration artifacts near structure boundaries, ignition timing predictions

remain within 10% error. This highlights the numerical robustness of the approach, particularly in capturing the dynamics of fire spread, which is the primary modeling objective.



(a) (b)

Figure 6.14: Example 2: Robustness test with mismatched mesh and footprint ($\Delta x = 3.24$ m). (a) Accumulated firebrand load; (b) Ignition time of each structure.

In conclusion, the proposed structure treatment ensures numerically accurate and resolution-independent modeling of firebrand-driven ignition and fire spread in idealized WUI layouts. While the approach does not fully correct for structural misrepresentation due to imperfect meshing, it significantly mitigates discretization-induced artifacts. The remaining challenge of capturing WUI geometry under coarse uniform grids may be addressed through future development of adaptive mesh algorithms, contingent upon the availability of higher-resolution spatial data.

6.4 Effect of Temporal Resolution on Fire Spread Predictions

The previous discussions focused on the effect of spatial resolution under a constant CFL condition. In this section, we investigate the robustness of the algorithm with respect to variations in the temporal resolution of the level-set solver. A series of simulations are performed using the same WUI layout shown in Figure 4.4. The grid resolution is fixed at $\Delta x = 10$ m, while the CFL number is varied across four values: 0.5, 1.0, 10.0, and 200. These correspond to time steps of

$\Delta t = 0.28$, 0.56, 5.59, and 111.86 s, respectively. Note that the CFL number is defined based on the local wind speed as $CFL = U_{\text{wind}}\Delta t/\Delta x$, and it is applied only to the level-set solver. The firebrand trajectory solver still uses $CFL_{\text{firebrand}} = 1$ for all cases, consistent with the scheme proposed in Section 4.2. Surface fire spread is deactivated in these tests, i.e., $ROS_{\text{surface}} = 0$. Therefore, we expect that the time step of the level-set solver defined by the CFL number should have minimal effect on the simulation results when $\Delta t_n \ll t_{\text{ign}}$, where t_{ign} is on the order of 1000 s in our cases.

Figure 6.15a presents the relative error in the accumulated firebrand load at each structure, referenced against the analytical solution defined in Eqn. 6.5, calculated as

$$\varepsilon_{N''} = \left| \frac{N''_{\text{simu}} - N''_{\text{analytical}}}{N''_{\text{analytical}}} \right|.$$

Even at CFL = 200, the maximum error is below 10%. For CFL $\leq$ 10, the error is generally limited to under 0.1%.

Figure 6.15b shows the relative error in the predicted time of arrival (TOA) of the first firebrand at each structure, referenced to the simulation with CFL = 0.5:

$$\varepsilon_{\text{TOA}} = \left| \frac{TOA_{\text{simu}} - TOA_{CFL=0.5}}{TOA_{CFL=0.5}} \right|.$$

The TOA error remains below 5% for fire spread distances up to 80 m when CFL $\leq$ 10. Larger deviations near the source are attributed to the hard-coded ignition time of the first structure at $t = \Delta t_n$, causing a propagating offset in TOA at neighboring cells. Beyond the maximum spotting distance of 80 m (defined under peak HRR), deviations diminish as ignition depends on

new firebrand sources rather than the initial one.

Figure 6.15d further validates this conclusion by presenting the relative error in ignition time, referenced against the CFL = 0.5 case. Ignition time errors remain under 0.1% when CFL ≤ 10, consistent with the fact that ignition delays (e.g., $\tau_{\text{ign,small}} = 42$ s, $t_{\text{ign,large}} = 300$ s) are much longer than firebrand transport times.

Figure 6.15c illustrates the firebrand accumulation history at the structure bounded by $x = 140-150$ m. Despite varying CFL values, all curves align well, further supporting the robustness of the transport algorithm.

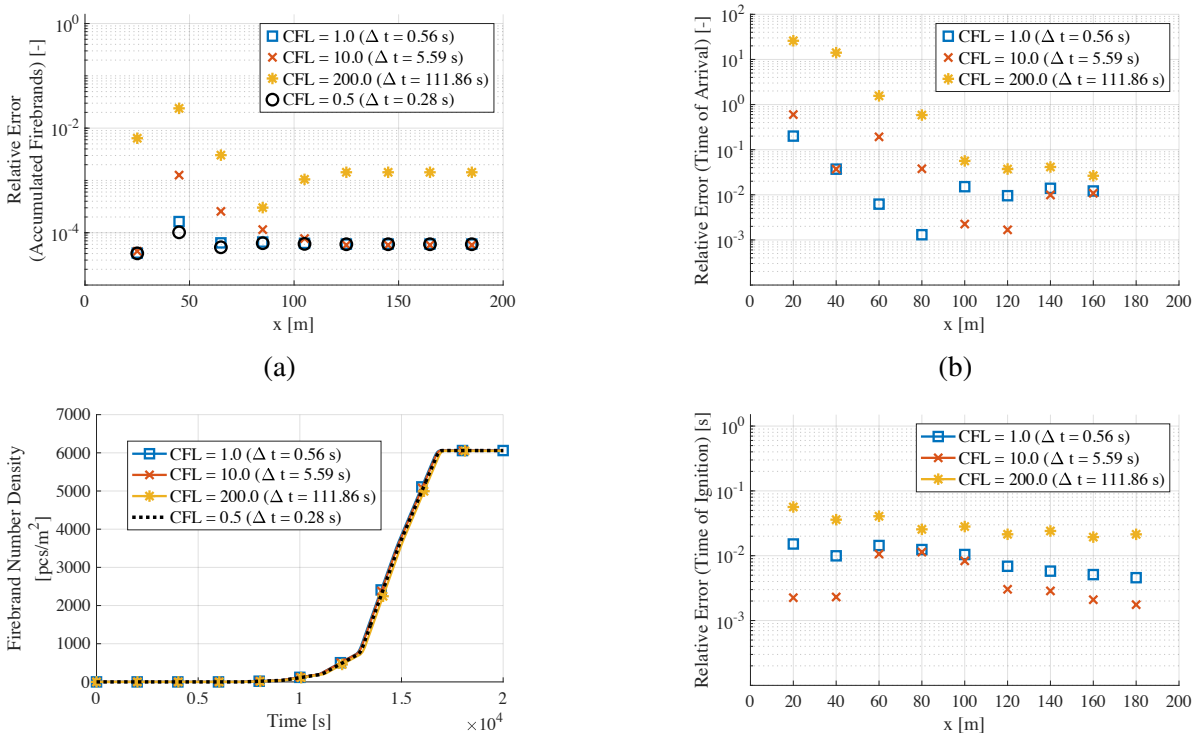


Figure 6.15: Evaluation of temporal resolution effects for firebrand-driven spread under varying CFL conditions. (a) Relative error of accumulated firebrand load; (b) Relative error in time of firebrand arrival (TOA); (c) Firebrand accumulation history at $x = 140-150$ m; (d) Relative error of ignition time.

Figure 6.16 provides a global perspective by plotting the mean absolute ignition time error

as a function of CFL:

$$\bar{\varepsilon} = \frac{1}{N} \sum_{i=1}^{N} \left| t_{\text{ign}}(x_i) - t_{\text{ign,CFL=0.5}}(x_i) \right| .$$

For CFL $\leq$ 50, the error remains below 0.1%. An unusually low error is observed at CFL = 5, likely due to the time step being an integer multiple of the CFL = 0.5 baseline. At CFL = 50 ($\Delta t = 28.0$ s), however, errors increase because the ignition timescale of 42 s is of similar magnitude, introducing statistical fluctuations in the smoldering-to-flaming (SFT) ignition model.

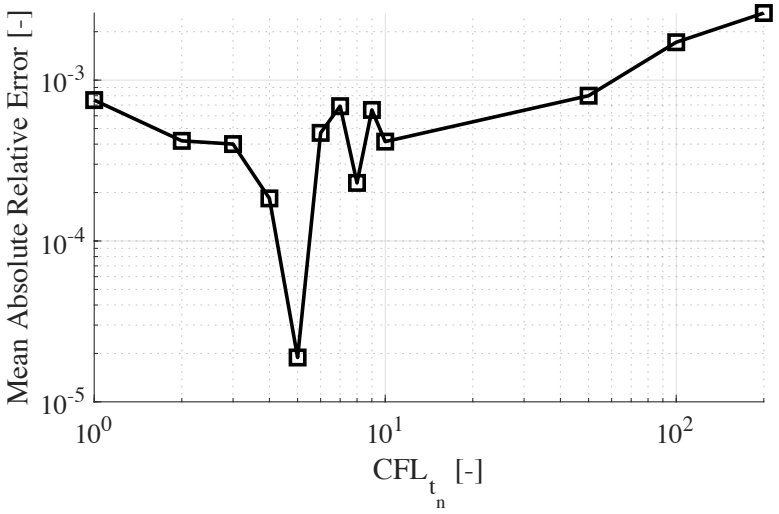


Figure 6.16: Mean absolute error of ignition time as a function of CFL.

In conclusion, the temporal integration scheme is robust within the implemented framework. In actual ELMFIRE simulations, the CFL constraint defined as

$$CFL_{t_n} = \frac{ROS_{\text{surface}} \Delta t_n}{\Delta x} \leq 0.5$$

is commonly employed for the level-set solver. Therefore, when the new firebrand models are activated—particularly in scenarios where the ignition timescale of firebrands at the target is significantly larger than the time step—this time constraint remains applicable. This stands in

contrast to the condition proposed in Chapter 4, where ignition occurs immediately upon deposition and a more restrictive CFL condition ($CFL_{t_n} = U_{wind}\Delta t_n/\Delta x < 0.8$) should be enforced.

It is important to note that this CFL condition applies exclusively to the level-set update. For the firebrand transport algorithm, the sub-time step $\Delta\tau$ continues to be constrained using $CFL_{\tau_n} = 1.0$, based on the local wind speed, to ensure consistent resolution of firebrand trajectories.

6.5 Model Insights from Parametric Analysis

This section investigates the physical implications predicted by the numerically accurate and robust firebrand-driven WUI fire spread algorithm. A series of simulations are designed to explore the critical conditions for firebrand-driven fire spread in the academic WUI configuration, focusing on the effects of the firebrand generation rate per unit heat release ($GR^*_{\text{structure}}$), inter-structure separation distance (d), and wind speed (U_{wind}).

6.5.1 Effect of Firebrand Generation Rate and Structure Separation

First, we examine the relationship between $GR^*_{\text{structure}}$ and d. Separation distances ranging from 2.5 m to 30 m (with 2.5 m increments) are tested against $GR^*_{\text{structure}}$ values from 5 to 50 $\text{pcs} \cdot \text{MW}^{-1} \cdot \text{s}^{-1}$. The structure footprint is fixed at $a = 10$ m, as in the baseline configuration. To accurately resolve structures and firebreaks, the grid resolution is set to $\Delta x = 2.5$ m. Consequently, the resolution-independent treatment described in Section 6.3 is adopted, treating each structure as a single unit for firebrand-related algorithms. The wind speed is held constant at 40 mph, and surface fire spread is disabled ($ROS_{\text{surface}} = 0$). All simulations enforce a CFL

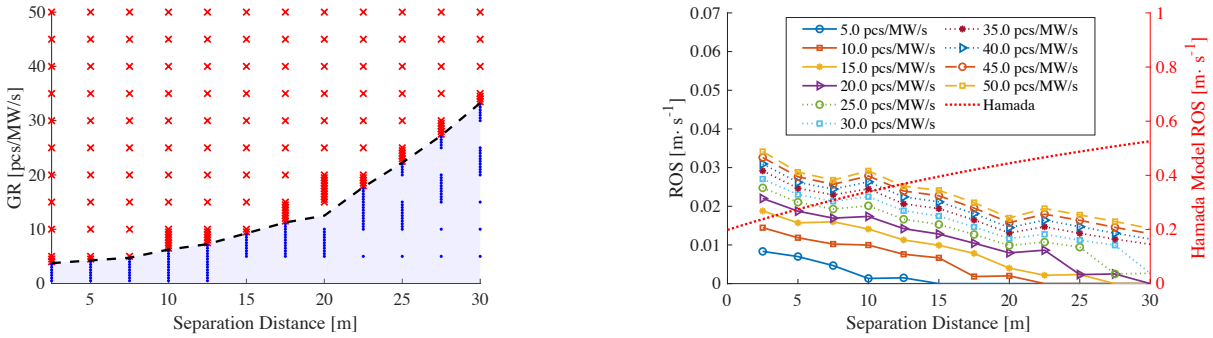
value of 0.5 (based on U_{wind}) for the level set solver.

Figure 6.17 presents the results. In Figure 6.17a, red crosses denote successful ignition and blue dots (in the shaded area) denote failure. The critical GR^* increases nonlinearly with d, ranging from approximately 2.5 to over 30 $\text{pcs} \cdot \text{MW}^{-1} \cdot \text{s}^{-1}$. Figure 6.17b shows that average ROS decreases with increasing d and decreasing GR^*, qualitatively consistent with field observations. These trends are physically expected: lower GR^* reduces the total number of emitted firebrands, and larger d increases the probability that downwind targets fall in the tail region of the firebrand deposition distribution. The surface fire spread rate (ROS_{surface}) predicted by the Hamada model is presented using the red dotted line. The Hamada model predicts ROS_{surface} values ranging from 0.2 to 0.6 $\text{m} \cdot \text{s}^{-1}$, which is approximately an order of magnitude higher than the firebrand-driven ROS. This is expected, as the Hamada model accounts for all fire spread mechanisms, including direct flame contact, thermal radiation, and potentially firebrands. It is worth noting that the Hamada model also predicts an increasing ROS_{surface} with increasing separation distance, which appears counterintuitive and physically unrealistic. In contrast, the WU-E model predicts a decreasing fire spread potential with increased separation as reported in Ref. [113].

6.5.2 Effect of Firebrand Generation Rate and Wind Speed

Next, we explore the relationship between $GR^*_{\text{structure}}$ and U_{wind}. Here, the separation distance is fixed at $d = 10$ m, and U_{wind} is varied from 10 mph to 150 mph. The $GR^*_{\text{structure}}$ is again varied between 5 and 50 $\text{pcs} \cdot \text{MW}^{-1} \cdot \text{s}^{-1}$, with other parameters identical to the previous section.

Figure 6.18 summarizes the results. In Figure 6.18a, a U-shaped trend emerges: the critical



(a) (b)

Figure 6.17: Effect of firebrand generation rate and structure separation distance on fire spread. (a) Ignition success/failure classification as a function of separation distance d and dimensionless firebrand generation rate $GR^*_{\text{structure}}$; (b) Average rate of spread (ROS) as a function of d and $GR^*_{\text{structure}}$. The Hamada model prediction for ROS_{surface} is shown as a red dotted line, scaled to the right-hand y-axis.

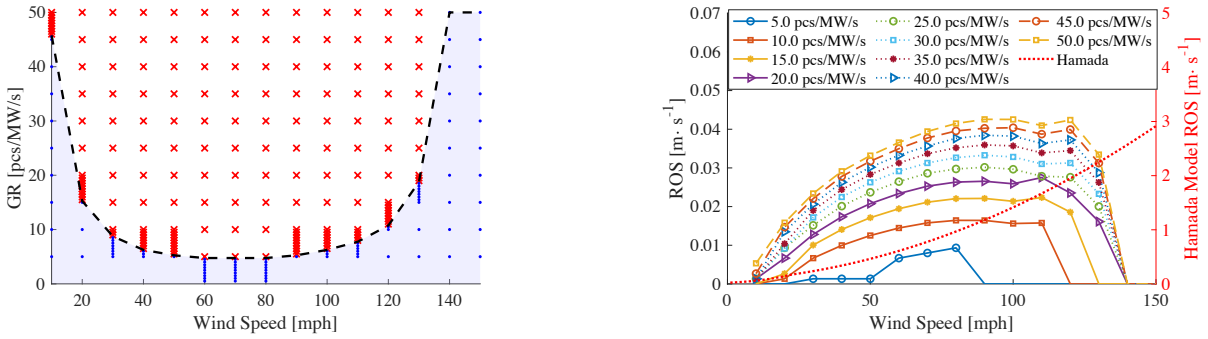
GR^* initially decreases with increasing wind speed, remains steady over the range 30–120 mph, and increases again at very high wind speeds. The observed U-shape results from two competing mechanisms:

- At low wind speeds, firebrands fail to reach distant targets due to a more concentrated (narrow) lognormal deposition PDF and reduced transport range. Additionally, lower wind speeds result in weaker near-surface airflow, which raises the ignition threshold (see Eqn. 5.12), making ignition less likely.

- At high wind speeds, while transport improves and more firebrands reach the target, the smoldering-to-flaming transition becomes less probable. According to the ignition model (Section 5.2), higher wind speeds increase the critical ψ, thus more firebrands are required to trigger ignition, which may offset the benefit of enhanced transport.

Figure 6.18b illustrates the corresponding ROS values, showing that fire spread is promoted by moderate wind speeds and high GR^*, but suppressed at very low and very high wind condi-

tions. Again, the surface fire spread rate (ROS_{surface}) predicted by the Hamada model is presented using the red dotted line, ranging from 0 to 3 $\mathrm{m} \cdot \mathrm{s}^{-1}$. These values are generally high because the model considers all modes of fire spread, including direct flame contact, radiation, and firebrand ignition. However, the extraordinarily high ROS_{surface} values predicted in the high wind speed regime—rising almost exponentially with wind speed—may indicate potential limitations of the Hamada model. In contrast, the WU-E model suggests a converged fire spread rate as wind speed increases [113].



(a) (b)

Figure 6.18: Effect of firebrand generation rate and wind speed on fire spread. (a) Ignition success/failure classification vs. U_{wind} and $GR^*_{\text{structure}}$; (b) Average ROS vs. U_{wind} and $GR^*_{\text{structure}}$; The Hamada model prediction for ROS_{surface} is shown as a red dotted line, scaled to the right-hand y-axis.

6.6 Conclusions

This chapter presented a series of numerical experiments, based on an academic one-dimensional WUI configuration, designed to evaluate the behavior and limitations of the firebrand-driven spread model when applied to urban fuels. Using MATLAB-based simulations, we identified several challenges that arise when directly applying the firebrand generation, transport, and ignition models developed in Chapters 3 through 5 to burning structures.

Specifically, we found that discretized structures—where heat release is defined per unit area—can lead to distorted firebrand transport and inconsistent ignition behavior when simulations are performed using different spatial resolutions. Additionally, coarse spatial resolution results in the misrepresentation of structure footprints and firebreaks, further degrading the numerical accuracy of firebrand behavior and fire spread predictions.

To mitigate these issues, a resolution-independent treatment for structural fuels was proposed. In this approach, each structure is treated as a single unit in terms of both heat release and firebrand generation/collection behavior. The method was shown to be numerically accurate and robust across a wide range of spatial and temporal resolutions. However, further improvements are needed—most notably, the incorporation of adaptive mesh refinement—to better represent community-scale fuel layouts and sub-grid-scale features such as roads, defensible zones (e.g., Zone 0), and heterogeneous fuel beds. Current implementations in ELMFIRE rely on structured meshes and publicly available input data with limited resolution (typically 30 m), which constrains the model's ability to resolve such features.

Despite these limitations, a parametric study was conducted to examine the model's predictive behavior under varying firebrand generation rates, inter-structure separation distances, and wind speeds. The results indicate that the model predicts a high likelihood of firebrand-driven propagation under moderate-to-high wind speeds (30–120 mph) and small separation distances between structures. These trends are physically interpretable based on the underlying firebrand transport and ignition mechanisms. However, it is important to note that the effective rates of spread derived in these tests reflect firebrand-only propagation, whereas real-world WUI fires typically involve multiple interacting mechanisms, including surface fire, radiation, and direct flame contact.

Lastly, we emphasize that while the results demonstrate physically consistent behavior, there is currently a lack of experimental or observational data suitable for direct validation. Future work should aim to bridge this gap through improved field observations, integration of high-resolution WUI datasets, and the adaptation of the proposed structural treatment into the full ELMFIRE framework. Ongoing work by the author and collaborators at UC Berkeley is actively addressing these challenges, although the error-mitigation methods proposed in this chapter have not yet been incorporated into the current version of ELMFIRE.

Chapter 7: Preparing GIS Inputs for Landscape-Scale Fire Spread Modeling in ELMFIRE

7.1 Overview of Input Data Requirements

The overarching objective of this study is to integrate the firebrand model developed in previous chapters into a physics-based, landscape-scale fire risk model—namely, ELMFIRE. This integration aims to support numerically accurate and physically realistic forecasts or hindcasts of real-world wildfire scenarios.

To achieve this goal, the current chapter is devoted to establishing a systematic protocol for deriving model input data from publicly available datasets and models. It outlines the necessary data and preparation workflows, while Chapter 8 evaluates the predictive performance of the firebrand model—coupled with the urban surface fire spread models introduced in Chapter 2—using the inputs developed in this chapter.

At the landscape scale, fire-prone regions often comprise a heterogeneous mix of wildlands and urban communities, where artificial structures coexist with vegetative biomass. ELMFIRE addresses this complexity by employing different fire spread models for distinct fuel types. For wildland environments, including grasslands, shrublands, and forests, ELMFIRE applies traditional Rothermel-based surface fire spread models (Section 2.1.1). For urban areas and wild-

land–urban interface (WUI) or intermix zones, a semi-physical surface fire spread model is used (Section 2.1.3).

To support model implementation, Table 7.1 summarizes the required inputs for each fire spread model in ELMFIRE, including those previously introduced in Table 2.1 for the Rothermel model.

Among the inputs to the Rothermel-based wildland fire spread models, fuel properties listed as Items 4 through 12 in Table 7.1 are well characterized in Ref. [114], which categorizes wildland fuels into 40 standardized models based on key physical and combustion properties. Consequently, a spatially resolved map of these fuel types is essential for ELMFIRE simulations.

In addition to fuel type, the Rothermel model requires dynamic information on fuel moisture, which varies over time and is influenced by environmental conditions. Specifically, moisture contents for 1-hour, 10-hour, and 100-hour dead fuel particles, as well as live herbaceous and live woody fuels, must be provided. This necessitates the integration of spatiotemporally resolved moisture maps into the simulation domain.

Canopy-related variables are also required for modeling crown fire behavior, including canopy height, bulk density, base height, and foliar moisture content. The critical canopy cover is treated as a constant and can be defined empirically by the user.

For the urban areas, the Hamada model [33], assuming a homogeneous community layout, requires the average building footprint, separation distance, and combustible material fraction. These parameters must be estimated at the community level to enable simulation.

Table 7.1: Overview of input variables required by ELMFIRE fire spread models

Rothermel Model	**Hamada Model**	**WU-E Model**	**Firebrand Model**
Fuel Properties:			
1. 1-hour dead fuel moisture 2. 10-hour dead fuel moisture 3. 100-hour dead fuel moisture 4. Live herbaceous fuel moisture 5. Live woody fuel moisture 6. Initial SAV ratio 7. Fuel particle density 8. Total mineral content 9. Effective mineral content 10. Fuel heat content 11. Dead fuel moisture of extinction 12. Fuel bed depth 13. Canopy cover 14. Critical canopy cover 15. Maximum crown fire spread rate 16. Canopy height 17. Canopy bulk density 18. Canopy base height 19. Foliar moisture content	1. Average footprint dimension 2. Average separation distance 3. Combustible material fraction	1. Design fire curve 2. Critical flux-time product for ignition 3. Combustible material fraction 4. Characteristic absorptivity	1. Firebrand generation rate per MW (Vegetation) 2. Firebrand generation rate per MW (Structure)
Meteorology:			
Wind speed and direction (perpendicular component)			
Topography:			
Slope angle, Slope aspect			

On the other hand, the WU-E model employs a detailed, physics-based framework that explicitly accounts for heterogeneity within wildland–urban interface (WUI) communities. This added modeling complexity requires structure-specific inputs, including design fire curves, critical flux–time products for ignition, combustible material fractions, and characteristic absorptivity values. Although a systematic classification of WUI structural materials is currently under development by collaborators at UC Berkeley, we adopt a homogeneous treatment for all structures in this study due to the absence of finalized classification schemes.

For firebrand showering, the current model requires two primary parameters related only to the generation phase: the particle number generation rate per unit energy release, specified separately for biomass and structural fuels. These parameters remain underexplored (see Chapter 3) and are therefore treated as tunable within the simulation framework to account for uncertainties and potential case-specific variation.

Beyond generation, additional variability exists in the firebrand transport and ignition phases. The current model applies Sardoy's formulation [83] for biomass-derived firebrands and Himoto's formulation [82] for structure-derived firebrands. However, they may be replaced with alternative or improved formulations as new data and models become available.

Ignition probability and ignition delay are also material-dependent properties. In principle, these variables should reflect differences in structural materials and their fire hardening treatments. However, due to the current lack of comprehensive classification data, we again apply a homogeneous treatment for ignition behavior across all structure types in this study, with the expectation that future model updates will incorporate more refined fuel-specific characterizations.

Lastly, for all the surface fire spread and firebrand models, wind data including direction and magnitude are required with sufficient resolution to capture evolving fire events. Meanwhile,

topographic inputs include slope angle and slope aspect are also required as they are used by the Rothermel model. Elevation is not currently used in ELMFIRE, while prior attempts to incorporate it for firebrand interception proved numerically unstable. Since this study focuses on short-range firebrand transport, elevation effects are considered negligible.

In summary, ELMFIRE inputs can be categorized into four groups:

1. **Categorical fuel data:** including standard fuel models (e.g., [114]) and building archetypes;
2. **Empirical parameters:** such as firebrand generation rate and critical canopy cover;
3. **Static GIS data:** including fuel maps, canopy properties, and topographic slope;
4. **Dynamic GIS data:** including wind fields and fuel moisture.

Since the first two categories are typically fixed within a single simulation (e.g., firebrand generation rate, critical canopy cover) or remain inflexible across scenarios (e.g., FBFM40 fuel characteristics), the remainder of this chapter focuses on developing a reproducible workflow for generating static and dynamic GIS inputs from publicly available data sources.

7.2 Static GIS Inputs: Topography and Wildland Fuels

7.2.1 Overview of the LANDFIRE project

The present study focuses on the contiguous United States; accordingly, the data sources and methods discussed herein are specific to U.S. territory conditions.

Since 2004, the U.S. Department of Agriculture Forest Service and the U.S. Department of the Interior have jointly operated the Landscape Fire and Resource Management Planning

Tools program, commonly known as LANDFIRE. The LANDFIRE program was established to produce a comprehensive, nationally consistent set of over 25 geospatial data layers—including vegetation, fuel types, topography, and disturbance history—at a spatial resolution of 30 meters. These datasets are designed to support wildfire response and long-term fire management planning across all land cover types and disciplines [141].

Of particular interest to this study are the LANDFIRE layers related to topography and fuel classification, as these are critical inputs to the fire spread and firebrand models implemented in ELMFIRE. The following sections provide a brief technical overview of these datasets, including their underlying methodologies and intended use cases.

Understanding the generation methods and assumptions behind LANDFIRE data is vital, as input quality is a primary source of uncertainty in landscape-scale fire risk modeling. A basic grasp of how these data products are created enhances the ability to interpret simulation outcomes and to identify or mitigate potential issues in future applications. Therefore, the following discussion will first introduce how the datasets was created and then the following section will address some necessary procedures to be applied to the dataset before feed into a simulation. Readers are encouraged to referenced materials for technical details about these method.

7.2.2 Topographic Inputs: Elevation, Slope, and Aspect

The topographic layers of primary interest in this study are elevation, slope, and slope aspect, due to their critical role in the fire spread models used within ELMFIRE.

These datasets are provided by the LANDFIRE 2016 Remap product, which was finalized in 2021. According to Ref. [141], elevation data for the contiguous United States were sourced

from the U.S. Geological Survey's 3D Elevation Program (3DEP), specifically the 1/3 arc-second Digital Elevation Model (DEM), which offers a native resolution of approximately 10 meters. This dataset was developed using a range of remote sensing technologies, including aerial Light Detection and Ranging (LiDAR).

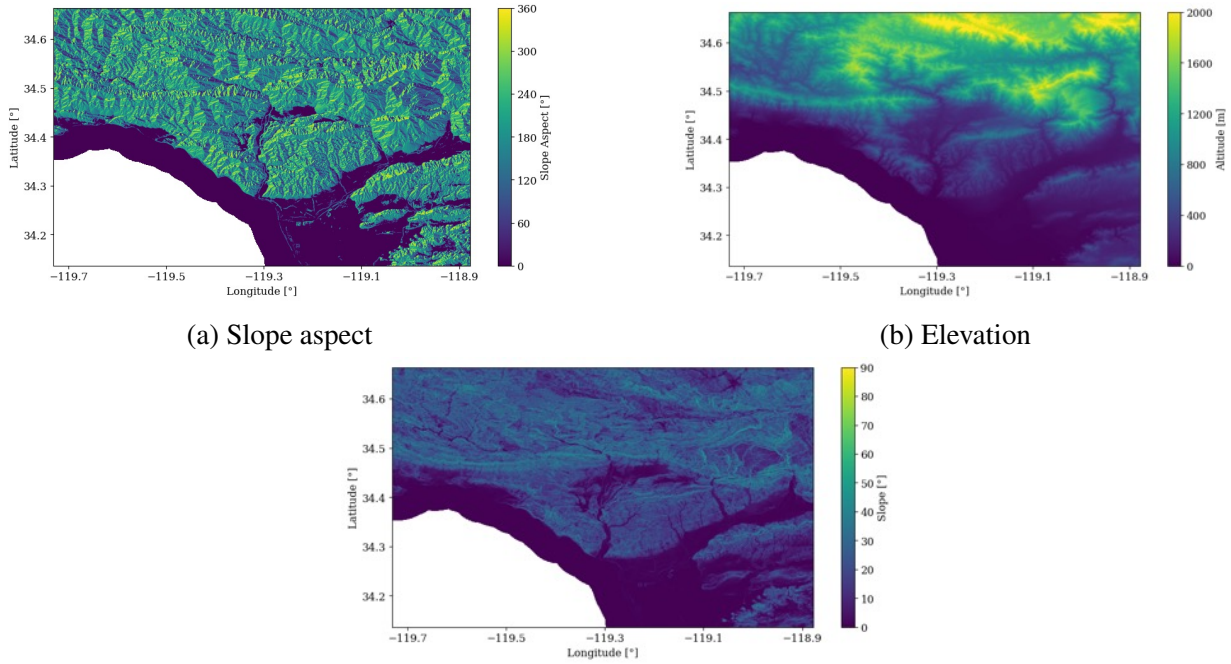
For Alaska and Hawaii, elevation data were obtained from alternative sources: Interferometric Synthetic Aperture Radar (IFSAR), with an approximate resolution of 5 meters, and the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), with a resolution of 1 arc-second (approximately 33 meters), respectively.

All elevation datasets were reprojected to the Albers Equal-Area Conic coordinate system (EPSG:5070) using bilinear interpolation and subsequently resampled to a uniform 30-meter resolution to ensure consistency with other LANDFIRE layers and compatibility with the ELMFIRE simulation framework.

Slope data—expressed both in degrees and as percent rise—were derived from the elevation raster. In the original 2016 Remap release, slope was calculated using the Geospatial Data Abstraction Library (GDAL, version 2.0.12) and used internally for vegetation mapping. A subsequent update released in 2020 recalculated slope using the ArcGIS geodesic method, which provides improved accuracy for large-scale geospatial datasets. In this version, slope values were capped at 400%. The updated slope dataset is publicly available and serves as the primary topographic input for ELMFIRE simulations in this study.

Similarly, slope aspect—the compass direction that a slope faces—was also derived from the elevation layer. In the 2016 version, aspect was calculated using GDAL with flat terrain assigned a default value of -1. The 2020 update applied the ArcGIS geodesic method to improve consistency with other topographic variables.

Figure 7.1 shows an example of topographic inputs from the LANDFIRE dataset. The selected region spans 77 km by 57 km and is resolved using 30-meter pixels. Figures 7.1a–7.1c illustrate slope aspect, elevation, and slope angle, respectively.



(a) Slope aspect

(b) Elevation

(c) Slope angle

Figure 7.1: Topographic inputs derived from the LANDFIRE 2016 Remap dataset for the Thomas Fire region (77 km $\times$ 57 km). Each raster is at 30-meter resolution: (a) slope aspect, (b) elevation, and (c) slope angle.

7.2.3 Wildland Fuel Classification and Canopy Properties

LANDFIRE provides two primary categories of fuel-related data critical for landscape-scale wildfire modeling: (1) surface fuel data, representing fuels located within approximately 6 feet (1.83 meters) of the ground surface, and (2) canopy fuel data, characterizing overstory vegetation structure.

To support ELMFIRE simulations, a spatially resolved map that classifies land cover types based on their fire behavior characteristics is required. LANDFIRE fulfills this need by assigning

each 30-meter grid cell a standardized fire behavior fuel model, using either the 40-category (FBFM40) or the older 13-category (FBFM13) classification system. The assignment process is rule-based, integrating various inputs such as Fuel Vegetation Type (FVT), Fuel Vegetation Cover (FVC), Fuel Vegetation Height (FVH), disturbance history, and regional biophysical settings. These rules were developed in collaboration with regional fire ecology experts and calibrated for consistency across the United States. Full technical documentation of this procedure is provided in Ref. [141].

The underlying vegetation data are derived from a combination of remote sensing and field-based sources, including Landsat satellite imagery, LiDAR canopy measurements, and field surveys cataloged in the LANDFIRE Reference Database (LFRDB). The LFRDB offers extensive ground-truth information on vegetation structure, composition, and distribution, which supports calibration and validation of satellite-derived layers. In addition to empirical observations, spatial modeling techniques and topographic variables are also incorporated to produce the final classified fuel model maps.

Figure 7.2 illustrates an example of the FBFM40 product for the Thomas Fire region, extracted from the LANDFIRE 2016 Remap dataset. The map displays fuel types using a color-coded scheme corresponding to the fuel model numbers defined in Ref. [114]. Of the 40 fuel types defined in the FBFM40 system, 27 are represented in this particular region.

Canopy fuel characteristics—specifically Canopy Base Height (CBH), Canopy Cover (CC), Canopy Bulk Density (CBD), and Canopy Height (CH)—are also essential inputs for modeling crown fire potential. These variables are generated through a combination of airborne LiDAR data and field-based measurements compiled in the LANDFIRE Reference Database (LFRDB). The derivation and processing methodologies for these canopy attributes are detailed

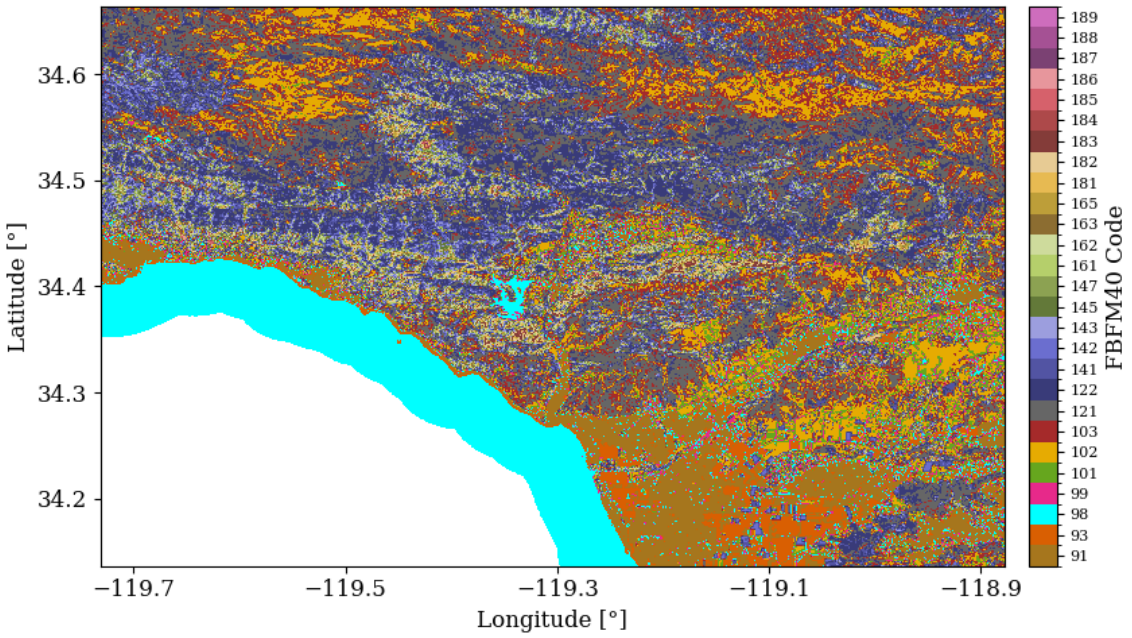


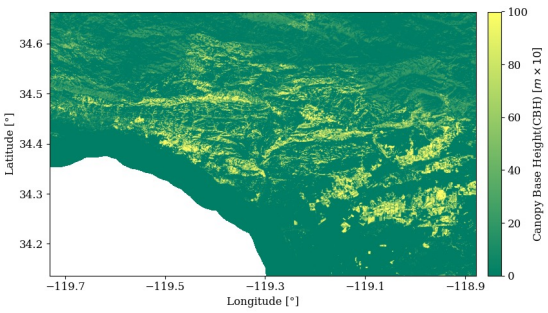
Figure 7.2: Fuel type classification (FBFM40) derived from the LANDFIRE 2016 Remap dataset for the Thomas Fire region. The map is shown at 30-meter resolution, covering a domain 77 km in length and 57 km in width.

in Ref. [141].

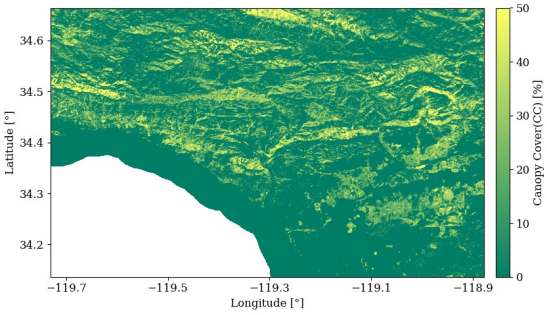
Figure 7.3 illustrates the spatial distributions of these canopy features for the Thomas Fire region, as derived from the LANDFIRE 2016 Remap dataset. Each layer is gridded at a 30-meter resolution.

7.2.4 Automated Access to LANDFIRE Fuel and Topography Data

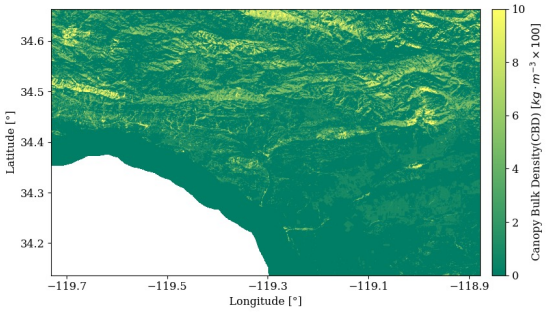
ELMFIRE is designed to align with established wildfire modeling frameworks such as FARSITE and FlamMap. Like these simulators, ELMFIRE operates on a two-dimensional structured grid and requires rasterized input data. Consequently, the raster datasets provided by the LANDFIRE project—available at a native resolution of 30 meters—can be readily integrated into ELMFIRE simulations.



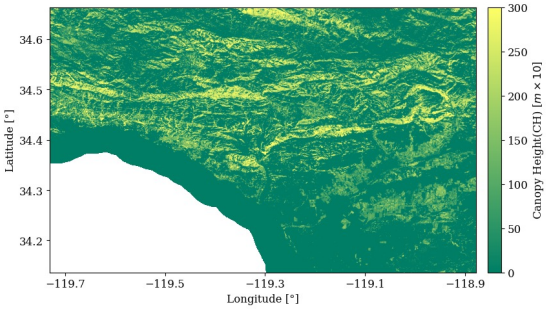
(a) Canopy Base Height (CBH)



(b) Canopy Cover (CC)



(c) Canopy Bulk Density (CBD)



(d) Canopy Height (CH)

Figure 7.3: Canopy fuel variables derived from the LANDFIRE 2016 Remap dataset for the Thomas Fire region (77 km by 57 km). All rasters are presented at 30-meter resolution.

To streamline the preparation of these raster inputs, we developed a suite of Python scripts for automated access, extraction, and formatting of LANDFIRE datasets. This section describes the features of these scripts related to fuel and topographic data; the following sections will cover additional input types.

The scripts are publicly available on GitHub and come with detailed documentation and usage instructions. Given a user-specified geographic extent and LANDFIRE version of interest, the scripts download the relevant datasets using the official `landfire` Python package maintained by the LANDFIRE project team. Each download returns a GeoTIFF file containing multiple data layers.

After download, the scripts extract individual layers from the multi-band GeoTIFF, and offer the option to either retain the default 30-meter resolution or resample the data to a custom resolution defined by the user. Additionally, all raster layers are reprojected to a user-specified coordinate reference system to ensure consistency across simulation inputs.

This automated pipeline facilitates reproducibility and simplifies the otherwise manual process of assembling high-quality geospatial inputs for ELMFIRE simulations.

7.2.5 High-Resolution Urban Fuel Mapping Using Microsoft Building Footprints

Although the LANDFIRE project incorporates infrastructure elements such as building footprints and roadways into its fuel classification products, its native resolution of 30 meters poses limitations for detailed modeling of wildland–urban interface (WUI) communities. Residential structures in WUI regions typically range from 10 to 30 meters in size, meaning that many

individual buildings are either misrepresented or omitted entirely in the coarser LANDFIRE fuel type maps.

Notably, LANDFIRE's urban fuel classification has utilized Microsoft's building footprint dataset to inform its mapping of developed areas. Building on this precedent, a natural extension for high-resolution, community-level fire modeling is to directly leverage the Microsoft building footprint data as a source for fine-scale urban fuel mapping.

In 2017, Microsoft released a dataset containing approximately 9.8 million high-quality building footprints with estimated building heights, covering 44 U.S. states. This dataset is open-source and publicly accessible. The building geometries were extracted from high-resolution aerial imagery, and heights were interpolated using digital terrain models. Additional technical details are available in the Microsoft Building Footprint Documentation [142].

The original dataset is provided in vector format, which theoretically offers infinitely high spatial resolution. However, vector data are not directly compatible with ELMFIRE, which requires all spatial input layers in raster format.

To address this, a preprocessing tool was developed by Maria Theodori at the University of California, Berkeley [143]. This tool automates the process of downloading the Microsoft building footprint data for a user-defined spatial extent and rasterizing the vector geometries to any desired spatial resolution. Rasterization enables seamless integration with ELMFIRE's grid-based framework.

Beyond generating binary raster maps indicating building presence, the tool extracts additional structural attributes from the vector data, including building height, footprint area, and separation distance. These parameters are then aggregated and exported as raster layers aligned with the simulation grid. While future development aims to generate a categorized fuel model for

urban areas, the current implementation distinguishes only between structures and non-structure fuels.

These raster layers provide essential structural information for simulating fire spread within WUI communities, where building-level details and spatial heterogeneity strongly influence fire behavior. The derived datasets support the application of both empirical models, such as the Hamada model [33], and physics-based models, such as the WU-E model.

This Microsoft-derived raster dataset has been fully integrated into the Python-based data preparation pipeline for ELMFIRE simulations. The pipeline performs automated downloading, rasterization, resampling, and reprojection of the building footprint data. The final product is a refined WUI fuel map that combines Microsoft building footprints with LANDFIRE vegetation fuel data, enabling high-resolution modeling of fire spread in complex WUI settings.

Figure 7.4 illustrates the modification of the original LANDFIRE-derived FBFM40 fuel type map using Microsoft building footprint polygons for a region affected by the Thomas Fire. The original 55 km by 77 km LANDFIRE FBFM40 dataset, provided at 30 m resolution, is reclassified into four classes for improved visualization. The comparison highlights the potential improvement in spatial accuracy for urban structure identification and enhanced simulation fidelity.

Figure 7.4b shows a zoomed-in view of the highlighted region from Figure 7.4a, where building footprints are represented by yellow polygons and structural fuels identified in the FBFM40 field are shown in brown. The preprocessing script first reclassifies all structure fuels as unburnable fuels (depicted in gray in Figure 7.4c), and subsequently reassigns burnable structural fuels by referring to the Microsoft building footprint polygons. Figures 7.4c and 7.4d present the results at different spatial resolutions, with the finely resolved case ($\Delta x = 10$ m,

Figure 7.4d) providing a more accurate representation of community layouts.

However, further improvements are necessary, particularly in the treatment of the ”gray” regions. Currently classified as unburnable surfaces, these areas may actually contain combustible materials that could play a significant role in enabling fire spread into communities. Future work will aim to refine the classification to better capture these transitional fuel zones.

7.3 Spatio-temporal Wind Inputs Acquisition

7.3.1 Sources and Requirements for Wind Data Inputs

Wind inputs are a critical component for driving both fire spread and firebrand transport in ELMFIRE simulations. To be compatible with the model, wind data must be rasterized and available at sufficient spatial and temporal resolutions. Moreover, since many fire spread models require wind estimates near the surface within the atmospheric boundary layer, the datasets must either directly represent or be translatable to near-surface conditions.

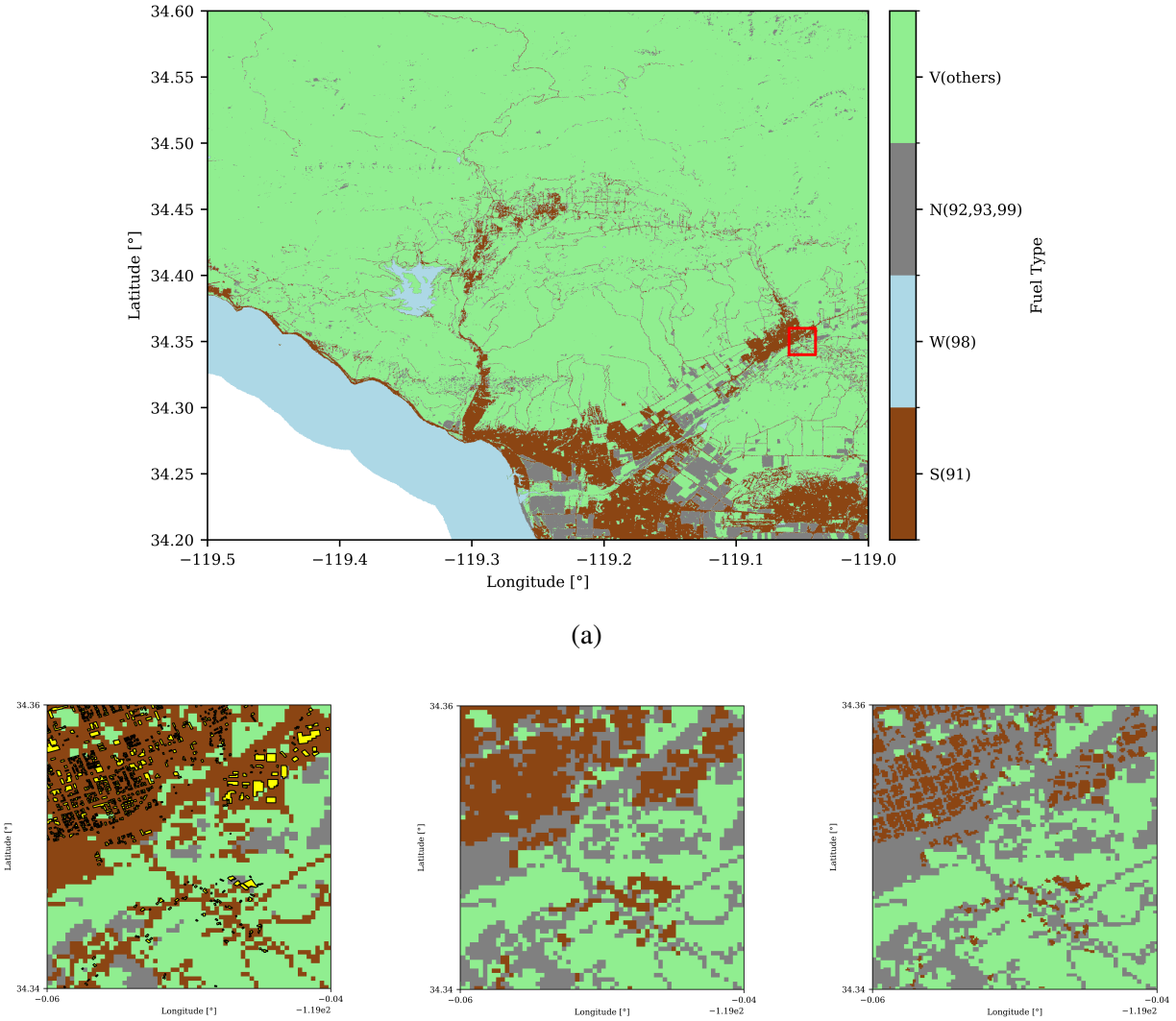
Wind data suitable for ELMFIRE generally originate from two primary sources:

1. Processed observational datasets, and

2. Physics-based numerical weather prediction (NWP) models.

In the context of the contiguous United States (CONUS), an example of high-resolution observation-based datasets include the Real-Time Mesoscale Analysis (RTMA) [139, 144]. Such datasets assimilate satellite, radar, and in-situ observations to produce short-term analyses that reflect atmospheric conditions at the time of analysis.

For forecasting applications, the Weather Research and Forecasting model—Advanced Re-



(a)

(b) (c) (d)

Figure 7.4: Integration of Microsoft OpenStreetMap building footprint data with LANDFIRE fuel maps for the Thomas Fire region (LANDFIRE 2016 Remap). The modification aims to improve structure representation and spatial resolution in WUI fire simulations. (a): Original LANDFIRE-derived fuel map (46.0 km × 44.5 km) reclassified for clarity—gray: non-burnable surfaces (FBFM40 codes 92, 93, 99), blue: water bodies (FBFM40 code 98), brown: structures (FBFM40 code 91), green: vegetation (remaining FBFM40 codes); (b): Zoomed-in view (4.5 km × 4.5 km) of the highlighted region in Panel (a) with Microsoft building footprint overlays (yellow polygons); (c), (d): Modified fuel maps (4.5 km × 4.5 km) at 30-meter and 10-meter resolution, respectively, demonstrating improved structural detail using the footprint data.

search WRF core (WRF-ARW) [145]—is one of the most widely used NWP systems. WRF solves the compressible, nonhydrostatic Euler equations using computational fluid dynamics (CFD) techniques to generate high-resolution atmospheric forecasts over customized domains. It typically employs observational analyses such as RTMA or reanalysis products for initialization and boundary conditions.

Several operational models, including the High-Resolution Rapid Refresh (HRRR) [146] and the North American Mesoscale (NAM) [147] model, are based on WRF and offer enhanced spatial resolution and tailored data assimilation strategies. These models are particularly beneficial for fire modeling due to their fine temporal resolution and near-surface wind outputs.

Given that this study focuses on validating ELMFIRE through the reconstruction of historical wildfire events, analysis data are preferred over forecast products. We utilize RTMA data as it is the dominant mesoscale analysis dataset in North America and provides a comprehensive historical archive dating back to 2011. This makes it suitable for reconstructing significant wildfires such as the Tubbs Fire, Camp Fire, and Thomas Fire. Accordingly, our input data pipeline includes automated acquisition and preprocessing of RTMA datasets.

7.3.2 Spatial Resolution and Wind Field Upscaling

Meteorological datasets such as RTMA and HRRR typically offer spatial resolutions of approximately 1–3 km. However, ELMFIRE simulations often operate at grid resolutions on the order of 10–100 meters. Direct application of coarse-resolution wind data can introduce significant uncertainties due to spatial mismatch. Therefore, we employ upscaling methods to derive fine-resolution wind fields suitable for ELMFIRE.

A simple resampling approach using bilinear interpolation may suffice in homogeneous terrains [148]. However, to account for terrain-driven wind variation, physics-informed upscaling methods should be implemented, including WRF nested grids and WindNinja.

While WRF allows high-resolution 3D wind field generation via nested domains, it is computationally expensive and not ideal for simulation areas of tens of kilometers discretized by grids on the scale of tens of meters. Alternatively, WindNinja [149, 150] is a more efficient tool that solves steady-state mass conservation equations to downscale coarse wind fields over complex terrain. It has also recently incorporated OpenFOAM solvers for momentum-conserving wind field refinement, though at a higher computational cost.

In this study, we apply WindNinja's default mass-conservation mode due to its efficiency and physical consistency. This method effectively bridges the scale gap between coarse atmospheric datasets and the fine-grid requirements of ELMFIRE.

7.3.3 Overview of WindNinja Upscaling Method

As previously noted, WindNinja employs a mass-consistent approach to refine coarse wind fields. This method solves an optimization problem formulated as:

$$L(\mathbf{x}, \mathbf{u}, \lambda) = \int_{\Omega} \Big[\alpha_1 (u - u_0)^2 + \alpha_2 (v - v_0)^2 + \alpha_3 (w - w_0)^2 + \lambda \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \Big] d\Omega \tag{7.1}$$

Here, $\mathbf{x} = (x, y, z)$ denotes the spatial coordinates, and $\mathbf{u} = [u(x, y, z), v(x, y, z), w(x, y, z)]$ is the refined wind velocity vector. The input coarse wind field is represented by (u_0, v_0, w_0). The first term of the integrand penalizes the deviation from the input wind field in a least-squares

sense. The weighting coefficients α_i ($i = 1, 2, 3$) control the relative importance of corrections in each velocity component. Typically, $\alpha_1 = \alpha_2$, while α_3 is assigned a smaller value to reflect atmospheric stratification and reduced vertical mixing.

The second term enforces mass conservation by introducing a Lagrange multiplier λ, which penalizes any divergence in the wind field. That is, the refined wind field satisfies the continuity condition $\nabla \cdot \mathbf{u} = 0$ within the computational domain Ω.

This mass-consistent approach produces physically realistic wind fields that preserve the structure of the input data while conforming to conservation principles. Further technical details can be found in Forthofer et al. [149] and Ratto et al. [151].

As indicated in Equation 7.1, the primary input to WindNinja's mass-conservation optimization problem is the initial wind field (u_0, v_0, w_0). When applying RTMA data to initialize this process, three technical challenges arise:

1. The RTMA dataset provides two-dimensional wind fields, containing only the horizontal components u_0 and v_0, with no information on the vertical component w_0.

2. RTMA data are available at relatively coarse spatial resolution (approximately 2.5 km), whereas WindNinja operates on high-resolution computational grids.

3. RTMA wind data are time-varying (hourly), but WindNinja is designed to accept static wind fields for each simulation run.

WindNinja addresses the first two issues internally. To approximate the vertical profile of wind, the model imposes a predefined boundary layer profile based on the Monin-Obukhov similarity theory or a logarithmic wind law. The spatial resolution mismatch is resolved using

bilinear interpolation, which projects the input 2D wind field (u_0, v_0) onto the high-resolution simulation grid.

The third issue—temporal variability—is handled externally by executing separate WindNinja simulations for each hourly RTMA snapshot. In this manner, WindNinja preserves the original temporal resolution of RTMA data while enhancing only the spatial resolution of the wind field.

An additional critical input for WindNinja is a high-resolution digital elevation model (DEM), which enables the refined wind field to reflect topographic influences via the conservation equations. In our implementation, elevation data are obtained from the LANDFIRE dataset, as detailed in Section 7.2.

WindNinja also requires specification of the dominant surface vegetation type—grass, shrub, or trees—unless the user provides detailed vegetation inputs in a format compatible with FARSITE. This vegetation classification can also be derived from the fuel model map obtained from LANDFIRE.

Finally, the model requires information about the height above ground level at which the input wind data are measured. In our case, RTMA provides wind speed and direction at 10 meters above the ground, which is directly compatible with WindNinja's input structure.

Given these input variables—(i) hourly wind speed and direction from RTMA, (ii) elevation from LANDFIRE, (iii) dominant vegetation type, and (iv) wind measurement height—it is possible to fully initialize WindNinja and upscale the wind fields from RTMA's native 2.5 km resolution to any desired resolution suitable for ELMFIRE simulations.

The input data generation scripts developed for this study are designed to automate the retrieval and preprocessing of RTMA wind data. Specifically, the scripts first download the RTMA

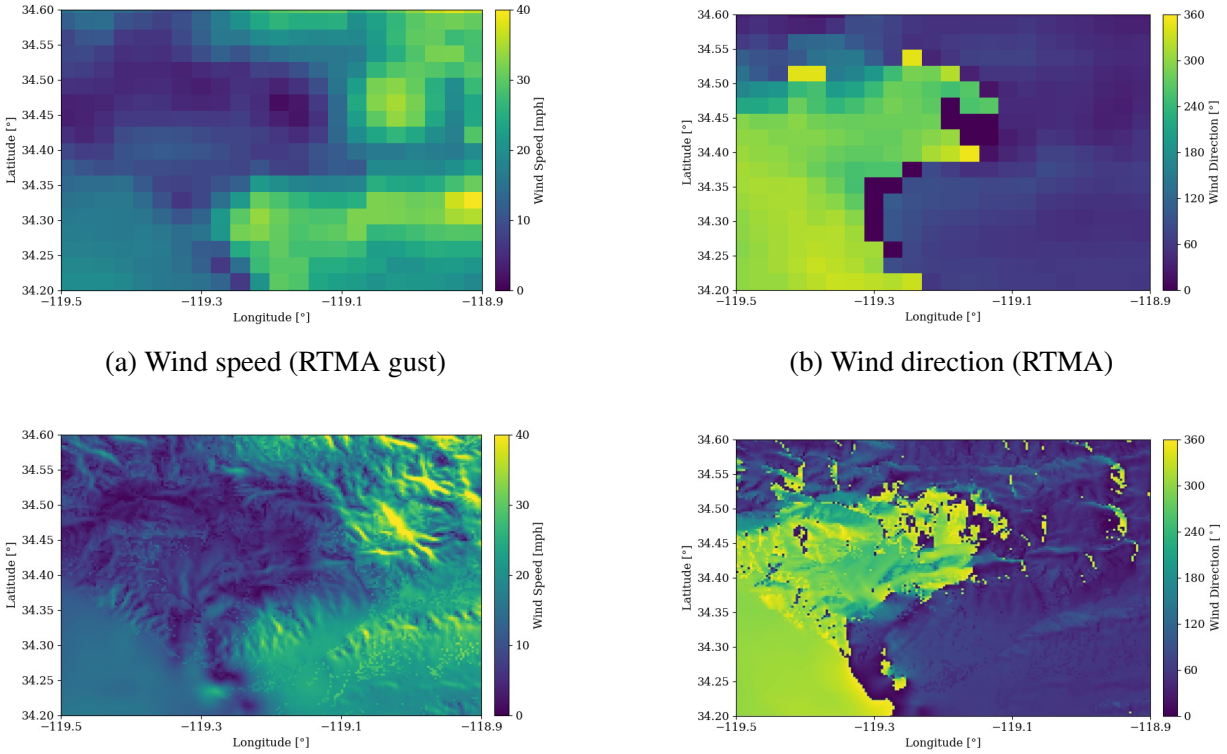
wind fields from the hosting server and then convert them into a format compatible with WindNinja.

It is important to note that ELMFIRE does not require the wind inputs to share the same spatial resolution as the topographical raster layers. In fact, meteorological input layers—such as wind and fuel moisture—may have coarser resolution than the topographic or fuel classification layers. This flexibility is useful, as one of the goals of this study is to evaluate ELMFIRE's performance under different wind field resolutions.

To support this objective, the preprocessing pipeline preserves both the original RTMA wind fields and the high-resolution wind fields produced through WindNinja upscaling. Comparison of fire spread simulations using raw RTMA versus refined wind inputs facilitates sensitivity analyses and model validation.

Figures 7.5a and 7.5b present the raw RTMA wind speed and direction fields at 2.5 km resolution, extracted over a 100 km by 85 km domain affected by the Thomas Fire at 2:00 UTC on December 5, 2017. In comparison, Figures 7.5c and 7.5d show the corresponding wind fields computed using WindNinja, following the preprocessing steps outlined above.

Compared to simple spatial interpolations of the RTMA data, the WindNinja-upscaled wind fields reflect terrain-induced flow patterns, capturing localized acceleration and directional shifts due to topographic variations. According to Ref. [152], such flow patterns can have a substantial impact on wildfire spread predictions and are thus critical for high-fidelity fire modeling. (Figures in the last version were incorrect. The new figures 7.5c and 7.5d are corrected and used in simulations in Chapter 8)



(a) Wind speed (RTMA gust)

(b) Wind direction (RTMA)

(c) Wind speed (WindNinja)

(d) Wind direction (WindNinja)

Figure 7.5: Comparison of raw RTMA wind fields and WindNinja-upscaled wind fields for the Thomas Fire region on December 5, 2017, at 2:00 UTC. (a) and (b) show RTMA wind speed (gust) and direction at 2.5 km resolution. (c) and (d) show WindNinja-upscaled wind speed and direction at 300 m resolution. The domain shown is approximately 85 $\times$ 100 km.

7.4 Fuel Moisture Mapping and Modeling

The final major class of GIS inputs required by ELMFIRE is fuel moisture, including both dead and live fuel moisture content. As discussed in Section 2.1, the surface fire spread models in ELMFIRE—particularly the Rothermel model—require the specification of dead fuel moisture for 1-hour, 10-hour, and 100-hour timelag fuels, as well as live fuel moisture for herbaceous and woody vegetation.

To the best of the author's knowledge, there are currently few publicly available GIS products that provide gridded raster layers of these specific fuel moisture quantities for the entire contiguous United States (CONUS). Among these, the Wildland Fire Assessment System (WFAS) provides recent-day fuel moisture estimates; however, historical archives of gridded products are limited or unavailable. While WFAS maintains historical records of weather station-based fuel moisture observations, these are often spatially sparse and therefore not directly suitable for landscape-scale simulations in ELMFIRE.

7.4.1 Overview of Fuel Moisture Models in NFDRS

Fuel moisture estimates provided through WFAS are derived using the modeling framework of the National Fire Danger Rating System (NFDRS) [153, 154]. The current implementation, NFDRS 4.0, computes dead and live fuel moisture based on physically and empirically derived models that leverage daily weather inputs.

For dead fuel moisture, NFDRS 4.0 incorporates the modeling framework developed by Nelson [155] and refined by Carlson et al. [156]. This physically based model predicts the moisture content of 1-hour, 10-hour, and 100-hour dead fuel classes based on key meteorological in-

puts including air temperature, relative humidity, precipitation, wind speed, downward shortwave radiation, and snow cover status. The Nelson model considers the energy balance and moisture transport within the fuel particles and the exchange with ambient environment, and solves the equilibrium condition for different types of fuel particles.

For live fuel moisture, NFDRS 4.0 adopts an extended version of the Growing Season Index (GSI) model [157], as improved by Dahal et al. [158]. The GSI model was originally designed to capture seasonal changes in vegetation phenology and canopy productivity across a broad range of global ecosystems. It predicts live fuel moisture based on surface weather observations, including daily minimum temperature, vapor pressure deficit, photoperiod, and rainfall—all of which can be obtained from gridded weather analysis datasets such as RTMA.

7.4.2 Implementation of NFDRS Model

To compute the spatial and temporal variation of fuel moisture within the simulation domain, this study applies the National Fire Danger Rating System (NFDRS) model, as implemented in the NFDRS4.0 codebase provided by the USDA Fire Lab [153]. The original implementation is written in `C++` and provides Python bindings via the Simplified Wrapper and Interface Generator (SWIG). This design allows for seamless integration of NFDRS routines into a Python-based data preprocessing pipeline for ELMFIRE.

The default implementation of the NFDRS model is designed to accept point-based meteorological inputs, typically sourced from weather station observations. In this study, however, spatially distributed inputs are derived from gridded weather analysis data provided by the Real-Time Mesoscale Analysis (RTMA) system. Core meteorological parameters required by the NF-

DRS model—such as air temperature, precipitation, wind speed, and atmospheric pressure—are directly available from RTMA products.

Several additional variables, however, must be derived either through post-processing or external sources. These include relative humidity (RH), downward shortwave radiation, slope angle, fuel type classification, and snow cover status. Relative humidity is calculated from specific humidity and temperature using thermodynamic relationships, while shortwave radiation is estimated based on cloud cover data. Topographic slope and fuel type are obtained from LANDFIRE datasets as described in Section 7.2. Snow cover is neglected in this study, as the selected case studies are located in regions and seasons where snow presence is assumed to be minimal.

Relative humidity (RH)

RTMA does not provide direct estimates of relative humidity. Instead, it reports *specific humidity* (SH), which is defined as the mass of water vapor per unit mass of moist air. To derive RH from SH, it is necessary to compute both the ambient vapor pressure e and the saturation vapor pressure e_s. The saturation vapor pressure e_s as a function of temperature T (in Kelvin) can be estimated from the Clausius–Clapeyron equation:

$$e_s = e_{s_0} \exp\left[\left(\frac{L_v(T)}{R_v}\right)\left(\frac{1}{T_0} - \frac{1}{T}\right)\right] \tag{7.2}$$

where e_{s_0} is the reference saturation vapor pressure at temperature T_0, $L_v(T)$ is the latent heat of vaporization, and R_v is the specific gas constant for water vapor.

For practical meteorological applications, the Tetens formula [159] provides a reliable ap-

proximation of Eqn. 7.2 for temperatures below $35\,^{\circ}$C:

$$e_s = 6.112 \times \exp\left(\frac{17.67\,T_{\mathrm{C}}}{T_{\mathrm{C}} + 243.5}\right) \times 100 \quad [\mathrm{Pa}] \tag{7.3}$$

where T_{C} is the air temperature in degrees Celsius.

To calculate the actual vapor pressure e from specific humidity, we apply the ideal gas law. Vapor pressure is related to vapor density χ via:

$$e = \frac{\chi\, R\, T}{MW_v} \tag{7.4}$$

where $\chi = m_v/V$ is the mass of water vapor per unit volume, MW_v is the molecular weight of water vapor, R is the universal gas constant, and T is the air temperature in Kelvin. Since specific humidity SH is defined as χ/ρ_{air}, and the density of moist air ρ_{air} can be estimated as:

$$\rho_{air} = \frac{e \times MW_v + MW_{air}\,(P - e)}{R\,T} \tag{7.5}$$

We have an empirical relationship for relative humidity using known quantities:

$$RH = \frac{SH \cdot P}{\varepsilon \cdot e_s + SH \cdot (P - e_s)} \times 100\% \tag{7.6}$$

where P is atmospheric pressure, $\varepsilon = MW_v/MW_{air}$ is the ratio of molecular weights (typically ≈ 0.622), SH is specific humidity and e_s is the saturation vapor pressure from Eqn. 7.3. This expression allows the conversion of specific humidity and pressure values into relative humidity on a gridded basis using RTMA inputs.

Downward shortwave radiation

The downward shortwave radiation (R_s) is not directly available from RTMA products. Instead, it is estimated using cloud cover data based on an empirical correlation developed by Perez et al. [160] and evaluated by Blankenau et al. [161]. The relationship is expressed as:

$$R_s = R_a \times \left[1 - 0.87 \times \left(\frac{\text{Cloud Cover (\%)}}{100}\right)^{1.9}\right] \tag{7.7}$$

Here, R_s is the estimated downward shortwave radiation ($W \cdot m^{-2}$), R_a is the extraterrestrial solar radiation ($W \cdot m^{-2}$), and cloud cover is expressed as a percentage between 0 and 100.

The extraterrestrial solar radiation R_a is computed using solar geometry as described in Ref. [162], following the expressions below:

$$\begin{aligned} R_a = {} & G_{\text{sc}} \times \\ & \left[1 + 0.033 \times \cos\left(\frac{2\pi d}{365}\right)\right] \times [\sin(\phi)\sin(\delta) + \cos(\phi)\cos(\delta)\cos(\omega_s)] \end{aligned} \tag{7.8}$$

where:

$$\begin{aligned} \delta &= 0.409 \times \sin\left(\frac{2\pi d}{365} - 1.39\right) \\ \omega_s &= 15 \times \left(\frac{t_s}{60} - 12\right) \times \frac{\pi}{180} \\ t_s &= t_{\text{st}} - 4(L_{\text{st}} - L_{\text{loc}}) + E(d) \\ E(d) &= 0.0172 + 0.4281\cos(B) - 7.3514\sin(B) - 3.3495\cos(2B) - 9.3712\sin(2B) \\ B &= (d-1) \times \frac{360}{365} \times \frac{\pi}{180} \end{aligned} \tag{7.9}$$

In these equations, $G_{\text{sc}} = 1367\, W \cdot m^{-2}$ is the solar constant; d is the day of the year; ϕ

is the site latitude in radians; δ is the solar declination angle; ω_s is the sunset hour angle; t_s is the solar time in minutes; t_{st} is the local standard time in minutes; L_{st} and L_{loc} are the standard meridian and the local longitude, respectively; $E(d)$ is the equation of time (in minutes); and B is the day angle (in radians). This formulation enables the estimation of R_a for any time and geographic location given the cloud cover data from RTMA.

Slope angle & Fuel type classification

While the LANDFIRE project provides GIS data for slope angle and fuel type classification, the NFDRS model requires these inputs in categorized formats. In particular, slope angle θ are discretized into five terrain classes: Class 1 corresponds to slopes between 0° and 25°; Class 2 includes slopes from 25° to 40°; Class 3 covers 40° to 55°; Class 4 ranges from 55° to 75°; and Class 5 includes all slopes greater than or equal to 75°. Accordingly, the continuous slope raster from LANDFIRE is reclassified into these five NFDRS slope classes prior to use in fuel moisture modeling.

Regarding fuel types, although the LANDFIRE project provides a classified fuel model map based on the FBFM40/FBFM13 systems, the NFDRS model employs a distinct fuel classification scheme. While users can customize fuel categories (fuel class M), the default NFDRS configuration classifies fuels into five broad categories: V, W, X, Y, and Z.

These NFDRS classes are derived from a separate 1-km resolution fuel map [163] developed for earlier versions of NFDRS, which originally included 21 unique fuel categories labeled A through U. In the updated NFDRS version, these 21 categories have been consolidated into the five aforementioned fuel classes using a fuel class reduction methodology based on the Energy Release Component (ERC) values [154]. This classification is summarized in Table 7.2, as described in Ref. [154]. Details of this classification can be find at Ref. [164].

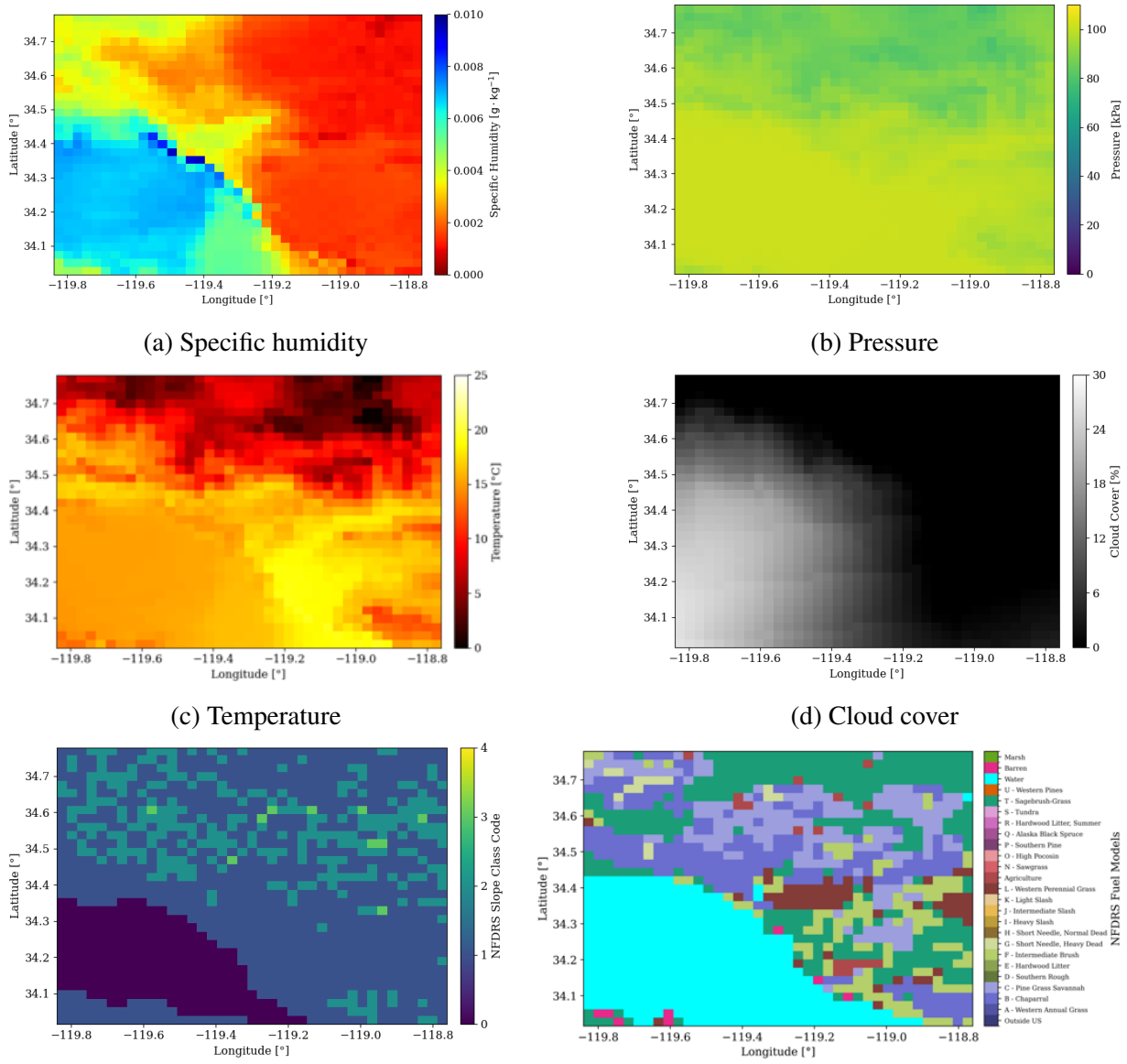
Table 7.2: Mapping between numeric fuel codes and NFDRS fuel classes

NFDRS2.0/3.0 fuel model class	**NFDRS4.0 fuel model class**
A, L	V
B, C, T	W
D, E, N, R	X
G, P	Y
F, H, I, J, K, O, Q, S, U	Z

Figure 7.6 presents the full set of required inputs to the NFDRS4 model, in addition to the wind field. All meteorological variables are extracted from the RTMA dataset. For the geographical data, the classified slope is processed from the corresponding LANDFIRE products, while the fuel map is reclassified from the original 21-category scheme to the 5 fuel categories required by the NFDRS model, following Ref. [163]. Both the slope and fuel map data are resampled to match the RTMA resolution (2.5 km) using nearest-neighbor interpolation for consistency, downscaling from their original resolutions of 30 m (slope) and 1 km (fuel), respectively.

The resulting dead fuel moisture fields are shown in Figure 7.7. Live fuel moistures are not shown, as constant values were derived for both herbaceous (60%) and woody (30%) live fuels throughout the domain. Because the fuel moisture fields are time-dependent and share the same temporal resolution as the RTMA inputs, Figure 7.7d presents the time history of the domain-averaged values, with shaded bounds corresponding to $\pm 1\sigma$ representing the spatial variation at each time step.

Although the colormaps in Figures 7.7a, 7.7b, and 7.7c are scaled for improved visualization, it is observed that both the temporal and spatial variations in fuel moistures are generally limited, except for the 1-hour dead fuel moisture, which exhibits larger spatial variability during certain hours of the day.



(a) Specific humidity

(b) Pressure

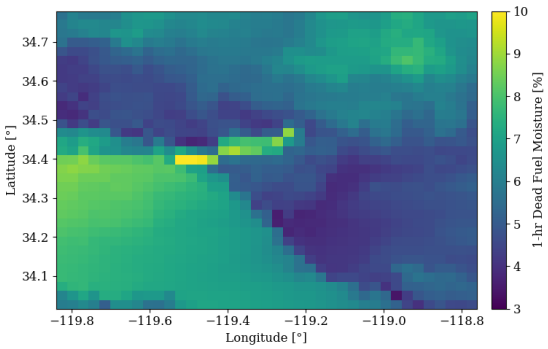
(c) Temperature

(d) Cloud cover

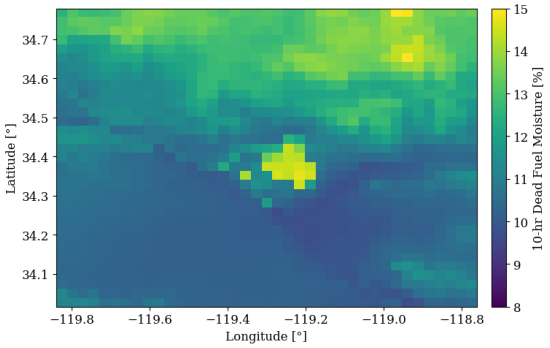
(e) Classified slope

(f) Reclassified fuel map

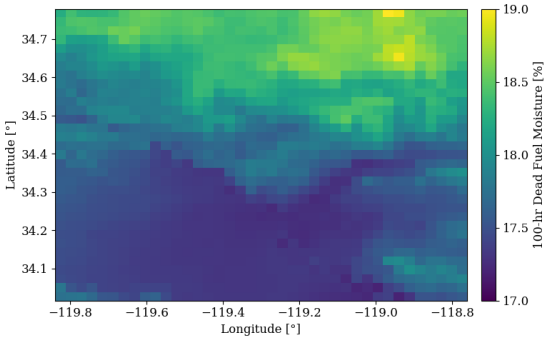
Figure 7.6: Necessary inputs for the NFDRS4 moisture field reconstruction (excluding the wind field). Examples are plotted for Thomas Fire region (85 × 100 km) on December 5, 2017, at 2:00 UTC. All inputs are resampled to match the RTMA resolution (2.5 km). Fuel map categories are reclassified following Table 7.2.



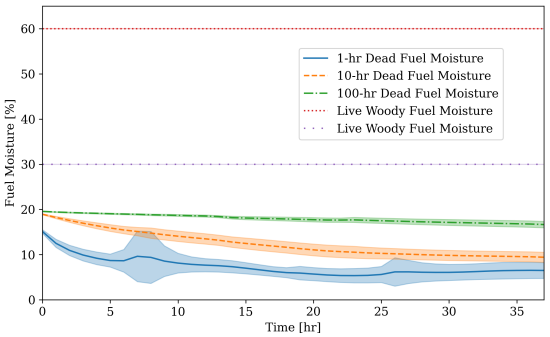
(a) 1-hour dead fuel moisture



(b) 10-hour dead fuel moisture



(c) 100-hour dead fuel moisture



(d) Time history of domain-averaged dead fuel moistures

Figure 7.7: Reconstructed dead fuel moisture fields using the NFDRS4 model. Subfigures (a)–(c) show the spatial distribution at the 20th hour of the simulation (85 $\times$ 100 km). Subfigure (d) shows the domain-averaged time evolution, with shaded bounds indicating $\pm 1\sigma$ spatial variation.

7.5 Conclusion

The complete preprocessing workflow for generating ELMFIRE-compatible inputs is illustrated in Figure 7.8. The process begins with acquiring raw datasets from the RTMA and LANDFIRE archives. RTMA meteorological data are used to calculate fuel moisture using the NFDRS model, while slope and fuel type information are obtained from the LANDFIRE and NFDRS datasets, respectively.

Since the NFDRS model expects slope and fuel inputs in specific categorized formats, the LANDFIRE slope map and the NFDRS fuel map are reclassified into classes required by NFDRS4.0, and reprocessed to align with the RTMA spatial resolution. After preparing all necessary inputs, the NFDRS model is executed as described in Section 7.4.2. The inputs and outputs associated with NFDRS are connected by green solid lines in Figure 7.8. Note that the resulting fuel moisture data maintain the original RTMA resolution of 2.5 km.

When finer-resolution wind fields are required, WindNinja is employed to downscale or upscale the wind inputs. RTMA wind speed and direction, along with the LANDFIRE elevation map, are used as inputs to WindNinja, following the procedure outlined in Section 7.3.3. The output wind field has a user-defined spatial resolution Δx_{wind}. Since ELMFIRE requires wind and moisture inputs to share the same resolution, the NFDRS-generated fuel moisture fields are resampled via bilinear interpolation to match the WindNinja output resolution, Δx_{wind}. All WindNinja-related steps are indicated by blue solid lines in the diagram.

Most LANDFIRE products can be directly used in ELMFIRE simulations. When resampling is necessary to match ELMFIRE's simulation resolution, the nearest-neighbor method is used for integer fields, while bilinear interpolation is applied to continuous (floating-point) fields.

These resampling and usage pathways are marked with green solid lines in Figure 7.8.

A key optional enhancement in this pipeline is the incorporation of high-resolution Microsoft Building Footprint data. In this case, the FBFM40/FBFM13 fuel type raster is first resampled to match the desired resolution using nearest-neighbor interpolation. Then, the Microsoft vector footprint data are rasterized and used to modify the fuel type map as described in Section 7.2.5. The modified fuel map is subsequently used in ELMFIRE simulations. Both the modified and unmodified fuel maps remain valid inputs. The steps related to Microsoft Footprint data are highlighted in orange in the diagram.

Optional components of the pipeline—specifically WindNinja-based wind field refinement and Microsoft Footprint integration—are enclosed in dashed borders in Figure 7.8. All other components and data layers enclosed in solid borders are essential for constructing a complete input dataset for ELMFIRE.

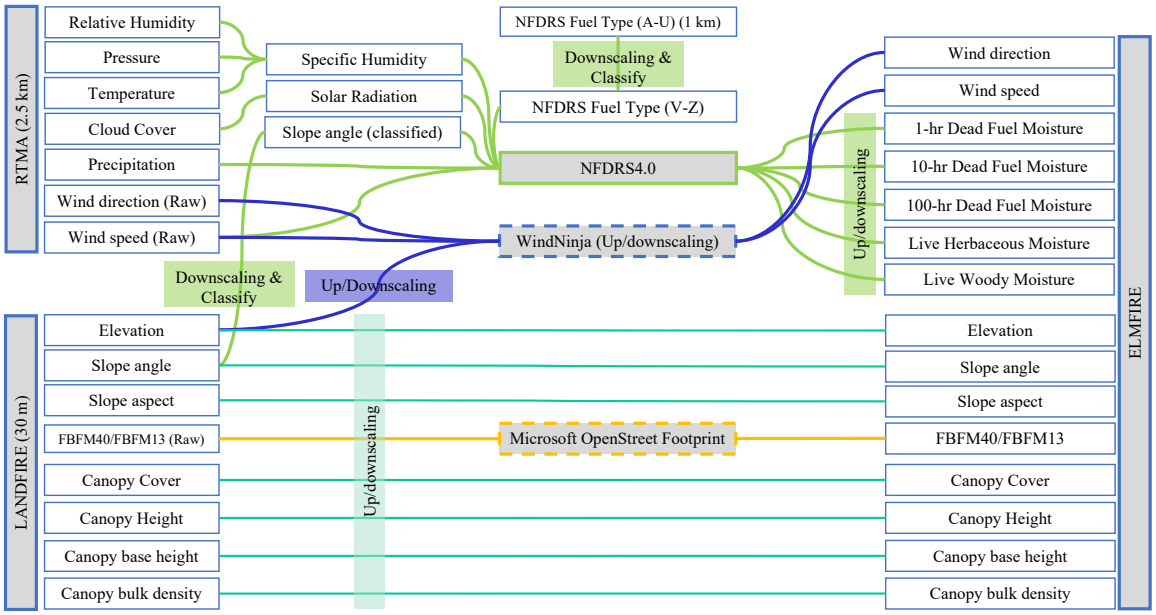


Figure 7.8: Schematic of ELMFIRE gridded input preparation workflow.

Chapter 8: Model Application: Historical Fire Reconstruction by ELMFIRE

This chapter presents a series of feasibility studies aimed at reconstructing historical fire events using the ELMFIRE simulator, with a primary focus on the 2017 Thomas Fire in California. The objective is to evaluate the performance of ELMFIRE's fire spread and spotting models under real-world conditions and to assess the effectiveness of recent model and data-driven improvements.

Up to this point, we have reviewed the key components of the ELMFIRE simulation framework. Chapter 2 introduced the level-set front tracking algorithm and surface fire spread models applicable to both wildland and urban fuels. Building on this foundation, we developed a physics-based firebrand modeling framework encompassing three essential phases: generation (Chapter 3), transport (Chapter 4), and ignition (Chapter 5). We also proposed a resolution-independent treatment for urban structural fuels (Chapter 6), though this method has not yet been incorporated into the operational ELMFIRE codebase.

In parallel, Chapter 7 described the pipeline used to acquire and preprocess open-source geospatial and meteorological datasets compatible with ELMFIRE. These include wind fields (e.g., RTMA), fuel maps (e.g., LANDFIRE), and elevation data. This preprocessing pipeline

ensures consistency in spatial resolution and coordinate reference systems, enabling seamless integration into the simulator.

Each of these modeling and data enhancements contributes incrementally to improved fire reconstruction capabilities. The baseline version of ELMFIRE supports fire spread through wildland fuels only, without accounting for spotting. Urban spread is subsequently enabled via the addition of the Hamada and WU-E models. While these surface fire spread models significantly expand coverage, they remain insufficient for capturing the full complexity of landscape-scale fire behavior. The addition of firebrand models—initially in the form of a Lagrangian particle system, and later as a numerically robust Eulerian formulation—improves the simulator's ability to reproduce short-range spotting and structure-to-structure ignition.

In addition to model selection, the resolution and accuracy of input data critically influence simulation performance. By default, ELMFIRE uses RTMA wind fields. The wind speed represents the value at the valid analysis time (i.e., the top of each hour) [144]. Wind gust fields [165] are also available in RTMA, referring to brief, sudden increases in wind speed—typically lasting less than 20 seconds—and are defined by fluctuations exceeding 10 mph (16 $\mathrm{km} \cdot \mathrm{h}^{-1}$) between peaks (i.e., the maximum instantaneous wind speed exceeding 25 knots since the last observation) and lulls (marked decreases in wind speed) [166]. These gust fields may better reflect fireline intensification events. Furthermore, fine-scale wind modeling tools such as WindNinja can provide locally resolved wind profiles, which are particularly valuable in complex terrain or WUI areas.

Fuel representation is another key driver of simulation fidelity. As discussed in Chapter 7, the LANDFIRE dataset often assigns similar fuel classifications to both combustible structures and nonflammable infrastructure such as roads, potentially leading to misrepresentation of firebreaks. The structure-fuel correction pipeline proposed in Section 7.2.5 provides a first-order

refinement by distinguishing building footprints and inert surfaces, thereby improving spatial realism in urban areas. Furthermore, as demonstrated through the sensitivity analyses in Chapter 6, the standard 30-meter resolution is frequently inadequate for capturing the fine-scale layout of WUI features. To address this, the present chapter also explores the impact of higher-resolution inputs (e.g., 10-meter resolution) on fire reconstruction accuracy.

The remainder of this chapter is organized as follows. Section 8.1 introduces the 2017 Thomas Fire and outlines the simulation configuration. Sections 8.2 through 8.10 present a series of simulations, illustrating the impact of progressively incorporating model and data improvements. Section 8.11 summarizes key findings and insights drawn from the simulation series. Finally, Section 8.12 concludes the chapter with a synthesis of lessons learned and recommendations for future work.

8.1 Introduction to the 2017 Thomas Fire

The Thomas Fire was the largest wildfire in modern California history at the time of its occurrence, ultimately burning 281,893 acres (approximately 1,141 km^2) across Ventura and Santa Barbara counties.

The fire began on December 4, 2017, and was first reported at 18:23 local time near the city of Santa Paula. Post-event investigations attributed the ignition to electrical faults in local utility infrastructure. Two separate ignition points were identified: one near a private residence along Koenigstein Road, an offshoot of California State Route 150 (approximate coordinates: 34.446, -119.118), and another in the Anlauf Canyon area (34.421, -119.083). The Anlauf Canyon ignition occurred around 18:20, followed by the Koenigstein Road ignition at approximately 19:30

the same evening [167, 168]. The location of the ignition points are visualized using asterisks in panel of of Figure 8.1.

Driven by unusually strong and persistent Santa Ana winds, the fire rapidly spread southwest from the initial ignition points. Wind gusts of 35–50 mph were recorded, with relative humidity as low as 3–10% during the first 24 hours [169].

The fire was fully contained on January 12, 2018, and officially declared extinguished on June 1, 2018, after no additional hotspots were detected [170]. As of 2025, the Thomas Fire ranks as the eighth-largest wildfire in California's modern history.

More than 1,000 structures were destroyed as the fire advanced into the cities of Ventura, Santa Paula, and Ojai. Initially spreading southwest, the fire later shifted northwest as wind patterns evolved. Figure 8.1 shows satellite-based hotspot detections from the Visible Infrared Imaging Radiometer Suite (VIIRS), onboard the Suomi NPP, NOAA-20, and NOAA-21 satellites. VIIRS provides 375 m spatial resolution and 14-hour temporal resolution for hotspot monitoring.

Figures 8.1(b) to (d) show accumulated VIIRS hotspots at three key time points: 2017-12-05 10:47 UTC (2:47 a.m. local), 2017-12-05 20:26 UTC (12:26 p.m.), and 2017-12-06 10:28 UTC (2:28 a.m.). Hotspots (red dots) are overlaid on a simplified fuel classification map consistent with Figure 7.4. In these maps, Ventura lies to the southwest (lower left), Santa Paula to the east (right), and Ojai in the north-central portion. Within just eight hours, the two separate ignitions merged and the fire front traveled over 30 km.

Observations indicate that while the fire encroached significantly into urban areas, it did not consume entire communities—making the Thomas Fire a strong validation case for evaluating firebrand ignition and partial urban fire spread.

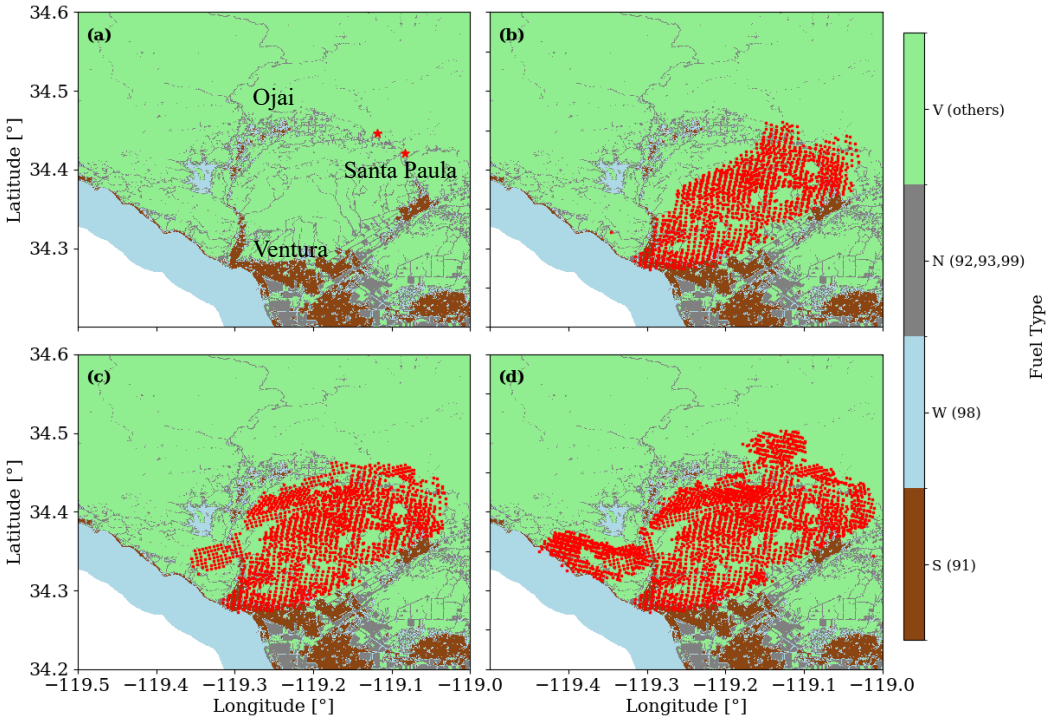


Figure 8.1: Accumulated VIIRS hotspot detections during the early stages of the Thomas Fire (extent: 42 km × 31 km). Red dots in Panels (b)–(d) indicate thermal anomalies derived from the VIIRS 375 m active fire product. Each panel overlays hotspot data on a simplified fuel classification map, consistent with Figure 7.4. Major population centers—Ventura (southwest), Santa Paula (east), and Ojai (north-central)—are labeled for reference in Panel (a), along with the initial fire locations (red stars). (a) 2017-12-04 03:00 UTC (7:00 p.m. local time); (b) 2017-12-05 10:47 UTC (2:47 a.m.); (c) 2017-12-05 20:26 UTC (12:26 p.m.); (d) 2017-12-06 10:28 UTC (2:28 a.m.).

Figure 8.2 shows a histogram of elevation distribution across the study area. Most elevations fall between approximately 50 and 900 meters above sea level, with the urban-occupied zones concentrated below 500 meters. This distribution is favorable for model testing because elevation (and hence slope) effects are minimal in most of the urban area. Since the firebrand and urban fire spread models in ELMFIRE currently do not consider terrain slope effects, the Thomas Fire provides a useful baseline for evaluating fire spread without topographic interference.

Figure 8.3 presents the distributions of wind speed, wind gusts, and wind direction from RTMA data during the early progression of the Thomas Fire. Although gust magnitudes vary con-

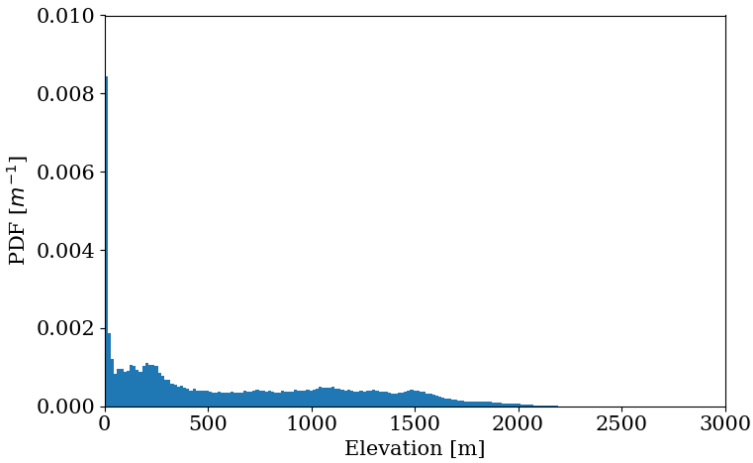
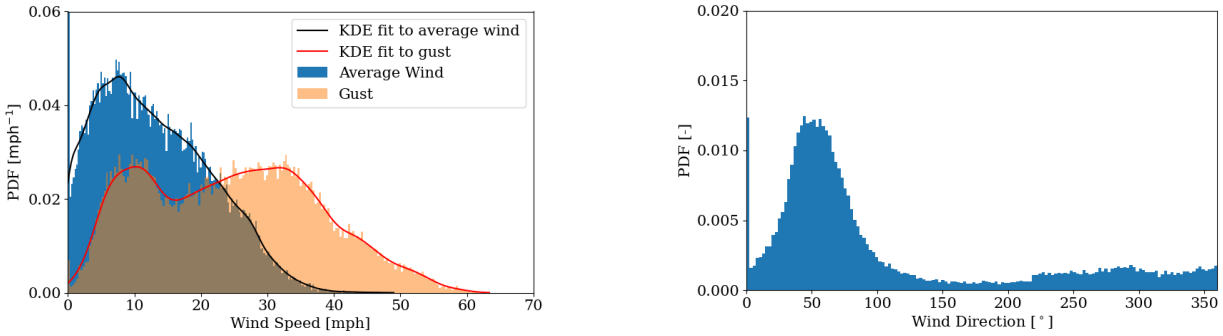


Figure 8.2: Elevation distribution within the Thomas Fire simulation domain. Most urban regions are located below 500 m.

siderably, the prevailing wind direction remained consistently toward the southwest (i.e., $< 90°$ from north), favoring downwind firebrand transport. This relatively consistent wind regime enables clearer interpretation of fire spread dynamics. However, it is worth noting that RTMA products may not fully capture localized, fire-induced wind, potentially contributing to discrepancies between observed and simulated fire behavior.



(a) (b)

Figure 8.3: Distribution of RTMA wind data over the first 38 hours of the Thomas Fire. Wind values were extracted over a domain approximately 82.5 km $\times$ 97.5 km, bounded by coordinates (34.757, -119.840) and (34.041, -118.761), centered on the region shown in Figure 8.1 with a 10 km buffer applied to each edge. (a) Distribution of default RTMA wind speed (blue bins, black Kernel density fit curve) and gust (orange bins, red Kernel density fit curve); (b) Histogram of wind direction in degrees from north.

In summary, the Thomas Fire features a large WUI region with relatively simple topography and wind conditions. Multiple ignitions, rapid spread, and evidence of firebrand-driven ignitions make it a compelling case for validating fire spread modeling capabilities in ELMFIRE. Combined with satellite observations and available local records, the event offers a good reference for evaluating model performance and the incremental impact of different model and data improvements.

In the remainder of this section, a series of simulation cases are presented to demonstrate the impact of different fire spread models and input data sources on large-scale fire behavior predictions. The baseline case evaluates the performance of the Rothermel model alone, applied within the level-set solver framework to simulate landscape-scale fire propagation.

Subsequent cases sequentially introduce additional modeling components. First, urban fire spread models—specifically the Hamada and WU-E models described in Chapter 2—are incorporated alongside the Rothermel model. Following this, the firebrand models are introduced into the simulation. The initial firebrand modeling approach is the original Lagrangian particle-based scheme, which is then compared against the improved Eulerian firebrand model proposed in this study. These cases are simulated using the most comprehensive surface fire spread configuration available in ELMFIRE (i.e., the Rothermel model enhanced by the WU-E model).

Next, a sensitivity analysis is performed to assess the influence of input data on simulation results. This includes evaluating the effects of different wind fields—namely raw RTMA winds, RTMA gust data, and WindNinja downscaled gust fields—as well as different fuel maps, comparing the standard LANDFIRE FBFM40 product against a modified fuel map derived from Microsoft Building Footprints and OpenStreetMap (OSM) data.

Finally, a high-resolution simulation using the Microsoft/OSM-modified fuel map upscaled

to 10 m resolution is presented. The results are compared to those from the default 30 m simulation to highlight the effects of spatial resolution on fire behavior prediction and to inform future directions for model development.

The main features and configurations of the simulations presented in this section are summarized in Table 8.1.

Table 8.1: Summary of simulation cases for Thomas Fire reconstruction.

Case ID	**Urban Fire**	**Firebrand**	**Wind Input**	**Fuel Map / Grid Resolution**
Case 1	None	None	RTMA (ws)	LANDFIRE / 30 m
Case 2	Hamada	None	RTMA (ws)	LANDFIRE / 30 m
Case 3	WU-E	None	RTMA (ws)	LANDFIRE / 30 m
Case 4	WU-E	Lagrangian	RTMA (ws)	LANDFIRE / 30 m
Case 5	WU-E	Eulerian	RTMA (ws)	LANDFIRE / 30 m
Case 6	WU-E	Eulerian	RTMA (gust)	LANDFIRE / 30 m
Case 7	WU-E	Eulerian	WindNinja (gust)	LANDFIRE / 30 m
Case 8	WU-E	Eulerian	WindNinja (gust)	Modified (MS/OSM) / 30 m
Case 9	WU-E	Eulerian	WindNinja (gust)	Modified (MS/OSM) / 10 m

8.2 Simulation 1: Baseline ELMFIRE Performance

We first evaluate the baseline performance of ELMFIRE using only the Rothermel model for wildland fire spread, with all urban fuels treated as inert. The simulation is configured at 30-meter resolution for both topography and land cover, with the level-set fire front tracking algorithm discretized on the same spatial scale. Meteorological inputs—including wind and fuel moisture—are originally provided at the native RTMA resolution (2.5 km) and interpolated to 30 m using bilinear spatial interpolation and linear temporal interpolation at the simulation time steps. The wind field used in this case corresponds to the RTMA wind speed product. Temporal resolution is governed by the local surface fire rate of spread, and the simulation time step is constrained such that the Courant–Friedrichs–Lewy (CFL) number does not exceed 0.45 (consistent

with the default selection in ELMFIRE applications such as in Refs. [71, 112]).

Land cover and topography data are obtained following the procedure in Section 7.2, using the LANDFIRE `2016_Remap` product—the most recent dataset available prior to the 2017 Thomas Fire. Wind fields are extracted from the RTMA archive covering 2017-12-05 02:00 UTC to 2017-12-06 15:00 UTC at hourly intervals. Fuel moisture fields, both dead and live components, are derived from these same RTMA data using the NFDRS4 model implementation described in Section 7.4.2.

Examples of these inputs are shown in the previous chapter, from Figure 7.1 through Figure 7.7, which provide visual context for the simulation domain and environmental conditions.

The fire is initialized at 2017-12-05 02:30 UTC (2017-12-04 18:30 local time) by setting the level-set variable $\phi = -1$ in two circular regions (each with a 1-km diameter) centered at the ignition points (34.446, -119.118) and (34.421, -119.083). All other grid cells are initialized with $\phi = 1$. Figure 8.4 presents the resulting fire spread predictions at time points aligned with VIIRS satellite overpasses.

Overall, the simulation captures the general southwestward spread of the fire but significantly underpredicts the rate of expansion during the early stages. The simulated fire front does not reach a scale comparable to the VIIRS-observed burn area until approximately 33 hours after ignition.

Importantly, because the simulation omits both urban fire spread and firebrand spotting, fire propagation halts at the wildland–urban interface and is blocked by infrastructural features such as roads (e.g., Highway CA-33). In contrast, the VIIRS data show that the fire continued to spread across urban areas and ignited spot fires across major barriers. A notable example is shown in panel (b), where a VIIRS-detected ignition appears across Highway CA-33—likely

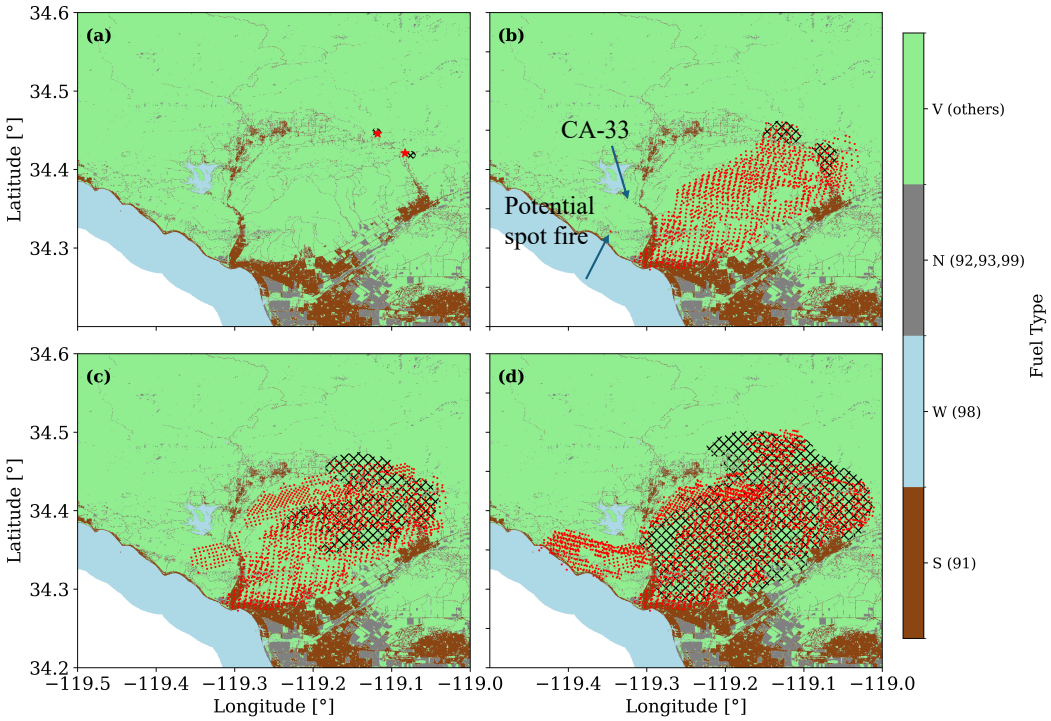


Figure 8.4: Baseline simulation (Simulation 1) results using only the Rothermel wildland fire spread model (the same extent to Figure 8.1). The simulation starts at 2017-12-05 02:30 UTC. Each panel corresponds to the approximate time of a VIIRS observation (see Figure 8.1), with simulation times at (a) 1 hr from 18:00 local time (03:00 UTC), (b) 9 hr (11:00 UTC), (c) 18 hr (20:00 UTC), and (d) 33 hr (2017-12-06 11:00 UTC). Burned areas ($\phi \leq 0$) are shown with a hatch pattern. A blue arrow in panel (b) marks a VIIRS-detected spot fire across Highway CA-33.

caused by long-range firebrand transport.

These findings underscore the importance of incorporating both urban fire spread and firebrand transport mechanisms for realistic simulation of fire dynamics in WUI-dominated landscapes.

8.3 Simulation 2: Improved ELMFIRE with Hamada Urban Spread Model

Now we turn to simulations in which urban fire spread is activated. We begin by evaluating the influence of the empirical Hamada model. This simulation retains the same configuration as Simulation 1, with the key difference being that urban fuels (category #91 in the raw LANDFIRE

FBFM40 fuel map) are now permitted to propagate fire. The rate of spread in these areas is determined using the Hamada model. It is important to note that fuel type 91 in the LANDFIRE classification includes not only structures but also inert infrastructure such as roads, which can lead to overrepresentation of burnable urban area in the raw fuel map.

In this test, spatially uniform parameters are assumed for the Hamada model across the entire domain. These include the average structure footprint and mean inter-structure separation distance, derived from the statistics shown in Figure 6.3. Specifically, we use a footprint dimension of $a = 16.24$ m and a separation distance of $d = 7.69$ m. All grid cells labeled FBFM91 are assumed to be fully covered by combustible materials, with a fuel fraction $f_b = 1.0$.

The resulting fire spread patterns are illustrated in Figure 8.5, using the same simulation time snapshots as those in Figure 8.4. Compared to the baseline case, the inclusion of the Hamada model substantially accelerates fire propagation during the first 9 hours. Notably, the fire crosses California State Route 150 (highlighted by the blue arrow in panel (b)), which previously slowed fire spread due to the absence of urban ignition. This propagation occurs because the rasterized representation of fuels in LANDFIRE allows for vegetative pathways that span across the road. While this early improvement better matches the observed fire spread, the model still underpredicts growth relative to VIIRS hotspots during the initial hours.

In the following 24 hours, however, the simulation indicates more aggressive growth than observed. The fire front rapidly expands to match—and in some cases surpass—the observed burn extent. By hour 33, the model predicts substantial intrusion into the Ventura and Ojai community along the southern and northern flanks of the fire, while VIIRS observations suggest only limited fire activity in these regions. This overprediction may result from: (1) overestimated fire spread in urban areas by the Hamada model; (2) misclassification of fuels in the LANDFIRE

map, particularly the grouping of roads and structures under a single urban category; or (3) the absence of representation of active suppression and containment efforts.

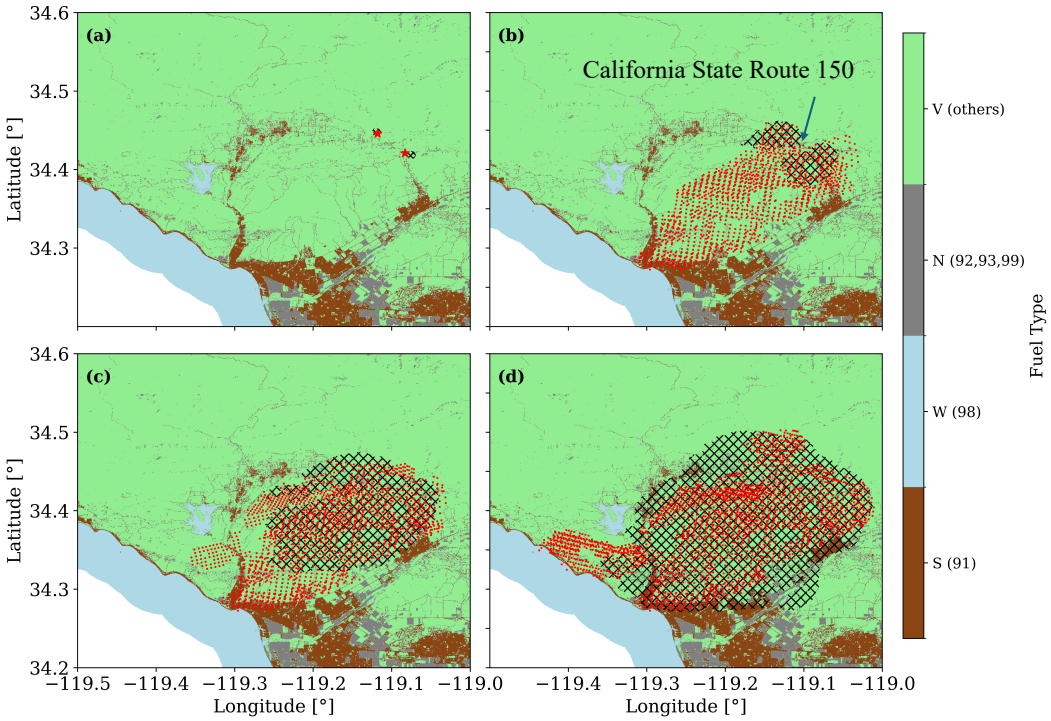


Figure 8.5: Simulation 2 results using the Hamada model for urban fire spread in combination with the Rothermel model for wildland fuels (the same extent to Figure 8.1). Time snapshots correspond to those in Figure 8.4. Panel (b) highlights the fire crossing California State Route 150 by hour 9, as indicated by a blue arrow. Compared to the baseline, early-stage spread is noticeably improved; however, overprediction is observed after 24 hours, especially near Ventura and Ojai.

8.4 Simulation 3: Improved ELMFIRE with WU-E Urban Spread Model

While the overprediction observed in Simulation 2 may stem from multiple sources—including coarse fuel classification, oversimplified assumptions, and unmodeled suppression effects—our focus here is to isolate the influence of the urban fire spread model itself. In this simulation, we replace the empirical Hamada model with the physics-based WU-E model introduced in Chapter 2. For consistency and simplicity, uniform parameters are assumed across all urban fuel cells.

Specifically, the structure footprint dimension and inter-structure separation distance are set to $a = 16.24$ m and $d = 7.69$ m, respectively, consistent with earlier simulations. These values are used in the parameterization of the elliptical flame geometry in Equations 2.84–2.86. The heat release rate per unit area for urban fuels follows the design fire curve shown in Figure 8.6, which reflects the default configuration proposed in Ref. [112]. This curve governs the radiative and convective heat flux components specified in Equations 2.81 and 2.83.

Other ignition-related parameters also adopt recommended default values from Ref. [112]: the critical flux–time product (FTP) required for ignition is set to 10,500 kJ·m^{-2}; the absorptivity of the target structure is assumed to be 0.89; and the non-combustible fraction α_c is set to 0.5.

One additional modification was made to address a numerical limitation identified in preliminary simulations: the Rothermel model often predicts fire intensities that are insufficient to ignite adjacent urban fuels under the WU-E framework. To overcome this, a crude but practical fix—originally proposed by the WU-E model developers—is adopted. Specifically, vegetative fuels located adjacent to urban fuels are assigned a fixed heat release rate of 250 kW·m^{-2} for the duration of the simulation. This modification only affects ignition dynamics when the adjacent urban cell has not yet ignited (i.e., when $\phi > -1$). Although simplistic, this treatment enables fire spread from vegetative fuels into urban zones and serves as a placeholder for future improvements, such as explicit modeling of Zone-0 fuels and associated design fire curves.

The resulting fire spread prediction is shown in Figure 8.7, using the same time slices as in previous simulations. Compared to the Hamada model (Simulation 2), the WU-E model produces a better match with VIIRS observations—particularly at the northern and southern flanks of the fire perimeter (compare panels (d) in Figures 8.5 and 8.7). This improvement underscores the importance of physically representative models in capturing urban fire dynamics.

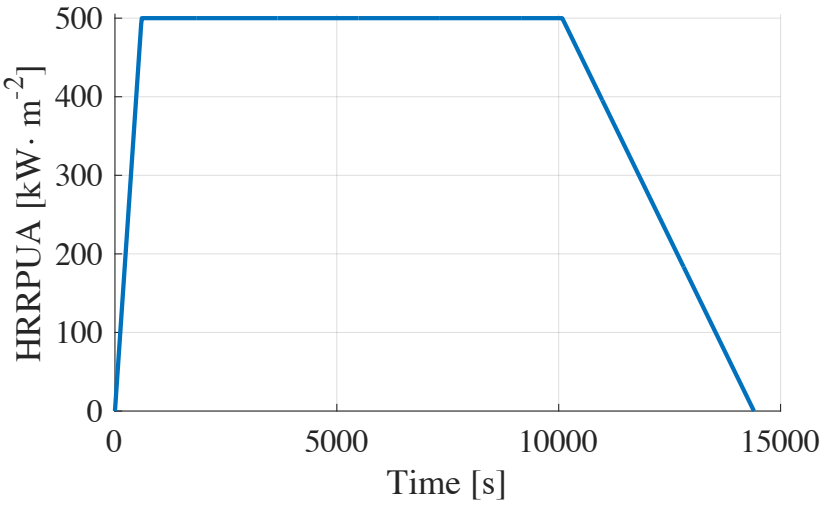


Figure 8.6: Design fire curve applied in the WU-E urban spread model for all structure fuels. The curve is used to compute heat fluxes and ignition conditions, following the default setup in Ref. [112]. The design fire curve in ELMFIRE is specified in units of $\mathrm{kW} \cdot \mathrm{m}^{-2}$, i.e., HRRPUA. Therefore, a 30-meter cell burns at a peak HRR of 450 MW, which may be inteprated as an overestimated number of buildings from 1 to 9 within the coarse cell.

Nonetheless, the simulation still underpredicts fire spread during the first 9 hours. From the fuel classification, it is evident that the early burn region is dominated by vegetative fuels, suggesting that this delay is likely not caused by urban model errors. Instead, the underestimated spread rate may result from: (1) the absence of firebrand-induced ignitions in vegetative zones; or (2) inaccurate fuel or meteorological inputs, such as wind speed or fuel moisture. These hypotheses will be addressed in subsequent simulations.

8.5 Simulation 4: Impact of Firebrands — Original ELMFIRE Lagrangian Firebrand Model

Building on the results of previous simulations, we now examine the influence of firebrand spotting using the original Lagrangian firebrand model implemented in ELMFIRE. This model explicitly simulates individual firebrand particles, tracking their emission, atmospheric transport,

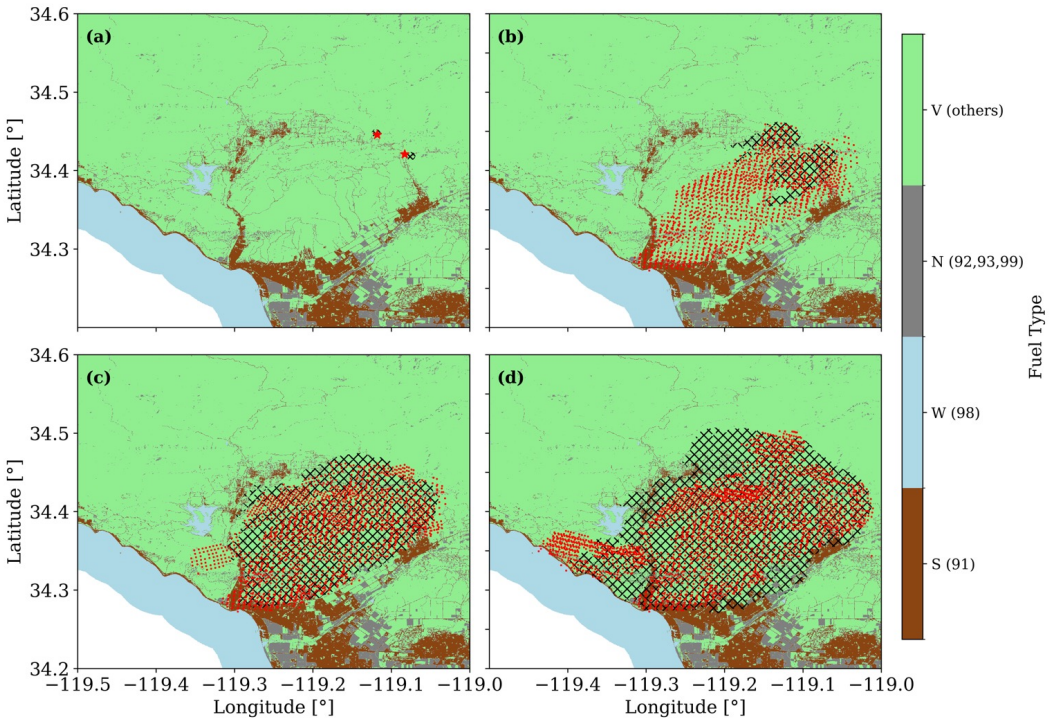


Figure 8.7: Simulation 3 results using the WU-E model for urban fire spread combined with the Rothermel model for wildland fuels (the same extent to Figure 8.1). Compared to the Hamada-based results (Figure 8.5), the predicted fire front shows better agreement with VIIRS observations at the flanks by hour 33. However, early underprediction suggests that firebrand ignition or input uncertainties may still limit simulation accuracy.

deposition, and ignition.

The configuration follows default parameters from Ref. [71] and recent Thomas Fire reconstructions in Ref. [112]. Key assumptions include:

- **Emission source:** 1% of crown fuel cells and 0.1% of surface fuel cells are randomly designated to emit firebrands.
- **Emission condition:** Firebrands are emitted only when the cell's fireline intensity exceeds 1000 kW·m^{-1} and $\phi < 0$ (i.e., ignition has occurred).
- **Number of firebrands:** Each qualifying cell emits exactly 10 firebrands at and only at the ignition time step.

- **Transport:** Trajectories are calculated using the Lagrangian integration approach described in Section 4.3.
- **Deposition and ignition:** Deposited firebrands immediately ignite the target cell with a probability of 1, and no ignition delay is applied—including no time lag between emission and deposition.

Travel distances are sampled from a lognormal distribution whose parameters are derived as:

$$\mu = \log\left(\frac{\mu_0^2}{\sqrt{\mu_0 \cdot \sigma_0 + \mu_0^2}}\right), \tag{8.1}$$

$$\sigma = \sqrt{\log\left(1 + \frac{\mu_0 \cdot \sigma_0}{\mu_0^2}\right)}, \tag{8.2}$$

where μ_0 is the expected mean spotting distance in meters, and σ_0 is fixed at 250 m. The mean distance μ_0 is parameterized by:

$$\mu_0 = \max\left(5 \cdot \dot{Q}'_{\text{fire}}{}^{0.2} \cdot U_{\text{wind}}^{0.4}, 1.0\right),$$

with $\dot{Q}'_{\text{fire}}$ representing the fireline intensity in kW·m^{-1}, and U_{wind} the wind speed at 20 ft (in mph). Figure 8.8 shows the resulting lognormal PDF when $\dot{Q}'_{\text{fire}} = 5000$ kW·m^{-1} and $U_{\text{wind}} = 40$ mph.

Note that this parameterization does not distinguish between wildland and urban fuels—unlike the treatment used in Chapter 6. For urban fuels, a nominal fireline intensity is computed based on the design fire curve (Figure 8.6), using:

$$\dot{Q}'_{\text{fire}} = HRRPUA(t) \cdot \Delta x.$$

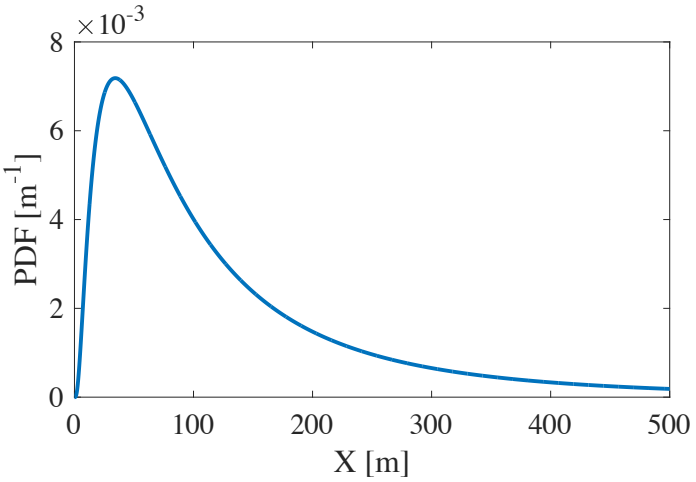


Figure 8.8: Prescribed lognormal probability density function for firebrand spotting distance in the original ELMFIRE Lagrangian model. Case shown: $\dot{Q}'_{\text{fire}} = 5000$ kW·m^{-1}, $U_{\text{wind}} = 40$ mph.

All other simulation settings remain consistent with Simulation 3. The resulting fire spread patterns are shown in Figure 8.9. Compared to Simulation 3, this model produces a more extensive burn area. However, even with firebrand spotting, the predicted burn extent at 9 hours remains significantly smaller than observed.

By hour 18, the fire catches up with observed hotspots, although it fails to capture the extrodinary area west of CA-33. This omission is likely due to the model's inability to simulate long-range spot fires, which are poorly represented in both the Lagrangian PDF (over 90% firebrands travel less than 350 m in Figure 8.8) and the upcoming Eulerian framework. By hour 33, the model overpredicts fire spread, particularly in the northern and southern urban flanks. Again, possible reasons include overestimated urban fuel presence due to raw LANDFIRE classification; lack of ignition delay or probabilistic treatment in firebrand ignition; or no representation of suppression or containment operations.

A customized diagnostic was added to ELMFIRE to quantify the influence of firebrand-induced ignitions. Figure 8.10a presents the distribution of all ignited cells across fuel categories

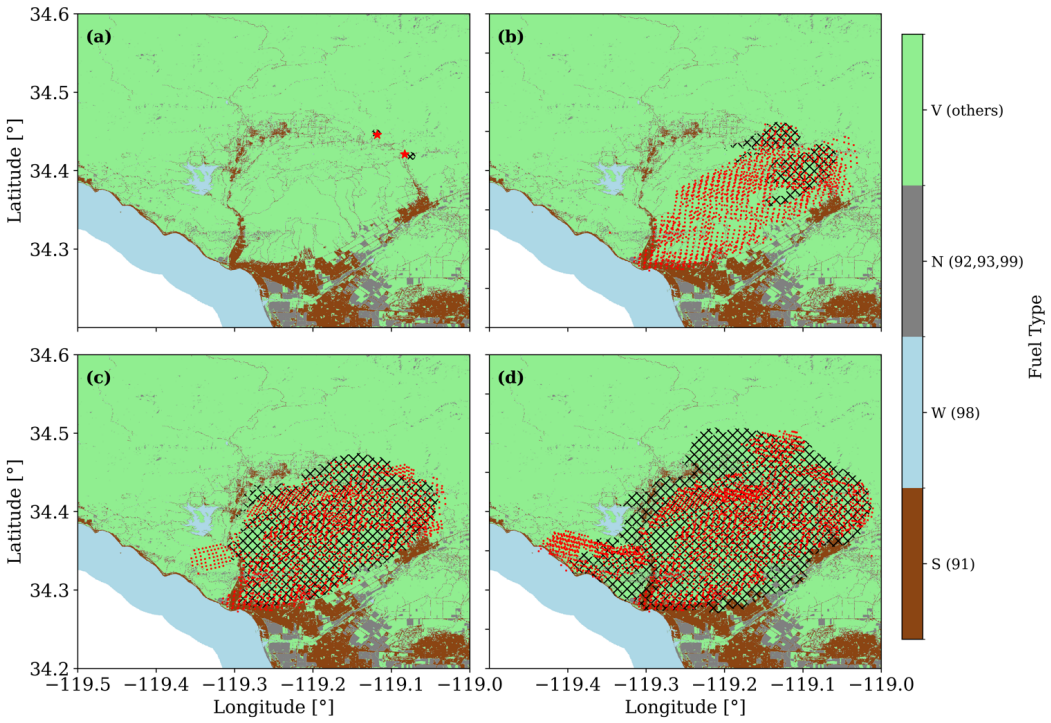
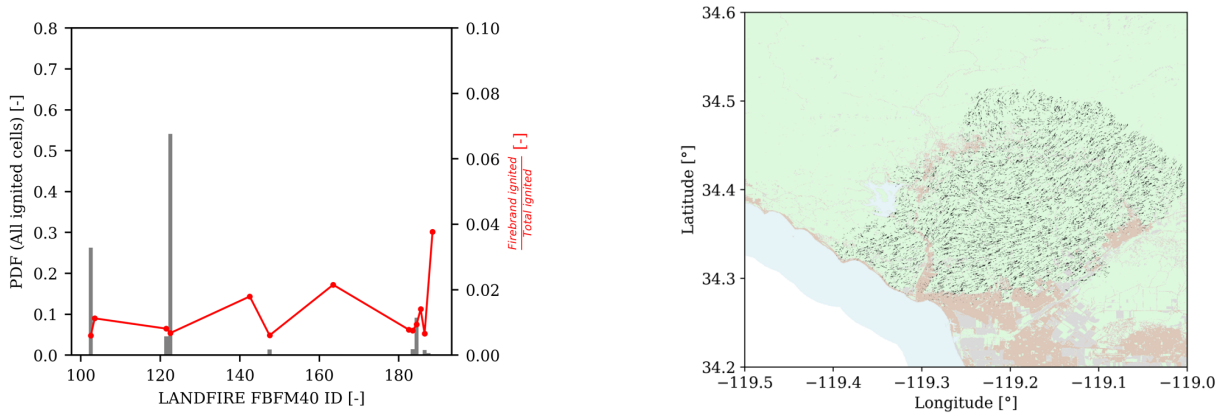


Figure 8.9: Simulation 4 results using the original ELMFIRE Lagrangian firebrand model, paired with the WU-E urban fire spread and Rothermel wildland model (the same extent to Figure 8.1). Although spread improves relative to Simulation 3, early growth remains underpredicted, and late-stage fire extent is overestimated in urban regions.

(gray bars), along with the proportion of firebrand-induced ignitions within each category (red line). It is found that the proportion of firebrand-ignited cells fluctuates around 1% across all fuel types, suggesting that the firebrand ignition model used here exhibits no strong bias toward any particular fuel category.

The spatial distribution of accumulated firebrand deposition is shown in Figure 8.10b, with cells receiving firebrands colored in black. As discussed in previous chapters, firebrand ignition is typically less consequential in wildland areas where high surface fire spread rates tend to overtake or subsume the effects of most spot fires. Consequently, firebrands are expected to exert a more significant influence in regions characterized by low rates of spread (ROS), such as timber-dominated landscapes or urbanized areas.

It is observed that the vast majority of cells receive fewer than 10 firebrand particles during the entire simulation period. Moreover, firebrand-ignited cells account for only approximately 0.7% of the total burned area at 33 hr. While this particle count is clearly insufficient to meet ignition thresholds based on physical heat flux or fuel load models, the results may be interpreted under a coarse assumption—i.e., each simulated firebrand particle implicitly represents a larger population of real firebrands. Even so, the deposition pattern remains spatially sparse and lacks sufficient resolution to meaningfully capture the spatial distribution of firebrands and local ignition dynamics.



(a) (b)

Figure 8.10: Firebrand accumulation and ignition diagnostics from Simulation 4 (same spatial extent as Figure 8.1). The firebrand deposition pattern is sparse and poorly resolved, with only 0.7% of the total burned area attributable to firebrand-induced ignitions. (a) Number of ignited cells by fuel type (gray bars) and the proportion attributed to firebrand ignition (red line); (b) Computational cells with firebrand deposition at 33 hr (black pixels).

In conclusion, although Simulation 4 shows improved agreement with satellite-derived burned area, this alignment may be coincidental and driven by physically unrealistic assumptions—namely, simplified ignition logic, deterministic deposition behavior, and the absence of ignition delay. The results are further influenced by customized assumptions, such as the arbitrary release of 10 firebrand particles from burning cells at the ignition time step. A more

physically grounded firebrand modeling framework is therefore essential to enable meaningful interpretation of fire spread mechanisms. The next simulation applies the Eulerian firebrand model developed in this study.

8.6 Simulation 5: Impact of Firebrands — Physics-Based Eulerian Firebrand Model

In this simulation, we implement the full physics-based fire spread framework developed in this study, integrating both surface fire spread models and an Eulerian firebrand transport model. Specifically, the simulation incorporates the Rothermel model for wildland fire spread, the WU-E model for urban and wildland–urban interface (WUI) fire spread, and the proposed Eulerian firebrand model featuring physically parameterized generation, transport, and ignition submodels.

The configuration of the WU-E model remains consistent with previous simulations, including prescribed structure dimensions, separation distances, and thermal response parameters. The firebrand model follows the setup presented in Section 6.1, with minor modifications made to align with current implementation constraints in ELMFIRE:

- **Firebrand generation:** The normalized generation rates (GR^*) are set to 33.3 pcs·MW^{-1}·s^{-1} for vegetative fuels and 10.0 pcs·MW^{-1}·s^{-1} for urban fuels. Although a fixed HRRPUA of 250 kW·m^{-2} is applied near the wildland–urban boundary to facilitate ignition of adjacent urban cells, this value is not used in the firebrand generation calculation. Instead, firebrand generation from vegetative cells is still governed by the Rothermel-predicted fireline intensity.

- **Firebrand transport:** Travel distance distributions are parameterized using Sardoy's formulation for vegetative fuels and Himoto's formulation for urban fuels.

- **Ignition delay:** The delay required to establish a fully developed fire is uniformly set to $t_{\text{ign,large}} = 300$ s for both fuel types. The smoldering-to-flaming transition time is set to $\tau_{\text{ign,small}} = 42$ s. These uniform settings deviate from the fuel-specific values originally proposed in Section 6.1, due to current software limitations in ELMFIRE, which only allows global ignition delay parameters. This restriction is expected to be resolved in future versions. Nonetheless, the simplification is expected to have minimal influence on fire spread dynamics in vegetated regions, where propagation is primarily driven by surface fire rather than firebrand ignition.

It is important to note that ELMFIRE has not yet implemented the subgrid treatments proposed in Chapter 6 to mitigate resolution-induced artifacts in urban fuel modeling. Consequently, each urban cell behaves as an independent emitter and recipient of firebrands. The heat release rate per urban cell is calculated as $HRRPUA \times \Delta x^2$ and used for evaluating firebrand generation rate and transport distance.

This simulation represents the most complete physics-based configuration evaluated in this chapter, integrating coupled multi-physics models for fire spread and spotting across both vegetative and built environments. Results from this simulation are presented in Figure 8.11. The predicted fire evolution shows strong similarity to Simulation 3, indicating that firebrand spotting is not the dominant driver of fire spread in the current simulation configuration. Due to the ignition delay mechanisms incorporated in the model, surface fire spread outpaces firebrand-induced ignition in vegetative fuels. However, given that a substantial portion of inert fuels (e.g., roads,

open spaces) are misclassified as burnable urban fuels in the LANDFIRE dataset, firebrands may still play a crucial role when these misrepresented fire barriers are correctly modeled. A dedicated simulation will be presented in a subsequent section to further investigate this effect.

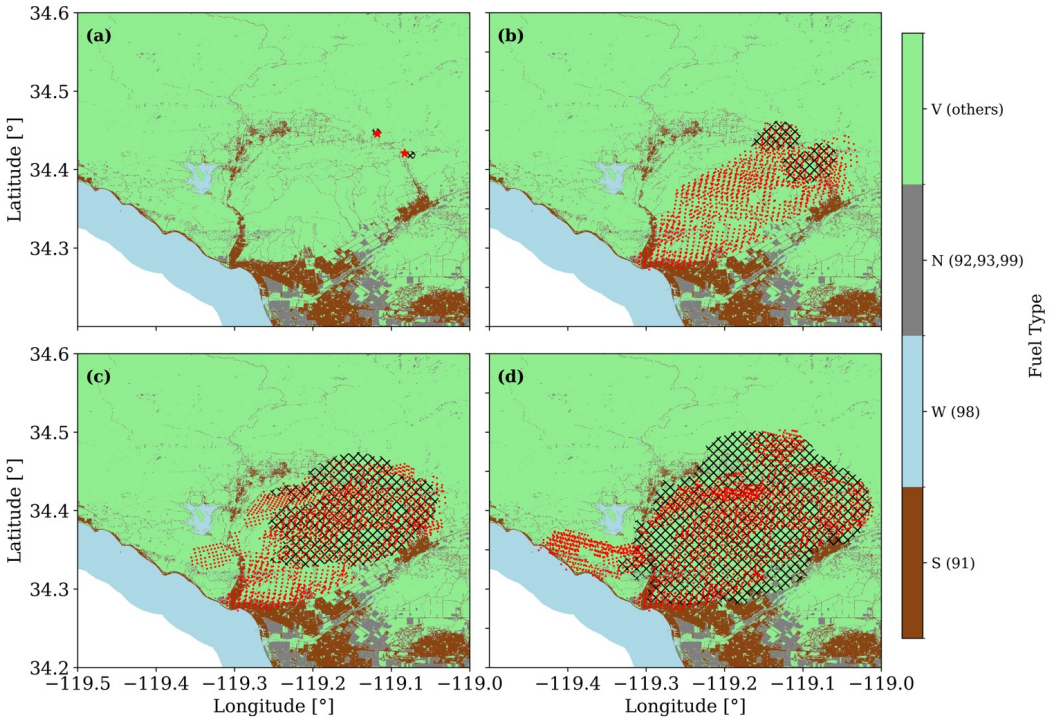


Figure 8.11: Simulation 5 results using the full physics-based configuration with coupled surface fire spread and Eulerian firebrand transport models (the same extent to Figure 8.1). Compared to Simulation 4, the overestimation of burned area is reduced due to the inclusion of ignition delay models. Urban spread is evident where firebrands breach misclassified fuel barriers.

Compared to Simulation 4, the predicted burn extent is significantly reduced, largely due to the physics-based ignition delay model, which more realistically limits the rate and spatial extent of firebrand-induced ignitions. Figure 8.12b shows the spatial distribution of accumulated firebrand number load at 33 hour. Unlike the sparse and under-resolved patterns seen in Simulation 4, the distribution is now fully resolved. Firebrand load (in pcs·m^{-2}) is visualized using a hot colormap.

Notably, a substantial portion of the domain—approximately 47.0% of the burned area at

33 hr—exhibits firebrand accumulations exceeding 10^4 pcs·m^{-2}. Urban regions show higher firebrand concentrations (appearing as yellow zones in the Figure 8.12b), while wildland areas exhibit lower accumulation levels (red zones near the center of the domain). This spatial distribution aligns with expected physical behavior: urban structures are governed by prescribed design fire curves that sustain high heat release rates over a multi-hour duration, thereby serving as prolonged firebrand sources. In contrast, wildland fuels typically exhibit short residence times—on the order of 100 s or less—corresponding to $\Delta x / ROS_{\text{surface}}$, and therefore emit fewer firebrands over their burning duration. As a result, urban and near-urban areas tend to receive a higher cumulative firebrand load than wildland regions.

Compared to Simulation 4, the predicted burn extent in Simulation 5 is significantly reduced, largely due to the incorporation of a physics-based ignition delay model. This modification more realistically constrains the rate and spatial extent of firebrand-induced ignitions. Figure 8.12b shows the spatial distribution of accumulated firebrand number load at hour 33. Unlike the sparse, under-resolved deposition patterns observed in Simulation 4, the firebrand distribution in Simulation 5 is continuous and fully resolved. Firebrand load (in pcs·m^{-2}) is visualized using a hot colormap.

Figure 8.12a highlights the ignition locations attributed to firebrands (red pixels), corresponding to 7.43% of the total ignited area. Figure 8.12d presents a histogram of fuel categories ignited by firebrands, showing that firebrand-induced ignitions predominantly occur in fuels with LANDFIRE FBFM codes exceeding 180—corresponding to timber fuels with very low surface fire spread rates ($0.007\ \mathrm{m} \cdot \mathrm{s}^{-}1$). Figure 8.12c maps these timber fuels and highlighted in red. The spatial correlation between these low-ROS fuels and firebrand ignitions suggests that firebrands play an important role in initiating fire spread in slow-burning fuel beds.

It is noted that the simulation does not show significant firebrand contributions to ignition in urban fuels. This outcome may be explained by the relatively high rate of spread predicted by the WU-E model, which likely overran any additive contribution from firebrands. Additionally, the LANDFIRE map's lack of resolution in capturing firebreaks (e.g., roads or Zone 0 defensible areas) results in contiguous burnable urban pixels, thereby reducing the necessity for firebrand-faciliated spread across barriers.

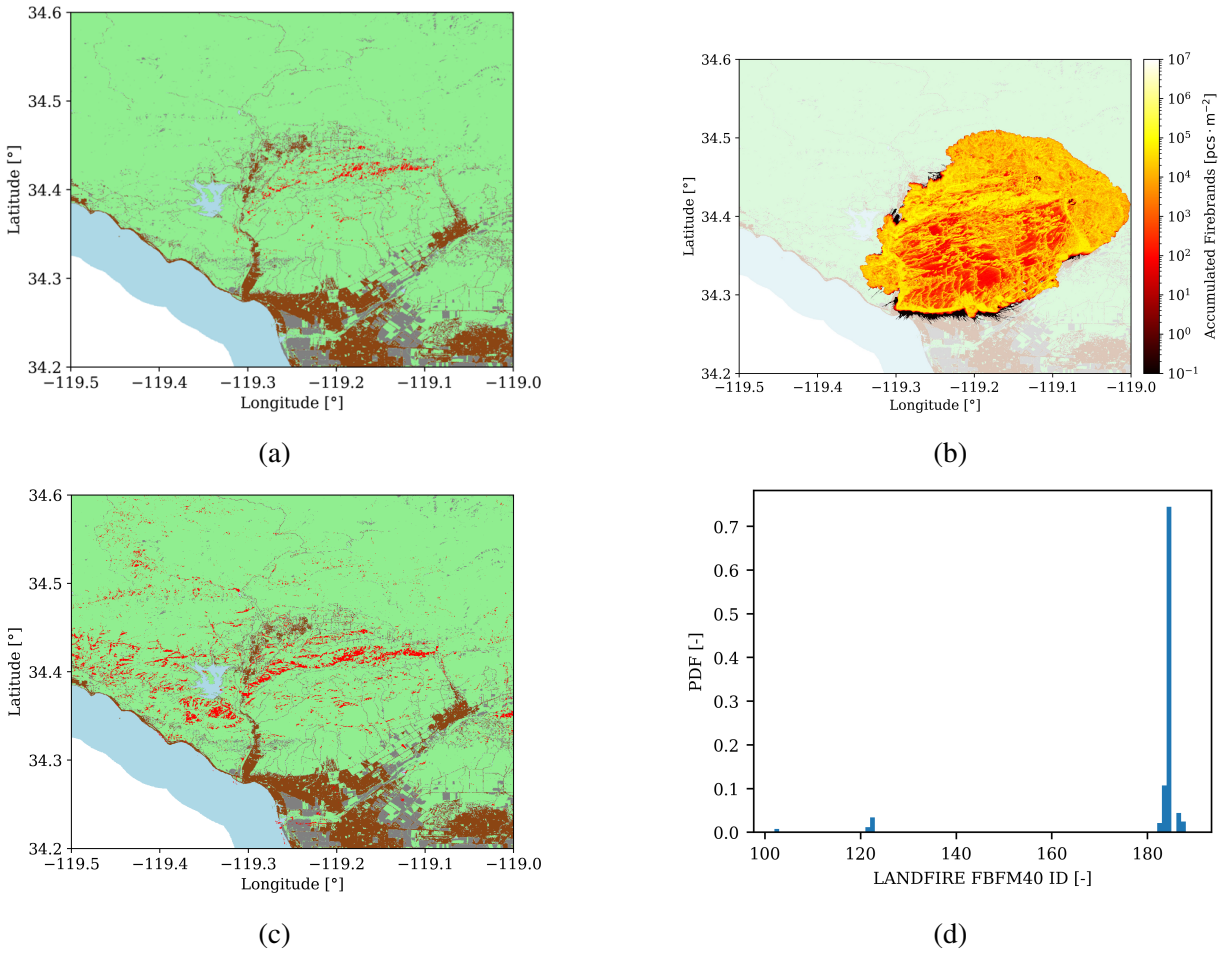


Figure 8.12: Firebrand accumulation and ignition statistics for Simulation 5 at 33 hr. (a) Firebrand-ignited pixels (red) at 33 hr; (b) Firebrand number load at 33 hr (pcs·m^{-2}); (c) Timber fuels (FBFM code $\geq$ 180) shown in red; (d) Distribution of fuel types ignited by firebrands. (a), (b) and (c) share the same extent to Figure 8.1.

8.7 Simulation 6: Impact of Inputs — Gust Wind-Induced Fire Spread

Previous simulations have demonstrated that even with the status-of-the-art fire spread and spotting models currently available in ELMFIRE, the predicted fire spread during the early phase of the Thomas Fire remains underrepresented. As a result, the following simulations aim to assess the influence of uncertain input fields—specifically, wind and urban fuel representation—on model performance. Other inputs, such as topography or static fuel classification, are not evaluated here as they are either derived from these primary fields or are considered to have lower uncertainty.

In this simulation, we evaluate the sensitivity of fire spread predictions to the choice of wind input. The RTMA wind speed field used in prior simulations is replaced with the gust wind field from the RTMA product. All other aspects of the model configuration and inputs remain identical to Simulation 5.

Note that the fuel moisture inputs are also retained from Simulation 5. This is because the NFDRS4 model used to estimate fuel moisture relies on the observed instantaneous meteorological variables, including wind speed. This setup ensures consistency in the inputs to the NFDRS4 model and provides a controlled basis for attributing observed differences in fire behavior primarily to the wind forcing used in the fire spread and spotting models, rather than to secondary changes in fuel moisture.

Gust wind fields capture short-term peak wind speeds and may more realistically represent the conditions responsible for extreme fire behavior, particularly in WUI contexts. As shown earlier in Figure 8.3a, gust winds exhibit a broader distribution and a higher mean compared to the default RTMA wind speed field. These elevated values have direct implications for both surface

fire spread and firebrand dynamics. In surface fire models (e.g., Rothermel and WU-E), the rate of spread is strongly correlated with wind speed. Similarly, firebrand transport models (e.g., Sardoy and Himoto) predict increased spotting distances with higher wind speeds, enhancing the potential for long-range ignition.

However, the relationship between wind speed and ignition probability is more complex. As discussed in Chapter 5, the ignition likelihood exhibits a non-monotonic U-shaped dependency on near-surface air velocity. For low wind speeds, ignition is inhibited due to insufficient convective flux. At very high wind speeds, convective cooling becomes dominant, again reducing ignition probability. This behavior is visualized in Figure 5.7b for example. Therefore, this test offers an opportunity to observe how enhanced wind fields influence the coupled dynamics of spread and spotting.

Figure 8.13 shows the predicted fire progression for Simulation 6, overlaid with VIIRS hotspot observations. Compared to Simulation 5, the fire spreads more rapidly in the first 9 hours following ignition, though it still underpredicts the burned area relative to the satellite data. By 18 and 33 hours, however, the simulation closely aligns with the observed fire perimeter. Although some overprediction is visible in the final frame—particularly in the southern and northern urban fringe—the model successfully captures the overall direction, extent, and morphology of the observed fire spread.

Figure 8.14b shows the spatial distribution of accumulated firebrand number load at 33 hours. As observed in previous simulations, higher firebrand concentrations are seen in urban areas—attributable to longer burning durations and elevated HRRs associated with structural fuels.

Figure 8.14a highlights the regions ignited by firebrands. Compared to Simulation 5, the

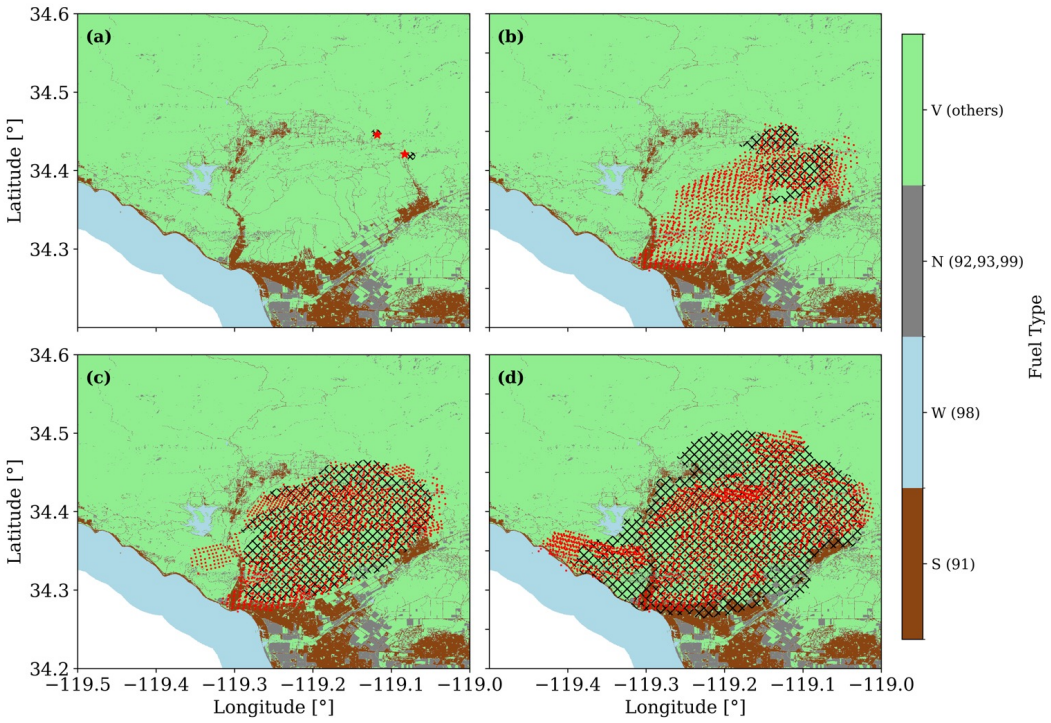
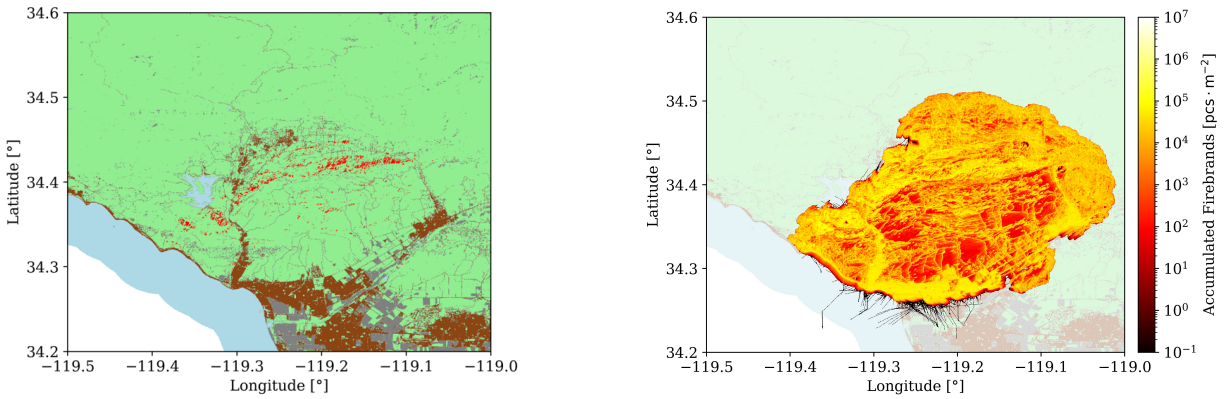


Figure 8.13: Simulation 6 results that use gust wind inputs(the same extent to Figure 8.1). Enhanced wind speeds result in more rapid early-phase spread and improved agreement with observations by 18 and 33 hours post-ignition.

proportion of firebrand-induced ignitions increases to approximately 7.97% of the total ignited area—an increase of about 0.5%. This increase underscores the influence of wind on spotting activity. Higher gust speeds not only enhance surface fire spread but also the firebrand generation, deposition, and ignition. These results suggest that incorporating gust wind inputs—rather than default RTMA wind speed—may provide a more realistic basis for simulating fire spread dynamics in ELMFIRE.



(a) (b)

Figure 8.14: Firebrand statistics from Simulation 6 using gust wind input. Increased wind speeds enhance both surface fire spread and firebrand transport, resulting in more ignitions and higher firebrand accumulation. (a) Firebrand-induced ignitions at 33 hr (red pixels); (b) Firebrand number load at 33 hr (pcs·m^{-2}).

8.8 Simulation 7: Impact of Inputs — Physically Downscaled Gust Wind-Induced Fire Spread

To further assess wind-related input sensitivity, we replace the RTMA gust wind field with a fine-scale (2.5 km to 0.3 km) wind field generated by the WindNinja solver. While finer resolutions (e.g., 30 m) are theoretically supported by WindNinja, they are not employed here due to memory constraints associated with the large computational domain. The downscaled wind field used in this simulation has already been visualized in Figures 7.5c and 7.5d.

All other configurations—including surface fire spread models, firebrand transport and ignition models, and fuel moisture fields—remain identical to those used in Simulations 5 and 6. Figure 8.15 presents the temporal progression of the simulated fire compared against VIIRS observations. Overall, the resulting burn area remains qualitatively similar to Simulation 6, though a slight overestimation is observed on the southern flank.

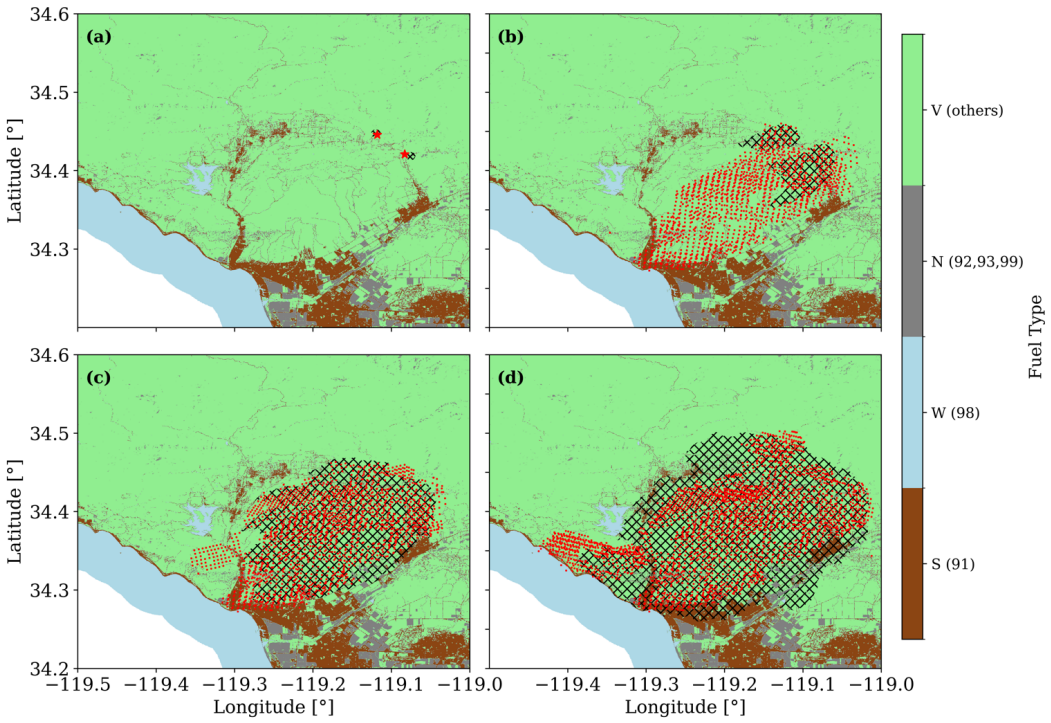
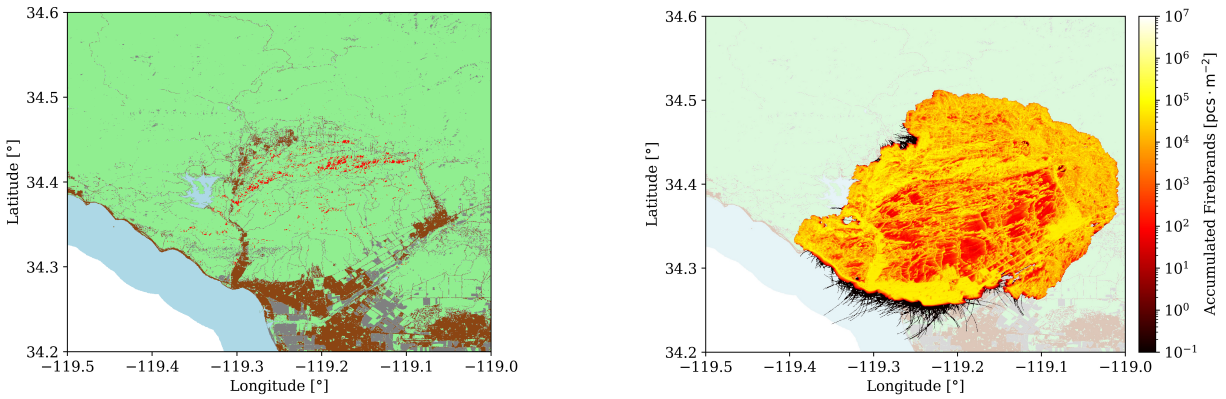


Figure 8.15: Simulation 7 results using WindNinja-generated gust wind field (upscaled to 300 m) for fire spread and spotting footprints (the same extent to Figure 8.1). Time snapshots correspond to the VIIRS observation windows shown previously. Overall burn extent remains similar to Simulation 6, with slightly larger spread observed in the southern flank at 33 hr.

Figure 8.16 presents the corresponding firebrand diagnostics. In Figure 8.16b, the spatial distribution of accumulated firebrand load shows no major deviation from Simulation 6. Similarly, firebrand ignition contributions, highlighted in Figure 8.16a, slightly decrease to 7.41% of the total ignited area—marginally lower than the 8.0% recorded in Simulation 6.

Interestingly, despite the reduced firebrand ignition fraction, the overall fire perimeter remains consistent with Simulation 6. This suggests that the general spread pattern is still dominated by surface fire mechanisms. The maintained extent of firebrand ignition across key barriers (e.g., Highway CA-33) further supports this conclusion.

This test suggests that the physically downscaled wind field provides only minor improvements in fire spread prediction compared to the simpler RTMA gust wind product. While Wind-



(a) (b)

Figure 8.16: Firebrand ignition statistics under WindNinja-driven gust winds. Compared to Simulation 6, firebrand-induced ignitions slightly decrease in proportion, but their spatial distribution remains similar—particularly near fire-prone infrastructure. (a) Firebrand-ignited cells (red) at 33 hr; (b) Accumulated firebrand load (pcs·m^{-2}) at 33 hr

Ninja may offer finer representation of terrain-induced wind variability, the primary wind-driven mechanisms for both surface fire spread and spotting appear similarly activated by the gust wind field. Therefore, from a simulation efficiency standpoint, the RTMA gust wind may offer an adequate alternative for capturing key fire dynamics in ELMFIRE simulations.

8.9 Simulation 8: Impact of Inputs — Improved Representation of Urban Area

We now examine the influence of improved urban fuel classification on fire spread predictions. In the baseline LANDFIRE dataset, fuel category #91 (”urban”) aggregates both combustible structures and inert surfaces (e.g., roads, parking lots), which may lead to overestimated fire propagation through WUI communities. To address this, we refine the fuel map using Microsoft building footprint data, as described in Section 7.2.5. The improved map distinguishes actual structural footprints from non-combustible surfaces, enabling more physically realistic simulations of fire spread through urban landscapes.

All other components of the simulation are held consistent with Simulation 5, including surface fire models, firebrand generation and transport algorithms, and ignition submodels. To maintain comparability with prior simulations using enhanced wind inputs, we apply the WindNinja upscaled gust wind field. Spatial resolution remains fixed at 30 m.

Figure 8.17 shows the progression of the simulated fire compared against VIIRS observations. Compared to Simulations 6 and 7, the refined urban classification significantly reduces the overprediction of fire spread—particularly on the southern flank, where much of the formerly burnable area is now correctly identified as non-combustible. As a result, fire propagation becomes better constrained in urban zones while maintaining consistency with observed behavior in wildland areas.

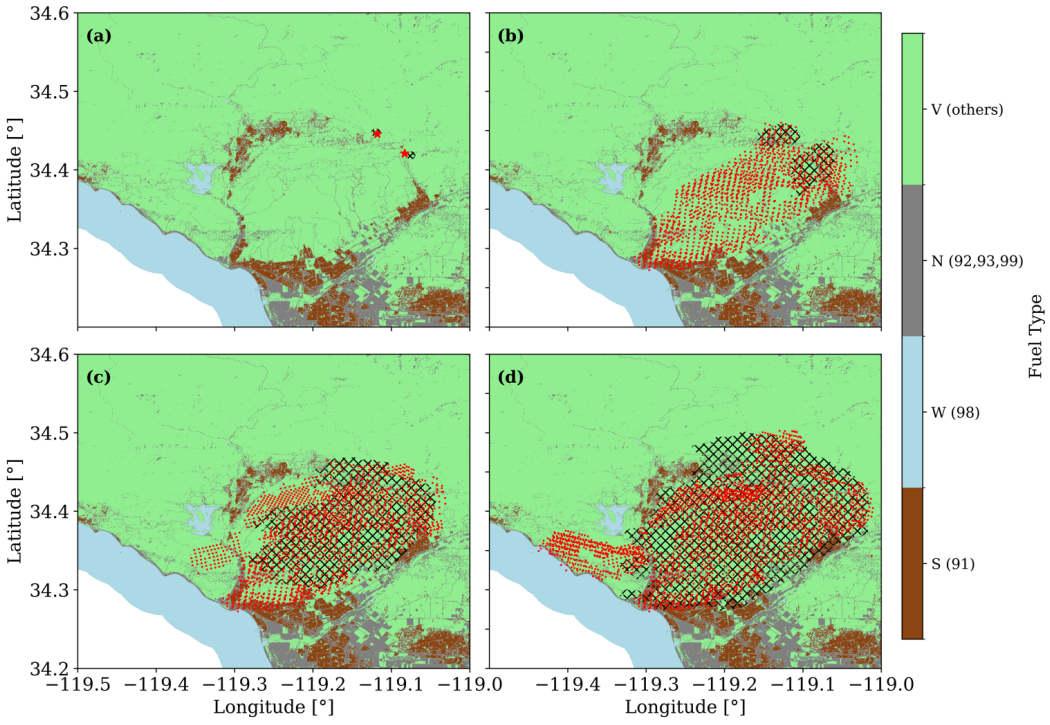
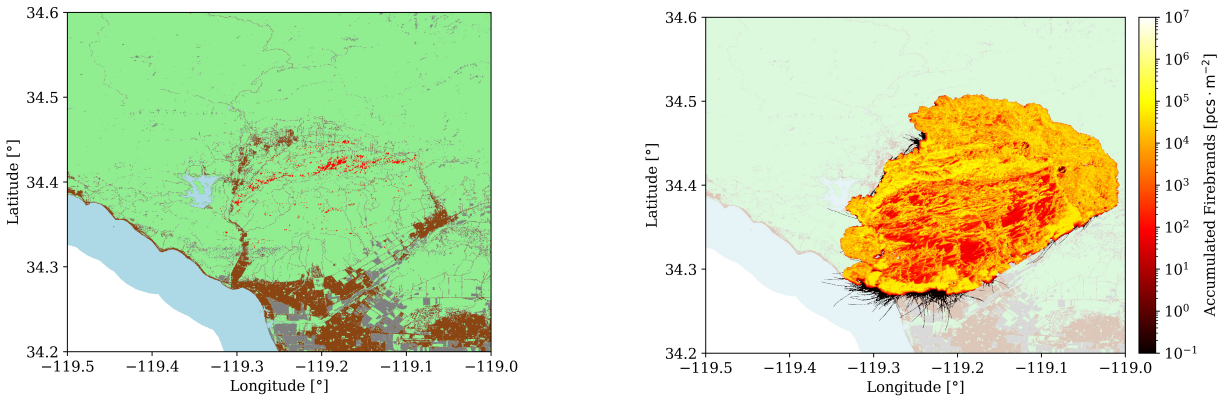


Figure 8.17: Simulation 8 results using improved urban representation derived from Microsoft building footprints(the same extent to Figure 8.1). Compared to prior simulations, the spread into the southern flank is substantially reduced due to refined classification of non-combustible surfaces.

Figure 8.18 presents the firebrand-related diagnostics from Simulation 8. The accumulated firebrand number load shown in Figure 8.18b remains qualitatively similar to that in Simulation 7, consistent with the fact that the same wind field and firebrand generation model are used. However, the proportion of area ignited by firebrands—highlighted in red in Figure 8.18a—increased slightly to 7.65% of the total ignited area, up from 7.41% in Simulation 7.

This increase suggests that firebrands play a more prominent role when fire barriers (e.g., non-combustible zones) are more accurately represented. With inert areas more clearly defined in the fuel map, firebrands become critical in bridging otherwise disconnected combustible patches. Thus, the combined dynamics of firebrand spotting and surface fire spread more accurately approximate the observed behavior of the Thomas Fire—particularly in WUI regions.



(a) (b)

Figure 8.18: Firebrand ignition diagnostics for Simulation 8 (the same extent to Figure 8.1). Refined fuel map leads to increased role of firebrand ignitions, particularly near barriers in the WUI. (a) Firebrand-ignited area at 33 hr (red pixels); (b) Accumulated firebrand number load at 33 hr (pcs·m^{-2}).

8.10 Simulation 9: Impact of Inputs — Highly Resolved Urban Representation (10 m Grid)

Building upon Simulation 8, this simulation investigates the effect of increasing spatial resolution in both fuel representation and input data. The urban fuel map is first resampled to 10-meter resolution and then modified using the Microsoft building footprint dataset, as described in Section 7.2.5. All other spatial inputs—such as topography and vegetation fuels—are similarly upscaled: integer-valued data (e.g., land cover categories) are resampled using the nearest-neighbor method, while continuous variables (e.g., elevation, moisture) use bilinear interpolation.

Fire spread and firebrand models remain unchanged from Simulation 8, and the wind field is interpolated to the 10-meter grid internally by ELMFIRE during simulation.

Figure 8.19 presents the progression of the burned area, analogous to previous simulations. Unlike earlier cases, however, this simulation was terminated at 24 hours due to excessive memory usage associated with the WU-E model at high resolution. As shown in panels (b) and (c), the burned area is significantly underpredicted compared to both the observations and the previous simulation.

This discrepancy may arise from several factors. First, the finer resolution introduces narrower firebreaks and more fragmented urban grids, reducing fuel continuity and slowing fire spread. Second, as analyzed in Chapter 6, discretized representations of urban fuels result in lower effective heat release rates per structure and internal delays associated with intra-structure fire spread. These effects collectively slowed the propagation rate. This simulation underscores the need for improved representation of Zone-0 fuels and motivates future work toward grid-

independence studies and structure-level treatments within the WU-E framework.

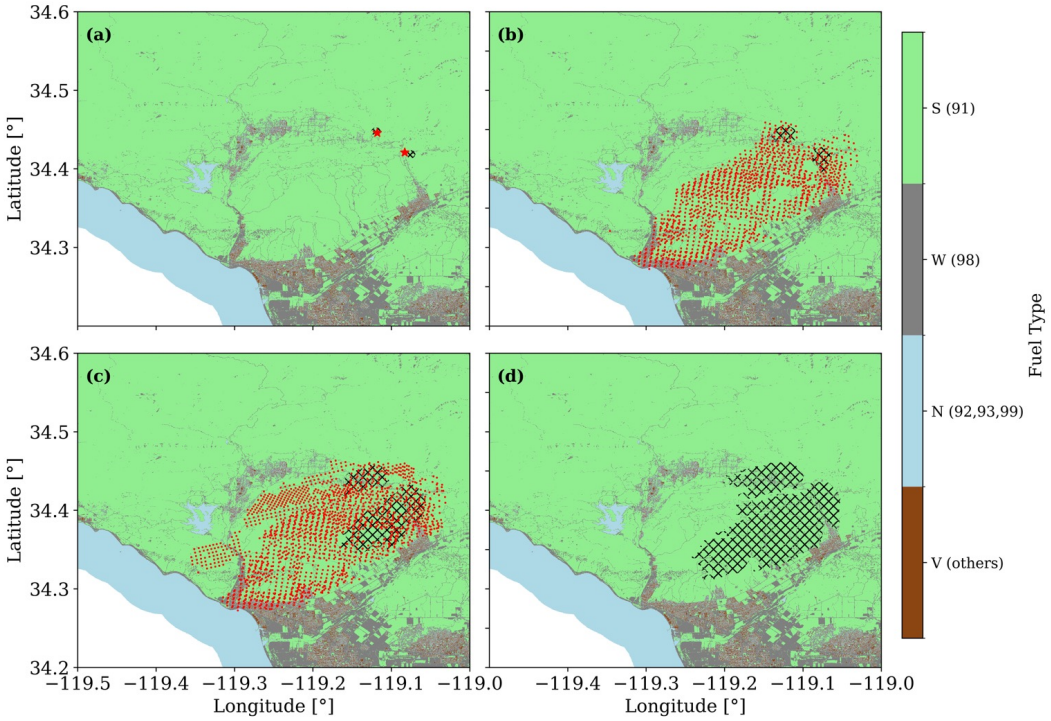
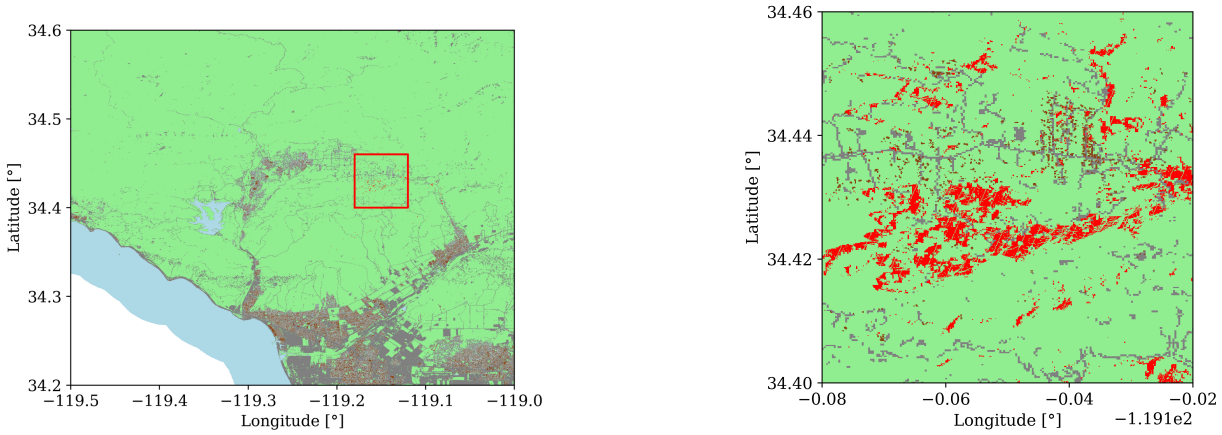


Figure 8.19: Simulation 9 results using 10-meter resolution with modified urban representation(the same extent to Figure 8.1). The simulation halted at 24 hours due to memory overload. Panels show fire progression at selected time snapshots. Compared to Simulation 8, the fire spread is significantly reduced, likely due to finer-scale firebreaks and reduced continuity in urban fuels.

To further characterize firebrand dynamics at high resolution, Figure 8.20 presents ignition results at 24 hours. Panel (a) shows firebrand-ignited areas across the domain, while panel (b) zooms into a region of high ignition density (approximately 5.5 km $\times$ 6.8 km), as indicated by the red bounding box in panel (a). Although the overall burned area remains underpredicted relative to observations, the firebrand ignition map reveals that firebrands serve as a dominant ignition mechanism in fragmented landscapes, enabling fire to bridge across fuel breaks and discontinuous urban zones.

This result highlights the model's capability to qualitatively capture realistic firebrand-induced spread mechanisms in WUI environments. Nonetheless, it also emphasizes the need

for better physical parameterization and higher-fidelity representations of small-scale urban fuel heterogeneity.



(a) (b)

Figure 8.20: Firebrand ignition results from Simulation 9 at 24 hr. Panel (a) shows the full-domain ignition pattern (the same extent to Figure 8.1). Panel (b) zooms in on a region (5.5 km $\times$ 6.8 km) where firebrands facilitate spread across fragmented fuels.

8.11 Discussion

The results of the above simulations are summarized in Figure 8.21, which presents the temporal evolution of burned area, calculated based on the total area where $\phi \leq 0$ at each time step.

With the exception of Simulations 1 and 9, most cases follow a similar three-phase progression: (1) a slow growth phase during the first 10 hours, (2) a rapid acceleration between 10–20 hours, and (3) a deceleration after 20 hours. These phases correspond to distinct stages of fire propagation as interpreted from the spatial patterns discussed earlier. During hours 0–10, the fire spreads from the ignition region across California State Route 150 and the adjacent WUI. From 10–20 hours, the fire advances into wildland regions dominated by fast-spreading fuels, resulting

in rapid area growth. After 20 hours, the fire encounters urban communities and crosses CA-33, where fuel fragmentation and lower spread rates slow the overall expansion.

This pattern indicates that surface fire spread remains the dominant mechanism driving the fire's growth. Simulations 3 and 5 exhibit nearly identical growth trends, consistent with their identical surface fire models and the relatively minor contribution of firebrands, which in Simulation 5 are shown to primarily influence low-ROS timber fuels.

Simulation 8 achieves a similar growth trajectory to Simulations 3 and 5, but for a different reason: the improved urban fuel classification (via Microsoft building footprints) corrects overestimated burnable area in the WUI. Despite using identical model physics, Simulations 6 and 7 display faster growth due to enhanced wind forcing—gust winds and downscaled WindNinja winds, respectively. These wind fields accelerate both surface spread and firebrand transport.

Interestingly, the Hamada-based urban spread model (Simulation 2) predicts a growth curve comparable to Simulations 3, 5, and 8 during the first 20 hours. However, it later catches up to the wind-enhanced cases (Simulations 6 and 7), suggesting potential overestimation of urban spread in the Hamada model under coarse fuel resolution.

Simulation 4 also shows rapid fire spread, despite not using strong winds. This can be attributed to the Lagrangian firebrand model, which lacks physical ignition thresholds and delay times. Its unfiltered, deterministic ignition mechanism enables more aggressive spread—especially in wildland areas—and leads to final burned areas similar to the wind-driven simulations.

Simulation 9, which uses high-resolution urban representation (10 m), substantially underpredicts fire growth due to reduced fuel continuity and internal delays in urban fire spread. Meanwhile, Simulation 1 underestimates fire growth because it disables both urban fire spread and firebrands entirely.

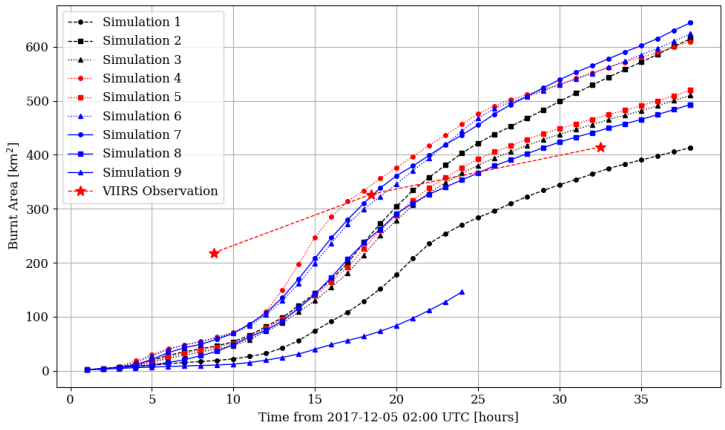


Figure 8.21: Summary of burned area over time for all simulation cases. The area is computed as the total number of grid cells with $\phi \leq 0$, scaled by grid cell area. Simulations exhibit three distinct growth phases: slow ignition and crossing of CA-150 (0–10 hr), rapid spread across wildlands (10–20 hr), and deceleration as fire reaches CA-33 and urban zones (20–33 hr).

The error of each simulation case is quantitatively evaluated using the Cohen's Kappa index [171], which measures the agreement between the simulated burn map and the reference (observed) fire perimeter, accounting for chance agreement. The metric evaluates the number of grid cells where both simulation and observation agree on either burned or unburned status.

Cohen's Kappa is defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{8.3}$$

where p_o is the observed agreement (i.e., the proportion of correctly predicted burned and unburned cells), and p_e is the expected agreement by chance, calculated as:

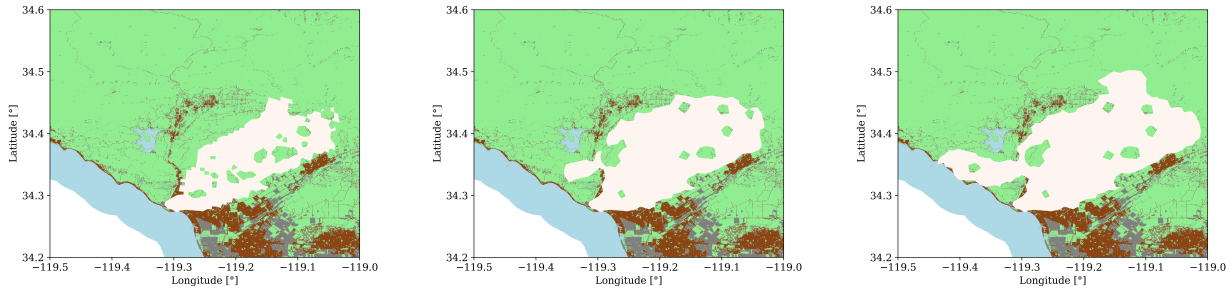
$$p_e = \frac{1}{N^2} \sum_{k=1}^{2} n_{k1} n_{k2}, \tag{8.4}$$

where N is the total number of simulated cells. The simulation results are classified into two categories: $k = 1$ for predicted burned cells and $k = 2$ for predicted unburned cells. n_{k1} represents the number of cells in category k that are actually burned, while n_{k2} represents the number of

cells in category k that are actually unburned.

To extract the reference burn perimeter from the VIIRS hotspot data, an alpha-shape concave hull algorithm is applied to the pointwise thermal anomaly detections. This method is implemented using the functionality provided by the `GeoPandas` Python package [172]. An alpha value of 0.03 (within a typical range of 0 to 1, where larger values result in fewer points being retained as polygon vertices) is selected empirically to best capture the observed burn extent at each VIIRS acquisition time.

The resulting alpha-shape concave hulls for each VIIRS snapshot are presented in Figure 8.22. These polygonal boundaries are then rasterized and compared to the simulation output at the corresponding times. The Cohen's Kappa index is computed for all time-matched simulation outputs, and the results are summarized in Figure 8.23. The approximate area enclosed by each concave hull—representing the observed cumulative burned area at each time step—is also computed and shown in Figure 8.21 using red dashed lines marked with stars.



(a) 2017-12-05 UTC 10:47 (b) 2017-12-05 UTC 20:26 (c) 2017-12-06 UTC 10:28

Figure 8.22: Concave hulls extracted from VIIRS hotspot data using an alpha-shape algorithm with $\alpha = 0.03$. Each panel corresponds to a distinct VIIRS acquisition time. These hulls are used as reference burned area for Cohen's Kappa calculation.

It is observed that all simulations, except for Simulation 9, achieve Cohen's Kappa values in the range of 0.7 to 0.8, indicating substantial agreement with the observed fire perimeters.

Among them, Simulations 4, 6, and 7 consistently demonstrate the highest performance across the evaluated time frames. However, it should be noted that when interpreting the result from Simulation 4, it may achieve high accuracy due to non-physical assumptions (e.g., instantaneous and deterministic firebrand ignition without delay).

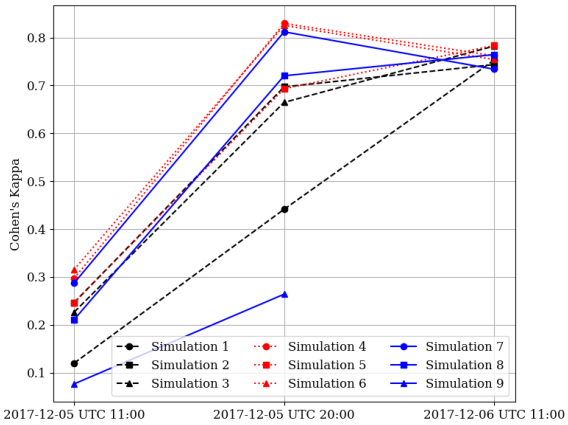


Figure 8.23: Cohen's Kappa index for all simulations at matching VIIRS observation times. Higher Kappa values indicate better agreement between simulated and observed fire perimeters, adjusted for chance agreement.

Simulations 6 and 7, which differ only in the source of the wind field (RTMA gust vs. WindNinja gust), exhibit nearly identical trends and outcomes, suggesting marginal effect of applying physics-based wind upscaling. Given current limitations in high-resolution fuel mapping and urban structure representation, the configuration used in Simulations 6 and 7 is recommended as the baseline for operational or future large-scale simulations. This includes the use of gust wind input, raw LANDFIRE fuels, WU-E model for WUI fire spread, and the new firebrand ignition framework developed in this study.

It is worth noting that while Simulations 3, 5, and 8 better predict the total burned area at at the final comparison time step, as shown in Figure 8.21, the Cohen's Kappa score is not significantly improved. This suggests that total burned area alone is insufficient as a metric for

evaluating simulation performance, as a model may match the correct total area but misrepresent the actual spatial extent of the burned region.

It is also noted that the underprediction of fire spread during the initial hours persists across all simulations. While this may partly stem from limitations in the implemented fire spread and firebrand models, another contributing factor could be the representation of the initial fire area. In the current setup, the initial level-set field (ϕ) is defined using a simplified approximation—two circular ignition zones with fixed radii centered at reported ignition points. However, in reality, the fire perimeter at ignition may be more irregular and influenced by local wind, fuel, and topography.

As a result, inaccuracies in the initial fire shape and extent may delay early fire growth in the simulation, especially if the ignition area is smaller or less representative than in reality. Future improvements could be achieved by initializing the simulation using more accurate fire perimeters derived from early satellite observations or ground-based incident reports. This would allow for better alignment between the simulated and observed fire progression from the onset of the event.

8.12 Conclusions

This chapter evaluates the performance of ELMFIRE in reconstructing the 2017 Thomas Fire through a series of systematic simulation studies. The fire spread models developed in earlier chapters were progressively incorporated to evaluate their impact, and simulation inputs were carefully prepared using the pipeline introduced in Chapter 7.

The results indicate that simulations relying solely on wildland fire spread models signifi-

cantly underestimate overall fire propagation. Incorporating fire spread mechanisms in urban regions—particularly the WUI—is essential for capturing realistic fire dynamics. Firebrand-driven ignition proves critical in enabling spread across inert barriers. However, the original Lagrangian firebrand model in ELMFIRE tends to overpredict burned areas due to its use of empirical parameterizations, lack of ignition delay, and uniform ignition probability.

The physics-based Eulerian firebrand model introduced in this study addresses many of these limitations by providing a spatially resolved firebrand number load field. This facilitates more physically grounded ignition predictions and enables the identification of firebrand-driven ignition in low-ROS fuel types. In urban areas, however, the influence of firebrands is diminished due to (1) recent modifications to the WU-E model that ensure robust structure-to-structure spread, and (2) underrepresentation of near-structure fuels (Zone-0) and firebreaks such as roads.

Simulations also reveal that RTMA gust wind fields provide better agreement with observed fire behavior than the default RTMA wind speed fields, particularly in the early stages of fire spread. Conversely, downscaled wind fields from WindNinja exhibit only marginal improvement. The treatment of urban fuels plays a central role in WUI fire spread prediction: the raw LANDFIRE product aggregates structures and inert surfaces, while high-resolution modifications improve realism but introduce new challenges, such as memory usage and over-discretization. As discussed in Chapter 6, a more robust algorithm that treats structures as integrated entities—and better represents Zone-0 fuels—is necessary for future enhancements.

In conclusion, the current ELMFIRE framework demonstrates reasonable fidelity in reconstructing the Thomas Fire. However, further improvements in firebrand modeling, wind field representation, and urban fuel treatment are required to advance predictive capabilities in WUI fire simulations.

Chapter 9: Conclusions and Future Work

9.1 Conclusions

This dissertation addresses the development of a physics-based firebrand-aided fire spread model for landscape-scale fire risk simulations. Existing landscape-scale fire spread models were first reviewed, with a particular focus on their underlying algorithms and physical submodels. While many of these models simulate fire spread primarily via traditional heat transfer mechanisms—such as radiation and direct flame contact—the influence of firebrands is often represented in a highly simplified manner. As a result, current firebrand treatments are frequently controlled by numerical parameters rather than physical parameters, limiting their predictive value.

To overcome these limitations, this work concentrates on extending the capabilities of ELMFIRE, a landscape-scale fire spread model originally developed by Chris Lautenberger. ELMFIRE employs a level-set method to track fire front propagation and supports rapid simulations using observed weather and geographical data, making it a suitable platform for real-time risk analysis and decision support.

A comprehensive review of the model's capabilities is provided in Chapter 2. The ELMFIRE framework includes the Rothermel model—modified by Albini—for surface fire spread in biomass fuels, along with the elliptical fire perimeter assumption introduced by Richards [60].

These models collectively define the fire spread behavior in wildland environments.

To simulate fire spread in urban settings, an empirical model developed by Hamada was first integrated into the framework. Although effective for preliminary studies, this model lacks a physical basis. Therefore, a physics-based alternative—the WU-E model—was later incorporated. The WU-E model developed by collaborators at the University of California at Berkeley captures the accumulation of radiative and convective heat fluxes in the wildland–urban interface (WUI) and predicts structure-to-structure ignition based on physical parameters. Its integration into the level-set framework allows for more realistic fire spread predictions in built environments.

Numerical integration techniques for the level-set method were also discussed, particularly in the context of coupling firebrand effects into the surface fire spread model. To support model development and verification, a one-dimensional MATLAB-based surrogate model was developed. This model simulates surface fire spread over flat terrain with heterogeneous fuels and time-varying wind conditions, and was used throughout the thesis to test and verify the firebrand generation, transport, and ignition modules developed in subsequent chapters. The numerical behavior of these newly introduced submodels was first evaluated using the MATLAB framework and then integrated into the full ELMFIRE codebase.

Building on this foundation, Chapters 3 through 5 systematically reviewed, developed, and verified models corresponding to the three phases of the firebrand lifecycle: generation, transport, and ignition. These phases collectively govern the behavior of firebrands in landscape-scale fire simulations and are critical to capturing the dynamics of short-range spotting, where firebrands may land in a flaming or smoldering state and pose a high risk of secondary ignitions.

In Chapter 3, a numerical framework was developed to characterize firebrand generation.

The generation process was defined through a particle number emission rate correlated with the local fuel heat release rate, leading to the introduction of a normalized firebrand generation rate per unit of heat release. This formulation is grounded in experimental observations that consistently report a strong correlation between fire intensity and firebrand production. Based on this premise, a numerical scheme was proposed to ensure that firebrand generation is mathematically consistent and independent of simulation time step—a key requirement for achieving numerical convergence. This resolution-independent treatment was applied to all components of the firebrand model throughout the thesis.

Due to the scarcity of direct experimental measurements, first-order estimations of the normalized generation rate were derived from existing empirical data. Characteristic values were defined separately for vegetation and structural fuels. Additionally, the emission duration—i.e., the effective time over which firebrands are emitted from a burning cell—was parameterized for each fuel type.

For vegetation fuels, the normalized generation rate was inferred from bench-scale experiments under the assumption of short flame residence times, typical of wildland surface fires. For structural fuels, empirical correlations from literature were used to estimate the normalized generation rate, with emission duration modeled according to a prescribed design fire curve to reflect the extended burning duration of structures.

Following the development of the firebrand generation model, Chapter 4 introduced and verified the firebrand transport model. This chapter focused on evaluating the numerical performance of transport schemes used in landscape-scale fire spread simulations. A series of idealized one-dimensional verification tests was designed under the assumption that all deposited firebrands immediately ignite spot fires (i.e., ignition probability equals one), thereby eliminating

uncertainties related to ignition and enabling direct comparison with exact reference solutions.

While most existing fire spread simulators adopt a discrete particle (Lagrangian) formulation—in which the trajectory of each firebrand is explicitly tracked—this study developed and implemented an Eulerian alternative. The Eulerian model represents the collective behavior of firebrands as a continuous concentration field, depositing firebrands proportionally based on source intensity, wind conditions, and distance-dependent probability distributions.

Compared to the Lagrangian approach, the Eulerian formulation demonstrated superior numerical performance, offering higher accuracy and significantly reduced computational cost for equivalent configurations. These advantages make the Eulerian model particularly well-suited for large-scale or high-resolution simulations, where particle-based methods often impose prohibitive memory and runtime requirements—especially when attempting to accurately resolve firebrand deposition patterns.

To ensure numerical accuracy, a wind-based Courant–Friedrichs–Lewy (CFL) condition is recommended: $CFL_{t_n} \leq 0.8$. For the Lagrangian formulation to achieve comparable accuracy under this condition, the particle generation rate must be sufficiently high: $GR_p \geq 10 \text{ pcs} \cdot \Delta t_n^{-1}$ for short emission durations ($t_{\text{Emission}} = \Delta t_n$), or $GR_p \geq 1 \text{ pcs} \cdot \Delta t_n^{-1}$ for longer emission durations ($t_{\text{Emission}} \gg \Delta t_n$).

Chapter 5 then focused on the development and verification of a numerically convergent ignition model suitable for landscape-scale fire spread simulations. The model assumes two consecutive ignition delays:

1. A delay associated with the occurrence of a small flame after firebrand deposition, governed by the smoldering-to-flaming transition (SFT). This process is modeled using an

empirically derived ignition probability that depends on both the accumulated firebrand mass per unit ground area and the local wind velocity. The SFT formulation is based on De Beer's work, supported by detailed pyrolysis simulations performed in ThermaKin.

2. A subsequent delay associated with the development of a fully self-sustaining flame, modeled using a t-square fire growth curve. This accounts for the time required for small flames to evolve into spreading fires.

Verification cases demonstrate that the ignition model is numerically convergent—i.e., its predictions are not sensitive to variations in different grid resolutions and time steps—provided that the CFL condition remains satisfied, as established in Chapter 4.

With the firebrand submodels established, Chapter 6 presented a series of numerical experiments using an academic one-dimensional WUI configuration to evaluate the behavior and limitations of firebrand-driven spread in urban environments. These MATLAB-based simulations revealed several challenges when directly applying the developed firebrand generation, transport, and ignition models to burning structures.

In particular, it was observed that discretized representations of structures—where heat release rate is defined per unit area—can lead to distorted firebrand trajectories and inconsistent ignition outcomes when simulations are performed at different spatial resolutions. At coarse resolutions, misrepresentation of structure footprints and firebreaks (e.g., roads or non-combustible zones) further degrades the numerical accuracy of firebrand behavior and the reliability of fire spread predictions.

To mitigate these issues, a resolution-independent treatment for structural fuels was proposed. In this framework, each structure is treated as a singular unit in terms of both heat release

and firebrand interaction (generation and collection). This approach was shown to be numerically robust and accurate across a range of spatial and temporal discretizations. Nonetheless, additional refinement—most notably the incorporation of adaptive mesh refinement (AMR)—is recommended to more accurately capture community-scale fuel layouts and sub-grid features such as defensible zones (e.g., Zone 0), narrow firebreaks, and heterogeneous urban fuel configurations.

Despite these limitations, a parametric study was conducted to assess the model's predictive behavior under varying firebrand generation rates, inter-structure separation distances, and wind conditions. The results indicated a high likelihood of firebrand-driven propagation under moderate to high wind speeds (30–120 mph) and short structural separation distances. These trends are consistent with physical expectations based on firebrand transport dynamics and ignition thresholds.

To enable realistic testing of the developed models within the ELMFIRE framework, a complete preprocessing pipeline for generating ELMFIRE-compatible inputs was developed and detailed in Chapter 7. This pipeline begins with acquiring raw geospatial and meteorological datasets from publicly available sources, specifically the Real-Time Mesoscale Analysis (RTMA) and LANDFIRE archives.

Meteorological variables from RTMA—including wind speed, wind direction, temperature, and humidity—are first used to calculate dead and live fuel moisture using the NFDRS4.0 model. Concurrently, slope and fuel type information are extracted from LANDFIRE and reclassified into the input categories required by NFDRS, which are also required for fuel moisture calculations.

While the native RTMA wind field can be directly adopted for ELMFIRE simulations,

higher-resolution wind fields are often desirable for improved spatial accuracy. For this purpose, the pipeline incorporates the WindNinja solver to upscale RTMA wind inputs. WindNinja uses LANDFIRE-derived elevation data and RTMA wind fields to produce terrain-informed wind outputs at a user-defined resolution. When WindNinja-generated wind fields are used, fuel moisture fields must be resampled to match the wind resolution—typically using bilinear interpolation for continuous variables and nearest-neighbor interpolation for categorical variables—to ensure spatial consistency across inputs.

An important optional enhancement—originally developed by Theodori [143]—involves the integration of high-resolution Microsoft Building Footprint data. In this workflow, the FBFM40 or FBFM13 raster is first resampled to the desired grid resolution, and Microsoft's vector-based building outlines are rasterized and overlaid to improve the classification of urban fuels. This step enhances the fidelity of fuel map inputs by enabling more accurate distinctions between combustible structures and inert surfaces such as roads or parking lots.

This full preprocessing pipeline, along with the developed firebrand and surface fire spread models, was applied in Chapter 8 to reconstruct the 2017 Thomas Fire. A series of ELMFIRE simulations were conducted to systematically evaluate the impact of model improvements and input configurations.

The results showed that simulations relying solely on wildland fire spread models significantly underestimated fire propagation. Incorporating fire spread mechanisms in urban regions—particularly within wildland–urban interface (WUI) communities—proved essential for capturing realistic fire behavior. Firebrand-induced ignition was shown to be critical for enabling fire to cross inert barriers. However, the original Lagrangian firebrand model in ELMFIRE tended to overpredict burned areas due to its use of empirical parameters, the absence of ignition delays,

and a uniform ignition probability. In contrast, the physics-based Eulerian firebrand model developed in this work addressed these limitations by generating a spatially resolved firebrand number load field. This enabled more accurate and physically interpretable modeling of firebrand-driven ignition processes.

Simulations further revealed that RTMA gust wind fields provided significantly better agreement with observed fire progression than 1-hour averaged winds, especially in the early stages of the fire. WindNinja-upscaled gust winds offered only marginal improvements. Finally, the representation of urban fuels was shown to be a central factor in fire spread prediction. While the raw LANDFIRE dataset conflates structures and inert surfaces, the use of Microsoft Building Footprints improved realism at the expense of greater computational cost and model discretization complexity.

In summary, this dissertation makes the following key contributions:

- Developed a numerically robust and physically interpretable firebrand modeling framework for use in landscape-scale fire simulators, comprising firebrand generation, transport, and ignition submodels.
- Introduced a physics-based Eulerian firebrand transport formulation that achieves higher accuracy and lower computational cost than conventional Lagrangian models.
- Proposed and verified a two-stage ignition model incorporating smoldering-to-flaming transition and t^2 flame growth, based on experimental and numerical data.
- Identified and addressed numerical inconsistencies associated with firebrand modeling in coarse-grid urban environments, and proposed resolution-independent treatments for structural fuels.

- Developed an autonomous preprocessing pipeline for ELMFIRE input generation using publicly available data (RTMA, LANDFIRE, NFDRS), with optional integration of WindNinja wind fields and Microsoft Building Footprints.
- Demonstrated the importance of urban fire spread models (WU-E), firebrand ignition, and improved urban representation in reconstructing historical fires, as shown in the Thomas Fire case study.
- Provided recommendations for input configurations—favoring gust wind over averaged wind, and Microsoft-based fuel maps over raw LANDFIRE products—based on simulation fidelity and computational feasibility.

9.2 Deficiencies and Future Work

While this dissertation emphasizes the development of a numerically robust firebrand modeling framework, several sources of physical uncertainty and modeling limitations remain. Addressing these issues will be critical to improving the physical accuracy, predictive power, and general applicability of the model. Future work should focus on the following areas:

Firebrand Modeling Deficiencies

1. **Uncertainty in Firebrand Generation Rates (GR^*):** The normalized firebrand generation rate requires further validation using controlled laboratory experiments or high-fidelity pyrolysis simulations. Current values are derived from limited literature sources and may not capture the variability across different fuels, burning conditions, or structural typologies. Establishing stronger and fuel-specific correlations between generation rate and local

heat release rate (HRR) remains a key research priority.

2. **Unvalidated PDF for Firebrand Travel Distance:** The current model uses the empirical formulations by Sardoy [83] and Himoto [82] to describe the probability distribution of firebrand travel distance for vegetative and structural fuels, respectively. However, these correlations have not been validated against real-world observations, primarily due to the lack of reliable field data. Future work should consider using CFD-based modeling approaches to evaluate the validity of these distributions under realistic fire conditions, particularly at spatial scales on the order of hundreds of meters.

3. **Oversimplified Firebrand Arrival Time:** The current transport model assumes deterministic arrival times based solely on wind trajectory. In reality, firebrand travel involves lofting, advection, and gravitational settling, each introducing delays. A more realistic treatment would prescribe a probability density function for arrival time that captures these dynamics.

4. **Uniform Particle Properties:** The model currently assumes all firebrands have the same mass and size. However, ignition probability depends on deposited mass rather than count alone. Future versions should incorporate a mass and size distribution of firebrands to more accurately estimate ground-level fuel loading and its ignition potential.

5. **Uniform Ignition Dynamics:** The ignition model presented in Chapter 5 is based on a smoldering-to-flaming transition (SFT) framework with empirically derived probability functions. Although grounded in De Beer's work and validated against ThermaKin simulations, the current implementation applies the same ignition behavior across all fuel types.

Future studies should investigate the sensitivity of simulation outcomes to SFT parameters and, if necessary, develop fuel-specific ignition models.

6. **Lack of Firebrand Consumption and Clustering Representation:** A firebrand consumption model was proposed but is not yet implemented in ELMFIRE due to limitations at coarse resolution. The assumption of uniform firebrand deposition within each grid cell may result in unrealistically high ignition thresholds. A subgrid-scale model that captures redistribution after deposition is needed to improve realism in operational-scale simulations.

Surface Fire Spread Model Limitations

1. **Edge Behavior of WU-E Model:** The current WU-E implementation lacks explicit treatment of Zone 0 fuels—vegetation or combustibles immediately adjacent to structures. This simplification limits model fidelity near wildland–urban boundaries and can either suppress or exaggerate spread into structures. Subgrid treatments or new perimeter algorithms are required to bridge this transition zone accurately.

2. **Treatment of Multidimensional Structural Fuels:** Due to grid discretization, WUI structures are often misrepresented in shape and connectivity. This misrepresentation limits the accuracy of predicted fire dynamics in built environments. Adaptive mesh refinement or structure-resolving algorithms are recommended to better capture per-building fire behavior and urban fire spread mechanisms.

Input Data and Resolution Constraints

1. **High-Resolution and Representative Fuel Maps:** The LANDFIRE fuel dataset used in this study aggregates diverse structural surfaces under a single fuel code (e.g., FBFM91), which includes both burnable structures and inert infrastructure. Future efforts should incorporate high-resolution fuel classification maps augmented with building and infrastructure data to better differentiate between combustible and non-combustible areas.

2. **Improved Meteorological Forcing:** The current use of RTMA data (2.5 km spatial and 1-hour temporal resolution) is insufficient to capture rapid fire-induced wind variability. Future studies should explore the use of numerical weather prediction models such as WRF, which can provide high-resolution wind and moisture fields in both space and time. For example, recent studies [173] have implemented WRF simulations with grid spacing as fine as 111 m. WRF does not impose a hard-coded limit on spatial resolution, which may allow flexible adaptation to fire-relevant scales. In addition, WRF produces mass- and momentum-consistent atmospheric fields, offering a more physically realistic representation of meteorological conditions during fire events.

Validation and Field-Scale Integration

While the proposed models demonstrate physically consistent behavior in both idealized test cases and the reconstruction of a historical fire (the 2017 Thomas Fire), experimental validation remains limited. Future validation efforts should benefit from:

- Well-instrumented field experiments that track firebrand generation, transport, deposition,

and ignition behavior;

- Controlled burns in WUI-like environments to study structure ignition dynamics and firebrand interactions;

- Integration of better-resolved satellite or aerial imagery to validate modeled fires.

In conclusion, this dissertation presents a foundational framework for physics-based firebrand modeling in landscape-scale fire simulations. Transitioning from numerically robust to physically validated models—particularly for firebrand dynamics and WUI fire spread—will require the integration of subgrid-scale representations, structure-resolving capabilities, and experimentally informed ignition and transport processes.

Before implementing new models or input datasets, it is essential to conduct comprehensive parametric studies to evaluate the model's sensitivity to key parameters and submodel assumptions. Such studies will help prioritize future development by identifying dominant sources of uncertainty and performance variability. Ultimately, targeted enhancements—including adaptive mesh refinement, stochastic ignition modeling, and dynamically coupled meteorological inputs—offer potential and promising pathways to improve both the physical fidelity and operational reliability of the current landscape scale wildland-urban interface fire risk models.

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